CHAPTER 33

Mensuration I: Area and Perimeter

INTRODUCTION

In this chapter, we shall be dealing with plane figures of various shapes by finding their sides, perimeters and areas.

Area:

The area of any figure is the amount of surface enclosed within its boundary lines. Area is always expressed in square units.

Units of Measuring Area

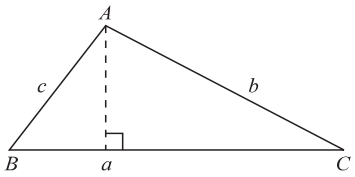
100 sq millimetres = 1 sq centimetre
100 sq centimetres = 1 sq decimetre
100 sq decimetres = 1 sq metre
100 sq metres = 1 sq decametre or arc
10,000 sq metres = 1 hectare
1,000,000 sq metres = 100 hectares
= 1 sq kilometre

Perimeter

The perimeter of a geometrical figure is the total length of the sides enclosing the figure.

SOME BASIC FACTS

1. Triangle



A *triangle* is a closed figure bounded by three sides. Here, ABC is a triangle.

The sides AB, BC and AC are denoted by c, a and b. respectivlely

Area of a Triangle (A)

(a) $A = \frac{1}{2}$ (base × height) $= \frac{1}{2}ah$

(b) $A = \sqrt{s(s-a)(s-b)(s-c)}$,

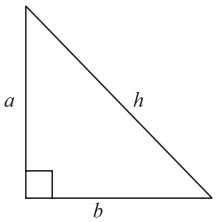
Where $s = \frac{1}{2}(a+b+c)$ or semi-perimeter of the triangle.

This formula is known as *Hero's formula*.

Perimeter (P) $= a + b + c = 2s$.

2. Right Angled Triangle

A triangle having one of its angles equal to 90° is called a *right-angled triangle*. The side opposite to the right angle is called the *hypotenuse*.



In a right angled triangle,

(Hypotenuse)^2 = Sum of the squares of sides

i.e., $h^2 = a^2 + b^2$.

Area (A) $= \frac{1}{2}$(product of the sides containing the right angle)

i.e., $A = \frac{1}{2}ab$.

Illustration 1: What is the area of a triangle having sides 3 m, 4 m and 5 m?

Solution: Let a = 3 m, b = 4 m, c = 5 m.

Then, $s = \frac{a+b+c}{2} = \frac{3+4+5}{2} = 6$ m.

$\therefore$ Area $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{6(6-3)(6-4)(6-5)}$

$= \sqrt{6\times3\times2\times1} = \sqrt{36} = 6$ m^2

Illustration 2: Find the area of a triangle whose base is 4.6 m and height is 67 cm.

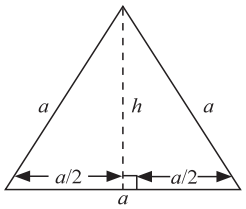
Solution: Area of the triangle $= \frac{1}{2}$(base × height)

$= \frac{1}{2}(4.6 \times 100 \times 67)$

$= 15410$ cm^2

3. Equilateral Triangle

A triangle whose all sides are equal is called an equilateral triangle.



Area (A) of an equilateral triangle

$$= \frac{\sqrt{3}}{4}(\text{side})^2 = \frac{\sqrt{3}}{4}a^2$$

Permeter (P) of an equilateral triangle

$$= 3 \times (\text{side}) = 3a$$

Altitude (h) of an equilateral triangle

$$= \frac{\sqrt{3}}{2} \times (\text{side}) = \frac{\sqrt{3}}{2}a.$$

In an equilateral triangle

$$\angle A = \angle B = \angle C = 60°.$$

Area (A) of an equilateral triangle

$$= \frac{(\text{altitude})^2}{\sqrt{3}} = \frac{h^2}{\sqrt{3}}.$$

Illustration 3: Find the area of an equilateral triangle each of whose sides measures 6 cm.

Solution: Area of the equilateral triangle

$$= \frac{\sqrt{3}}{4}(\text{side})^2 = \frac{\sqrt{3}}{4} \times 36 = 9\sqrt{3} \text{ cm}^2.$$

Illustration 4: Length of the side of an equilateral triangle is $\frac{4}{\sqrt{3}}$ cm. Find its height.

Solution: Height of the equilateral triangle

$$= \frac{\sqrt{3}}{2} \times (\text{side})$$

$$= \frac{\sqrt{3}}{2} \times \frac{4}{\sqrt{3}} = 2 \text{ cm.}$$

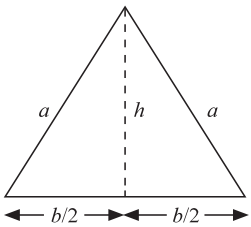
Illustration 5: Height of an equilateral triangle is 4 $\sqrt{3}$ cm. Find its area.

Solution: Area of the equilateral triangle

$$= \frac{(\text{altitude})^2}{\sqrt{3}} = \frac{4\sqrt{3}\times4\sqrt{3}}{\sqrt{3}} = 16\sqrt{3} \text{ cm}^2.$$

4. Isosceles Triangle

A triangle whose two sides are equal is called an *isosceles triangle*.



Area (A) of an isosceles triangle

$$= \frac{b}{4}\sqrt{4a^2 - b^2}$$

Perimeter (P) of an isosceles triangle

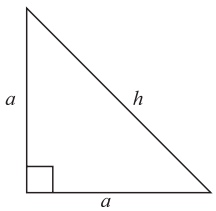
$$= (2a + b)$$

Height (h) of an isosceles triangle

$$= \frac{1}{2}\sqrt{4a^2 - b^2}.$$

5. Isosceles Right-angled Triangle

An *isosceles right-angled triangle* has two sides equal with equal sides making 90° to each other.



Hypotenuse $(h) = \sqrt{2a}$

Area $(A) = \frac{1}{2}a^2$

Perimeter $(P) = 2a + \sqrt{2a} = \sqrt{2}a(\sqrt{2}+1)$

$= h(\sqrt{2}+1)$.

If the perimeter of an isosceles triangle is P and the base is b, then the length of the equal sides is $\left(\frac{P-b}{2}\right)$.

If the perimeter of an isosceles triangle is P and the length of equal sides is a, then base is $(P - 2a)$.

Illustration 6: An isosceles right-angled triangle has two equal sides of length 6 m each. Find its area.

Solution: Area $= \frac{1}{2}$ (equal side)$^2 = \frac{1}{2}(6)^2 = 18$ m^2.

Illustration 7: The perimeter of an isosceles triangle is 80 cm. If the length of the equal sides is 15 cm, find the length of the base.

Solution: Length of the base $= P - 2a$

$= 80 - 2(15) = 50$ cm.

Illustration 8: The perimeter of an isosceles triangle is 42 cm. If the base is 16 cm, find the length of the equal sides.

Solution: The length of the equal sides $= \frac{P-b}{2}$.

$= \frac{42-16}{2} = \frac{26}{2} = 13$ cm.

Illustration 9: If the base of an isosceles triangle is 10 cm and the length of the equal sides is 13 cm, find its area.

Solution: Area of the isosceles triangle

$= \frac{b}{4}\sqrt{4a^2-b^2}$

$= \frac{10}{4}\sqrt{4\times(13)^2-(10)^2}$

$= \frac{10}{4}\sqrt{676-100} = \frac{10}{4}\times 24 = 60$ cm^2.

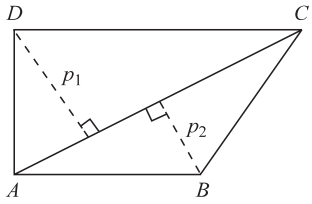
6. Quadrilateral

A closed figure bounded by four sides is called a *quadrilateral*.

It has four angles included in it.

The sum of these four angles is 360°.

i.e., $\angle A + \angle B + \angle C + \angle D = 360°$.



Area (A) of a quadrilateral

$= \frac{1}{2} \times$ one diagonal $\times$ (sum of the perpendiculars to it from opposite vertices)

$= \frac{1}{2}d(p_1 + p_2)$

Notes

If the lengths of four sides and one of its diagonals are known, then,

A = Area of ΔADC + Area of ΔABC.

The special cases of quadrilateral are parallelogram, rectangle, square, rhombus, trapezium, etc., which are discussed below separately.

7. Parallelogram

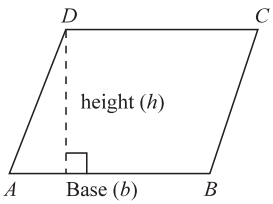
A quadrilateral in which opposite sides are equal and parallel is called a parallelogram.

The diagonals of a parallelogram bisect each other.

Area (A) of a parallelogram

= base $\times$ altitude corresponding to the base

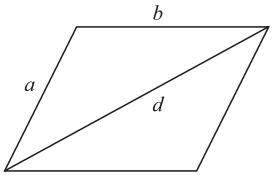
$= b \times h$



Area (A) of a parallelogram

$= 2\sqrt{s(s-a)(s-b)(s-d)}$

where a and b are adjacent sides, d is the length of the diagonal connecting the ends of the two sides and $s = \frac{a+b+d}{2}$.



In a parallelogram, the sum of the squares of the diagonals = 2 (the sum of the squares of the two adjacent sides),

i.e., $d_1^2 + d_2^2 = 2(a^2 + b^2)$.

Perimeter (P) of a parallelogram

$$= 2(a + b),$$

where a and b are adjacent sides of the parallelogram.

Illustration 10: One side of a parallelogram is 15 cm and the corresponding altitude is 5 cm. Find the area of the parallelogram.

Solution: Area of the parallelogram

$$= \text{base} \times \text{corresponding altitude}$$

$$= 15 \times 5 = 75 \text{ cm}^2.$$

Illustration 11: In a parallelogram, the lengths of the adjacent sides are 11 cm and 13 cm, respectively. If the length of one diagonal is 20 cm then, find the length of the other diagonal.

Solution: We have,

$$d_1^2 + d_2^2 = 2(a^2 + b^2)$$

$$\Rightarrow \quad (20)^2 + d_2^2 = 2(11^2 + 13^2)$$

$$\Rightarrow \quad d_2^2 = 2(121 + 169) - 400 = 180.$$

$$\therefore \quad d_2 = \sqrt{180} = 13.4 \text{ m (approx.)}$$

Illustration 12: Find the area of a quadrilateral of whose diagonal is 38 cm long and the lengths of perpendiculars from the other two vertices are 31 cm and 19 cm, respectively.

Solution: Area of the quadrilateral

$$= \frac{1}{2} d(p_1 + p_2)$$

$$= \frac{1}{2} \times 38 \times (31 + 19)$$

$$= 19 \times 50 = 950 \text{ cm}^2.$$

Illustration 13: Find the area of a parallelogram whose two adjacent sides are 130 m and 140 m and one of the diagonals is 150 m long.

Solution: Here, a = 130 m, b = 140 m and d = 150 m

$$\therefore \quad s = \frac{a+b+d}{2} = \frac{130+140+150}{2} = \frac{420}{2} = 210 \text{ m}$$

$\therefore$ Area of the parallelogram

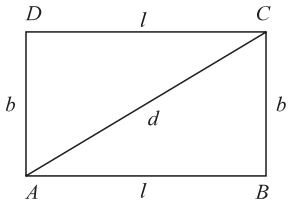
$$= 2\sqrt{s(s-a)(s-b)(s-a)}$$

$$= 2\sqrt{210(210-130)(210-140)(210-150)}$$

$$= 2\sqrt{210 \times 80 \times 70 \times 60} = 2 \times 8400 \text{ m}^2.$$

8. Rectangle

A *rectangle* is a quadrilateral with opposite sides equal and all the four angles equal to 90°.



The diagonals of a rectangle bisect each other and are equal.

(a) Area (A) of a rectangle = length × breadth

$$= l \times b$$

or,

Area of a rectangle = $(l \times \sqrt{d^2 - l^2})$,

if one side (l) and diagonal (d) are given.

or,

Area of a rectangle = $\left(\frac{p^2}{8} - \frac{d^2}{2}\right)$,

if perimeter (P) and diagonal (d) are given.

(b) Perimeter (P) of a rectangle

= 2(length + breadth)

= 2(l + b).

or,

Perimeter of a rectangle = $2(l + \sqrt{d^2 - l^2})$,

if one side (l) and diagonal (d) are given.

(c) Diagonal of a rectangle

$$= \sqrt{(\text{length})^2 + (\text{breadth})^2}$$

$$= \sqrt{l^2 + b^2}$$

(d) If area (A) and perimeter (P) of a rectangle are given, then,

length of a rectangle $= \left(\sqrt{\frac{P^2}{16} - A} + \frac{P}{4}\right)$

and,

breadth of a rectangle $= \left(\frac{P}{4} - \sqrt{\frac{P^2}{16} - A}\right)$

Illustration 14: Find the diagonal of a rectangle whose sides are 8 cm and 6 cm.

Solution: Diagonal of the rectangle

$= \sqrt{l^2 + b^2}$

$= \sqrt{8^2 + 6^2} = \sqrt{64 + 36}$

$= 10$ cm.

Illustration 15: Find the perimeter of a rectangle of length 12 m and breadth 6 m.

Solution: Perimeter of the rectangle

$= 2(l + b)$

$= 2(12 + 6) = 36$ m.

Illustration 16: Calculate the area of a rectangular field whose length is 12.5 cm and breadth is 8 cm.

Solution: Area of the rectangular field

$= l \times b$

$= 12.5 \times 8 = 100$ cm^2.

Illustration 17: Calculate the area of a rectangular field whose one side is 16 cm and the diagonal is 20 cm.

Solution: Area of the rectangular field

$= (l \times \sqrt{d^2 - l^2})$

$= (16 \times \sqrt{20^2 - 16^2})$

$= 16 \times 12 = 192$ cm^2.

Illustration 18: A rectangular carpet has an area of 120 m^2 and perimeter of 46 m. Find the length of its diagonal.

Solution: We have,

Area of rectangle $= \left(\frac{P^2}{8} - \frac{d^2}{2}\right)$

$\Rightarrow \quad 120 = \frac{46^2}{8} - \frac{d^2}{2}$

$\Rightarrow \quad 46^2 - 4d^2 = 120 \times 8$

$\Rightarrow \quad 4d^2 = 2116 - 960 = 1156$

$\therefore \quad d = \sqrt{289} = 17$ m.

Illustration 19: The perimeter of a rectangle is 82 cm and its area is 400 m^2. Find the length and breadth of the rectangle.

Solution: Length of the rectangle

$= \left(\sqrt{\frac{P^2}{16} - A} + \frac{P}{4}\right)$

$= \left(\sqrt{\frac{(82)^2}{16} - 400} + \frac{82}{4}\right)$

$= (4.5 + 20.5) = 25$ m.

Breadth of the rectangle

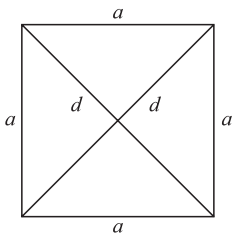
$= \left(\frac{P}{4} - \sqrt{\frac{P^2}{16} - A}\right)$

$= \left(\frac{82}{4} - \sqrt{\frac{(82)^2}{16} - 400}\right)$

$= (20.5 - 4.5) = 16$ m.

9. Square

A square is a quadrilateral with all sides equal and all the four angles equal to 90°.



The diagonals of a square are equal and bisect each other at 90°.

(a) Area (A) of a square

$= a^2$, i.e., (side)^2

$= \frac{d^2}{2}$, i.e., $\frac{(\text{diagonal})^2}{2}$

$= \frac{P^2}{16}$, i.e., $\frac{(\text{perimeter})^2}{16}$

(b) Perimeter (P) of a square

$= 4a$, i.e., $4 \times$ side

$= \sqrt{16 \times \text{area}}$

$= 2\sqrt{2}d$, i.e., $2\sqrt{2} \times$ diagonal

(c) Length (d) of the diagonal of a square

$= \sqrt{2a}$, i.e., $\sqrt{2} \times$ side

$= \sqrt{2 \times \text{area}}$

$= \dfrac{P}{2\sqrt{2}}$, i.e., $\dfrac{\text{Perimeter}}{2\sqrt{2}}$.

Illustration 20: If the area of a square field is 6050 m^2, find the length of its diagonal.

Solution: Length of the diagonal of the square field

$= \sqrt{2 \times \text{area}}$

$= \sqrt{2 \times 6050}$

$= \sqrt{12100}$, i.e., 110 m.

Illustration 21: Find the area of a square with perimeter 48 m.

Solution: Area of the square

$= \dfrac{(\text{Perimeter})^2}{16}$

$= \dfrac{(48)^2}{16} = \dfrac{48 \times 48}{16} = 3 \times 48 = 144 \text{ m}^2$.

Illustration 22: Find the diagonal of a square field whose side is 6 m.

Solution: Length of the diagonal

$= \sqrt{2} \times$ side

$= 6\sqrt{2}$ m.

Illustration 23: In order to fence a square, Ramesh fixed 36 poles. If the distance between the two poles is 6 m, then find the area of the square so formed.

Solution: Perimeter of the square

$= 36 \times 6 = 216$ m.

$\therefore$ Area of the square $= \left(\dfrac{\text{Perimeter}}{4}\right)^2$

$= \dfrac{16\sqrt{2}}{2\sqrt{2}} = 54 \times 54$

$= 2916 \text{ m}^2$.

Illustration 24: Perimeter of a square field is $16\sqrt{2}$ cm. Find the length of its diagonal.

Solution: We have,

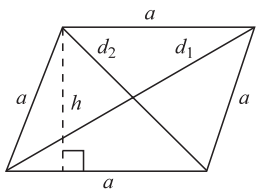
Perimeter of the square field $= 2\sqrt{2} \times$ diagonal

$\Rightarrow \quad 16\sqrt{2} = 2\sqrt{2} \times$ diagonal

$\therefore$ Length of the diagonal $= \dfrac{16\sqrt{2}}{2\sqrt{2}} = 8$ cm.

10. Rhombus

A *rhombus* is a quadrilateral whose all sides are equal.



The diagonals of a rhombus bisect each other at 90°.

(a) Area (A) of a rhombus

$= a \times h$, i.e., base $\times$ height

$= \dfrac{1}{2}d_1 \times d_2$, i.e., $\dfrac{1}{2} \times$ product of its diagonals

$= d_1 \times \sqrt{a^2 - \left(\dfrac{d_1}{2}\right)^2}$,

since, $d_2^2 = 4\left[a^2 - \left(\dfrac{d_1}{2}\right)^2\right]$

$= d_1 \times \sqrt{\left(\dfrac{\text{Perimeter}}{4}\right)^2 - \left(\dfrac{d_1}{2}\right)^2}$,

since, $d_2^2 = 4\left[\left(\dfrac{\text{Perimeter}}{4}\right)^2 - \left(\dfrac{d_1}{2}\right)^2\right]$

(b) Perimeter (P) of a rhombus

$= 4a$, i.e., $4 \times$ side

$= 2\sqrt{d_1^2 + d_2^2}$,

where d_1 and d_2 are two diagonals.

(c) Side (a) of a rhombus

$= \dfrac{1}{2}\sqrt{d_1^2 + d_2^2}$.

Illustration 25: The area of a rhombus is 156 m^2. If one of its diagonals is 13 m^2 then, find the length of the other diagonal.

Solution: Area of rhombus = $\frac{1}{2}(d_1 \times d_2)$

$\Rightarrow \quad 156 = \frac{1}{2}(13 \times d_2)$

$\Rightarrow \quad d_2 = \frac{2\times156}{13} = 24$ m.

Illustration 26: Find the area of a rhombus whose one side is 13 cm and one diagonal is 24 cm.

Solution: Area of rhombus = $d_1 \times \sqrt{a^2 - \left(\frac{d_1}{2}\right)^2}$

$= 24 \times \sqrt{(13)^2 - \left(\frac{24}{2}\right)^2}$

$= 24 \times \sqrt{169-144}$

$= 24 \times 5$

$= 120$ cm^2.

Illustration 27: If the perimeter of a rhombus is 73 cm and one of its diagonals is 27.5 cm then, find the other diagonal and the area of the rhombus.

Solution: One side of rhombus (a) = $\frac{73}{4} = 18.25$ cm.

$\therefore$ Other diagonal (d_2) = $2 \times \sqrt{a^2 - \left(\frac{d_1}{2}\right)^2}$

$= 2 \times \sqrt{(18.25)^2 - \left(\frac{27.5}{2}\right)^2}$

$= 24$ cm.

$\therefore$ Area of rhombus = $\frac{1}{2}\times d_1 \times d_2$

$= \frac{1}{2}\times 24 \times 27.5$

$= 330$ cm^2.

Illustration 28: In a rhombus, the lengths of two diagonals are 18 m and 24 m. Find its perimeter.

Solution: Perimeter of the rhombus

$= 2 \times \sqrt{d_1^2 + d_2^2}$

$= 2 \times \sqrt{(18)^2 + (24)^2}$

$= 2 \times \sqrt{324+576}$

$= 2 \times \sqrt{900} = 60$ m.

Illustration 29: Find the side of a rhombus, one of whose diagonals measure 4 m and the other 3 m.

Solution: Side of the rhombus

$= \frac{1}{2}\times\sqrt{d_1^2 + d_2^2}$

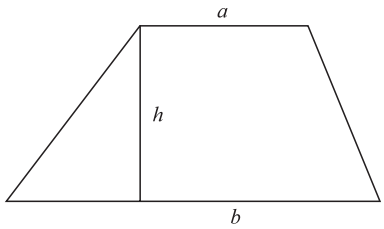
$= \frac{1}{2}\times\sqrt{(4)^2 + (3)^2}$

$= \frac{1}{2}\times 25$, i.e., 12.5 m.

11. Trapezium (Trapezoid)

A *trapezium* is a quadrilateral whose any two opposite sides are parallel.

Distance between the parallel sides of a trapezium is called its height.



(a) Area (A) of a trapezium

= $\frac{1}{2}$ × (sum of the parallel sides) × perpendicular distance between the parallel sides

i.e., $\frac{1}{2} \times (a + b) \times h$.

$= \frac{a+b}{l}\sqrt{s(s-l)(s-c)(s-d)}$,

where, $l = b - a$ if $b > a$

$= a - b$ if $a > b$

and, $s = \frac{c+d+l}{2}$

(b) Height (h) of the trapezium

$= \frac{2}{l}\sqrt{s(s-l)(s-c)(s-d)}$

$= \left(\frac{2A}{a+b}\right)$

Illustration 30: Find the area of a trapezium having parallel sides 65 m and 44 m and distance between them being 20 m.

Solution: Area of the trapezium

$$= \frac{1}{2} \times (a + b) \times h$$

$$= \frac{1}{2} \times (65 + 44) \times 20$$

$$= 1100 \text{ cm}^2.$$

Illustration 31: The parallel sides of a trapezium are 24 m and 52 m. If its other two sides are 26 m and 30 m, then what is the area of the trapezium?

Solution: Area of the trapezium

$$= \frac{a+b}{l}\sqrt{s(s-l)(s-c)(s-d)}\,.$$

Here, $a = 24$, $b = 52$, $c = 26$, $d = 30$, $l = b - a = 28$,

$$s = \frac{c+d+l}{2} = \frac{26+30+28}{2} = 42.$$

$\therefore$ Area of the trapezium

$$= \frac{24+52}{28}\sqrt{42(42-28)(42-26)(42-30)}$$

$$= \frac{76}{28} \times \sqrt{42 \times 14 \times 16 \times 12}$$

$$= \frac{76 \times 336}{28} = 912 \text{ m}^2.$$

Illustration 32: The two parallel sides of a trapezium of area 180 cm^2 measure 28 cm and 12 cm. What is the height of the trapezium?

Solution: Height of the trapezium

$$= \left(\frac{2A}{a+b}\right) = \left(\frac{2 \times 180}{28+12}\right) = \frac{360}{40} = 9 \text{ cm}.$$

12. Walls of a Room

Area of four walls of a room

= 2(length + breadth) × height

Illustration 33: Find the cost of painting the walls of a room of 6 m long, 5 m broad and 4 m high at ₹7.50 per m^2.

Solution: Area of 4 walls of the room

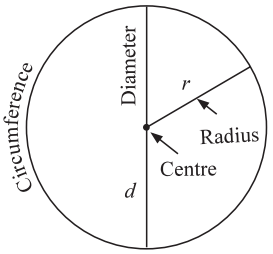
= 2(length + breadth) × height

$$= 2(6 + 5) \times 4 = 88 \text{ m}^2.$$

$\therefore$ Cost of painting = 88 × 7.50 = ₹660.

12. Circle

A *circle* is the path travelled by a point which moves in such a way that its distance from a fixed point remains constant.



The fixed point is known as *centre* and the fixed distance is called the *radius*.

(a) Circumference or perimeter of a circle

$$= 2\pi r = \pi d,$$

where r is radius and d is diameter of the circle

(b) Area of a circle

$= \pi r^2$, r is radius

$= \frac{\pi d^2}{4}$, d is diameter

$= \frac{c^2}{4\pi}$, c is circumference

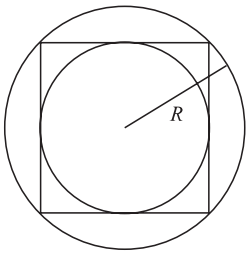
$= \frac{1}{2} \times$ circumference × radius

(c) Radius of a circle $= \sqrt{\frac{\text{Area}}{\pi}}$

$$= \frac{\text{Perimeter or circumference}}{2\pi}$$

(d) Ratio of the areas of the two circles is

$$= \frac{\text{Area of circle circumscribing the square}}{\text{Area of circle inscribed in the square}} = \frac{2}{1}.$$



(e) Ratio of the area of the two squares is

$$= \frac{\text{Area of square circumscribing the circle}}{\text{Area of square inscribed in the circle}} = \frac{2}{1}.$$

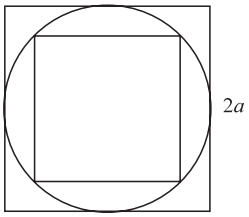


Illustration 34: What is the radius of a circular plot whose circumference is 176 m?

Solution: $r = \dfrac{\text{Circumference}}{2\pi}$

$= \dfrac{176}{2\times\frac{22}{7}} = \dfrac{176\times 7}{2\times 22} = 28$ m.

Illustration 35: A circular plot covers an area of 154 m^2. How much wire is required for fencing the plot?

Solution: Area of the plot $= \pi r^2 = 154$

i.e., $r^2 = 154 \times \dfrac{7}{22} = 49$

$\therefore \quad r = 7$ m

$\therefore$ Length of the wire $= 2\pi r = 2 \times \dfrac{22}{7}\times 7 = 44$ m.

Illustration 36: Find the length of a rope by which a buffalo must be tethered in order that she may be able to graze an area of 9856 m^2.

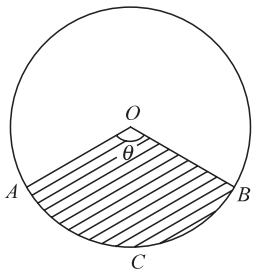
Solution: The required length of the rope

$= r = \sqrt{\dfrac{\text{Area}}{\pi}}$

$= \sqrt{\dfrac{9856\times 7}{22}} = \sqrt{3136} = 56$ m.

Sector

A *sector* is a figure enclosed by two radii and an arc lying between them.



For sector AOB,

Arc $AB = \dfrac{2\pi r\theta}{360}$,

where r = radius and $\angle AOB = \theta$

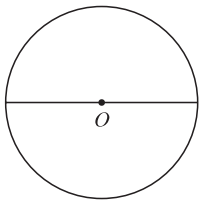
Area of the sector $ACBO$

$= \dfrac{1}{2} \times (\text{arc } AB) \times \text{radius}$

$= \dfrac{\pi(\text{radius})^2\theta}{360}$.

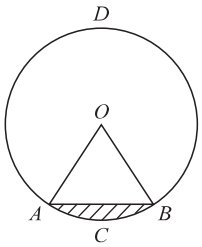
Semi-Circle

A *semi-circle* is a figure enclosed by a diameter and the part of the circumference cut off by it.



Segment

A *segment* of a circle is a figure enclosed by a chord and an arc which it cuts off.



Notes

Any chord of a circle which is not a diameter (such as AB), divides the circle into two segments, one is greater and the other is less than a semi-circle.

Area of segment ACB

= area of sector $ACBO$ – area of ΔOAB

and area of segment ADB

= area of circle – area of segment ACB

Illustration 37: If a piece of wire 20 cm long is bent into an arc of a circle subtending an angle of 60° at the centre, find the radius of the circle.

Solution: Length of the arc $= \dfrac{2\pi r\theta}{360}$

$\Rightarrow \quad 20 = \dfrac{2\pi r\times 60}{360} \Rightarrow r = \dfrac{20\times 360}{60\times 2\times\pi} = \dfrac{60}{\pi}$ cm.

Illustration 38: Find the area of sector of a circle whose radius is 14 cm and the angle at the centre is 60°.

Solution: Area of thesector $= \frac{\pi(\text{radius})^2\theta}{360}$

$= \frac{22\times14\times14\times60}{7\times360}$

$= \frac{22\times2\times14}{6} = 102\frac{2}{3}$ cm^2.

Illustration 39: Find the area of sector of a circle whose radius is 10 cm and the length of the arc is 13 cm.

Solution: Area of the sector $= \frac{1}{2}\times$ (length of arc) $\times$ radius

$= \frac{1}{2} \times 13 \times 10$

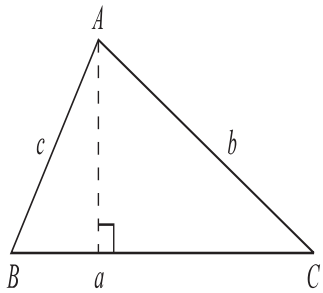
$= 65$ cm^2.

Polygon

A *polygon* is a plane figure enclosed by four or more straight lines.

Regular Polygon

If all the sides of a polygon are equal, it is called a *regular polygon.*



All the interior angles of a regular polygon are equal.

For a regular polygon:

Sum of exterior angles = 2π

Sum of interior angles = $(n - 2)\pi$

Number of diagonals in a polygon = $\frac{n(n-3)}{2}$

Perimeter $(P) = n \times a$,

where, n = number of sides

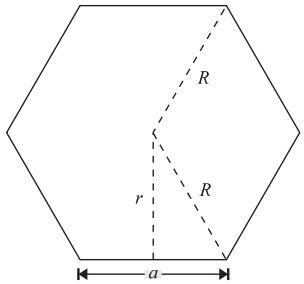
and, a = length of each side

Each interior angle = $\frac{n-2}{n}\times\pi$

Each exterior angle = $\frac{2\pi}{n}$

Area = $\frac{1}{2} \times P \times r = \frac{1}{2} \times n \times a \times r$,

where, r is radius of the circle drawn inside the polygon touching its sides.



$= \frac{1}{2} \times n \times a \times \sqrt{R^2 - \left(\frac{a}{2}\right)^2}$,

where, R is radius of the circle drawn outside the polygon touching its sides.

$= \frac{na^2}{4}\cot\left(\frac{\pi}{n}\right)$.

Area of a regular hexagon = $\frac{3\sqrt{3}}{2}$ (side)^2

Area of a regular octagon = $2(\sqrt{2}+1)$ (side)^2.

Illustration 40: Find the side of a regular hexagon whose area is $48\sqrt{3}$ cm^2.

Solution: Area of a regular hexagon

$= \frac{3\sqrt{3}}{2} \times (\text{side})^2$

$\Rightarrow \quad 48\sqrt{3} = \frac{3\sqrt{3}}{2} \times (\text{side})^2$

$\Rightarrow \quad (\text{side})^2 = 32$

$\therefore$ Side of the hexagon = $4\sqrt{2}$ cm.

Illustration 41: Find the area of a regular octagon whose side measures $\sqrt{2}$ cm.

Solution: Area of regular octagon

$= 2(\sqrt{2}+1)\times(\text{side})^2$

$= 2(\sqrt{2}+1)\times(\sqrt{2})^2$

$= 4(\sqrt{2}+1)$ cm^2.

Illustration 42: Find the sum of interior angles of a regular polygon of 10 sides. Also, find the value of each interior angle.

Solution: Sum of interior angles $= (n - 2) \times \pi$

$= (10 - 2) \times \pi$

$= 8\pi.$

Also, value of each interior angle

$$= \left(\frac{n-2}{n}\right)\times\pi$$

$$= \left(\frac{10-2}{10}\right)\pi = \frac{4\pi}{5}.$$

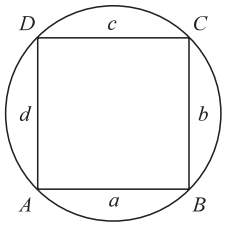
Illustration 43: Find the sum of all the exterior angles of a regular polygon of 12 sides. And, also find the value of each exterior angle.

Solution: Sum of exterior angles $= 2\pi$

Also, value of each exterior angle $= \frac{2\pi}{n} = \frac{2\pi}{12} = \frac{\pi}{6}.$

Cyclic Quadrilateral

A quadrilateral whose vertices lie on the circumference of the circle is called a *cyclic quadrilateral*.



For a cyclic quadrilateral

- Area $= \sqrt{s(s-a)(s-b)(s-c)(s-d)}$,

  where,

  $$s = \frac{a+b+c+d}{2}$$

- $\angle A + \angle B + \angle C + \angle D = 2\pi$
- $\angle A + \angle C = \angle B + \angle D = \pi.$

SHORT-CUT METHODS

01 If the length and the breadth of a rectangle are increased by x% and y% respectively, then the area of the rectangle will increase by $\left(x + y + \frac{xy}{100}\right)$%

Explanation:

Area of the original rectangle is

$$A = l \times b.$$

Area of the new rectangle is

$$A' = l\left(\frac{100+x}{100}\right)\times b\left(\frac{100+y}{100}\right)$$

$$= lb\left(\frac{100+x}{100}\right)\left(\frac{100+y}{100}\right)$$

$$= A\left(\frac{100+x}{100}\right)\left(\frac{100+y}{100}\right)$$

$$\therefore \quad \frac{A'}{A} - 1 = \left(\frac{100+x}{100}\right)\left(\frac{100+y}{100}\right) - 1$$

or, $$\frac{A'-A}{A} = \frac{(100+x)(100+y)-(100)^2}{(100)^2}$$

or, $$\left(\frac{A'-A}{A}\right)\times 100$$

$$= \frac{(100)^2 + 100(x+y) + xy - (100)^2}{(100)}$$

$$\therefore \quad \text{Percentage increase in area} = \frac{100(x+y)+xy}{100}\%$$

or, $$\left(x + y + \frac{xy}{100}\right)\%$$

Notes

If any of x or y decreases, we put negative sign.

Illustration 1: The length and the breadth of a rectangle are increased by 20% and 5%, respectively. Find the percentage increase in its area.

Solution: Percentage increase in the area of rectangle

$$= \left(x + y + \frac{xy}{100}\right)\%$$

$$= \left(20 + 5 + \frac{20\times 5}{100}\right)\% = 20\%$$

02 If the length of a rectangle is increased by x%, then its breadth will have to be decreased by $\left(\frac{100x}{100+x}\right)$% in order to maintain the same area of rectangle.

Explanation:

Percentage increase in area of rectangle

$$= \left(x + y + \frac{xy}{100}\right)\%$$

or, $$0 = \left(x + y + \frac{xy}{100}\right)$$

or, $$-x = y\left(1 + \frac{x}{100}\right) \quad \text{or,} \quad y = -\left(\frac{100x}{100 + x}\right)$$

–ve sign indicates decrease.

Therefore, breadth must be decreased by $\left(\frac{100x}{100 + x}\right)\%$ in order to maintain the same area.

Illustration 2: The length of a rectangle is increased by 25%. By what per cent should its breadth be decreased so as to maintain the same area?

Solution: The breadth must be decreased by

$$= \left(\frac{100x}{100 + x}\right)\% = \left(\frac{100 \times 25}{100 + 25}\right)\%, \text{ i.e., } 20\%$$

03 If each of the defining dimensions or sides of any two-dimensional figure (triangle, rectangle, square, circle, quadrilateral, pentagon, hexagon, etc.) is changed by x%, its area changes by $x\left(2 + \frac{x}{100}\right)\%$

Illustration 3: If the radius of a circle is decreased by 10%, what is the percentage decrease in its area?

Solution: Here, $x = -10$ (–ve sign indicates decrease)

$\therefore$ Percentage change in area $= x\left(2 + \frac{x}{100}\right)\%$

$$= -10\left(2 - \frac{10}{100}\right)\%$$

$$= (-10)\left(\frac{19}{10}\right)\% = -19\%$$

$\therefore$ Area of the circle decreases by 19%

04 If all the sides of a quadrilateral are increased (or decreased) by x%, its diagonals also increase (or decrease) by x%

Illustration 4: The length and the two diagonals of a rectangle are decreased by 5% each. What is the percentage decrease in its breadth?

Solution: Since the length and the two diagonals decreased by 5% each, the breadth also must decrease by 5%.

05 If each of the defining dimensions or sides of any two-dimensional figures are increased (or decreased) by x%, its perimeter also increases (or decreases) by x%

Illustration 5: If all the sides and diagonals of a square are increased by 8% each, then find the percentage increase in its perimeter.

Solution: The perimeter also increases by 8%

06 If the ratio of the areas of two squares be a:b, then the ratio of their sides, ratio of their perimeters and the ratio of their diagonals, each will be in the ratio $\sqrt{a} : \sqrt{b}$.

Illustration 6: Ratio of the areas of two squares is 16:9. Find the ratio of their diagonals.

Solution: The ratio of their diagonals

$$= \sqrt{a} : \sqrt{b}$$
$$= \sqrt{16} : \sqrt{9}, \text{ i.e., } 4:3.$$

07 If the diagonal of a square increases by x times, then the area of the square becomes x^2 times.

Illustration 7: The diagonal of a square is doubled. How many times will the area of the new square become?

Solution: The area of the new square will become x^2 times, i.e., $(2)^2 = 4$ times.

08 Standard Properties of Diagonals of Quadrilaterals

Quadrilateral	*Meet at right angles*	*Bisect each other*	*Equal to each other*	*Bisect angle at vertex*
Square	√	√	√	√
Rectangle	×	√	√	×
Parallelogram	×	√	×	×
Rhombus	√	√	×	√
Trapezium	×	×	may or may not be	×

09 Carpeting the Floor of a Room

If the length and the breadth of a room are l and b, respectively, and a carpet of width w is used to cover the floor, then the required length of the carpet

$$= \frac{l \times b}{w}.$$

Illustration 8: How many metres of a carpet of 12 cm wide will be required to cover the floor of a room which is 600 cm long and 420 cm broad? Also, calculate the amount required in carpeting the floor if the cost of carpet is ₹15 per metre.

Solution: Length of the carpet

$$= \frac{l \times b}{w}$$

$$= \frac{600 \times 420}{12}$$

$$= 21000 \text{ cm, i.e., } 210 \text{ m.}$$

The amount required for carpeting the floor

$$= 15 \times 210 = ₹3150.$$

10 Number of Square Tiles Required for Flooring

If the length and the breadth of a room are l and b respectively, then the least number of square tiles required to cover the floor

$$= \frac{l \times b}{\text{H.C.F. } (l, b)}$$

Also, the size of the largest tile so that the tiles exactly fit

$$= \text{H.C.F. } (l, b).$$

Illustration 9: A hall of length 24 cm and breadth 20 m is to be paved with equal square tiles. What will be the size of the largest tile so that the tiles exactly fit and also find the number of tiles required.

Solution: Size of the largest possible square tile

$$= \text{H.C.F. } (l, b)$$

$$= \text{H.C.F. } (24, 20) = 4 \text{ m}$$

Number of tiles required

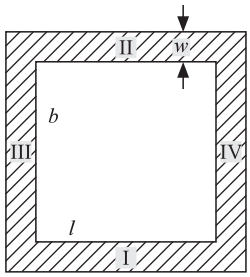
$$= \frac{l \times b}{\text{H.C.F. } (l, b)}$$

$$= \frac{24 \times 20}{4} = 120 \text{ tiles.}$$

11 Path around a Rectangular Space

(a) A rectangular garden l m long and b m broad is sorrounded by a path of w m wide. The area of the path is given by

$$= 2w(l + b + 2w) \text{ m}^2.$$



Explanation:

Area of part I = Area of part II

$$= (l + 2w)w \text{ m}^2$$

Area of part III = Area of part IV

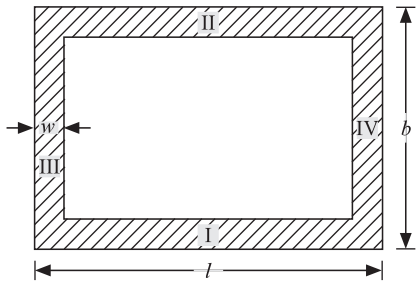
$$= bw \text{ m}^2.$$

$\therefore$ Total area of the path $= 2[(l + 2w)w + bw]$

$$= 2w(l + b + 2w) \text{ m}^2.$$

(b) A rectangular garden l m long and b m broad is sorrounded by a path of w m wide constructed inside it along its boundary. The area of the path is given by

$$= 2w(l + b - 2w) \text{ m}^2.$$



Explanation:

Area of part I = Area of part II

$$= lw \text{ m}^2$$

Area of part III = Area of part IV

$$= (b - 2w)w \text{ m}^2$$

$\therefore$ Total area of the path

$$= 2[lw + (b - 2w)w]$$

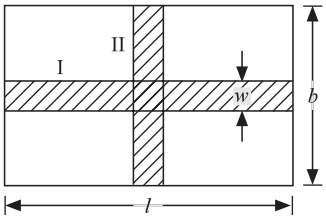
$$= 2w(l + b - 2w) \text{ m}^2.$$

(c) A rectangular park is l m long and b m broad. Two paths inside the park of w m wide and they are perpendicular to each other. The area of the paths

$$= w(l + b - w) \text{ m}^2$$

Also, the area of the park minus the paths

$$= (l - w)(b - w) \text{ m}^2$$



Explanation:

Total area of the path

= Area of path I + Area of path II − Area of common central part

$= lw + bw - w^2$

$= w(l + b - w)$ m^2.

$\therefore$ Area of the park minus the paths

$= [lb - w(l + b - w)]$

$= lb - lw - w(b - w)$

$= l(b - w) - w(b - w)$

$= (l - w)\ (b - w)$ m^2.

Notes

1. Clearly, from the figure, the area of the paths does not change on shifting their location as long as they are perpendicular to each other.
2. For a square park, take $l = b$ in all the results derived above.

Illustration 10: A rectangular park 18 m × 12 m, is sorrounded by a path of 4 m wide. Find the area of the path.

Solution: The area of the path

$= 2w(l + b + 2w)$

$= 2 \times 4(18 + 12 + 2 \times 4) = 304$ m^2.

Illustration 11: A park is square in shape with side 18 m. Find the area of the pavement of 3 m wide to be laid all around it on its inside.

Solution: Area of the pavement

$= 2w(l + b - 2w)$

$= 2 \times 3(18 + 18 - 2 \times 3)$ (Here, $l = b = 18$)

$= 180$ m^2.

Illustration 12: A playground measures 27 m × 13 m. From the centre of each side a path 2 m wide goes across to the centre of the opposite side. Calculate the area of the path and the cost of constructing it at ₹4 per m^2.

Solution: Area of the path

$= w(l + b - w) = 2(27 + 13 - 2)$

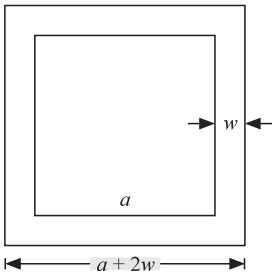
$= 76$ m^2.

$\therefore$ Cost = $4 \times 76 = ₹304$.

12 Square Room Surrounded by a Verandah

(a) A square room of side a is sorrounded by a verandah of width w on the outside of the square room. If the area of the verandah is A, then the area of the room is given by

$$\left(\frac{A - 4w^2}{4w}\right)^2.$$



Explanation:

Area of the room = a^2.

Area of the (room + verandah) = $(a + 2w)^2$.

$\therefore$ Area (A) of the verandah = $(a + 2w)^2 - a^2$

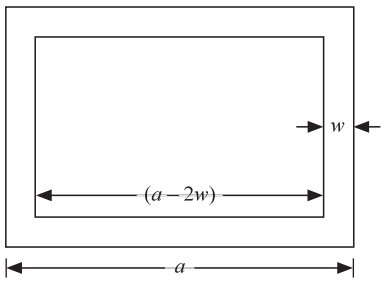
$= (4aw + 4w^2)$

or, $a = \dfrac{A - 4w^2}{4w}$

$\therefore$ Area of the room = $a^2 = \left(\dfrac{A - 4w^2}{4w}\right)^2$.

(b) A square room of side a is surrounded by a verandah of width w on its inside. If the area of the verandah is A, then the area of the room is given by

$$\left(\frac{A + 4w^2}{4w}\right)^2.$$



Explanation:

Area (A) of the verandah = $a^2 - (a - 2w)^2$

$= 4aw - 4w^2$

$= 4w(a - w)$

or, $a = \frac{A}{4w} + w = \frac{A+4w^2}{4w}$

$\therefore$ Area of the room $= a^2 = \left(\frac{A+4w^2}{4w}\right)^2$.

Illustration 13: A square field is surrounded by a path of 2 m wide on its outside. The area of the path is 72 m^2. What is the area of the field?

Solution: Area of the field

$$= \left(\frac{A-4w^2}{4w}\right)^2$$

$$= \left(\frac{72-4\times 2^2}{4\times 2}\right)^2 = 49 \text{ m}^2.$$

Illustration 14: A square room has a verandah of area 24 m^2 and width 1 m all around it on its inside. Find the area of the room.

Solution: Area of the room

$$= \left(\frac{A+4w^2}{4w}\right)^2$$

$$= \left(\frac{24+4\times 1^2}{4\times 1}\right)^2 = 49 \text{ m}^2.$$

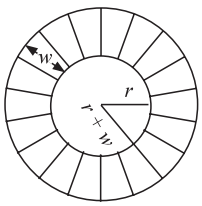
13 (a)A circular ground of radius r has a pathway of width w around it on its outside. The area of the circular pathway is given by
$= \pi w(2r + w)$.

Explanation:

Area of the circular ground $= \pi r^2$

Area of the circular ground + pathway

$$= \pi(r + w)^2 = \pi r^2 + 2\pi rw + \pi w^2.$$



$\therefore$ Area of circular the pathway

$$= (\pi r^2 + 2\pi rw + \pi w^2) - \pi r^2$$

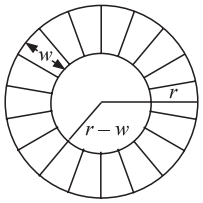
$$= \pi w(2r + w).$$

(b) A circular ground of radius r has a pathway of width w around it on its inside. The area of the circular pathway is given by

$$= \pi w(2r - w).$$

Explanation:

Area of circular the ground $= \pi r^2$



Area of the circular ground – pathway

$$= \pi (r - w)^2$$

$$= \pi r^2 - 2\pi rw + \pi w^2$$

$\therefore$ Area of the circular pathway

$$= \pi r^2 - (\pi r^2 - 2\pi rw + \pi w^2)$$

$$= \pi w(2r - w).$$

Illustration 15: A circular ground of radius 16 m has a path of width 2.8 m around it on its outside. Find the area of the path.

Solution: The area of the circular path

$$= \pi\, w(2r + w)$$

$$= \frac{22}{7} \times 2.8 \times (2 \times 16 + 2.8)$$

$$= 8.8 \times (32 + 2.8) = 306.2 \text{ m}^2.$$

Illustration 16: A circular park of radius 22 m has a path of width 1.4 m around it on its inside. Find the area of the path.

Solution: The area of the circular path

$$= \pi\, w(2r - w)$$

$$= \frac{22}{7} \times 1.4 \times (2 \times 22 - 1.4)$$

$$= 4.4 \times (44 - 1.4) = 187.45 \text{ m}^2.$$

14 If the area of a square is a cm^2, then the area of the circle formed by the same perimeter is $\left(\frac{4a}{\pi}\right)$ cm^2.

Explanation:

Area of the square $= a$.

$\therefore$ Side of the square $= \sqrt{\text{Area}} = \sqrt{a}$.

$\therefore$ Perimeter of the square $= 4\sqrt{a}$.

Given, Circumference of the circle = Perimeter of the square

$\Rightarrow$ $2\pi r = 4\sqrt{a}$

$\therefore$ Radius of the circle $(r) = \dfrac{4\sqrt{a}}{2\pi} = \dfrac{2\sqrt{a}}{\pi}$

$\therefore$ Area of the circle $= \pi r^2 = \pi \left(\dfrac{2\sqrt{a}}{\pi}\right)^2 = \dfrac{4a}{\pi}$ cm^2.

Illustration 17: If the area of a square is 33 cm^2, then find the area of the circle formed by the same perimeter.

Solution: Required area of the circle

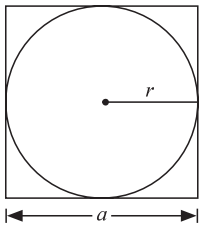
$$= \frac{4a}{\pi}$$

$$= \frac{4\times 33}{\pi} = \frac{4\times 33\times 7}{22} = 42 \text{ cm}^2.$$

15 The area of the largest circle that can be inscribed in a square of side a is $\dfrac{\pi a^2}{4}$.

Explanation

Clearly, from the figure, the diameter of the inscribed circle equals the side of the square i.e., $D = a$.



Area of the circle $= \dfrac{\pi D^2}{4}$

$\therefore$ Area of the inscribed circle $= \dfrac{\pi a^2}{4}$.

Illustration 18: Find the area of largest circle inscribed in a square of side 112 cm.

Solution: The area of the largest circle

$$= \frac{\pi a^2}{4}$$

$$= \frac{22\times 112\times 112}{7\times 4} = 9856 \text{ cm}^2.$$

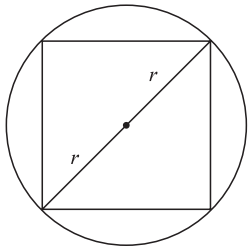
16 Area of a square inscribed in a circle of radius r is $2r^2$ and the side of the square is $\sqrt{2}\,r$.

Explanation:

Clearly, from the figure, diagonal of the inscribed square is equal to the diameter of the circle, i.e., $2r$.

$\therefore$ Area of square = (diagonal)^2

$$= \frac{1}{2}(2r)^2 = 2r^2.$$



Also, side of the square $= \sqrt{\text{Area}} = \sqrt{2r^2} = \sqrt{2}r$.

Illustration 19: Find the side of the square inscribed in a circle whose circumference is 308 cm.

Solution: Circumference of the circle $(2\pi r) = 308$

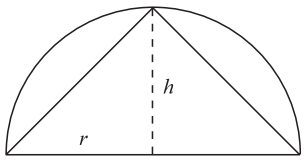
$\Rightarrow r = \dfrac{308}{2\pi} = \dfrac{308\times 7}{2\times 22} = 49$ cm.

$\therefore$ Side of the inscribed square $= \sqrt{2}\,r = 49\sqrt{2}$ cm.

17 The area of the largest triangle inscribed in a semi-circle of radius r is r^2.

Explanation:

Clearly, from the figure, the largest triangle inscribed in a semi-circle is an isosceles triangle with diameter as its base and radius as its height.



Area of the triangle $= \dfrac{1}{2} \times$ base $\times$ height

$$= \frac{1}{2} \times 2r \times r = r^2.$$

18 The number of revolutions made by a circular wheel of radius r in travelling a distance d is given by

$$= \left(\frac{d}{2\pi r}\right).$$

Explanation:
Circumference of the wheel = $2\pi r$

In travelling a distance $2\pi r$, the wheel makes 1 revolution.

$\therefore$ In travelling a distance d, the wheel makes $\frac{22}{7}$ revolutions.

Illustration 20: The diameter of a wheel is 2 cm. If it rolls forward covering 10 revolutions, find the distance travelled by it.

Solution: Radius of the wheel = 1 cm.

The distance travelled by the wheel in 10 revolutions

$= 10 \times 2\pi r$

$= 10 \times 2 \times \frac{22}{7} \times 1 = 62.8$ cm.

EXERCISE-I

1. The area of a triangle whose sides are 15 m, 16 m and 17 m is:
(a) $24\sqrt{4}$ m^2 (b) $24\sqrt{3}$ m^2
(c) $24\sqrt{21}$ m^2 (d) None of these

2. The area of a right-angled triangle with base 6 m and hypotenuse 6.5 m is:
(a) 7.5 m^2 (b) 9.5 m^2
(c) 8.5 m^2 (d) None of these

3. The length of each side of a triangle is 12 cm. The height of the triangle is:
(a) $3\sqrt{2}$ m (b) $6\sqrt{3}$ m
(c) $6\sqrt{2}$ (d) None of these

4. The area of a triangular lawn is 1600 m^2. If one side is 64 m long and the other two sides are equal in length, the length of each equal side is:
(a) 60.37 m (b) 59.36 m
(c) 60.36 cm (d) None of these

5. Three sides of a triangular field are 20 m, 21 m and 29 m long, respectively. The area of the field is:
(a) 215 m^2 (b) 230 m^2
(c) 210 m^2 (d) None of these

6. The hypotenuse and the semi-perimeter of a right triangle are 20 cm and 24 cm, respectively. The other two sides of the triangle are:
(a) 16 cm, 12 cm
(b) 20 cm, 12 cm
(c) 20 cm, 16 cm
(d) None of these

7. The sides of a triangle are in the ratio 3:4:5. If its perimeter is 36 cm, then the area of the triangle is:
(a) 57 m^2 (b) 54 m^2
(c) 56.5 m^2 (d) None of these

8. For a triangle whose sides are 50 m, 78 m and 112 m, respectively, then the length of the perpendicular from the opposite angle on the side 112 m is:
(a) 45 m (b) 35 m
(c) 30 m (d) None of these

9. A ladder is resting with one end in contact with the top of a wall of height 12 m and the other end on the ground is at a distance 5 m from the wall. The length of the ladder is:
(a) 13 m (b) 17 m
(c) 16 m (d) None of these

10. A ladder is placed so as to reach a window 63 cm high. The ladder is then turned over to the opposite side of the street and is found to reach a point 56 cm high. If the ladder is 65 cm long, then the width of the street is:
(a) 59 cm (b) 39 cm
(c) 49 cm (d) None of these

11. If the area of a triangle with base x is equal to the area of a square with side x, then the altitude of the triangle is:
(a) $\frac{x}{2}$ (b) x
(c) $2x$ (d) $3x$

12. If the area of a triangle is 150 m^2 and the ratio of the base and the height is 3:4, then find its height.
(a) 25 m (b) 35 m
(c) 20 m (d) None of these

13. The base of a triangular field is three times its height. If the cost of cultivating the field at ₹36.72 per hectare is ₹495.72, then find its base and height.
(a) 950 m, 350 m (b) 800 m, 500 m
(c) 900 m, 300 m (d) None of these

14. If the sides of a triangle are doubled, its area:
(a) remains same (b) is doubled
(c) becomes 4 times (d) Can't say

15. Two sides of a triangular field are 85 m and 154 m, respectively, and its perimeter is 324 m. The cost of ploughing the field at the rate of ₹10 per m^2 is:
(a) ₹27720 (b) ₹37620
(c) ₹26750 (d) None of these

16. The area of an equilateral triangle, each of whose sides measures $2\sqrt{3}$ cm is:
(a) $5\sqrt{3}$ m^2 (b) $4\sqrt{3}$ m^2
(c) $3\sqrt{3}$ cm^2 (d) None of these

17. If the height of an equilateral triangle is $2\sqrt{3}$ cm, then the length of its side is:
(a) 4 cm (b) 6 cm
(c) 5 cm (d) None of these

18. If the perimeter of an equilateral triangle is 12 m, then find its area.
(a) $3\sqrt{4}$ m (b) $4\sqrt{3}$ m
(c) $5\sqrt{3}$ m (d) None of these

19. The height of an equilateral triangle whose perimeter is 24 cm, is:
(a) $4\sqrt{3}$ cm (b) $3\sqrt{4}$ cm
(c) $5\sqrt{3}$ cm (d) None of these

20. The perimeter of a right angled triangle is 90 cm and its hypotenuse is 39 cm. Find its other sides.
(a) 30 cm, 10 cm (b) 36 cm, 15 cm
(c) 48 cm, 20 cm (d) None of these

21. The perimeter of an isosceles triangle is 306 m and each of the equal sides is $\frac{5}{8}$ of the base. Find the area.
(a) 3648 cm (b) 3468 m^2
(c) 3846 cm (d) None of these

22. Find the area of an isosceles right-angled triangle whose hypotenuse is 8 cm.
(a) 32 m^2 (b) 24 cm^2
(c) 16 cm^2 (d) None of these

23. The perimeter of an isosceles triangle is equal to 14 cm; the lateral side and the base is in the ratio 5:4. The area of the triangle is:
(a) $3\sqrt{21}$ cm^2 (b) $2\sqrt{21}$ cm^2
(c) $4\sqrt{21}$ cm^2 (d) None of these

24. If all the sides of a triangle are increased by 200%, then the area of the triangle will increase by:
(a) 400% (b) 600%
(c) 800% (d) None of these

25. A plot of land is in the shape of a right-angled isosceles triangle. The length of the hypotenuse is $50\sqrt{2}$ m. The cost of fencing is ₹3 per metre. The total cost of fencing the plot will be:
(a) Less than ₹300
(b) Less than ₹500
(c) More than ₹500
(d) None of these

26. If the area of an equilateral triangle is equal to the area of an isosceles triangle whose base and equal sides are 16 cm and 10 cm respectively, then the side of the equilateral triangle is:
(a) 10.5 cm (b) 9.5 cm
(c) 12.5 cm (d) None of these

27. If the perimeter of a right-angled isosceles triangle is $4\sqrt{2} + 4$ m, then the hypotenuse is:
(a) 8 m (b) 6 m
(c) 4 m (d) None of these

28. The two adjacent sides of a parallelogram are 60 m and 40 m and one of the diagonals is 80 m long. The area of the parallelogram is:
(a) $600\sqrt{15}$ m^2 (b) $800\sqrt{25}$ m^2
(c) $700\sqrt{15}$ m^2 (d) None of these

29. One side of the parallelogram is 14 cm. Its distance from the opposite side is 16 cm. The area of the parallelogram is:
(a) 234 cm^2 (b) 324 cm^2
(c) 224 cm^2 (d) None of these

30. A field is in the shape of a parallelogram. Its adjacent sides and one diagonal are 65 m, 119 m and 156 m, respectively. Find the cost of gravelling it at the rate of ₹10 per m^2.
(a) ₹81400 (b) ₹71400
(c) ₹91400 (d) None of these

31. One side of a parallelogram is 10 m and the corresponding altitude is 7 m. The area of the parallelogram is:
(a) 70 m^2 (b) 60 m^2
(c) 80 m^2 (d) None of these

32. The adjacent sides of a parallelogram are 8 m and 5 m. The distance between the longer sides is 4 m. The distance between the shorter sides is:
(a) 4.6 m (b) 6.4 m
(c) 8.6 m (d) None of these

33. The area of a quadrilateral is 420 m^2 and the perpendiculars drawn to one diagonal from the

opposite vertices are 18 m and 12 m. Then, the length of the diagonal is:

(a) 32 m (b) 24 m
(c) 28 m (d) None of these

34. The area of a parallelogram is 72 cm^2 and its altitude is twice the corresponding base. The length of the base is:

(a) 6 cm (b) 8 cm
(c) 4 cm (d) None of these

35. The area of a parallelogram is 240 cm^2 and its height is 12 cm. The base of the parallelogram is:

(a) 24 cm (b) 20 cm
(c) 28 cm (d) None of these

36. In a quadrilateral *ABCD*, the sides, *AB*, *BC*, *CD*, *DA* measure 20 m, 13 m, 17 m and 10 m, respectively and the diagonal *AC* is 21 m. The area of the quadrilateral is:

(a) 210 m^2 (b) 220 m^2
(c) 240 m^2 (d) None of these

37. If the two diagonals of a parallelogram are 72 cm and 30 cm respectively, then find its perimeter.

(a) 156 cm (b) 164 cm
(c) 172 cm (d) None of these

38. If the base of a parallelogram is $(x + 4)$, altitude to the base is $(x - 3)$ and the area is $(x^2 - 4)$, then the actual area is equal to:

(a) 64 sq units (b) 48 sq units
(c) 60 sq units (d) None of these

39. In a parallelogram, the lengths of adjacent sides are 12 cm and 14 cm, respectively. If the length of one diagonal is 16 cm, then find the length of the other diagonal.

(a) 24.8 cm (b) 20.6 cm
(c) 22.4 cm (d) None of these

40. Find the perimeter of a circular plot which occupies an area of 154 m^2.

(a) 54 m (b) 44 m
(c) 22 m (d) 11 m

41. The perimeter of a circle is equal to that of a square. Compare their areas.

(a) 14:11 (b) 25:12
(c) 24:7 (d) 22:7

42. The length of a rectangle is thrice its breadth and its perimeter is 96 m. The area of the rectangle is:

(a) 288 m^2 (b) 442 m^2
(c) 438 m^2 (d) 432 m^2

43. A cow is tied by a rope at the corner of a rectangular field. If the length of the rope is 14 m, then the area of the field which the cow could graze is:

(a) 77 m^2 (b) 308 m^2
(c) 23 m^2 (d) 154 m^2

44. The wheel of a scooter has diameter 70 cm. How many revolutions per minute must the wheel make so that the speed of the scooter is kept 66 Km/h?

(a) 400 (b) 600
(c) 500 (d) 800

45. 2 small circular parks of diameters 16 m, 12 m are to be replaced by a bigger circular park. What would be the radius of this new park, if the new park occupies same space as the two small parks?

(a) 10 (b) 15
(c) 20 (d) 25

46. A rectangular park is 65 m long and 50 m wide. 2 cross paths each 2 m wide are to be constructed parallel to the sides. If these paths pass through the centre of the rectangle and cost of construction is ₹17.25 per m^2, then find the total cost involved in the construction.

(a) ₹2265.59 (b) ₹1772.45
(c) ₹3898.50 (d) ₹8452.32

47. The area of a trapezium is 2500 m^2. One of its parallel sides is 75 m. If the distance between the two parallel sides is 40 m, then find the length of the other parallel side.

(a) 20 m (b) 30 m
(c) 40 m (d) 50 m

48. The length of a rectangle exceeds its breadth by 3 cm. If the numerical values of the area and the perimeter of the rectangle are equal, then the breadth is:

(a) 1 cm (b) 2 cm
(c) 3 cm (d) 3.5 cm

49. If the ratio of the areas of two squares is 9:1, then the ratio of their perimeters is:

(a) 9:1 (b) 3:4
(c) 3:1 (d) 1:3

50. A square field with side 30 m is surrounded by a path of uniform width. If the area of the path is 256 m^2, then its width is:

(a) 16 m (b) 14 m
(c) 4 m (d) 2 m

51. A rope by which a calf is tied is increased from 12 m to 23 m. How much additional grassy ground shall it graze?
(a) 1120 m^2 (b) 1250 m^2
(c) 1210 m^2 (d) 1200 m^2

52. Four circular cardboard pieces, each of radius 7 cm are placed in such a way that each piece touches two other pieces. The area of the space enclosed by the four pieces is:
(a) 21 cm^2 (b) 42 cm^2
(c) 84 cm^2 (d) 168 cm^2

53. The length and breadth of a rectangular field are in the ratio 5:3. If the cost of cultivating the field at 25 paise per square metre is ₹6000, then find the dimensions of the field:
(a) 250 m, 100 m (b) 50 m 30 m
(c) 200 m, 120 m (d) Cannot be determined

54. The cost of carpeting a room 5 m wide with carpet at ₹3.50 per m^2 is ₹105. The length of the room is:
(a) 3.5 m (b) 5 m
(c) 6 m (d) 6.5 m

55. The length of a rectangular field is twice its breadth. If the rent of the field at ₹3500 per hectare is ₹28000, then find the cost of surrounding it with fence at ₹5 per metre.
(a) ₹6000 (b) ₹7000
(c) ₹6500 (d) ₹8000

56. The area of a rectangular field is 27000 m^2. The ratio of its length and breadth is 6:5. The length and breadth of the field are respectively:
(a) 180 m, 150 m (b) 200 m, 150 m
(c) 180 m, 120 m (d) 150 m, 100 m

57. The area of a sector of a circle of radius 5 cm, formed by an arc of length 3.5 cms, is:
(a) 35 cm^2 (b) 17.5 cm^2
(c) 8.75 cm^2 (d) 55 cm^2

58. The length of a plot is double its width. If a square piece of land of area 150 m^2 occupies $\frac{1}{3}$ area of the plot, then what is the length of the plot?
(a) 15 m (b) 7.5 m
(c) 30 m (d) 10 m

59. A wire is in the form of a semi-circle of 7 cm radius. The length of the wire will be:
(a) 25 cm (b) 36 cm
(c) 5 cm (d) 69 cm

60. A circular road runs round a circular ground. If the difference between the circumferences of the outer circle and the inner circle is 66 m, the width of the road is:
(a) 21 m (b) 10.5 m
(c) 7 m (d) 5.25 m

61. If the area of a square is 50 sq units, then the area of the circle drawn on its diagonal is:
(a) 25 π sq units (b) 50 π sq units
(c) 100 π sq units (d) None of these

62. A rectangular sheet of cardboard is of 4 cm × 2 cm. If a circle of greatest possible area is cut from it, then the area of remaining portion is:
(a) $(2 - \pi)$ cm^2 (b) $(4 - \pi)$ cm^2
(c) $(8 - \pi)$ cm^2 (d) $(16 - \pi)$ cm^2

63. What is the radius of a circle, to the nearest cm, whose area is equal to the sum of the areas of the three circles of radii 22 cm, 19 cm and 8 cm, respectively?
(a) 17 cm (b) 30 cm
(c) 29 cm (d) 19 cm

64. The length of a rectangle is increased by 33.33%. By what per cent should the width be decreased to maintain the same area?
(a) 25% (b) 33.33%
(c) 22.5% (d) None of these

65. If the area of a square is equal to the area of a rectangle 6.4 m long and 2.5 m wide, then each side of the square measures:
(a) 8 m (b) 5.4 m
(c) 3.8 m (d) 4 m

66. The diameters of two concentric circles are 8 cm and 10 cm. The area of the region between them is:
(a) π cm^2 (b) 3π cm^2
(c) 6π cm^2 (d) 9π cm^2

67. The length of a rectangular room is 4 m. If it can be partitioned into two equal square rooms, what is the length of the partition in metres?
(a) 1 (b) 2
(c) 4 (d) Data inadequate

68. A rectangular carpet has an area of 120 m^2 and a perimeter of 46 m. The length of its diagonal is:
(a) 15 m (b) 16 m
(c) 17 m (d) 20 m

69. The side of a square is 22 m. What is the radius of the circle whose circumference is equal to the perimeter of the square?

(a) 28 m (b) 3.5 m
(c) 14 m (d) 7 m

70. A piece of wire 132 cm long is bent successively in the shapes of an equilateral triangle, a square, a regular hexagon, and a circle. Then, which has the largest surface area?
(a) Equilateral triangle
(b) Square
(c) Circle
(d) Regular hexagon

71. If the radius of one circle is twelve times the radius of another, how many times does the area of the greater contain the area of the smaller?
(a) 12 (b) 72
(c) 144 (d) 96

72. If the circumference of a circle is equal to the perimeter of a square, what is the ratio of the area of the circle to the area of the square?
(a) 22:7 (b) 14:11
(c) 11:7 (d) 4:1

73. The length of a rectangular plot of land is three times as much as its breadth. A playground measuring 1200 ft^2 occupies $\frac{1}{4}$ of the total area of the plot. What is the length of the plot in feet?
(a) 40 (b) 360
(c) 120 (d) Data inadequate

74. There are two squares s_1 and s_2. The ratio of their areas is 4:25. If the side of s_1 is 6 cm, what is the side of s_2?
(a) 20 cm (b) 15 cm
(c) 5 cm (d) 12 cm

75. The radius of the wheel of a vehicle is 70 cm. The wheel makes 10 revolutions in 5 seconds. The speed of the vehicle is:
(a) 29.46 Km/h (b) 31.68 Km/h
(c) 36.25 Km/h (d) 32.72 Km/h

76. A rectangular carpet has an area of 60 m^2. Its diagonal and longer side together equal 5 times the shorter side. The length of the carpet is:
(a) 5 m (b) 13 m
(c) 14.5 m (d) 12 m

77. A playground has the shape of a rectangle with two semi-circles on its smaller sides as diameters, added outside. If the sides of the rectangle are 36 m and 24.5 m, then the area of the playground is: $\left(\text{use } \pi = \frac{22}{7}\right)$
(a) 2259.529 m^2 (b) 1353.625 m^2
(c) 1139.523 m^2 (d) None of these

78. A man runs around a circle of 50 m radius at a speed of 12 Km/h. Find the time taken by him for going around it ten times:
(a) 10 minutes (b) 12.5 minutes
(c) 15.7 minutes (d) None of these

79. A room 5 m × 8 m is to be carpeted leaving a margin of 10 cm from each wall. If the cost of the carpet is ₹18 per m^2, then the cost of carpeting the room will be:
(a) ₹702.60 (b) ₹691.80
(c) ₹682.46 (d) ₹673.92

80. The area of a big rectangle is equal to the area of a small rectangle. If the length of the big rectangle is equal to the length of the small rectangle and the width of big rectangle is 2 m, what is the width of a small rectangle?
(a) $\frac{1}{3}$ m (b) 1 m
(c) 2 m (d) Cannot be determined
(e) None of these

81. If the radius of a circle is reduced by 40%, then its circumference is reduced by:
(a) 60% (b) 40%
(c) 35% (d) 45%

82. A figure consists of a square of side 'a' m with semi-circles drawn on the outside of the square. The area (in m^2) of the figure so formed will be:
(a) a^2 (b) $a^2 + 2\pi a^2$
(c) $4\pi a^2$ (d) $a^2 + \frac{\pi a^2}{2}$

83. The area of a square field is 6050 m^2. How much time it will take to reach from one of its corner to the opposite corner at the rate of 10 m in every 30 seconds?
(a) $5\frac{1}{2}$ minutes (b) 11 minutes
(c) 22 minutes (d) 110 minutes

84. If a regular hexagon is inscribed in a circle of radius r, then its perimeter is:
(a) $3r$ (b) $6r$
(c) $9r$ (d) $12r$

85. 2 poles 15 m and 30 m high stand upright in a playground. If their feet be 36 m apart, find the distance between their tops.

(a) 36 m (b) 39 m
(c) 15 m (d) None of these

86. The length of a rectangular hall is $\frac{4}{3}$ of its width. If the area of the hall is 300 m^2, what is the difference between the length and the breadth?
(a) 15 m (b) 20 m
(c) 3 m (d) 5 m
(e) 6 m

87. Each side of an equilateral triangle is increased by 1.5%. The percentage increase in its area is:
(a) 1.5% (b) 3%
(c) 4.5% (d) 5.7%

88. A rope, by which a horse is tied, is increased from 12 to 23 m. How much additional ground will it be able to graze?
(a) 1315 m^2 (b) 765 m^2
(c) 1210 m^2 (d) 1012 m^2

89. If a diagonal of a square is doubled, how does the area of the square change?
(a) Becomes 4-fold (b) Becomes 3-fold
(c) Becomes 2-fold (d) None of these

90. If the sides of a rectangle are increased by 20%, the percentage increase in its perimeter is:
(a) 80 (b) 40
(c) 20 (d) None of these

91. A circle road runs around a circular garden. If the difference between the circumference of the outer circle and the inner circle is 44 m, then find the width of the road.
(a) 4 m (b) 7 m
(c) 3.5 m (d) 7.5 m

92. The length of a rectangular field is double its width. Inside the field there is a square-shaped pond 8 m long. If the area of the pond is $\frac{1}{8}$ of the area of the field, what is the length of the field?
(a) 32 m (b) 64 m
(c) 16 m (d) 20 m
(e) None of these

93. A circular disc of area 0.49π m^2 rolls down a length of 1.76 Km. The number of revolutions it makes is:
(a) 300 (b) 400
(c) 600 (d) 4000

94. If area of a triangle whose base is 6 cm is equal to the area of a circle of radius 6 cm, then find the height of this triangle:
(a) 10 cm (b) 22 cm
(c) 12 cm (d) 18 cm

95. The length of a ladder exactly equals the height of a wall. If the ladder is placed on a 2-feet tall stool placed 10 feet away from the wall, its tip just touches the top of the wall. The height of the wall in feet is:
(a) 15 (b) 26
(c) 28 (d) 32

96. If the diagonal of a square is doubled to make another square, the area of the new square will:
(a) Become 4-fold (b) Become 2-fold
(c) Become 6-fold (d) Become 8-fold

97. The area of a circle is 154 cm^2. The length of an arc of the circle which subtends an angle of 45° at the centre is:
(a) 11 cm (b) 5.5 cm
(c) 7 cm (d) None of these

98. A lawn is in the form of a triangle having its base and height in the ratio 2:3. The area of the lawn is $\frac{1}{12}$ hectare. Find the base and height of the lawn.
(a) 55 m, 34 m (b) 50 m, $33\frac{1}{3}$ m
(c) 50 m, 35 m (d) Data inadequate

99. A rectangular farm has to be fenced on one long side, one short side and the diagonal. If the cost of fencing is ₹10 per m, the area of the farm is 1200 m^2 and the short side is 30 m long, how much would the job cost?
(a) ₹700 (b) ₹1200
(c) ₹1400 (d) ₹1500

100. The diameter of a circle is 105 cm less than the circumference. What is the diameter of circle?
(a) 44 cm (b) 46 cm
(c) 48 cm (d) 49 cm

101. A garden is 24 m long and 14 m wide. There is a path 1 m wide outside the garden along its sides. If the path is to be constructed with square marble tiles 20 cm × 20 cm, then find the number of tiles required to cover the path:
(a) 1800 (b) 200
(c) 2000 (d) 2150

102. If the length of the diagonal of a rhombus is 80% of the length of the other diagonal, the area of the rhombus is how many times the square of the length of the longer diagonal?

(a) $\frac{4}{5}$ (b) $\frac{2}{5}$

(c) $\frac{3}{4}$ (d) $\frac{1}{4}$

103. *ABCD* is a trapezium in which $AB \parallel CD$ and $AB = 2CD$. If its diagonals intersect each other at *O*, the ratio of the area of triangle *AOB* and *COD* is:

(a) 1:2 (b) 2:1
(c) 1:4 (d) 4:1

104. The ratio of the corresponding sides of two similar triangles is 3:4. The ratio of their areas is:

(a) 4:3 (b) 3:4
(c) 9:16 (d) $\sqrt{3}:2$

105. The area of the circle inscribed in an equilateral triangle of side 24 cm is:

(a) 24 π cm^2 (b) 36 π cm^2
(c) 48 π cm^2 (d) 18 π cm^2

106. The radius of wheel is 1.4 decimetre. How many times does it revolve during a journey of 0.66 Km?

(a) 375 (b) 750
(c) 1500 (d) 3000

EXERCISE-2
(BASED ON MEMORY)

1. If the length of one side and the diagonal of a rectangle are 7 cm and 25 cm respectively, then find its perimeter (in cm).

(a) 124 (b) 36
(c) 62 (d) 72

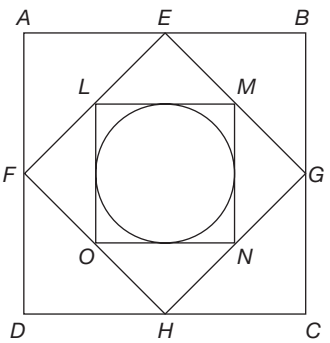
[SSC CHSL (10+2) Tier-I CBE, 2018]

2. Find the perimeter (in cm) of a semicircle od radius 28 cm.

(a) 288 (b) 144
(c) 121 (d) 242

[SSC CHSL (10+2) Tier-I CBE, 2018]

3. In the given figure. ABCD is a square. EFGH is a square formed by joining mid points of sides of ABCD. LMNO is a square fanned by joining mid points of sides of EFGH. A circle is. Inscribed inside EFGH. If area of circles is 38.5 cm^2 then what is the area (in cm^2) of square ABCD?



(a) 98 (b) 196
(c) 122.5 (d) 171.5

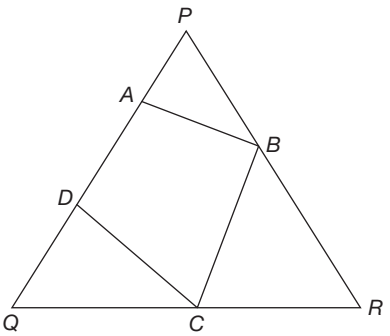
[SSC CGL Tier-II CBE, 2018]

4. ABC is a triangle. AB = 5 cm. AC = $\sqrt{41}$ cm. and BC = 8 cm. AD is perpendicular to BC. What is the area (in cm^2) of triangle ABD?

(a) 12 (b) 6
(c) 10 (d) 20

[SSC CGL Tier-II CBE, 2018]

5. In the given figure, PQR is a triangle and quadrilateral ABCD is inscribed in it QD = 2 cm. QC = 5cm. CR = 3 cm. BR = 4 cm. PB = 6cm. What is the area (In cm^2) of the quadrilateral ABCD?



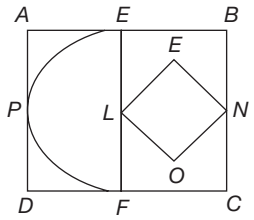
(a) $\frac{(23\sqrt{21})}{4}$ (b) $\frac{(15\sqrt{21})}{4}$

(c) $\frac{(17\sqrt{21})}{5}$ (d) $\frac{(23\sqrt{21})}{5}$

[SSC CGL Tier-II CBE, 2018]

6. In the given figure, ABCD is a square of side 14 cm. E and F are mid-points of sides AB and DC

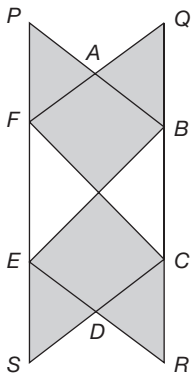
respectively. EPF is a semicircle whose diameter is EF. LMNO is square. What is the area (In cm^2) of the shaded region?



(a) 108.5 (b) 94.5
(c) 70 (d) 120

[SSC CGL Tier-II CBE, 2018]

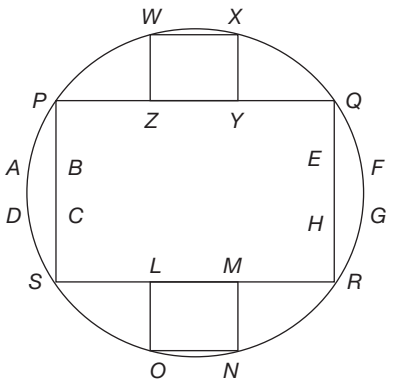
7. In the given figure, ABCDEF is a regular hexagon whose side is 6 cm. APF, QAB, DCR and DES are equilateral triangles. What is the area (In cm^2.) of the shaded region?



(a) $24\sqrt{3}$ (b) $18\sqrt{3}$
(c) $72\sqrt{3}$ (d) $36\sqrt{3}$

[SSC CGL Tier-II CBE, 2018]

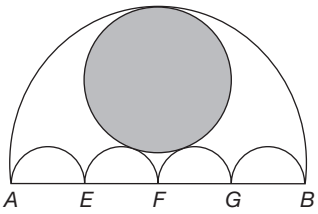
8. In the given figure radius of a circle is $14\sqrt{2}$ cm. PQRS is a square. *EFGH*, *ABCD*. *WXYZ* and *LMNO* are four identical squares. What Is the total area (In cm^2) of all the small squares?



(a) 31.36 (b) 125.44
(c) 62.72 (d) 156.8

[SSC CGL Tier-II CBE, 2018]

9. In the given figure, *AB AE*. *EF*, *FG* and *GB* are semicircles. *AB* = 56 cm. and *AE* = *EF* = *FG* = *GB*. What is the area (In cm) of the shaded region?



(a) 414.46 (b) 382.82
(c) 406.48 (d) 394.24

[SSC CGL Tier-II CBE, 2018]

10. The length and breadth of a rectangle are increased by 10% and 20% respectively. What will be the percentage increase in the area of rectangle?

(a) 30% (b) 32%
(c) 28% (d) 33%

[SSC CHSL (10+2) Tier-I CBE, 2018]

11. A circular wire of length 168 cm is cut and bent in the form of a rectangle whose sides are in the ratio of 5:7. What is the length (in cm) of the diagonal of the rectangle?

(a) $\sqrt{4127}$ (b) $\sqrt{3137}$

(c) $\sqrt{1813}$ (d) $\sqrt{3626}$

[SSC CAPFs ASI & Delhi Police SI, 2017]

12. ΔABC is similar to ΔPQR. Length of AB is 16 cm and length of the corresponding side PQ is 9 cm. If area of ΔABC is 1024sq cm, what is the area of ΔPQR?

(a) 768 sq. cm. (b) 32 sq. cm
(c) 324 sq. cm. (d) 128 sq. cm

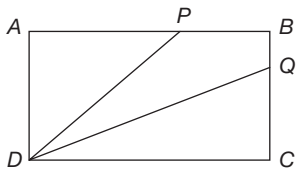
[SSC CHSL (10+2) Tier-I CBE, 2017]

13. In ΔABC, a line parallel to side BC cuts the sides *AB* and *AC* at points *D* and E respectively and also point *D* divides *AB* In the ratio of 1:4. If area of ΔABC is 200 cm^2, then what is the area (in cm^2) of quadrilateral *DECB*?

(a) 192 (b) 50
(c) 120 (d) 96

[SSC CAPFs ASI & Delhi Police SI, 2017]

14. In the given figure, PB is one-third of AB and BQ is one-third of BC. If the area of BPDQ is 20 cm^2 then what is the area (in cm^2) of ABCD?



(a) 45 (b) 30
(c) 40 (d) 60

[SSC CAPFs ASI & Delhi Police SI, 2017]

15. The area of a circle is same as the area of a square. What is the ratio of the diameter of the circle and diagonal of the square?

(a) $1:\sqrt{\pi}$ (b) $2:\sqrt{\pi}$
(c) $\sqrt{2}:\sqrt{\pi}$ (d) 1: π

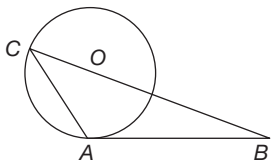
[SSC CAPFs ASI & Delhi Police SI, 2017]

16. In ΔABC, AD and AE are bisectors of $\angle BAC$ and $\angle BAD$ respectively. If $\angle BAE = 30°$, $AE = 9$ cm and $EC = 15$ cm, what is the area (In cm^2) of ΔAEC?

(a) 36 (b) 54
(c) 72 (d) 216

[SSC CAPFs ASI & Delhi Police SI, 2017]

17. In the given figure, triangle ABC is drawn such that AB is tangent to a circle at A whose radius is 10 cm and BC passes through centre of the circle. Point C lies on the .circle. If BC = 36 cm and AB = 24 cm, then what is the area (in cm^2) of triangle ABC?



(a) 134.5 (b) 148
(c) 168 (d) 180

[SSC CGL Tier-I CBE, 2017]

18. The tangents drawn at points A and B of a circle with centre O, meet at P If ∠AOB = 120° .and AP = 6 cm, then what is the area of triangle APB (in cm^2)?

(a) $6\sqrt{3}$ (b) $8\sqrt{3}$
(c) 9 (d) $9\sqrt{3}$

[SSC CGL Tier-I CBE, 2017]

19. D and E are points on sides and AC of ΔABC. DE is parallel to BC. If.AD: DB = 2: 3, what is the ratio of area of ΔADE and area of quadrilateral BDEC?

(a) 4: 21 (b) 4: 25
(c) 4: 29 (d) 4: 9

[SSC CHSL (10+2) Tier-I CBE, 2017]

20. Three circles of radius 21 cm are placed in such a way· that each circle touches the other two. What is the area of the portion enclosed by the three circles? (In sq. cm)

(a) $441\sqrt{3}-693$ (b) $882\sqrt{3}-693$
(c) $882\sqrt{3}-462$ (d) $441\sqrt{3}-462$

[SSC CAPFs ASI & Delhi Police SI, 2017]

21. If the area of a square is increased by 44%, retaining its shape as a square, each of its sides increases by:

(a) 19% (b) 21%
(c) 22% (d) 20%

[SSC CAPFs (CPO) SI & ASI Delhi Police, 2017]

22. A cylindrical vessel of height 5 cm and radius 4 cm is completely filled with sand. When this sand is poured out it forms a right circular cone of radius 6 cm. what will be the height of this cone?

(Take $\pi + \frac{22}{7}$)

(a) 6.67 cm (b) 2.22 cm
(c) 3.33 cm (d) 1.67 cm

[SSC CHSL (10+2) Tier-I (CBE), 2017]

23. A solid metallic sphere of radius 21 cm is melted and recast into a cone with diameter of the base as 21 cm. What is the height (in cm) of the cone?

(a) 336 (b) 112
(c) 224 (d) 66

[SSC CAPFs ASI & Delhi Police SI, 2017]

24. Three equal circles of unit radius touch one another. Then the area of the circle circumscribing in the three circles is

(a) $6\pi(2+\sqrt{3})^2$ (b) $\frac{\pi}{6}(2+\sqrt{3})^2$
(c) $\frac{\pi}{3}(2+\sqrt{3})^2$ (d) $3\pi(2+\sqrt{3})^2$

[SSC CPO, 2016]

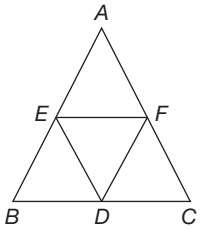
25. In Δ ABC and Δ PQR ∠B = ∠Q, ∠C = ∠R. M is the mid-point of side QR. If AB:PQ = 7:4, then $\frac{\text{area}\,(\Delta ABC)}{\text{area}(\Delta PMR)}$ is

(a) $\frac{1}{2}$ (b) 1

(c) $\frac{1}{3}$ (d) $\frac{1}{4}$

[SSC CAPFs (CPO) SI & ASI, Delhi Police, 2016]

26. If D, E and F are the mid-points of the sides of an equilateral triangle ABC, then the ratio of the area of triangle DEF and DCF is:

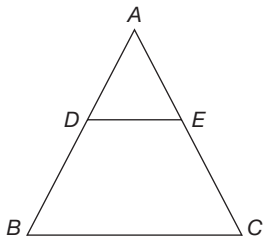


(a) 1.1:1 (b) 1:1.1

(c) 0.9:1 (d) 1:1

[SSC CGL Tier-I (CBE), 2016]

27. In figure, $DE \parallel BC$. If DE = 3cm BC = 6 cm and area of ΔADE = 15 sq. cm, then the area of ΔABC is



(a) 75 sq. cm. (b) 45 sq. cm.

(c) 30 sq. cm. (d) 60 sq. cm.

[SSC CGL Tier-I CBE, 2016]

28. The length and breadth of a rectangular place of a land are in a ratio 5: 3. The owner spent ₹6000 for surrounding it from all sides at ₹7.50 per metre. The difference between its length and breadth is

(a) 50 metre (b) 100 metre

(c) 150 metre (d) 250 metre

[SSC CGL Tier-II (CBE), 2016]

29. The length of the two parallel sides of a trapezium are 16m and 20m respectively. If its height is 10m its area in square metre is

(a) 360 (b) 260

(c) 240 (d) 180

[SSC CGL Tier-I (CBE), 2016]

30. Three medians AD, BE and CF of ΔABC intersect at G. The area of ΔABC is 36 sq. cm. Then the area of Δ CGE is

(a) 12 sq. cm. (b) 6 sq. cm.

(c) 9 sq. cm. (d) 18 sq. cm.

[SSC CGL Tier-I (CBE), 2016]

31. A rectangular plot has a concrete path running in the middle of the plot parallel to the breadth of the plot. The rest of the plot is used as a lawn, which has an area of 240 sq m. If the width of the path is 3m and the length of the path is greater than its breadth by 2m, what is the area of the rectangular plot? (in sq m)

(a) 255 (b) 168

(c) 288 (d) 360

(e) 224

[LIC, 2015]

32. The sides of a triangle having area 7776 sq. cm are in the ratio 3:4:5. The perimeter of the triangle is

(a) 400 cm (b) 412 cm

(c) 424 cm (d) 432 cm

[SSC, 2015]

33. In ΔABC, a line through A cuts the side BC at D such that BD:DC = 4:5. If the area of ΔABD = 60 cm^2, then the area of ΔADC is

(a) 50 cm^2 (b) 60 cm^2

(c) 75 cm^2 (d) 90 cm^2

[SSC, 2015]

34. The area of an isosceles trapezium is 176 cm^2 and the height is $2/11^{th}$ of the sum of its parallel sides. If the ratio of the length of the parallel sides is 4 : 7, then the length of a diagonal (in cm) is

(a) 28 (b) 24

(c) $\sqrt{137}$ (d) $2\sqrt{137}$

[SSC, 2015]

35. In trapezium ABCD, AB || CD and AB = 2CD. Its diagonals intersect at O. If the area of Δ AOB = 84 cm^2, then the area of Δ COD is equal to

(a) 21 cm^2 (b) 42 cm^2

(c) 72cm^2 (d) 26 cm^2

[SSC, 2015]

36. The perimeter of a rhombus is 60 cm and one of its diagonal is 24 cm. The area (in sq.cm) of the rhombus is

(a) 432 (b) 216

(c) 108 (d) 206

[SSC, 2015]

37. ABCD is a cyclic quadrilateral. AB and DC when produced meet at P, if PA = 8 cm, PB = 6 cm, PC = 4 cm, then the length (in cm) of PD is

(a) 8 (b) 10
(c) 12 (d) 6

[SSC, 2015]

38. Two circles touch externally. The sum of their areas is 130π sq cm and the distance between their centres is 14cm. The radius of the smaller circle is:

(a) 3 cm (b) 4 cm
(c) 5 cm (d) 2 cm

[SSC, 2015]

39. A circular swimming pool is surrounded by a concrete wall 4m wide. If the area of the concrete wall surrounding the pool is 11/25 that of the pool, then the radius (in m) of the pool is:

(a) 16 (b) 20
(c) 30 (d) 8

[SSC, 2015]

40. AB and CD are two parallel chords of a circle lying on the opposite side of the centre and the distance between them is 17cm. The length of AB and CD are 10cm and 24 cm respectively. The radius (in cm) of the circle is:-

(a) 18 (b) 15
(c) 13 (d) 9

[SSC, 2015]

41. A conical iron piece having diameter 28cm and height 30cm is totally immersed into the water of a cylindrical vessel, resulting in the rise of water level by 6.4cm. The diameter, in cm, of the vessel is:-

(a) 35 (b) 32
(c) 3.5 (d) $\frac{35}{2}$

[SSC, 2015]

42. If the altitude of an equilateral triangle is $12\sqrt{3}$ cm, then its area would be:

(a) 12 cm^2 (b) 72 cm^2
(c) $36\sqrt{3}$ cm^2 (d) $144\sqrt{3}$ cm^2

[SSC, 2015]

43. *ABCD* is a trapezium with *AD* and *BC* parallel sides. *E* is a point on *BC*. The ratio of the area of *ABCD* to that of *AED* is:

(a) $\frac{\overline{AD}}{\overline{BC}}$ (b) $\frac{\overline{BE}}{\overline{EC}}$
(c) $\frac{\overline{AD}+\overline{BE}}{\overline{AD}+\overline{CE}}$ (d) $\frac{\overline{AD}+\overline{BC}}{AD}$

[SSC, 2014]

44. In an equilateral triangle of side 24 cm, a circle is inscribed touching its sides. The area of the remaining portion of the triangle is $\left(\sqrt{3}=1.732\right)$:

(a) 98.55 cm^2 (b) 100 cm^2
(c) 101 cm^2 (d) 95 cm^2

[SSC, 2014]

45. Perimeter of a rhombus is 2*p* units and sum of length of diagonals is *m* units, then area of the rhombus is:

(a) $\frac{1}{4}m^2p$ sq units

(b) $\frac{1}{4}mp^2$ sq units

(c) $\frac{1}{4}(m^2-p^2)$ sq units

(d) $\frac{1}{4}(p^2-m^2)$ sq units

[SSC, 2014]

46. Two sides of a plot measuring 32 *m* and 24 *m* and the angle between them is a perfect right angle. The other two sides measure 25 *m* each and the other three angles are not right angles. The area of the plot in m^2 is:

(a) 768 (b) 534
(c) 696.5 (d) 684

[SSC, 2014]

47. A is the centre of circle whose radius is 8 and B is the centre of a circle whose diameter is 8. If these two circles touch externally, then the area of the circle with diameter AB is:

(a) 36 π (b) 64 π
(c) 144 π (d) 256 π

[SSC, 2014]

48. The perimeters of a circle, a square and an equilateral triangle are same and their areas are C, S and T respectively. Which of the following statement is true?

(a) C = S = T (b) C > S > T
(c) C < S < T (d) S < C < T

[SSC, 2014]

49. A wire of length 44 cm is first bent to form a circle and then rebent to form a square. The difference of the two enclosed areas is

(a) 44 cm^2 (b) 33 cm^2
(c) 55 cm^2 (d) 66 cm^2

[SSC, 2014]

50. ABCD is a parallelogram in which diagonals AC and BD intersect at O. If E, F, G and H are the mid points of AO, DO, CO and BO respectively, then the ratio of the perimeter of the quadrilateral EFGH to the perimeter of parallelogram ABCD is

(a) 1 : 4 (b) 2 : 3
(c) 1 : 2 (d) 1 : 3

[SSC, 2014]

51. On decreasing each side of an equilateral triangle by 2 cm, there is a decrease of $4\sqrt{3}$ cm^2 in its area. The length of each side of the triangle is

(a) 6 cm (b) 8 cm
(c) 3 cm (d) 5 cm

[SSC, 2014]

52. A circular wire of diameter 112 cm is cut and bent in the form of a rectangle whose sides are in the ratio of 9 : 7. The small side of the rectangle is

(a) 67 cm (b) 87 cm
(c) 77 cm (d) 97 cm

[SSC, 2014]

53. A parallelogram has sides 60 m and 40 m and one of its diagonals is 80 m long. Its area is

(a) $400\sqrt{15}$ m^2 (b) $450\sqrt{15}$ m^2
(c) $500\sqrt{15}$ m^2 (d) $600\sqrt{15}$ m^2

[SSC, 2014]

54. A lawn is in the form of a rectangle having its breadth and length in the ratio 3:4. The area of the lawn is $\frac{1}{12}$ hectare. The breadth of the lawn is:

(a) 25 metres (b) 50 metres
(c) 75 metres (d) 100 metres

[SSC, 2013]

55. The area of a rectangle is thrice that of a square. The length of the rectangle is 20 cm and the breadth of the rectangle is $\frac{3}{2}$ times that of the side of the square. The side of the square (in cm) is:

(a) 10 (b) 20
(c) 30 (d) 60

[SSC, 2013]

56. The diagonals of a rhombus are 12 cm and 16 cm. The length of one side is:

(a) 8 cm (b) 6 cm
(c) 10 cm (d) 12 cm

[SSC, 2013]

57. The diameter of a circular wheel is 7 m. How many revolutions will it make in travelling 22 Km?

(a) 100 (b) 400
(c) 500 (d) 1000

[SSC, 2013]

58. The area of an equilateral triangle is $9\sqrt{3}$ m^2. The length (in m) of the median is:

(a) $2\sqrt{3}$ (b) $3\sqrt{3}$
(c) $3\sqrt{2}$ (d) $2\sqrt{2}$

[SSC, 2013]

59. How many tiles, each 4 decimetres square, will be required to cover the floor of a room 8 m long and 6 m broad?

(a) 200 (b) 260
(c) 280 (d) 300

[SSC, 2013]

60. The area of the circumcircle of an equilateral triangle is 3π cm^2. The perimeter of the triangle is:

(a) $3\sqrt{3}$ cm (b) 9 cm
(c) 18 cm (d) 3 cm

[SSC Assistant Grade III, 2013]

61. In ΔABC, $\angle A = 90°$ and $AD \perp BC$ where D lies on BC. If $BC = 8$ cm, $AC = 6$ cm, then $\Delta ABC : \Delta ACD$ is:

(a) 4:3 (b) 25:16
(c) 16:9 (d) 25:9

[SSC Assistant Grade III, 2013]

62. The sides of a triangle are in the ratio $\frac{1}{4}:\frac{1}{6}:\frac{1}{8}$ and its perimeter is 91 cm. The difference of the length of the longest side and that of the shortest side is:

(a) 19 (b) 20
(c) 28 (d) 21

[SSC Assistant Grade III, 2013]

63. The perimeter of an isosceles right-angled triangle is $2p$ cm. Its area is:

(a) $(3+2\sqrt{2})p$ cm^2 (b) $(3-2\sqrt{2})p^2$ cm^2
(c) $(2-\sqrt{2})p$ cm^2 (d) $(2+\sqrt{2})p^2$ cm^2

[SSC Assistant Grade III, 2013]

64. The ratio between the areas of two circles is 4:7. What will be the ratio of their radii?

(a) $2:\sqrt{7}$ (b) 4:7

(c) 16:49 (d) $4:\sqrt{7}$

[SSC Assistant Grade III, 2013]

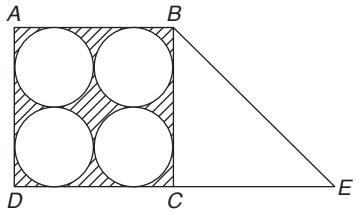
65. The perimeter of a non-square rhombus is 20 cm. One of its diagonals is 8 cm. The area of the rhombus is:

(a) 28 cm^2 (b) 20 cm^2

(c) 22 cm^2 (d) 24 cm^2

[SSC Assistant Grade III, 2013]

Directions (66–67): Study the following diagram to answer the questions.



66. If the diameter of each circle is 14 cm and $DC = CE$, the area of ΔBDE is:

(a) 784 cm^2 (b) 748 cm^2

(c) 874 cm^2 (d) 441 cm^2

(e) None of these

[IBPS PO/MT, 2013]

67. The area of the shaded region of square $ABCD$ is:

(a) 186 cm^2 (b) 168 cm^2

(c) 188 cm^2 (d) 441 cm^2

(e) None of these

[IBPS PO/MT, 2013]

68. What is the area of a given right-angled triangle?

I. The length of the hypotenuse is 5 cm.

II. The perimeter of the triangle is four times that of itsbase.

III. Oneof the angles of the triangle is 60°.

(a) Only II (b) Only III

(c) Either II or III (d) Both I and III

(e) Question cannot be answered even with the information in all three statements

[IBPS PO/MT, 2013]

69. The sides of a triangle are 50 cm, 78 cm and 112 cm. The smallest altitude is:

(a) 20 cm (b) 30 cm

(c) 40 cm (d) 50 cm

[SSC Assistant Grade III, 2012]

70. In a triangle ABC, $AB + BC = 12$ cm, $BC + CA = 14$ cm and $CA + AB = 18$ cm. Find the radius of the circle (in cm) which has the same perimeter as the triangle.

(a) $\frac{5}{2}$ (b) $\frac{7}{2}$

(c) $\frac{9}{2}$ (d) $\frac{11}{2}$

[SSC, 2012]

71. A playground is in the shape of a rectangle. A sum of ₹1,000 was spent to make the ground usable at the rate of 25 paisa per m^2. The breadth of the ground is 50 m. If the length of the ground is increased by 20 m, what will be the expenditure in rupees at the same rate per m^2?

(a) 1,250 (b) 1,000

(c) 1,500 (d) 2,250

[SSC, 2012]

72. The lengths of three medians of a triangle are 9 cm, 12 cm and 15 cm. The area (in cm^2) of the triangle is:

(a) 24 (b) 72

(c) 48 (d) 144

[SSC, 2012]

73. A circle and a rectangle have the same perimeter. The sides of the rectangle are 18 cm and 26 cm. The area of the circle is $\left[\text{Take } \pi = \frac{22}{7}\right]$

(a) 125 cm^2 (b) 230 cm^2

(c) 550 cm^2 (d) 616 cm^2

[SSC, 2012]

74. The area of a circle is increased by 22 cm^2 when its radius is increased by 1 cm. The original radius of the circle is:

(a) 3 cm (b) 5 cm

(c) 7 cm (d) 9 cm

[SSC, 2012]

75. The sum of all interior angles of a regular polygon is twice the sum of all its exterior angles. The number of sides of the polygon is:

(a) 10 (b) 8

(c) 12 (d) 6

[SSC, 2012]

76. If the diagonals of a rhombus are 8 and 6, then the square of its size is:

(a) 25 (b) 55

(c) 64 (d) 36

[SSC, 2012]

77. The area of the square inscribed in a circle of radius 8 cm is:

(a) 256 cm^2 (b) 250 cm^2
(c) 128 cm^2 (d) 125 cm^2

[SSC, 2012]

78. The area of a square is 1444 square metres. The breadth of a rectangle is $\frac{1}{4}$ the side of the square and the length of the rectangle is thrice its breadth. What is the difference between the area of the square and the area of the rectangle?

(a) 1152.38 m^2 (b) 1169.33 m^2
(c) 1181.21 m^2 (d) 1173.25 m^2
(e) None of these

[IBPS PO/MT, 2012]

79. What would be the cost of laying a carpet on a floor which has its length and breadth in the respective ratio of 32:21 and where its perimeter is 212 feet, if the cost per square foot of laying the carpet is ₹2.5?

(a) ₹6720 (b) ₹5420
(c) ₹7390 (d) Cannot be determined

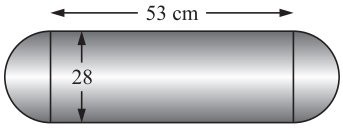
[Rajasthan Grameen Bank PO, 2011]

80. The circumference of 2 circles is 83 m and 220 m, respectively. What is the difference between the area of the larger circle and the smaller circle?

(a) 3422 m^2 (b) 3242 m^2
(c) 3244 m^2 (d) None of these

[Corporation Bank PO, 2011]

81. What is the area of the following figure?



(a) 2504 cm^2 (b) 1940 cm^2
(c) 2100 cm^2 (d) Cannot be determined

82. The length of a rectangle is twice the diameter of a circle. The circumference of the circle is equal to the area of a square of side 22 cm. What is the breadth of the rectangle if its perimeter is 668 cm?

(a) 24 cm (b) 26 cm
(c) 52 cm (d) Cannot be determined

[Union Bank of India PO, 2011]

83. A square is of area 200 m^2. A new square is formed in such a way that the length of its diagonal is $\sqrt{2}$ times of the diagonal of the given square. Then the area of the new square formed is:

(a) $200\sqrt{2}$ m^2 (b) $400\sqrt{2}$ m^2
(c) 400 m^2 (d) 800 m^2

[SSC, 2011]

84. If the length of a rectangular field is increased by 20% and the breadth is reduced by 20%, the area of the rectangle will be 192 m^2. What is the area of the original rectangle?

(a) 184 m^2 (b) 196 m^2
(c) 204 m^2 (d) 225 m^2
(e) None of these

[SBI Associates Banks PO, 2011]

85. Inside a square plot, a circular garden is developed which exactly fits in the square plot and the diameter of the garden is equal to the side of the square plot which is 28 m. What is the area of the space left out in the square plot after developing the garden?

(a) 98 m^2 (b) 146 m^2
(c) 84 m^2 (d) 168 m^2
(e) None of these

[SBI Associates Banks PO, 2011]

86. What is the cost of flooring a rectangular hall?

Statements:

I. The length of the rectangle is 6 meters.

II. The breadth of the rectangle is 2/3 of its length.

III. The cost of flooring the area of 100 cm^2 is ₹45.

(a) Only I and III
(b) Only II and III
(c) All I, II and III
(d) Question cannot be answered even with data in all three statements.
(e) None of these

[SBI Associates Banks PO, 2011]

87. The length of a rectangle is 3/5 of the side of a square. The radius of a circle is equal to side of the square. The circumference of the circle is 132 cm. What is the area of the rectangle if the breadth of the rectangle is 8 cm?

(a) 112.4 cm^2 (b) 104.2 cm^2
(c) 100.8 cm^2 (d) Cannot be determined
(e) None of these

[IOB PO, 2011]

88. The smallest side of a right-angled triangle is 8 cm less than the side of a square of perimeter 56 cm. The second largest side of the right-angled triangle is 4 cm less than the length of the rectangle of area 96 cm^2 and breadth 8 cm. What is the largest side of the right-angled triangle?

(a) 20 cm (b) 12 cm
(c) 10 cm (d) 15 cm
(e) None of these

[IOB PO, 2011]

89. The sum of the circumference of a circle and the perimeter of a square is equal to 272 cm. The diameter of the circle is 56 cm. What is the sum of the areas of the circle and the square?

(a) 2464 cm^2 (b) 2644 cm^2
(c) 3040 cm^2 (d) Cannot be determined
(e) None of these

[Allahabad Bank PO, 2011]

90. The largest and the second largest angles of a triangle are in the ratio of 4: 3. The smallest angle is half the largest angle. What is the difference between the smallest and the largest angles of the triangle?

(a) 30° (b) 60°
(c) 40° (d) 20°
(e) None of these

[Allahabad Bank PO, 2011]

91. The ratio of the 3 angles of a quadrilateral is 13:9:5. The value of the 4 angle of the quadrilateral is 36°. What is the difference between the largest and the second smallest angles of the quadrilateral?

(a) 104° (b) 108°
(c) 72° (d) 96°
(e) None of these

[Allahabad Bank PO, 2011]

92. The circumference of two circles is 88 metres and 220 metres respectively. What is the difference between the area of the larger circle and that of the smaller circle?

(a) 3422 m^2 (b) 3242 m^2
(c) 3244 m^2 (d) 3424 m^2
(e) None of these

[Corporation Bank PO, 2011]

93. The area of a square is 196 cm^2. whose side is half the radius of a circle. The circumference of the circle is equal to breadth of a rectangle, if perimeter of the rectangle is 712 cm. What is the length of the rectangle?

(a) 196 cm^2 (b) 186 cm^2
(c) 180 cm^2 (d) 190 cm^2

[OBC PO, 2010]

94. The sides of a right-angled triangle forming right angle are in the ratio 5:12. If the area of the triangle is 270 cm^2, then the length of the hypotenuse is:

(a) 39 cm (b) 42 cm
(c) 45 cm (d) 51 cm

[SSC, 2010]

95. If the measures of a diagonal and the area of a rectangle are 25 cm and 168 cm^2 respectively, what is the length of the rectangle?

(a) 31 cm (b) 24 cm
(c) 17 cm (d) 27 cm

[SSC, 2010]

96. A General, while arranging his men, who were 6000 in number, in the form of a square, found that there were 71 men leftover. How many were arranged in each row?

(a) 73 (b) 77
(c) 87 (d) 93

[SSC, 2010]

97. If the length of a rectangle is increased by 10% and its breadth is decreased by 10%, then its area:

(a) decreases by 1% (b) increases by 1%
(c) decreases by 2% (d) remains unchanged

[SSC, 2010]

98. If the length of a rectangle is increased in the ratio 6:7 and its breadth is diminished in the ratio 5:4 then its area will be diminished in the ratio:

(a) 17:16 (b) 15:14
(c) 9:8 (d) 8:7

[SSC, 2010]

99. 3 horses are tethered at 3 corners of a triangular plot of land having sides 20 m, 30 m and 40 m each with a rope of length 7 m. The area (in m^2) of the region of this plot, which can be grazed by the horses, is:

$\left(\text{Use } \pi = \frac{22}{7}\right)$

(a) $\frac{77}{3}$ (b) 75
(c) 77 (d) 80

[SSC, 2010]

100. A wire, when bent in the form of a square, encloses a region of area 121 cm^2. If the same wire is bent into the form of a circle, then the area of the circle is:

$\left(\text{Use } \pi = \frac{22}{7}\right)$

(a) 150 cm^2 (b) 152 cm^2
(c) 154 cm^2 (d) 159 cm^2

[SSC, 2010]

101. The ratio of the area of a sector of a circle to the area of the circle is 1:4. If the area of the circle is 154 cm^2, the perimeter of the sector is:

(a) 20 cm (b) 25 cm
(c) 36 cm (d) 40 cm

[SSC, 2010]

102. The sum of the areas of the 10 squares, the lengths of whose sides are 20 cm, 21 cm, ..., 29 cm respectively is:

(a) 6085 cm^2 (b) 8555 cm^2
(c) 2470 cm^2 (d) 11025 cm^2

[SSC, 2010]

103. The angles of a quadrilateral are in the ratio of 2:4:7:5. The smallest angle of the quadrilateral is equal to the smallest angle of a triangle. One of the angles of the triangle is twice the smallest angle of the triangle. What is the second largest angle of the triangle?

(a) 80° (b) 60°
(c) 120° (d) Cannot be determined
(e) None of these

[Central Bank of India PO, 2010]

104. The area of a square is 1024 cm^2. What is the ratio of the length to the breadth of a rectangle whose length is twice the side of the square and breadth is 12 cm less than the side of this square?

(a) 5:18 (b) 16:7
(c) 14:5 (d) 32:5
(e) None of these

[Central Bank of India PO, 2010]

105. The perimeter of a square is equal to twice the perimeter of a rectangle whose dimensions are: length 8 cm and breadth 7 cm. What is the circumference of a semicircle whose diameter is equal to the side of the square?

(Rounded off of the decimal place)

(a) 38.57 cm (b) 23.57 cm
(c) 42.46 cm (d) 47.47 cm
(e) None of these

[Punjab and Sind Bank PO, 2010]

106. The circumferences of two circles are 132 metres and 176 metres respectively. What is the difference between the area of the larger circle and that of the smaller circle?

(a) 1048 m^2 (b) 1076 m^2
(c) 1078 m^2 (d) 1090 m^2
(e) None of these

[Indian Bank PO, 2010]

ANSWER KEYS

EXERCISE–1

1. (c)	**2.** (a)	**3.** (b)	**4.** (b)	**5.** (c)	**6.** (a)	**7.** (b)	**8.** (c)	**9.** (a)	**10.** (c)	**11.** (c)	**12.** (c)	**13.** (c)
14. (c)	**15.** (a)	**16.** (c)	**17.** (c)	**18.** (b)	**19.** (a)	**20.** (b)	**21.** (b)	**22.** (c)	**23.** (b)	**24.** (c)	**25.** (c)	**26.** (a)
27. (c)	**28.** (a)	**29.** (c)	**30.** (b)	**31.** (a)	**32.** (b)	**33.** (c)	**34.** (a)	**35.** (b)	**36.** (a)	**37.** (a)	**38.** (c)	**39.** (b)
40. (b)	**41.** (a)	**42.** (d)	**43.** (d)	**44.** (c)	**45.** (a)	**46.** (c)	**47.** (d)	**48.** (c)	**49.** (c)	**50.** (d)	**51.** (c)	**52.** (b)
53. (c)	**54.** (c)	**55.** (a)	**56.** (a)	**57.** (c)	**58.** (c)	**59.** (b)	**60.** (b)	**61.** (a)	**62.** (c)	**63.** (b)	**64.** (a)	**65.** (d)
66. (d)	**67.** (b)	**68.** (c)	**69.** (c)	**70.** (c)	**71.** (c)	**72.** (b)	**73.** (c)	**74.** (b)	**75.** (b)	**76.** (d)	**77.** (b)	**78.** (c)
79. (d)	**80.** (a)	**81.** (b)	**82.** (d)	**83.** (a)	**84.** (b)	**85.** (b)	**86.** (d)	**87.** (a)	**88.** (c)	**89.** (a)	**90.** (c)	**91.** (b)
92. (a)	**93.** (b)	**94.** (c)	**95.** (b)	**96.** (a)	**97.** (b)	**98.** (b)	**99.** (b)	**100.** (d)	**101.** (c)	**102.** (b)	**103.** (d)	**104.** (c)
105. (c)	**106.** (b)											

EXERCISE–2

1. (c)	**2.** (b)	**3.** (b)	**4.** (a)	**5.** (c)	**6.** (c)	**7.** (c)	**8.** (a)	**9.** (d)	**10.** (b)	**11.** (d)	**12.** (c)	**13.** (a)
14. (d)	**15.** (c)	**16.** (b)	**17.** (c)	**18.** (d)	**19.** (a)	**20.** (a)	**21.** (d)	**22.** (b)	**23.** (a)	**24.** (c)	**25.** (c)	**26.** (d)
27. (d)	**28.** (b)	**29.** (d)	**30.** (b)	**31.** (a)	**32.** (d)	**33.** (c)	**34.** (d)	**35.** (a)	**36.** (b)	**37.** (a)	**38.** (a)	**39.** (b)
40. (c)	**41.** (a)	**42.** (d)	**43.** (d)	**44.** (a)	**45.** (c)	**46.** (d)	**47.** (a)	**48.** (b)	**49.** (b)	**50.** (c)	**51.** (d)	**52.** (c)
53. (d)	**54.** (a)	**55.** (a)	**56.** (c)	**57.** (d)	**58.** (b)	**59.** (d)	**60.** (b)	**61.** (c)	**62.** (d)	**63.** (b)	**64.** (a)	**65.** (d)
66. (a)	**67.** (b)	**68.** (d)	**69.** (b)	**70.** (b)	**71.** (a)	**72.** (b)	**73.** (d)	**74.** (a)	**75.** (d)	**76.** (a)	**77.** (c)	**78.** (d)
79. (a)	**80.** (d)	**81.** (c)	**82.** (b)	**83.** (c)	**84.** (e)	**85.** (d)	**86.** (c)	**87.** (c)	**88.** (c)	**89.** (c)	**90.** (c)	**91.** (d)
92. (e)	**93.** (c)	**94.** (a)	**95.** (b)	**96.** (b)	**97.** (a)	**98.** (b)	**99.** (c)	**100.** (c)	**101.** (b)	**102.** (a)	**103.** (b)	**104.** (e)
105. (a)	**106.** (c)											

EXPLANATORY ANSWERS

EXERCISE-1

1. (c) Let, a = 15 m, b = 16 m, c = 17 m.

Then, $s = \frac{a+b+c}{2} = \frac{15+16+17}{2}$ = 24 m.

$\therefore$ Area $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{24(24-15)(24-16)(24-17)}$

$= \sqrt{24\times 9\times 8\times 7}$

$= \sqrt{12096} = 24\sqrt{21}$ m^2.

2. (a) Height of the triangle $= \sqrt{(6.5)^2 - 6^2}$

$= \sqrt{42.25-36}$

$= \sqrt{6.25}$ = 2.5 m

$\therefore$ Area $= \frac{1}{2}$ (base $\times$ height)

$= \frac{1}{2} \times 6 \times 2.5 = 7.5$ m^2.

3. **(b)** Each side = 12 cm

Then, $s = \frac{12+12+12}{2} = 18$.

Area $= \sqrt{18(18-12)(18-12)(18-12)}$

$= \sqrt{18\times6\times6\times6} = \frac{1}{2} \times 12 \times$ height

or, height $= \frac{36\sqrt{3}}{6} = 6\sqrt{3}$ cm.

4. **(b)** Let, the length of equal sides be x.

Then, $s = \frac{x+x+64}{2} = x + 32$.

Area = 1600 m^2.

$= \sqrt{(x+32)(x+32-x)(x+32-x)(x+32-64)}$

$= \sqrt{(x+32)\times32\times32\times(x-32)}$

or, $1600 = 32\sqrt{x^2-32^2}$

$\Rightarrow \sqrt{x^2-32^2} = 50$

or, $x^2 = 32^2 + 50^2 = 1024 + 2500 = 3524$.

$\therefore$ $x = 59.36$ m.

5. **(c)** Here, $s = \frac{20+21+29}{2} = \frac{70}{2} = 35$ m.

$\therefore$ Area $= \sqrt{35(35-20)(25-21)(35-29)}$

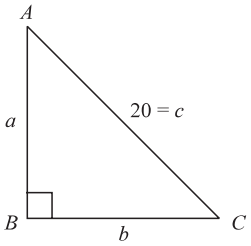
$= \sqrt{35\times15\times14\times6}$

$= \sqrt{5^2\times7^2\times3^2\times2^2}$

$= 5 \times 7 \times 3 \times 2 = 210$ m^2.

6. **(a)** We have, $a + b + 20 = 2 \times 24$

$\therefore$ $a + b = 28$ …(1)



By Pythagores Theorem, $a^2 + b^2 = 400$

$\because$ $(a + b)^2 = 28^2$

$\Rightarrow ab = \frac{28^2-(a^2+b^2)}{2} = \frac{28^2-400}{2} = 192$

$\therefore$ $a - b = \sqrt{(a+b)^2-4ab}$

$= \sqrt{784-4\times192} = 4$ …(2)

Solving (1) and (2), we get

$a = 16$ cm and $b = 12$ cm.

7. **(b)** The sides of the triangle are

$a = \frac{3}{12} \times 36 = 9$ cm, $b = \frac{4}{12} \times 36 = 12$ cm

$c = \frac{5}{12} \times 36 = 15$ cm

$s = \frac{a+b+c}{2} = \frac{36}{2} = 18$ cm

$\therefore$ Area $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{18(18-9)(18-12)(18-15)}$

$= \sqrt{18\times9\times6\times3}$

$= \sqrt{9^2\times2^2\times3^2} = 9 \times 2 \times 3 = 54$ cm^2.

8. **(c)** We have, $2s = 50 + 78 + 112 = 240$

$\therefore$ $s = 120$.

$\therefore$ Area $= \sqrt{120(120-50)(120-78)(120-112)}$

$= \sqrt{120\times70\times42\times8}$

$= \sqrt{2^3\times5\times3\times7\times5\times2\times2\times7\times3\times2^3}$

$= 2^4 \times 5 \times 3 \times 7 = 1680$ m^2

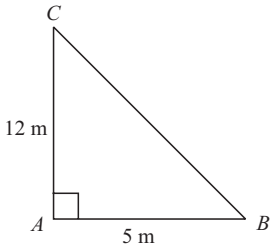
$\therefore$ $\frac{1}{2} \times 112 \times h = 1680$,

where h is the length of perpendicular

$\therefore$ $h = \frac{1680\times2}{112} = 30$ m.

9. **(a)** Let, BC be the ladder.

Then, $BC = \sqrt{12^2+5^2} = 13$ m.

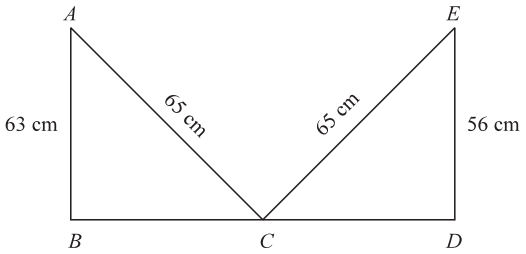


10. **(c)** $BC = \sqrt{(65)^2-(63)^2} = 16$

$CD = \sqrt{(65)^2-(56)^2} = 33$.

$\therefore$ Width of the street

$= 16 + 33 = 49$ cm.



11. **(c)** Given: $\frac{1}{2} \times x \times h = x^2$

$\therefore$ $h = 2x$.

12. (c) Let, the base be $3x$.

Then, the height is $4x$.

Given: $\frac{1}{2} \times 3x \times 4x = 150 \Rightarrow x = 5$.

$\therefore$ Base = $3 \times 5 = 15$ m and height = $4 \times 5 = 20$ m.

13. (c) Area of field = $\frac{495.72}{36.72} \times 10000$

$= 135000 \text{ m}^2$

Let, the height be x.

Then, base = $3x$.

We have, $\frac{1}{2} \times x \times 3x = 135000$

$\Rightarrow x^2 = 90000$ or $x = 300$.

$\therefore$ Height = 300 m and base = $3 \times 300 = 900$ m.

14. (c) Let, the original sides be a, b, c, then

$s = \frac{1}{2}(a + b + c)$

and area of the triangle = $\sqrt{s(s-a)(s-b)(s-c)}$

For the new triangle, the sides are $2a$, $2b$, $2c$.

Then, $S = \frac{1}{2}(2a + 2b + 2c) = a + b + c = 2s$.

$\therefore$ Area of new triangle

$= \sqrt{S(S-2a)(S-2b)(S-2c)}$

$= \sqrt{2s(2s-2a)(2s-2b)(2s-2c)}$

$= \sqrt{16s(s-a)(s-b)(s-c)}$

$= 4\sqrt{s(s-a)(s-b)(s-c)}$

$= 4 \times$ (area of original triangle).

15. (a) The third side of the triangle

$= 324 - (85 + 154) = 85$ m.

Also, $s = \frac{a+b+c}{2} = \frac{324}{2} = 162$.

$\therefore$ Area of the triangle

$= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{162(162-85)(162-85)(162-154)}$

$= \sqrt{162 \times 77 \times 77 \times 8} = 2772 \text{ m}^2$.

$\therefore$ The cost of ploughing the field

$= 2772 \times 10 = ₹27720$.

16. (c) The area of the equilateral triangle

$= \frac{\sqrt{3}}{4}(\text{side})^2 = \frac{\sqrt{3}}{4} \times (2\sqrt{3})^2$

$= 3\sqrt{3} \text{ cm}^2$.

17. (a) Height of the equilateral triangle = $\frac{\sqrt{3}}{2} \times$ (side)

$\Rightarrow 2\sqrt{3} = \frac{\sqrt{3}}{2} \times$ (side)

$\therefore$ Side of the equilateral triangle = 4 cm.

18. (b) $3 \times$ (side) $= 12 \Rightarrow$ side = 4 m.

$\therefore$ Area of equilateral triangle = $\frac{\sqrt{3}}{4}(\text{side})^2$

$= \frac{\sqrt{3}}{4} \times 16 = 4\sqrt{3}$ m.

19. (a) $3 \times$ (side) $= 24 \Rightarrow$ side = 8 cm.

$\therefore$ Height of the equilateral triangle

$= \frac{\sqrt{3}}{2} \times (\text{side}) = \frac{\sqrt{3}}{2} \times 8 = 4\sqrt{3}$ cm.

20. (b) Let, x and $(51 - x)$ be the other two sides of the triangle

Then, $x^2 + (51 - x)^2 = 39^2$

$\Rightarrow x^2 + 2601 - 102x + x^2 = 1521$

$\Rightarrow x = \frac{51 \pm \sqrt{441}}{2} = \frac{51 \pm 21}{2} = 36, 15$.

$\therefore$ The other two sides are 36 cm and 15 cm.

21. (b) Given: $a = \frac{5}{8}b$.

Now, perimeter of isosceles triangle = $2a + b$

$\Rightarrow 306 = 2 \times \frac{5}{8}b + b$ or, $b = 136$.

$\therefore a = \frac{5}{8}b = \frac{5}{8} \times 136 = 85$.

$\therefore$ Area of isosceles triangle = $\frac{b}{4}\sqrt{4a^2 - b^2}$

$= \frac{136}{4}\sqrt{4 \times (85)^2 - (136)^2}$

$= 34 \times 102 = 3468 \text{ m}^2$.

22. (c) We have, hypotenuse = $\sqrt{2}\,a = 8 \Rightarrow a = 8/\sqrt{2}$.

$\therefore$ Area of isosceles triangle = $\frac{1}{2}a^2$

$= \frac{1}{2} \times \frac{64}{2} = 16 \text{ cm}^2$.

23. (b) Let, the lateral side = $5x$ and the base = $4x$.

Then, $5x + 5x + 4x = 14 \Rightarrow x = 1$.

$\therefore$ The sides of the triangle are 5 cm, 5 cm and 4 cm.

$\therefore$ Area of the isosceles triangle

$= \frac{b}{4}\sqrt{4a^2 - b^2}$

$= \frac{4}{4}\sqrt{4(5)^2 - (4)^2} = \sqrt{84} = 2\sqrt{21}$ cm^2.

24. (c) Area $= \sqrt{s(s-a)(s-b)(s-c)} = A$

where $s = \frac{a+b+c}{2}$ and a, b, c are sides of the triangle.

When the sides are increased by 200%, the sides become $3a$, $3b$ and $3c$.

$s_1 = \frac{3a+3b+3c}{2} = 3\frac{(a+b+c)}{2} = 3s.$

$A_1 = \sqrt{s_1(s_1-3a)(s_1-3b)(s_1-3c)}$

$= \sqrt{3s.3(s-a).3(s-b).3(s-c)}$

$= 9\sqrt{s(s-a)(s-b)(s-c)} = 9A.$

$\therefore$ Increase in area $= 9A - A = 8A$ or, 800%

25. (c) Let, the length of each of equal sides of the triangle be x m.

Then, $x^2 + x^2 = (50\sqrt{2})^2 = 5000$.

$\Rightarrow 2x^2 = 5000 \Rightarrow x = 50.$

$\therefore$ Perimeter of the triangle $= 50 + 50 + 50\sqrt{2}$

$= 100 + 50 \times 1.4146 = 170.73$ m.

$\therefore$ Cost of fencing $= ₹(170.73 \times 3) = ₹512.19$.

26. (a) Area of isosceles triangle $= \frac{b}{4}\sqrt{4a^2 - b^2}$

$= \frac{16}{4}\sqrt{4\times(10)^2 - (16)^2} = 4 \times 12 = 48$ cm^2

Given: Area of equilateral triangle = 48.

$\Rightarrow \frac{\sqrt{3}}{4}(\text{side})^2 = 48$

$\Rightarrow$ Side of equilateral triangle = 10.5 cm.

27. (c) Perimeter of a right-angled isosceles triangle

$= (\sqrt{2}+1) \times$ hypotenuse

$\Rightarrow 4\sqrt{2} + 4 = (\sqrt{2} + 1) \times$ hypotenuse

$\Rightarrow$ hypotenuse = 4 m.

28. (a) Here, $a = 60$, $b = 40$ and $d = 80$

$\therefore s = \frac{a+b+d}{2} = \frac{60+40+80}{2} = 90.$

$\therefore$ Area of the parallelogram

$= 2\sqrt{s(s-a)(s-b)(s-d)}$

$= 2\sqrt{90(90-60)(90-40)(90-80)}$

$= 2\sqrt{90\times30\times50\times10}$

$= 600\sqrt{15}$ m^2.

29. (c) Area of the parallelogram

$= 16 \times 14 = 224$ cm^2.

30. (b) We have, $s = \frac{a+b+d}{2}$

$= \frac{65+119+156}{2} = 170.$

$\therefore$ Area of parallelogram

$= 2\sqrt{s(s-a)(s-b)(s-d)}$

$= 2\sqrt{170(170-65)(170-119)(170-156)}$

$= 2\sqrt{170\times51\times105\times14}$

$= 2 \times 3570 = 7140$ m^2.

$\therefore$ Cost of gravelling $= 7140 \times 10 = ₹71400$.

31. (a) The area of the parallelogram

$= 10 \times 7 = 70$ m^2.

32. (b) Area of the parallelogram $= 8 \times 4 = 32$ m^2.

Distance between the shorter sides $= \frac{32}{5} = 6.4$ m.

33. (c) Area of quadrilateral $= \frac{1}{2}d(p_1 + p_2)$

$\Rightarrow 420 = \frac{1}{2}\times d \times (18+12)$

$\Rightarrow d = \frac{420\times2}{30} = 28$ m.

34. (a) Let, the base be x and altitude be $2x$.

Then, $x \times 2x = 72 \Rightarrow x^2 = 36 \Rightarrow x = 6.$

35. (b) Let, the base be x.

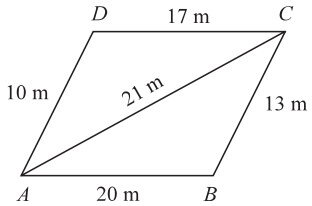
Area of the parallelogram = base $\times$ altitude

$\Rightarrow 240 = x \times 12$

$\therefore x = \frac{240}{12} = 20$ cm.

36. (a) Area of quadrilateral $ABCD$

= Area of ΔADC + Area of ΔABC, where,



Area of ΔABC

$= \sqrt{s(s-AB)(s-BC)(s-d)}$

$\left(s = \frac{AB+BC+d}{2} = \frac{20+13+21}{2} = 27\right)$

$= \sqrt{27(27-20)(27-13)(27-21)}$

$= \sqrt{27\times7\times14\times6} = 126 \text{ m}^2.$

Area of ΔADC

$= \sqrt{s(s-AD)(s-DC)(s-d)}$

$\left(s = \frac{AD+DC+d}{2} = \frac{10+17+21}{2} = 24\right)$

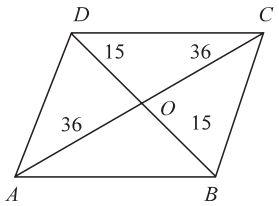
$= \sqrt{24(24-10)(24-17)(24-21)}$

$= \sqrt{24\times14\times7\times3} = 84 \text{ m}^2.$

Here, the area of quadrilateral $ABCD$

$= 126 + 84 = 210 \text{ m}^2.$

37. (a) Since $AC = 72$ and $BD = 30$



$\therefore \quad OB = OD = \frac{BD}{2} = \frac{30}{2} = 15$

and, $OA = OC = \frac{AC}{2} = \frac{72}{2} = 36$

$\therefore \quad AB = BC = CD = DA = \sqrt{36^2+15^2}$

$= \sqrt{1296+225}$

$= \sqrt{1521} = 39.$

$\therefore$ Perimeter of the parallelogram $= 4 \times 39$

$= 156$ cm.

38. (c) Area of the parallelogram

= base × altitude

$\Rightarrow \quad (x^2 - 4) = (x + 4) \times (x - 3)$

$\Rightarrow \quad x^2 - 4 = x^2 + 4x - 3x - 12$

$\Rightarrow \quad x = 8.$

$\therefore$ Area of the parallelogram

$= x^2 - 4 = (8)^2 - 4 = 60$ sq units.

39. (b) We have,

$d_1^2 + d_2^2 = 2(a^2 + b^2)$

$\Rightarrow \quad (16)^2 + d_2^2 = 2(12^2 + 14^2)$

$\Rightarrow \quad d_2^2 = 2(144 + 196) - 256 = 424.$

$\therefore \quad d^2_2 = \sqrt{424} = 20.6$ cm.

40. (b) $\pi r^2 = 154 \text{ m}^2$

$\Rightarrow r^2 = \frac{7}{22} \times 154 \text{ m}^2 = 49 \text{ m}^2$

$\therefore \quad r = 7$ m

Perimeter $= 2 \times \frac{22}{7} \times 7 \text{ m} = 44$ m.

41. (a) If x be the side of the square and r be the radius of the circle, then,

$4x = 2\pi r$

or, $\quad x = \frac{\pi r}{2}$

Now, $\pi r^2 :: x^2 = \pi r^2 : : \frac{\pi^2 r^2}{4}$ or, $4:\pi$

$= 4 : : \frac{22}{7}$ or, 14:11.

42. (d) Let, the width of the rectangle be x m

Then length $= 3x$ m

Perimeter $= 2(x + 3x) = 96$

$\Rightarrow \quad 8x = 96$ or, $x = 12$

Area $= 12 \times 36 = 432 \text{ m}^2.$

43. (d) As the cow is tied at the corner of a rectangular field, it will graze the area of the field enclosed between the two sides of the rectangle.

$= \frac{1}{4}(\pi \times 14 \times 14)$

$= \frac{1}{4} \times \frac{22}{7} \times 14 \times 14$

$= 154 \text{ m}^2.$

44. (c) The distance covered by the wheel in one minute

$= \frac{66\times1000\times100}{60} = 110000$ cm

The distance covered by the wheel in one revolution

= The circumference of the wheel

$= 2\pi r = 2 \times \frac{22}{7} \times \frac{70}{2} = 220$ cm

$\therefore$ Number of revolutions of the wheel

$= \frac{110000}{220} = 500.$

45. (a) $\pi(8)^2 + \pi(6)^2 = \pi r^2$

$\therefore \quad r^2 = 64 + 36$

$r = 10$

46. (c) Area of two paths $= 2 \times (65 + 50 - 2) = 226 \text{ m}^2$

Cost of construction $= 226 \times 17.25$

$= ₹3898.50.$

47. (d) $\frac{1}{2}(75 + x) \times 40 = 2500$

$\Rightarrow \quad 75 + x = 125$

$\Rightarrow \quad x = 50$

$\therefore$ The other parallel side = 50 m.

48. (c) Let, the breadth be x, then length $= x + 3$

Given: $2(x + x + 3) = x(x + 3)$

$\Rightarrow \quad 4x + 6 = x^2 + 3x$

$\Rightarrow \quad 4x + 6 = x^2 + 3x$

$\Rightarrow \quad x^2 - x - 6 = 0$

$\therefore \quad x = \dfrac{1 \pm \sqrt{1+24}}{2} = \dfrac{1 \pm 5}{2}$

$\therefore \quad x = 3, -2$

$\therefore$ Breadth = 3 cm.

49. (c) Let, x, y be the sides of squares

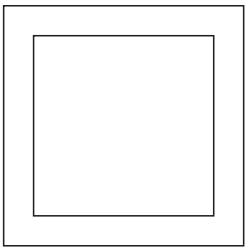
$\dfrac{x^2}{y^2} = \dfrac{9}{1} \Rightarrow \dfrac{x}{y} = \dfrac{3}{1}$

$\therefore$ The ratio of perimeters is $4x:4y$

i.e., $x:y = 3:1$.

50. (d) Let, x (in metres) be the width of the path

Side of outer square = $30 + 2x$



$\therefore$ Area of path = $(30 + 2x)^2 - 30^2$

$\therefore \quad (30 + 2x)^2 - 30^2 = 256$

$\Rightarrow \quad 4x^2 + 120x - 256 = 0$

$\Rightarrow \quad x^2 + 30x - 64 = 0$

$\Rightarrow \quad (x - 2)(x + 32) = 0$

$\therefore \quad x = 2 \qquad (\because x < 0)$

51. (c) Additional grassy ground grazed

$= \pi(23^2 - 12^2)$ m^2

$= \dfrac{22}{7} \times 35 \times 11$

$= 1210$ m^2.

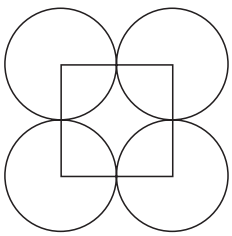
52. (b) Required area enclosed

$= 14 \times 14 - 4 \times$ (area of a quadrant)

$= \left[196 - 4 \times \dfrac{22}{7} \times 7 \times 7 \times \dfrac{90}{360}\right]$ cm^2

$= (196 - 154)$ cm^2

$= 42$ cm^2.



53. (c) Area = $\dfrac{6000 \times 100}{25} = 24000$ m^2

Let, the length be $5x$ and breadth be $3x$

$\therefore \quad 5x \times 3x = 24000$

$\therefore \quad x = \sqrt{1600} = 40$

$\therefore$ Length = $5 \times 40 = 200$ m, breadth = 3×40

$= 120$ m.

54. (c) Area of the carpet = $\dfrac{105}{3.50} = 30$ m^2

Area of the room = 30 m^2

Width = 5 m

$\therefore$ Length = $\dfrac{30}{5} = 6$ m.

55. (a) Area of the rectangular field

$= \dfrac{28000}{3500} = 8$ hectares

$2x \times x = 80000$

$\therefore \quad x = \sqrt{40000} = 200$

$\therefore$ Breadth = 200 m

Length = 400 m

Perimeter = $2(400 + 200) = 1200$ m

$\therefore$ Cost of fencing = $1200 \times 5 =$ ₹6000.

56. (a) Let, the length and breadth be $6x$ and $5x$ m respectively. Then

$6x \times 5x = 27000$ or $30x^2 = 27000$

$\Rightarrow \quad x^2 = 900$ or $x = 30$

Hence, length of the field = 180 m and width = 150 m.

57. (c) Area of sector = $\left(\dfrac{1}{2} \times \text{arc length} \times \text{radius}\right)$ cm^2

$= \left(\dfrac{1}{2} \times 3.5 \times 5\right)$ cm^2

$= 8.75$ cm^2.

58. (c) Area of the plot = $150 \times 3 = 450$ m^2

If the length of the plot be x m then breadth

$= \dfrac{x}{2}$ m

$\therefore \quad x \times \dfrac{x}{2} = 450$

or, $x^2 = 900$

or, $x = 30$ m.

59. (b) Perimeter of circle = $2 \times \dfrac{22}{7} \times 7$

$= 44$ cm

Perimeter of half circle = 22 cm

The length of the wire = $22 + 14 = 36$ cm.

60. **(b)** $2\pi R - 2\pi r = 66$

$2\pi(R - r) = 66$

$2 \times \frac{22}{7} \times (R - r) = 66$

or, $R - r = 66 \times \frac{7}{22} \times \frac{1}{2} = \frac{21}{2} = 10.5$ m.

61. **(a)** One side of the square $= \sqrt{50} = 5\sqrt{2}$

Length of diagonal $= 5\sqrt{2} \times \sqrt{2} = 10$

Radius of circle $= 5$

$\therefore$ Area of the circle $= 25\pi$ units.

62. **(c)** Area of remaining portion

= Area of rectangle − Area of circle

$= 4 \times 2 - \pi \times (1)^2 = (8 - \pi)$ cm^2.

63. **(b)** $r = \sqrt{22^2 + 19^2 + 8^2}$ cm

$= \sqrt{484 + 361 + 64}$ cm

$= \sqrt{909}$ cm $= 30$ cm

64. **(a)** Area of rectangle $= l \times b$

Let, the new width be b_1

Then, $l \times b = \frac{4}{3} l \times b_1$

$\therefore \quad b_1 = \frac{3}{4} b = 0.75b$

Thus, there should be a reduction of 25% in the width.

65. **(d)** Area of rectangle $= 6.4 \times 2.5 = 16$ m^2

According to question:

Area of square = Area of the rectangle

$\therefore$ Area of square $= 16$ m^2

$\therefore$ Side of the square $= 4$ m.

66. **(d)** $\pi \times 5 \times 5 - \pi \times 4 \times 4$

$= 25\pi - 16\pi = 9\pi$ cm^2.

67. **(b)** The length of the room $= 4$ m

Since it can be partitioned into two equal square rooms, the two equal parts can only be of 2 m each.

68. **(c)** Let length $= a$ m and breadth $= b$ m

Then, $2(a + b) = 46$

or, $a + b = 23$ and $ab = 120$

$\therefore$ Diagonal $= \sqrt{a^2 + b^2} = \sqrt{(a+b)^2 - 2ab}$

$= \sqrt{(23)^2 - 2 \times 120} = \sqrt{289} = 17$ m.

69. **(c)** Perimeter of the square $= 4 \times 22$ m $= 88$ m

$2\pi r = 88$ m

$\Rightarrow \quad r = 88 \times \frac{7}{22 \times 2}$ m $= 14$ m.

70. **(c)** Area of an equilateral $\Delta = \frac{\sqrt{3}}{4} a^2$

$[3a = 132 \ \therefore \ a = 44]$

$= \frac{\sqrt{3}}{4} \times 44 \times 44$

$= 838.312$ m^2.

Area of square $= a^2 = 33 \times 33$

$= 1089$ m^2 $\left[a = \frac{132}{4} = 33\right]$

Area of regular hexagon

$= \frac{3\sqrt{3}a^2}{2} = \frac{3\sqrt{3} \times 22 \times 22}{2}$. $\left[a = \frac{132}{6} = 22\right] = 1257.47$

Area of circle $= \pi r^2$

$= \frac{22}{7} \times 21 \times 21$ $\left[r = \frac{132}{2\pi} = 21\right] = 1386$

$\therefore$ Circle has largest surface area.

71. **(c)** Let, r be the radius of smaller circle

Radius of larger circle $= 12r$

$\frac{\text{Area of larger circle}}{\text{Area of smaller circle}} = \frac{\pi(12r)^2}{\pi r^2} = \frac{144}{1}$

$\therefore$ Area of large circle contains the area of smaller circle 144 times.

72. **(b)** If a be the side of the square and r be the radius of the circle, then,

$2\pi r = 4a$

or, $r = \frac{4a}{2\pi} = \frac{2a}{\pi}$

$\therefore$ Area of the circle $= \pi r^2$ and,

area of the square $= a^2$

$\therefore$ Area of the circle:Area of the square

$= \pi r^2 : a^2$

$= \frac{\pi\left(\frac{2a}{\pi}\right)^2}{a^2} = \frac{\pi \frac{4a^2}{\pi^2}}{a^2}$

$= \frac{4}{\pi} = \frac{4 \times 7}{22} = 14:11.$

73. **(c)** Let, the length and breadth of the plot be $3x$ and x feet, respectively.

Total area of the plot $= 4 \times 1200 = 4800$ ft^2.

$\therefore \quad x \times 3x = 4800 \Rightarrow x = 40$ ft

$\therefore$ Length $= 3 \times 40 = 120$ ft.

74. **(b)** Ratio of the areas = 4:25

Ratio of the sides = 2:5

If the side of s_1 is 2 cm, then side of s_2 is 5 cm.

If the side of s_1 is 6 cm, the side of s_2 is $\frac{5}{2} \times 6 = 15$ cm.

CHAPTER 34

Mensuration II: Volume and Surface Area

INTRODUCTION

Solids

A *solid* is a figure bounded by one or more surfaces. It has three dimensions, namely, length, breath or width, and thickness or height. The plane surfaces that bind it are called its *faces*.

The *volume* of any solid figure is the amount of space enclosed within its bounding faces. It is measured in cubic units, e.g., m^3, cm^3, etc.

The area of the plane surfaces that bind the solid is called its *surface area*.

For any regular solid,

Number of faces + Number of vertices

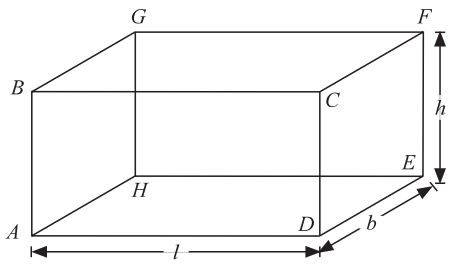
= Number of edges + 2.

We discuss here some important three-dimensional figures and the formulae associated with them.

Cubic

It is ta solid figure which has six rectangular faces. It is also called *rectangular parallelopiped.*

BASIC FORMULAE



If l, b and h denote the length, breadth and height of the cuboid, and d denotes the body diagonal (AF or BE or DG or CH), then

(i) Volume = $l \times b \times h = \sqrt{A_1 \times A_2 \times A_3}$,

where, A_1 = area of base or top,

A_2 = area of one side face, and

A_3 = area of other side face.

(ii) Total surface area = $2(lb + bh + lh)$

$= (l + b + h)^2 - d^2$

(iii) Diagonal of cuboid = $\sqrt{l^2 + b^2 + h^2}$

Notes

1. For painting the surface area of a box or to know how much tin sheet is required for making a box, we use formula (ii).
2. To find how much a box contains or how much space a box shall occupy, we use formula (i). To find the length of the longest pole to be placed in a room, we use formula (iii).
3. The rise or fall of liquid level in a container

$$= \frac{\text{Total volume of objects submerged or taken out}}{\text{Cross-sectional area of container}}.$$

Illustration 1: Find the volume and the total surface area of a cuboid whose dimensions are 25 m, 10 m and 2 m.

Solution: Here, l = 25 m, b = 10 m and h = 2 m.

Volume of the cuboid = $l \times b \times h$

$= 25 \times 10 \times 2$

$= 500\ m^3$

Total surface area of the cuboid

$= 2\ (lb + bh + lh)$

$= 2\ (25 \times 10 + 10 \times 2 + 25 \times 2)$

$= 2\ (250 + 20 + 50)$

$= 640\ \text{m}^2$.

Illustration 2: Find out the length of the longest bamboo that can be placed in a room which is 12 m long, 9 m broad and 8 m high.

Solution: Length of the bamboo

= length of the diagonal of the room

$= \sqrt{12^2 + 9^2 + 8^2}$

$= \sqrt{289} = 17$ m.

Illustration 3: The area of a side of a box is 120 cm^2. The area of the other side of the box is 27 cm^2. If the area of the upper surface of the box is 60 cm^2, then find out the volume of the box.

Solution: Volume to the box

$= \sqrt{\text{area of base} \times \text{area of one face} \times \text{area of the other face}}$

$= \sqrt{60 \times 120 \times 72}$

$= \sqrt{518400} = 720\ \text{cm}^3$.

Illustration 4: The sum of length, breadth and height of a cuboid is 12 cm. Find out the total surface area of the cuboid.

Solution: Total surface area

$= (\text{Sum of all three sides})^2 - (\text{Diagonal})^2$

$= 12^2 - 8^2 = 144 - 64 = 80\ \text{cm}^2$.

Cube

It is a special type of cuboid in which each face is a square.

For a cube, length, breadth and height are equal and is called, the edge of the cube,

If a be the edge of a cube, then

(i) Volume of the cube = $(\text{edge})^3 = a^3$

(ii) Total surface area of the cube = $6\ (\text{edge})^2 = 6a^2$

(iii) Diagonal of the cube = $\sqrt{3}a$ (edge) = $\sqrt{3}a$

(iv) Volume of the cube = $\left(\frac{\text{diagonal}}{\sqrt{3}}\right)^3 = \left(\frac{d}{\sqrt{3}}\right)^2$

$= \left(\sqrt{\frac{\text{Surface area}}{6}}\right)^3$

(v) Total surface area of the cube

$= 2\ (\text{diagonal})^2 = 2d^2$

(vi) For two cubes

(a) Ratio of volumes = $(\text{ratio of sides})^3$

(b) Ratio of surface areas = $(\text{Ratio of sides})^2$

(c) $(\text{Ratio of surface areas})^3 = (\text{Ratio of volumes})^2$.

Illustration 5: Find out the volume, surface area and the diagonal of a cube, each of whose sides measures 4 cm.

Solution: Volume of the cube = $a^3 = (4)^3 = 64\ \text{cm}^3$.

Surface area of the cube = $6a^2 = 6\ (4)^2 = 96\ \text{cm}^2$.

Diagonal of the cube = $\sqrt{3}a = 4\sqrt{3}$ cm.

Illustration 6: The surface area of a cube is 216 cm^2. Find out its volume.

Solution: Volume of the cube

$= \left(\sqrt{\frac{\text{Surface area}}{6}}\right)^3$

$= \left(\sqrt{\frac{216}{6}}\right)^3 = (6)^3 = 216\ \text{cm}^3$.

Illustration 7: The diagonal of a cube is $8\sqrt{3}$ cm. Find out its total surface area and volume.

Solution: We have,

Diagonal of cube = $\sqrt{3}$ (edge)

$\therefore$ Edge of cube = $\frac{\text{Diagonal of cube}}{\sqrt{3}}$

$= \frac{8\sqrt{3}}{\sqrt{3}} = 8$ cm.

Total surface area = $6\ (\text{edge})^2 = 6(8)^2$

$= 384\ \text{cm}^2$.

Volume of cube = $(\text{edge})^3 = (8)^3 = 512\ \text{cm}^3$.

Illustration 8: If the volumes of two cubical blocks are in the ratio of 8:1, then what will be the ratio of their edges?

Solution: We have,

Ratio of volumes = $(\text{Ratio of sides})^3$

Since, ratio of volumes = 8:1, i.e., $2^3:1^3$

$\therefore$ ratio of sides = 2:1.

Illustration 9: Volumes of the two cubes are in the ratio of 1:9. Find the ratio of their surface areas.

Solution: $(\text{Ratio of the surface areas})^3$

$= (\text{Ratio of volumes})^2$

$\therefore$ Ratio of surface areas = $\sqrt[3]{1:81} = 1:3\ (3)^{1/3}$.

Illustration 10: Sides of two cubes are in the ratio of 2:3. Find out the ratio of their surface areas.

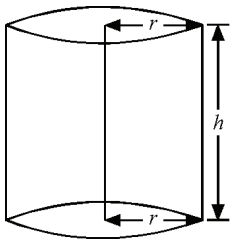
Solution: Ratio of surface areas

= (Ratio of sides)^2

= $(2:3)^2$ = 4:9.

Right Circular Cylinder

A *right circular cylinder* is a solid with circular ends of equal radius and the line joining their centres perpendicular to them. This is called, axis of the cylinder. The length of the axis is called, the height of the cylinder.



Notes

Take a rectangular sheet of paper and role it lengthwise or breadthwise in a round way, you will get a cylinder, i.e., a cylinder is generated by rotating a rectangle by fixing one of its sides.

If r is the radius of base and h is the height of the cylinder, then

(i) Volume of cylinder

= Area of the base × height

= $\pi r^2 \times h = \pi r^2 h$ cubic units

(ii) Area of the curved surface

= Circumference of the base × height

= $2\pi r \times h = 2\pi rh$ sq units

(iii) Area of the total surface

= Area of the curved surface

+ Area of the two circular ends

= $2\pi rh + 2\pi r^2$

= $2\pi r (h + r)$ sq units.

(iv) For two cylinders,

When radii are equal

(a) Ratio of volumes = Ratio of heights

(b) Ratio of volumes

= Ratio of curved surface areas

(c) Ratio of curved surface areas

= Ratio of heights

When heights are equal

(a) Ratio of volumes = (Ratio of radii)^2

(b) Ratio of volumes

= (Ratio of curved surface areas)^2

(c) Radii of curved surface areas

= Ratio of radii

When volumes are equal

(a) Ratio of radii = $\sqrt{\text{Inverse ratio of heights}}$

(b) Ratio of curved surface areas

= Inverse ratio of radii

(c) Ratio of curved surface areas

= $\sqrt{\text{Ratio of heights}}$

When curved surface areas are equal

(a) Ratio of radii = Inverse ratio of heights

(b) Ratio of volumes = Inverse ratio of heights

(c) Ratio of volumes = Ratio of radii

(v) For a cylinder

(a) Ratio of radii = (Ratio of curved surfaces) × (Inverse ratio of heights)

(b) Ratio of heights = (Ratio of curved surfaces) × (Inverse ratio of radii)

(c) Ratio of curved surfaces

= (Ratio of radii) × (Ratio of heights).

Illustration 11: The diameter of the base of a right circular cylinder is 28 cm and its height is 10 cm. Find out the volume and area of the curved surface of the cylinder.

Solution: Radius of the base = $\frac{28}{2}$ = 14 cm.

Volume of the cylinder = $\pi r^2 h$

= $\frac{22}{7} \times 14 \times 14 \times 10$

= 6160 cm^3.

Area of the curved surface = $2\pi rh$

= $2 \times \frac{22}{7} \times 14 \times 10$

= 880 cm^2.

Illustration 12: A cylinder of height 21 cm has base of radius 4 cm. Find out the total surface area of the cylinder.

Solution: Total surface area = $2\pi r (h + r)$

= $2 \times \frac{22}{7} \times 4 \times (21 + 4)$

= $\frac{4400}{7} = 628 \frac{4}{7}$ cm^2.

Illustration 13: A rectangular piece of paper is 71 cm long and 10 cm wide. A cylinder is formed by rolling the paper along its breadth. Find out the volume of the cylinder. $\left[\text{Take } \pi = \frac{355}{113}\right]$

Solution: Circumference of the paper = Breadth of the paper

$\Rightarrow \quad 2\pi r = 10$

$\Rightarrow \quad r = \frac{10}{2\pi} = \frac{10 \times 113}{2 \times 355} = \frac{113}{71}$ cm.

As the length of the paper becomes the height of the cylinder,

$\therefore$ Volume of the cylinder

$= \pi r^2 l$

$= \frac{355}{113} \times \frac{113}{71} \times \frac{113}{71} \times 71 = 565 \text{ cm}^3.$

Illustration 14: Two circular cylinders of equal volume have their heights in the ratio of 9:16. Find out the ratio of their radii.

Solution: Ratio of radii $= \sqrt{\text{inverse ratio of heights}}$

$= \sqrt{16:9} = 4:3.$

Illustration 15: Two circular cylinders of equal volume have their heights in the ratio of 16:25. Find out the ratio of their curved surface areas.

Solution: Ratio of curved surface areas

$= \sqrt{\text{Ratio of heights}} = \sqrt{16:25} = 4:5.$

Illustration 16: Two circular cylinders of equal volume have their radii in the ratio of 4:9. Find out the ratio of their curved surface areas.

Solution: Ratio of curved surface areas

= inverse ratio of radii = 9:4.

Illustration 17: Two circular cylinders of equal heights have their radii in the ratio of 2:5. Find out the ratio of their volumes.

Solution: Ratio of volumes = (Ratio of radii)^2 = 4:25.

Illustration 18: Two circular cylinders of equal heights have their curved surface areas in the ratio of 3:5. Find out the ratio of their volumes.

Solution: Ratio of volumes

= (Ratio of curved surface areas)^2

= 9:25.

Illustration 19: Two circular cylinders of equal curved surface areas have their heights in the ratio of 4:7. Find out the ratio of their volumes.

Solution: Ratio of volumes = Inverse ratio of heights

$= \frac{1}{4} : \frac{1}{7} = 7:4.$

Illustration 20: Two circular cylinders of equal curved surface areas have their heights in the ratio of 4:5. Find out the ratio of their volumes.

Solution: Ratio of volumes = Inverse ratio of heights

$= \frac{1}{4} : \frac{1}{5} = 5:4.$

(vi) If the ratio of heights and the ratio of radii of two right circular cylinders are given, then

Ratio of curved surface areas = (ratio of radii) (ratio of heights).

Illustration 21: If the heights and the radii of two right circular cylinders are in the ratio 2:3 and 4:5, respectively. Find out the ratio of their curved surface areas.

Solution: Ratio of curved surface areas = (ratio of radii) (ratio of heights)

= (4:5) (2:3) = 8:15.

(vii) If the ratio of heights and the ratio of curved surface areas of two right circular cylinders are given, then

Ratio of radii = (ratio of curves surface areas) (inverse ratio of heights).

Illustration 22: The heights and curved surface areas of two right circular cylinders are in the ratio 3:4 and 5:8, respectively. Find out the ratio of their radii.

Solution: Ratio of radii = (ratio of curved surface areas) (inverse ratio of heights)

$= (5:8) \left(\frac{1}{3} : \frac{1}{4}\right) = (5:8)\ (4:3) = 5:6.$

(viii) If the ratio of radii and the ratio of curved surface areas of two right circular cylinders are given, then

Ratio of heights = (ratio of curved surface areas) (inverse ratio of radii)

Illustration 23: The radii of two right circular cylinders are in the ratio of 3:4 and their curved surface areas are in the ratio of 5:6. Find out the ratio of their heights.

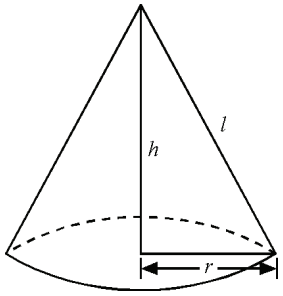
Solution: Ratio of heights = (ratio of curved surface areas) (inverse ratio of radii)

$= (5:6) \left(\frac{1}{3} : \frac{1}{4}\right)$

$= (5:6)\ (4:3) = 10:9.$

Right Circular Cone

A *right circular cone* is a solid obtained by rotating a right-angled triangle around its height.



If r = radius of base; h = height,

l = slant height = $\sqrt{h^2 + r^2}$, then

(i) Volume of cone

$= \frac{1}{3} \times$ area of the base $\times$ height

$= \frac{1}{3} \times \pi r^2 h$ cubic units

(ii) Area of curved surface = πrl

$= \pi r\sqrt{h^2 + r^2}$ sq. units

(iii) Total surface area of cone

= Area of the base + area of the curved surface

$= \pi r^2 + \pi rl = \pi r (r + l)$ sq units.

(iv) For two cones

(a) When volumes are equal

Ratio of radii = $\sqrt{\text{inverse ratio of heights}}$

(b) When radii are equal

Ratio of volumes = Ratio of heights

(c) When heights are equal

Ratio of volumes = (ratio of radii)^2

(d) When curved surface areas are equal

Ratio of radii = inverse ratio of slant heights.

Illustration 24: Find out the slant heights of a cone whose volume is 1232 cm^3 and radius of the base is 7 cm.

Solution: Volume of the cone = $\frac{1}{3}\pi r^2 h = 1232$

$\Rightarrow \quad h = \frac{1232 \times 3}{\pi r^2} = \frac{1232 \times 3 \times 7}{22 \times 7 \times 7} = 24$ cm.

Slant height l is given by the relation

$l = \sqrt{h^2 + r^2}$

$= \sqrt{(24)^2 + (7)^2} = \sqrt{576 + 49}$

$= \sqrt{625} = 25$ cm.

$\therefore$ Slant height of the cone is 25 cm.

Illustration 25: A tent is of diameter 12 m at the base and its height is 8 m.

(i) Find the slant height; and

(ii) The canvas required in m^2.

How many persons can the tent accommodate, at the most, if each person requires 18 m^3 of air?

Solution: Diameter of the base of a conical tent = 12 m.

$\therefore$ Radius (r) = $\frac{12}{2}$ = 6 m and its height (h) = 8 m.

(i) Slant height (l) = $\sqrt{r^2 + h^2} = \sqrt{6^2 + 8^2}$

$= \sqrt{64 + 36} = \sqrt{100}$ = 10 m.

(ii) Area of canvas required

$= \pi \times r \times l$

$= \frac{22}{7} \times 6 \times 10 = 188.57$ m

(iii) Volume of conical portion

$= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 6 \times 6 \times 8 = 301.71$ m^3.

Space required for each person = 18 m^3.

$\therefore$ Number of persons that can be accommodated

$= \frac{301.71}{18} = 16.$

Illustration 26: The height of a cone is 21 cm and radius of its base is 28 cm. Find out its total surface area.

Solution: We have, r = 28 cm and h = 21 cm.

Slant height (l) = $\sqrt{r^2 + h^2} = \sqrt{(28)^2 + (21)^2}$

$= \sqrt{1225} = 35$ cm.

Total surface area = $\pi r (l + r)$

$= \frac{22}{7} x \ 28 \times (35 + 28)$

$= \frac{22}{7} \times 28 \times 63$

$= 5544$ cm^2.

Illustration 27: Two right circular cones of equal curved surface areas have their slant heights in the ratio of 3:5. Find out the ratio of their radii.

Solution: Ratio of radii = inverse ratio of slant heights

$= \frac{1}{3} : \frac{1}{5} = 5:3.$

Illustration 28: Two right circular cones of equal volumes have their heights in the ratio of 4:9. Find out the ratio of their radii.

Solution: Ratio of radii $= \sqrt{\text{inverse ratio of heights}}$

$$= \sqrt{\frac{1}{4}:\frac{1}{9}} = \sqrt{9:4} = 3:2.$$

Illustration 29: Two right circular cones of equal heights have their radii in the ratio of 1:3. Find out the ratio of their volumes.

Solution: Ratio of volumes = (Ratio of radii)^2

$$= (1:3)^2 = 1:9.$$

(v) If the ratio of volumes and the ratio of heights of two right circular cones (or cylinders) are given, then

Ratio of radii

$= \sqrt{(\text{ratio of volumes})(\text{inverse ratio of heights})}$

$= \sqrt{(3:2)(8:3)} : \sqrt{4:1} = 2:1.$

Illustration 30: The volumes of two cones are in the ratio 3:2 and their heights in the ratio 3:8. Find out the ratio of their radii.

Solution: Ratio of radii

$= \sqrt{(\text{ratio of volumes})(\text{inverse ratio of heights})}$

$= \sqrt{(3:2)(8:3)} = \sqrt{4:1} = 2:1.$

(vi) If the ratio of heights and the ratio of diameters (or radii) of two right circular cones (or cylinders) are given, then

Ratio of volumes = (ratio of radii)^2 × (ratio of heights).

Illustration 31: The heights of two cones are in the ratio of 5:3 and their radii is in the ratio 2:3. Find out the ratio of their volumes.

Solution: Ratio of volumes

$= (\text{ratio of radii})^2 \times (\text{ratio of heights})$

$= (2:3)^2 \times (5:3)$

$= \frac{4}{9} \times \frac{5}{3} = 20:27.$

(vii) If the ratio of radii (or diameter) and the ratio of volumes of two right circular cones are given, then ratio of heights

= (inverse ratio of radii)^2 (ratio of volumes).

Illustration 32: The volumes of two cones are in the ratio of 1:4 and their diameters are in the ratio of 4:5. Find out the ratio of their heights.

Solution: Ratio of heights

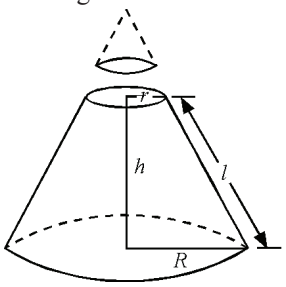
$= (\text{inverse ratio of diameters})^2 \times (\text{ratio of volumes})$

$= \left(\frac{1}{4}:\frac{1}{5}\right)^2 (1:4) = (5:4)^2\ (1:4)$

$= \frac{25}{16} \times \frac{1}{4} = 25:64.$

Frustum of a Right Circular Cone

A cone with some of its top portion cut off is called, the *frustum* of the original cone.



If R = Radius of the base of frustum
r = Radius of the top of the frustum
h = Height of the frustum
l = Slant height of the frustum, then

(a) Slant height = $\sqrt{h^2 + (R-r)^2}$ units

(b) Area of the curved surface = $\pi\ (R + r)\ l$ sq. units

(c) Total surface area of the frustum
$= \pi\ [(R^2 + r^2) + l\ (R + r)]$ sq. units

(d) Volume of the frustum = $\frac{\pi h}{3}(R^2 + r^2 + Rr)$ cu units.

Illustration 33: A reservoir is in the shape of a frustum of a right circular cone. It is 8 m across at the top and 4 m across the bottom. It is 6 m deep. Find out the area of its curved surface, total surface area and also its volume.

Solution: Here, $R = 4$, $r = 2$ and $h = 6$.

$\therefore$ Slant height $(l) = \sqrt{h^2 + (R-r)^2}$

$= \sqrt{(6)^2 + (4-2)^2} = \sqrt{40}$.

$\therefore$ Area of the curved surface

$= \pi\ (R + r)\ l$

$= \frac{22}{7}(4 + 2)\sqrt{40}$

$= 18.8 \times 6.3 = 118.4$ m

Total surface area $= \pi\ [(R^2 + r^2) + l\ (R + r)]$

$= \frac{22}{7}[(4^2 + 2^2) + \sqrt{40}\ (4 + 2)]$

$= \frac{22}{7}(20 + 6\sqrt{40}) = 181.6\ \text{m}^2$

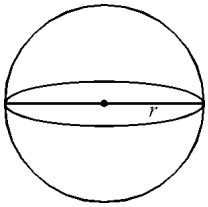
Volume of the frustum $= \frac{\pi h}{3}(R^2 + r^2 + Rr)$

$= \frac{22}{7} \times \frac{6}{3}(4^2 + 2^2 + 4 \times 2)$

$= \frac{44}{7}(16 + 4 + 8) = 176\ \text{m}^3.$

Sphere

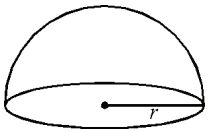
A *sphere* is the solid figure formed by revolving a semi-circle on its diameter.



The mid-point of the diameter is called, centre of the sphere, and the radius of the semi-circle is called, the radius of the sphere.
If r = radius of the spheres, then

(i) Volume of sphere = $\frac{4}{3}\pi r^3$ cubic units

(ii) Surface area = $4\pi r^2$ sq units.



(iii) Volume of hemisphere = $\frac{2}{3}\pi r^2$ cubic units

(iv) Area of curved surface = $2\pi r^2$ sq units of hemisphere

(v) Total surface area of hemisphere = $3\pi r^2$ sq units.

Illustration 34: Diameter of a sphere is 28 cm. Find out its surface area and volume.

Solution: Radius of the sphere (r) = $\frac{28}{2}$ = 14 cm.

$$\text{Surface area} = 4\pi r^2 = 4 \times \frac{22}{7} \times 14 \times 14$$
$$= 2464 \text{ cm}^2.$$

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3$$
$$= \frac{4}{3} \times \frac{22}{7} \times 14 \times 14 \times 14$$
$$= 11498.6 \text{ cm}^3.$$

Illustration 35: Find out the volume, curved surface area and total surface area of a hemisphere of radius 21 cm.

Solution: Volume of the hemisphere

$$= \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 21 \times 21 \times 21 = 19494 \text{ cm}^3.$$

Curved surface area = $2\pi r^2$

$$= 2 \times \frac{22}{7} \times 21 \times 21$$
$$= 2772 \text{ cm}^2.$$

Total surface area = $3\pi r^2$

$$= 3 \times \frac{22}{7} \times 21 \times 21$$
$$= 4158 \text{ cm}^2.$$

(vi) For two spheres

(a) (Ratio of radii)^2 = Ratio of surface areas
(b) (Ratio of radii)^3 = Ratio of volumes
(c) (Ratio of surface areas)^3
= (Ratio of volumes)^2.

Illustration 36: The radii of two spheres are in the ratio of 2:3. What is the ratio of their surface areas?

Solution: Ratio of surface areas = (ratio of radii)^2
= $(2:3)^2$ = 4:9.

Illustration 37: The surface areas of two spheres are in the ratio 1:2. Find out the ratio of their volumes.

Solution: We have,

(Ratio of surface areas)^3 = (Ratio of volumes)^2

$\Rightarrow$ $(1:2)^3$ = (Ratio of volumes)^2

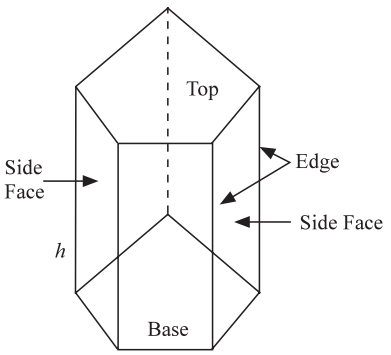
$\therefore$ Ratio of volumes = $\sqrt{1:8}$ = $1:2\sqrt{2}$.

Illustration 38: The radii of two spheres are in the ratio of 2:45. Find out the ratio of their volumes.

Solution: Ratio of volumes = (Ratio of radii)^3
= $(2:5)^3$ = 8:125.

Prism

A solid having top and bottom faces identical and side faces rectangular is a prism.



In a prism with a base of n sides;
Number of vertices = $2n$
and Number of faces = $n + 2$.
Volume of the prism = area of base × height
Lateral surface area = perimeter of base × height
Total surface area = 2 × Base area
+ Lateral Surface area.

Illustration 39: Find out the volume and the total surface area of a triangular prism whose height is 30 m and the sides of whose base are 21 m, 20 m and 13 m, respectively.

Solution: Perimeter of base = 21 + 20 + 13 = 54 m.
height = 30 m.

$$\text{Area of base} = \sqrt{s(s-a)(s-b)(s-c)}$$
$$= \sqrt{27(27-21)(27-20)(27-13)}$$
$$= \sqrt{27\times6\times7\times14} = 126 \text{ m}^2.$$

$\therefore$ Volume of the prism = area of base × height
= 126 × 54 = 6804 m^3.

Also, surface area of the prism
= 2 × Base area + lateral surface area
= 2 Base area + perimeter of base × height
= 2 × 126 + 54 × 30 = 1872 m^2.

SOLIDS INSCRIBED/CIRCUMSCRIBING OTHER SOLIDS

1. If a largest possible sphere is circumscribed by a cube of edge 'a' cm, then the radius of the sphere $= \frac{a}{2}$.

Illustration 40: Find out the volume of largest possible sphere circumscribed by a cube of edge 8 cm.

Solution: Radius of the sphere $= \frac{a}{2} = \frac{8}{2} = 4$ cm.

$\therefore$ Volume of the sphere $= \frac{4}{3}\pi r^3$

$$= \frac{4}{3}\times\frac{22}{7}\times4\times4\times4$$
$$= 268.1 \text{ cm}^3.$$

2. If a largest possible cube is inscribed in a sphere of radius 'a' cm, then the edge of the cube $= \frac{2a}{\sqrt{3}}$.

Illustration 41: Find out the surface area of largest possible cube inscribed in a sphere of radius 4 cm.

Solution: Edge of the cube $\frac{2a}{\sqrt{3}} = \frac{2\times4}{\sqrt{3}} = \frac{8}{\sqrt{3}}$.

$\therefore$ Surface area of the cube = 6 (edge)^2

$$= 6 \times \frac{64}{3}$$
$$= 128 \text{ cm}^2.$$

3. If a largest possible sphere is inscribed in a cylinder of radius 'a' cm and height 'h' cm, then

$$\text{radius of the sphere} = \begin{cases} a \text{ for } h > a \\ \frac{h}{2} \text{ for } a > h \end{cases}$$

Illustration 42: Find out the surface area of largest possible sphere inscribed in a cylinder of radius 14 cm and height 17 cm.

Solution: Radius of the sphere = 14 cm ($\because$ $h > a$)

$\therefore$ Surface area of sphere $= 4\pi r^2$

$$= 4 \times \frac{22}{7}\times14\times14$$
$$= 2464 \text{ cm}^2.$$

4. If a largest possible sphere is inscribed in a cone of radius 'a' cm and slant height equal to the diameter of the base, then radius of the sphere $= \frac{a}{\sqrt{3}}$.

Illustration 43: Find out the surface area of largest possible sphere inscribed in a cone of radius 21 cm and slant height equal to the diameter of the base.

Solution: Radius of the sphere $= \frac{a}{\sqrt{3}} = \frac{21}{\sqrt{3}}$ cm.

$\therefore$ Surface area of the sphere $= 4\pi r^2$

$$= 4 \times \frac{22}{7}\times\frac{21}{\sqrt{3}}\times\frac{21}{\sqrt{3}}$$
$$= 1848 \text{ cm}^2.$$

5. If a largest possible cone is inscribed in a cylinder of radius 'a' cm and height 'h' cm, then radius of the cone = a and height = h.

Illustration 44: Find out the volume of largest possible cone inscribed in a cylinder of radius 6 cm and height 14 cm.

Solution: Radius of the cone (r) = 6 cm.
and height of the cone (h) = 14 cm.

$\therefore$ Volume of the cone $= \frac{1}{3}\pi r^2 h$

$$= \frac{1}{3}\times\frac{22}{7}\times6\times6\times14$$
$$= 528 \text{ cm}^2.$$

6. If a largest possible cube is inscribed in a hemisphere of radius 'a' cm, then the edge of the cube = $a\sqrt{\frac{2}{3}}$.

Illustration 45: Find out the length of the diagonal of largest possible cube inscribed in a hemisphere of radius $4\sqrt{2}$ cm.

Solution: Edge of the cube $= a\sqrt{\frac{2}{3}} = 4\sqrt{2} \times \sqrt{\frac{2}{3}} = \frac{8}{\sqrt{3}}$ cm.

$\therefore$ Diagonal of the cube = $\sqrt{3}$ (edge)

$= \sqrt{3} \times \frac{8}{\sqrt{3}} = 8$ cm.

SHORT-CUT METHODS

01 If all three measuring dimensions of a sphere, cuboid, cube, cylinder or cone are increased or decreased by x%, y% and z% respectively, then the volume of the figure will increase or decrease by

$$\left(x + y + z + \frac{xy + yz + zx}{100} + \frac{xyz}{100^2}\right)\%$$

For cuboid, the three measuring dimensions are length, breadth and height.

For cube, all three measuring dimensions are equal, i.e., $x = y = z$.

For sphere also, (or diameter) all three measuring dimensions are equal and is given by radius, i.e., $x = y = z = r$.

For cylinder or a cone two measuring dimensions are equal to radius and third measuring dimension is height

i.e., $x = y = r$ and $z = h$.

Illustration 46: The length, breadth and height of a cuboid are increased by 5%, 10% and 20%, respectively. Find out the percentage increase in its volume.

Solution: Here, $x = 5$, $y = 10$ and $z = 20$.

$\therefore$ Percentage increase in volume

$$= \left[x + y + z + \frac{xy + xz + yz}{100} + \frac{xyz}{(100)^2}\right]\%$$

$$= \left[5 + 10 + 20 + \frac{(5\times10) + (5\times20) + (10\times20)}{100} + \frac{5\times10\times20}{(100)^2}\right]\%$$

$$= \left(35 + \frac{350}{100} + \frac{1000}{(100)^2}\right)$$

$= (35 + 3.5 + 0.1)\%$

$= 38.6\%$

Illustration 47: The sides of a cube are decreased by 10% each. Find out the percentage change in its volume.

Solution: Here, $x = y = z$.

$\therefore$ Percentage change in volume

$$= \left[3x + \frac{3x^2}{100} + \frac{x^3}{(100)^2}\right]\%$$

$$= \left[3(-10) + \frac{3(-10)^2}{100} + \frac{(-10)^3}{(100)^2}\right]\%$$

$= (-30 + 3 - 0.1)\% = -27.1\%$

−ve sign indicates decrease in volume, that is, there is a decrease in volume by 27.1%

Illustration 48: The diameter of a sphere is increased by 20%. What is the percentage increase in its volume?

Solution: Percentage increase in volume

$$= \left[3x + \frac{3x^2}{100} + \frac{x^3}{(100)^2}\right]\% \quad [\text{Here, } x = y = z]$$

$$= \left[3\times20 + \frac{3(20)^2}{100} + \frac{(20)^3}{(100)^2}\right]\%$$

$= (60 + 12 + 0.8)\% = 72.8\%$

Illustration 49: The radius of a right circular cylinder is decreased by 5%, but its height is increased by 10%. What is the percentage change in its volume?

Solution: Here, $x = y = -15$ and $z = 10$.

$\therefore$ Percentage change in volume

$$= \left[-5 - 5 + 10 + \frac{(-5)(-5) + (-5)(10) + (-5)(10)}{100} + \frac{(-5)(-5)(10)}{(100)^2}\right]\%$$

$= (0 - 0.75 + 0.025)\% = -0.725\%$

Therefore, volume decrease by 0.725%

Illustration 50: Each of the radius and the height of a cone is increased by 25%. Find out the percentage increase in volume.

Solution: Here, $x = y = 25$ and $z = 25$.

$\therefore$ Percentage increase in volume

$$= \left[25+25+25+\frac{25\times25+25\times25+25\times25}{100}+\frac{25\times25\times25}{(100)^2}\right]\%$$

$= (75 + 18.75 + 1.56)\% = 95.3\%$

02 If the two measuring dimensions which are included in the surface area of a sphere, cuboid, cube, cylinder or cone are increased or decreased by x% and y%, then the surface area of the figure will increase or decrease by

$$\left(x+y+\frac{xy}{100}\right)\%$$

Note that in case of percentage increase, values of x, y and z are positive, and in case of percentage decrease, values of x, y and z are negative.

Illustration 51: Each edge of a cube is increased by 20%. What is the percentage increase in its surface area?

Solution: Here, $x = y = 20$.

$\therefore$ Percentage increase in surface area

$$= \left(x+y+\frac{xy}{100}\right)\%$$

$$= \left(20+20+\frac{20\times20}{100}\right)\%$$

$= (40 + 4)\% = 44\%$

Illustration 52: The radius of a hemisphere is decreased by 10%. Find out the percentage change in its surface area.

Solution: Here, $x = y = -10$.

$\therefore$ Percentage change in surface area

$$= \left(x+y+\frac{xy}{100}\right)\%$$

$$= \left(-10-10+\frac{(-10)(-10)}{100}\right)\%$$

$= (-20 + 1)\% = -19\%$

Therefore, surface area of hemisphere decreases by 19%

Illustration 53: The radius of a right circular cone is increased by 25% and slant height is decreased by 30%. Find out the percentage change in curved surface area of the cone.

Solution: Here, $x = 25$ and $y = -30$.

$\therefore$ Percentage change in curved surface area

$$= \left(x+y+\frac{xy}{100}\right)\%$$

$$= \left[25-30+\frac{(25)(-30)}{100}\right]\%$$

$= (-5 - 7.5)\% = -12.5\%$

Therefore, curved surface area decreases by 12.5%

Illustration 54: The radius and height of a cylinder are increased by 10% and 20%, respectively. Find out the percentage increase in its surface area.

Solution: Here, $x = 10$ and $y = 20$.

$\therefore$ Percentage increase in surface area

$$= \left(x+y+\frac{xy}{100}\right)\%$$

$$= \left(10+20+\frac{10\times20}{100}\right)\% = 32\%$$

03 If a sphere of radius R is melted to form smaller spheres each of radius r, then

The number of smaller spheres

$$= \frac{\text{Volume of the bigger sphere}}{\text{Volume of the smaller sphere}}$$

$$= \left(\frac{R}{r}\right)^3.$$

Illustration 55: Find out the number of lead balls of radius 1 cm each that can be made from a sphere of radius 4 cm.

Solution: Number of lead balls $= \left(\frac{R}{r}\right)^3 = \left(\frac{4}{1}\right)^3 = 64.$

04 If by melting n spheres, each of radius r, a big sphere is made, then

Radius of the big sphere $= r\cdot\sqrt[3]{n}$.

Illustration 56: If by melting 8 spheres, each of radius 5 cm, a big sphere is made, what will be the radius of the big sphere?

Solution: Radius of the big sphere $= r\cdot\sqrt[3]{n}$

$= 5\cdot\sqrt[3]{8} = 5\cdot2 = 10$ cm.

05 If a cylinder is melted to form smaller spheres each of radius r, then

The number of small spheres $= \frac{\text{Volume of cylinder}}{\text{Volume of 1 sphere}}$.

Illustration 57: How many bullets can be made out of a loaded cylinder 24 cm high and 5 cm diameter, each bullet being 2 cm in diameter?

Solution: Number of bullets = $\frac{\text{Volume of cylinder}}{\text{Volume of 1 sphere}}$.

$$= \frac{\pi \times \frac{5}{2} \times \frac{5}{2} \times 24}{\frac{4}{3} \times \pi \times 1 \times 1 \times 1} = 450.$$

06 If a sphere of radius r is melted and a cone of height h is made, then

Radius of the cone = $2 \times \sqrt{\frac{r^3}{h}}$.

or,

If a cone of height h is melted and a sphere of radius r is made, then

Radius of the cone = $2 \times \sqrt{\frac{r^3}{h}}$.

Illustration 58: A solid cone of copper of height 3 cm is melted and a solid sphere of radius 3 cm is made. Then what is the diameter of the base of the cone?

Solution: Radius of the base of the cone

$$= 2 \times \sqrt{\frac{r^3}{h}} = 2 \times \sqrt{\frac{3^3}{3}} = 6$$

$\therefore$ Diameter of the base of the cone = 2×6

= 12 cm.

Illustration 59: If a solid cone of copper of height 2 cm is melted and a solid sphere of radius 2 cm is made, what is the diameter of the base of the cone?

Solution: Radius of the base of the cone

$$= 2 \times \sqrt{\frac{r^3}{h}} . = 2 \times \sqrt{\frac{(2)^3}{2}}$$

$= 2 \times 2 = 4$ cm.

$\therefore$ Diameter of the base of the cone = $2 \times 4 = 8$ cm.

EXERCISE-I

1. A tank, 16 m long and 23 m wide contains water. How many cubic m of water must be rushed into it to make the surface rise by $16\frac{2}{3}$ cm?

(a) 48 m^3 (b) 40 m^3
(c) 32 m^3 (d) 42 m^3

2. The outer dimensions of a closed box are 12 cm by 10 cm by 8 cm. If the box is made of wood 1 cm thick, find out the capacity of the box.

(a) 360 cm^3 (b) 480 cm^3
(c) 240 cm^3 (d) 560 cm^3

3. A cistren of dimensions 2.4 m × 2.0 m × 1.5 m takes 2 hours 30 minutes to get filled with water. The rate at which water flows into the cistern is:

(a) 0.48000 cu.m/h (b) 800 cu.m/min
(c) 800 cu.m/sec (d) None of these

4. The area of three adjacent faces of a rectangular box are p, q and r square cm. The volume of the box is given by:

(a) $(p + q + r)$ cm^3 (b) $\sqrt{pqr}$ cm^3
(c) $(pqr)^{1/3} cm^3$ (d) pqr cm^3

5. A reservoir, 30 m long, and 15 m broad, is filled with water. How many gallons of water must be taken out to lower the level of water by 4 m?

(a) 342000 gallons (b) 364200 gallons
(c) 324000 gallons (d) 386400 gallons

6. How many bricks, each measuring 250 cm by 12.5 cm by 7.5 cm, will be required to build a 5 m long, 3 m high and 20 cm thick wall?

(a) 1480 (b) 1280
(c) 1680 (d) 1480

7. Find out the cost of the log of wood measuring $15\frac{1}{2}$ m by $2\frac{3}{4}$ m by $1\frac{1}{3}$ m at ₹45 per cm^3.

(a) ₹4257.50 (b) ₹4005.00
(c) ₹4207.50 (d) ₹4357.50

8. How many bricks are required to build a 15 m long 3 m high and 50 cm thick wall, if each brick measures 25 cm by 12 cm by 6 cm.

(a) 16500 (b) 14500
(c) 12500 (d) 10500

9. Find the diagonal of a cuboid whose dimensions are 12 m by 10 m by 8 m.

(a) 18 m (b) 17.5 m
(c) 17 m (d) 16.5 m

10. The outer dimensions of a closed wooden box of 1 cm thick are 12 cm by 10 cm by 8 cm. Find out the cost of the wood required to make the box if 1 cm^3 of wood costs ₹3.00.

(a) ₹1440 (b) ₹1640
(c) ₹1840 (d) ₹2040

11. 3 equal cubes are placed adjacently in a row. Find out the ratio of the total surface area of the new cuboid to that of the sum of the surface areas of the three cubes:

(a) 3:5 (b) 4:5
(c) 6:7 (d) 7:9

12. The diagonal of a cubical box is $\sqrt{300}$ cm. Find out the surface area:

(a) $600\sqrt{3}$ cm^2 (b) 600 cm^2
(c) 1200 cm^2 (d) $900\sqrt{3}$ cm^2

13. An iron cube of 10 cm sides is hammered into a rectangular sheet of thickness 0.5 cm. If the sides of the sheet be in the ratio 1:5, the sides (in cm) are:

(a) 110 cm, 50 cm (b) 20 cm, 100 cm
(c) 40 cm, 200 cm (d) None of these

14. The length of a room is 12 m, width 8 m, and height 6 m. How many boxes will it hold if each is allowed 1.5 cubic metre of space?

(a) 864 (b) 506
(c) 384 (d) 436

15. A 3.3 m high room is half as long again as it is wide and its volume is $123\frac{3}{4}$ m^3. Find out its length and breadth.

(a) 7.5 m, 6 m (b) 8 m, 5 m
(c) 7.5 m, 5 m (d) 8.5 m, 5 m

16. A tank 3 m long, 2 m wide and 1.5 m deep is dug in a field 22 m long and 14 m wide. If the earth dug out is evenly spread out over the field, the level of the field will rise by nearly:

(a) 0.299 cm (b) 0.29 mm
(c) 2.98 cm (d) 4.15 cm

17. A school room is to be built to accommodate 70 children, so as to allow 2.2 m^2 of floor and 11 m^3 of space for each child. If the room be 14 m long, what must be its breadth and height?

(a) 12 m, 5,5 m (b) 11 m, 5 m
(c) 13 m, 6 m (d) 11 m, 4 m

18. If 210 m^3 of sand be thrown into a tank 12 m long and 5 m wide, find how much the water will rise?

(a) 3.5 m (b) 4 m
(c) 7 m (d) Data inadequate

19. If the length, breadth and height of a rectangular parallelopiped are in the ratio 6:5:4 and if total surface area is 33,300 m^2, then the length, breadth and height of parallelopiped (in cm) respectively are:

(a) 90, 85, 600 (b) 90, 75, 70
(c) 85, 75, 60 (d) 90, 75, 60

20. A m^3 of metal weighing 90 Kg is rolled into a square bar 9 metre long. An exact cube is cut off from the bar. How much does it weigh?

(a) $5\frac{2}{3}$ Kg (b) $6\frac{1}{3}$ Kg
(c) $3\frac{1}{3}$ Kg (d) $4\frac{2}{3}$ Kg

21. How many cubes, each of surface 24 cm^2, can be made out of a cube of edge measure 1 metre?

(a) 165000 (b) 125000
(c) 180000 (d) 155000

22. 3 solid cubes whose edges are 6, 8 and 10 cm respectively, are melted and formed into a single cube. If there be no loss of metal in the process, find out the edge of the new cube.

(a) 16 cm (b) 10 cm
(c) 14 cm (d) 12 cm

23. If a cube with its edge 6 cm is melted and smaller cubes with edge 2 cm each are formed, then how many cubesare formed?

(a) 39 (b) 24
(c) 27 (d) 21

24. How many small cubical blocks of side 5 cm can be cut from a cubical block whose each edge measures 20 cm?

(a) 56 (b) 48
(c) 64 (d) 52

25. Surface area of a cube is 600 cm^2. Find out the length of its diagonal.

(a) $15\sqrt{3}$ (b) $12\sqrt{3}$
(c) $10\sqrt{3}$ cm (d) None of these

26. A rectangular tank is 30 m long and 20 m broad. Water is being flown into it through a square pipe of side 5 cm. What is the speed of water if the level of water in the tank rises by 1 m in 8 hours?

(a) 30 Km/h (b) 36 Km/h
(c) 40 Km/h (d) None of these

27. Calculate the number of bricks, each measuring 25 cm × 15 cm × 8 cm, required to construct a wall with its dimension 19 m × 4 m × 5 m, when 10% of its volume is occupied by mortar.

(a) 4000 (b) 8000
(c) 7000 (d) 6000

28. 3 cubes of metal whose edges are in the ratio 3:4:5 are melted into a single cube, the length of whose

diagonal is $48\sqrt{3}$ m. Calculate the edges of the three cubes.

(a) 24 m, 32 m, 40 m (b) 40 m, 32 m, 24 m
(c) 30 m, 22 m, 18 m (d) 48 m, 36 m, 24 m

29. A cube of lead with edges measuring 6 cm each is melted and recasted into 27 equal cubes. The length of the edge of the new cube is:

(a) 3 cm (b) 4 cm
(c) 2 cm (d) 1.5 cm

30. The volume of a cube is 729 cm^3. The total surface area of the cube is:

(a) 216 cm^2 (b) 384 cm^2
(c) 486 cm^2 (d) 512 cm^2

31. 2 cubes have volumes in the ratio 1:27. The ratio of the area of the face of one to that of the other is:

(a) 1:2 (b) 1:3
(c) 1:6 (d) 1:9

32. 2 cubes, each of side 12 cm, are joined end-to-end. The surface area of the resulting cuboid is:

(a) 1240 cm^2 (b) 1440 cm^2
(c) 2250 cm^2 (d) 4252 cm^2

33. The perimeter of one face of a cube is 20 cm. Its volume is:

(a) 1009 cm^3 (b) 525 cm^3
(c) 320 cm^3 (d) 125 cm^3

34. A cube of edge 3 cm of iron weighs 12 gm. What is the weight of a similar cube of iron whose edge is 12 cm?

(a) 768 gm (b) 678 gm
(c) 964 gm (d) 864 gm

35. The weight of a solid cube of iron of 1 cm edge is 17 gm. What should be the weight with a similar cube of edge 3 cm?

(a) 449 gm (b) 459 gm
(c) 469 gm (d) 4390 gm

36. A cubic metre of silver weighing 900 Kg is rolled into a 16 m long square bar. Find out the weight of an exact cube cut off from it.

(a) 14 Kg $62\frac{1}{2}$ gm (b) 30 Kg
(c) 10 Kg (d) 7 Kg 50 gm

37. A 4 cm cube is cut into 1 cm cubes. What is the ratio of the surface area of small cubes to that of the large cube?

(a) 1:16 (b) 2:3
(c) 4:1 (d) 6:1

38. A large cube is formed from the material obtained by melting three smaller cubes of 3, 4 and 5 cm side. What is the ratio of the total surface areas of the smaller cubes and the large cube?

(a) 2:1 (b) 3:2
(c) 25:18 (d) 27:20

39. How many small cubes, each of 96 cm^2 surface area, can be formed from the material obtained by melting a larger cube with 384 cm^2 surface area?

(a) 8 (b) 5
(c) 800 (d) 8000

40. A cubical metallic tank whose each edge measures 30 cm, is completely filled with water. If 2,7 litres water is taken out of it, what will be the depth of the remaining water in the tank?

(a) 37 cm (b) 27 cm
(c) 17 cm (d) None of these

41. Find out the weight of a hollow cylindrical lead pipe 28 cm long and $\frac{1}{2}$ cm thick. Its internal diameter is 8 cm. $\left(\text{Weight of 1 } cm^3 \text{ of lead is 11.4 g, } \pi = \frac{22}{7}\right)$

(a) 3.762 Kg (b) 4.562 Kg
(c) 7.462 Kg (d) 6.762 Kg

42. Volume of the cylinder is 1650 m^3. whereas the surface area of its base is $78\frac{4}{7}$ m^2. Find out the height of the cylinder.

(a) 2.1 m (b) 7.5 m
(c) 21 m (d) 14 m

43. 1496 cm^3 of a metal is used to cast a pipe of length 28 cm. If the internal radius of the pipe is 8 cm, then the outer radius of the pipe is:

(a) 7 cm (b) 9 cm
(c) 10 cm (d) 12 cm

44. The base of a of 10 cm high solid cylinder is a semi-circle of radius 7 cm. Its total surface (in cm^2) is $\left(\text{Use } \pi = \frac{22}{7}\right)$

(a) 154 (b) 176
(c) 514 (d) None of these

45. A sphere is melted to form a cylinder whose height is $4\frac{1}{2}$ times its radius; What is the ratio of radii of sphere to the cylinder?

(a) 3:2 (b) 4:3
(c) 3:5 (d) 2:3

46. The radius of a cylinder is the same as that of a sphere. Their volumes are equal. The height of the cylinder is:

(a) 4/3 times its radius. (b)2/3 times its radius.
(c) equal to its radius. (d)equal to its diameter.

47. A 12 m deep well with internal diameter 3.5 m is dug up. The earth from it is spread evenly to form a platform 10.5 m by 8.8 m. Determine the height of the platform.

(a) 2.25 m (b) 3.25 m
(c) 1.25 m (d) 4.25 m

48. The curved surface of a well is 264 sq m and its capacity is 924 m^3. What is the diameter and the depth of the well?

(a) 8 m (b) 9 m
(c) 4.5 m (d) 6 m

49. The sum of the radius of the base and the height of a solid cylinder is 37 m. If the total surface area of the cylinder be 1628 m^2, find out the volume:

(a) 4620 m^3 (b) 4630 m^3
(c) 4520 m^3 (d) 4830 m^3

50. A brick measures 20 cm by 10 cm by $7\frac{1}{2}$ cm. How many bricks will be required for constructing a 25 m long, 2 m high and $\frac{3}{4}$ m thick wall?

(a) 25000 (b) 35000
(c) 20000 (d) 45000

51. The height of a right circular cylinder is 6 m. 3 times the sum of the areas of its two circular faces is twice the area of its curved surface. The radius of the base is:

(a) 4 m (b) 2 m
(c) 6 m (d) 1.5 m

52. The radius of the cylinder is made twice as large. How should the height be changed so that the volume remains the same?

(a) $\frac{1}{2}$ × height of two cylinders

(b) $\frac{1}{4}$ × height of original cylinder

(c) $\frac{1}{4}\ \pi r^2$

(d) None of these

53. A right cylinder and a right circular cone have the same radius and the same volume. The ratio of the height of the cylinder to that of the cone is:

(a) 3:5 (b) 2:5
(c) 3:1 (d) 1:3

54. 2 cans have the same height equal to 21 m. One can is cylindrical, the diameter of whose base is 10 cm. The other can has square base of side 10 cm. What is the difference in between their capacities?

(a) 350 cm^2 (b) 450 cm^3
(c) 250 cm^2 (d) None of these

55. A roller is 120 cm long and has diameter 84 cm. If it takes 500 complete revolutions to level a play ground, then determine the cost of levelling at the rate of 30 paise per m^2. $\left(\text{Use } \pi = \frac{22}{7}\right)$

(a) ₹475.40 (b) ₹375.45
(c) ₹375.20 (d) ₹475.20

56. The circumference of one end of a frustum of a right circular cone is 48 cm and of the other end is 34 cm. If the height of the frustrum is 10 cm, its volume (in cm^3) is:

(a) 5400 (b) 1350
(c) 2700 (d) 4050

57. Find out the amount of concrete required to erect a concrete pillar whose circular base will have a perimeter 8.8 m and whose curved surface is 17.6 m. $\left(\pi = \frac{22}{7}\right)$

(a) $12\frac{4}{25}\ m^3$ (b) $12\frac{3}{25}\ m^3$

(c) $12\frac{1}{2}\ m^3$ (d) $12\frac{8}{25}\ m^3$

58. Sum of the length, width and depth of a cuboid is s and its diagonal is d. Its surface area is:

(a) s^2 (b) d^2
(c) $s^2 - d^2$ (b) $s^2 + d^2$

59. A cylindrical tower is 5 m in diameter and 14 m high. The cost of white washing its curved surface at 50 paise per m^2 is:

(a) ₹90 (b) ₹97
(c) ₹100 (d) ₹110

60. A solid piece of iron of dimensions 49 × 33 × 24 cm is moulded into a sphere, The radius of the sphere is:

(a) 35 cm (b) 21 cm
(c) 29 cm (d) None of these

61. How many coins, 2 mm thick and 1.5 cm in diameter, should be melted in order to form a right circular cylinder its base diameter 6 cm and height 8 cm?

(a) 640 (b) 540
(c) 740 (d) 840

62. A hemisphere is made of lead. Its of radius is 6 cm; It cast into a right circular cone of 75 cm height. The radius of the base of the cone is:

(a) 1.4 cm (b) 2.4 cm
(c) 1.6 cm (d) 3.2 cm

63. A solid cylinder has a total surface area of 231 cm^2. Its curved surface area is (2/3) of the total surface area. Find out the volume of the cylinder.

(a) 270 cm^3 (b) 269.5 cm^3
(c) 256.5 cm^3 (d) 289.5 cm^3

64. It is required to design a circular pipe such that water flowing through it at a speed of 7 m per min fills a tank of capacity 440 cubic m in 10 min. The inner radius of the pipe should be:

(a) 2 m (b) $\sqrt{2}$ m
(c) $\frac{1}{2}$ m (d) $\frac{1}{\sqrt{2}}$ m

65. From a solid right circular cylinder with height 10 cm and radius of the base 6 cm; a right circular cone of the same height and base is removed. The volume (in cm^3) of the remaining solid is:

(a) 377 (b) 754.3
(c) 1131 (d) None of these

66. The radii of two cylinders are in the ratio 2:3. The ratio their height is 5:3. The ratio of their volume is:

(a) 20:27 (b) 10:9
(c) 18:13 (d) 9:20

67. The capacity of a tank, in the form of a cylinder, is 6160 m^3. If the diameter of its base is 28 m, find out the cost of painting its inner curved surface at the rate of ₹2.8 per m^2. $\left(\text{Use } \pi = \frac{22}{7}\right)$

(a) 2464 (b) 2664
(c) 3064 (d) 2864

68. The sum of the radius of the base and the height of a solid cylinder is 37 m. If the total surface area of the solid cylinder is 1628 m^2, then the circumference of its base and the volume of the cylinder are:

(a) 68 m; 7875 m^3 (b) 52 m; 5825 m^3
(c) 44 m; 4620 m^3 (d) 30 m; 3859 m^3

69. A rectangular piece of paper is 22 cm long and 10 cm wide. A cylinder is formed by rolling the paper along its length. The volume of the cylinder is:

(a) 225π cm^3 (b) 385 cm^3
(c) 25π cm^3 (d) None of these

70. The ratio of total surface area to lateral surface area of a cylinder whose radius is 80 cm and height 20 cm, is:

(a) 2:1 (b) 3:1
(c) 4:1 (d) 5:1

71. A right cylindrical vessel is full with water. How many right cones having the same diameter and height as those of right cylinder will be needed to store that water?

(a) 2 (b) 3
(c) 4 (d) 5

72. A cylindrical bucket is 72 cm high and 28 cm in diameter and is full of water. This water is emptied in a rectangular tank whose length and breadth are 66 cm and 28 cm, respectively. What will be the height of the water level in the tank?

(a) 36 cm (b) 48 cm
(c) 24 cm (d) 22 cm

73. A cylindrical iron rod is 70 cm long. The diameter of its end portion is 2 cm. What is its weight, reckoning a cm^3 of iron to weigh 10 grams?

(a) 4 Kg (b) 4.2 Kg
(c) 2.2 Kg (d) Data inadequate

74. If the radius of a cylinder is doubled and the height is halved, then what would be ratio between the new curved surface area and the previous curved surface area of the cylinder:

(a) 1:1 (b) 2:1
(c) 3:2 (d) 2:3

75. A cylindrical jar of diameter 24 cm contains water to a height of 30 cm. A spherical steel ball is dropped into the jar and the level of the water rises by 67.5 mm. The diameter of the ball is:

(a) 16 cm (b) 15 cm
(c) 20 cm (d) 18 cm

76. The material of a solid cone is converted into the shape of solid cylinder of equal radius. If the height of the cylinder is 5 cm, then what is the height of the cone?

(a) 25 cm (b) 15 cm
(c) 20 cm (d) 10 cm

77. The volume of solid cylinder whose diameter of the base is 14 mm and length 25 mm is 3850 mm^3. If length of the cylinder is doubled, but the diameter is halved, then what will be the volume of the resulting cylinder?

(a) 1172 mm^3 (b) 1925 mm^3
(c) 3850 mm^3 (d) 7700 mm^3
(e) None of these

78. A monument has 50 cylindrical pillars each of diameter 50 cm and height 4 m. What will be the labour charges for cleaning these pillars at the rate of 50 paise per m^2? (Use $\pi = 3.14$):

(a) ₹237 (b) ₹257
(c) ₹157 (d) ₹353

79. The radius of a cylinder is made twice large. How should the height be changed, so that its volume remains unchanged?

(a) $\frac{1}{4}$ of original (b) $\frac{1}{3}$ of original
(c) $\frac{1}{2}$ of original (d) $\frac{1}{8}$ of original

80. A spherical ball of lead, 3 cm in diameter, is melted and re-cast into three spherical balls. The diameter of 2 of these are 1.5 cm and 2 cm, respectively. The diameter of the third ball is:

(a) 2.66 cm (b) 2.5 cm
(c) 3 cm (d) 3.5 cm

81. A cone and a cylinder having the same area of the base have also the same area of curved surfaces. If the height of cylinder be 2 m, find out the slant height of the cone:

(a) 3 m (b) 3.5 m
(c) 4.5 m (d) 4 m

82. The radii of a cylinder and a cone are equal. If the height of the cylinder is equal to the slant height of the cone then the ratio of the curved surfaces of the cylinder and the cone is:

(a) 1:1 (b) 2:1
(c) 3:1 (d) 4:1

83. From a cubical block of wood of side 1 m, a cylinder of the largest possible volume is curved out. The volume (in m^3) of the remaining wood is:

(a) $\frac{3}{14}$ (b) $\frac{5}{14}$
(c) $\frac{1}{2}$ (d) $\frac{2}{7}$

84. The radius of the base of a solid cylinder is r cm and its height is 3 cm. It is re-casted into a cone of same radius, the height of the cone will be:

(a) 3 cm (b) 6 cm
(c) 9 cm (d) 27 cm

85. 2 cm of rain has fallen on a Km^2 of land. Assuming that 50% of the raindrops could have been collected and contained in a pool having a 100 m × 10 m base, by what level would the water level in the pool have increased?

(a) 15 m (b) 20 m
(c) 10 m (d) 25 m

86. The perpendicular height of a conical tent is $4\frac{2}{3}$ m and the diameter of its base is 6 m. If 11 persons can sleep in this tent, find how many average cu m of air each person gets?

(a) 2 cu m (b) 4 cu m
(c) 6 cu m (d) 8 cu m

87. The circumference of the base of a 9 m high conical tent is 44 m. The volume of the air contained in it is:

(a) 462 m^3 (b) 452 m^3
(c) 472 m^3 (d) 512 m^3

88. A conical vessel of base radius 2 cm and height 3 cm is filled with kerosene. This liquid leaks through a hole in the bottom and collects in a cylindrical jar of radius 2 cm. The kerosene level in the jar is:

(a) 1.5 cm (b) π cm
(c) 1 cm (d) 3 cm

89. A hollow cylinder of height 3 cm, is re-casted into a solid cylinder. If the external and internal radii of the hollow cylinder are 4.3 cm and 1.1 cm, respectively. What will be the radius of the solid cylinder?

(a) 2.8 cm (b) 2.4 cm
(c) 3.2 cm (d) 4.8 cm

90. A solid consists of a circular cylinder with an exact fitting right circular cone placed on the top. The height of the cone is h. If the total volume of the solid is three times the volume of the cone, then the height of the cylinder is:

(a) $2h$ (b) $4h$

(c) $\frac{2h}{3}$ (d) $\frac{3h}{3}$

91. A well of 11.2 m diameter is dug 8 m deep. The earth taken out has been spread all around it to a width of 7 cm to form a circular embankment. Find out the height of the embankment.

(a) 304.8 m^2 (b) 400.4 m^2
(c) 408.4 m^2 (d) 412.4 m^2

92. The curved surface of a circular cylinder of height h and the slant surface of the cone of slant height '$2h$' having the same circular base are in the ratio of:

(a) 1:1 (b) 1:2
(c) 3:2 (d) 1:3

93. The material of a cone is converted into the shape of a cylinder of equal radius. If the height of the cylinder is 5 cm, the height of the cone is:

(a) 10 cm (b) 15 cm
(c) 18 cm (d) 24 cm

94. A right circular cone is exactly fitted inside a cube in such a way that the edges of the base of the cone are touching the edges of one of the faces of the cube and the vertex is on the opposite face of the cube. If the volume of the cube is 343 c.c, then what approximately is the volume of the cone?

(a) 90 c.c. (b) 75 c.c.
(c) 80 c.c. (d) 85 c.c.

95. A solid cone is 25 cm high and the radius of its base is 50 cm. It is melted and re-cast into a solid sphere. Determine the surface area of the sphere.

(a) 8757.28 cm^2 (b) 5877.42 cm^2
(c) 7857.14 cm^2 (d) None of these

96. The radius and height of right circular cone are in the ratio 5:12. If its volume is 314 $\frac{3}{7}$ m^3. Find out the radius of the cone.

(a) 5 m (b) 8 m
(c) 12 m (d) 6 m

97. A cone, a hemisphere and a cylinder stand on equal bases and have the same height. The ratio of their volumes are:

(a) 1:2:2 (b) 1:2:3
(c) 1:2:4 (d) 2:3:4

98. Find out the length of the canvas 2 m in width required to make a conical tent 12 m in diameter and 6.3 m in slant height:

(a) 118.8 m (b) 62.4 m
(c) 59.4 m (d) 112.4 m

99. The radius of a cylinder is doubled and the height is halved, what is the ratio between the new volume and the previous volume?

(a) 3:1 (b) 2:3
(c) 2:1 (d) 1:3

100. A circus tent is cylindrical to a height of 3 m and conical above it. If its diameter is 105 m and slant height of the conical portion is 53 m, then calculate the length of the canvas 5 m wide to make the tent.

(a) 1857 m (b) 1647 m
(c) 1947 m (d) 1847 m

101. If base radius of a cone is increased by 20% and its slant height is doubled, then by how much per cent will the area of its curved surface be increased?

(a) 140% (b) 160%
(c) 130% (d) 180%

102. The radius of the base of conical tent is 5 cm. If the tent is 12 m high, then area of the canvas required in making the tent is:

(a) 60 π m^2 (b) 300 π m^2
(c) 90 π m^2 (d) None of these

103. A cone of height 7 cm and base radius 3 cm is carved from a rectangular block of wood 10 cm × 5 cm × 2 cm. The percentage % wood wasted is:

(a) 34% (b) 46%
(c) 54% (d) 66%

104. The diameter and slant height of a conical tomb are 28 m and 50 m, respectively. The cost of white washing its curved surface at the rate of 80 paise per m^2 is:

(a) ₹2640 (b) ₹1760
(c) ₹264 (d) ₹176
(e) None of these

105. A rectangular sheet of area 264 cm^2 and width 11 cm is rolled along its breadth to make a hollow cylinder. The volume of the cylinder is:

(a) 231 c.c. (b) 230 c.c.
(c) 235 c.c. (d) 234 c.c.

106. A cylinder and a cone have their heights in the ratio 2:3 and the radii of their bases in the ratio 3:4. Find out the ratio of their volumes.

(a) 1:9 (b) 2:9
(c) 9:8 (d) 1:8

107. If the height of a cone is doubled, then its volume is increased by:

(a) 100% (b) 200%
(c) 300% (d) 400%

108. 3 cubes of side 3, 4 and 5, respectively, are melted to form into new cube. The side of the new cube is:

(a) 5 cm (b) 6 cm
(c) 6.5 cm (d) 7 cm

109. The height and base radius of a cone are each increased by 100%. The volume of the cone now becomes:

(a) double the original.
(b) 4 times the original.
(c) 3 times the original.
(d) 8 times the original.

110. If the radius of a sphere is doubled, then its volume is increased by:

(a) 100% (b) 200%
(c) 700% (d) 800%

111. The diameter of a sphere is 6 cm. It is melted and drawn into a wire of diameter 0.2 cm. Find out the length of the wire.

(a) 24 m (b) 28 m
(c) 36 m (d) 32 m

112. A cone-shaped circular tent is 9 m high and the circumference of its circular base is 44 m. How much air is contained in the tent? $\left(\text{Use } \pi = \frac{22}{7}\right)$

(a) 362 m^3 (b) 462 m^3
(c) 562 m^3 (d) 662 m^3

113. If the radius of a sphere is doubled, then its surface area is increased by:

(a) 100% (b) 200%
(c) 300% (d) 50%

114. The height of a cylinder is decreased by 8%, keeping its radius unchanged. What is the percentage change in its volume?

(a) 8% increase (b) 12% decrease
(c) 8% decrease (d) None of these

115. The radius of a cylinder is increased by 20%, keeping its height unchanged. What is the percentage increase in its volume?

(a) 33% (b) 44%
(c) 22% (d) None of these

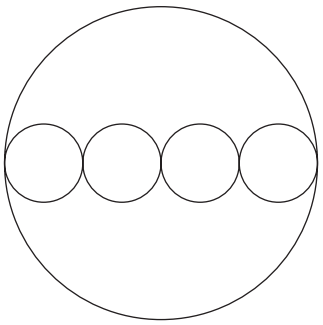
EXERCISE-2
(BASED ON MEMORY)

1. The radius of base of solid cone is 9 cm. and its height is 21 cm. It cut into 3 parts by two cuts which are parallel to its base. The cuts are at height of 7 cm. and 14 cm. from the base respectively. What is the ratio of curved surface area of top, middle and bottom parts respectively?

(a) 1: 4: 8 (b) 1: 3: 5
(c) 1: 3: 9 (d) 1: 6: 12

[SSC CGL Tier-II CBE, 2018]

2. A solid cylinder has radius of base 14 cm. and height 15 cm. identical cylinders are cut from each base as shown in the given figure. Height of small cylinder is 5 cm. what is the total surface area (In cm^2) of the remaining part?



(a) 3740 (b) 3432
(c) 3124 ` (d)42818

[SSC CGL Tier-II CBE, 2018]

3. Find the curved surface area (in cm^2) of a hemisphere of diameter 28 cm.

(a) 1152 (b) 1024
(c) 956 (d) 1232

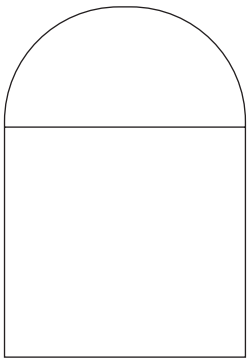
[SSC CHSL (10+2) Tier-I CBE, 2018]

4. The ratio of curved surface area and volume of a cylinder is 1:7. The ratio of total surface area and volume is 187:770. What is the respective ratio of its base radius and height?

(a) 5:8 (b) 4:9
(c) 3:7 (d) 7:10

[SSC CGL Tier-II CBE, 2018]

5. A hemisphere is kept on top of a cube. Its front view is shown in the given figure. The total height of the figure is 221 cm. The ratio of curved surface area of hemisphere and total surface area of cobe is 11:42. What is the total volume (in cm) of figure?



(a) 3318.33 (b) 3462.67
(c) 3154.67 (d) 3248.33

[SSC CGL Tier-II CBE, 2018]

6. 10 identical solid spherical balls of radius 3 cm. are melted to form a single sphere. In this process 20% of solid is wasted. What is the radius (in cm) of the biggest sphere?

(a) 24 (b) 12
(c) 8 (d) 6

[SSC CGL Tier-II CBE, 2018]

7. A right circular cylinder has height as 18 cm and radius as 7 cm. The cylinder is cut in the three equal parts (by 2 cuts parallel to base). What is the percentage increase in total surfacearea?

(a) 62 (b) 56
(c) 48 (d) 52

[SSC CGL Tier-II CBE, 2018]

8. Radius of base of a hollow cone is 8 cm. and its height is 15 cm. A sphere of largest radius is put inside the cone. What is the ration of base of cone to the radius of sphere?

(a) 5:3 (b) 4:1
(c) 2:1 (d) 7:3

[SSC CGL Tier-II CBE, 2018]

9. The Total surface area of a hemisphere is 41.58 sq. cm. Find its curved surface area.

(a) 27.72 sq. cm. (b) 55.44 sq. cm.
(c) 9.24 sq. cm. (d) 13.86 sq. cm.

[SSC CGL Tier-I CBE, 2017]

10. What is the curved surface area (In cm^2) of a cylinder having radius of base as 14 cm and height as 10 cm?

(a) 440 (b) 880
(c) 220 (d) 1320

[SSC CAPFs ASI & Delhi Police SI, 2017]

11. The perimeter of base of a right circular cone is 132 cm. If the height of the cone is 72 cm. then what is the total surface area (In cm^2) of the cone?

(a) 6600 (b) 6336
(c) 4224 (d) 5784

[SSC CAPFs ASI & Delhi Police SI, 2017]

12. A solid right circular cone of radius 4 cm and height 7 cm is put inside a cylindrical vessel of radius 5 cm and height 8 cm. How much water in cube cm will be required to fill the cylindrical vessel completely?

(a) 1022.48 cubic cm (b) 1533.72 cubic cm
(c) 511.24 cubic cm (d) 255.62 cubic cm

[SSC CGL Tier-I CBE, 2017]

13. The cross section of a canal is in the shape of an isosceles trapezium which is 4 metre wide art the bottom and 5 metre wide at the top. If the depth of the canal is 2 metre and it is 120 metre long, what is the maximum capacity of this canal?

(a) 2160 cubic metre (b) 3240 cubic metre
(c) 4320 cubic metre (d) 1080 cubic metre

[SSC CHSL (10+2) Tier-I CBE, 2017]

14. The curved surface area of a hemisphere is 27.72 square cm and volume is 19.404 cubic cm, find its radius.

(a) 4.2 cm. (b) 2.1 cm.
(c) 0.7 cm. (d) 1.05 cm.

[SSC CGL Tier-I CBE, 2017]

15. What is the volume (in cm^3) of a right pyramid of height 12 cm and having a square base whose diagonal is $6\sqrt{2}$ cm?

(a) 864 (b) 432
(c) 144 (d) 288

[SSC CAPFs ASI & Delhi Police SI, 2017]

16. A solid cone of height 24 cm and having radius of base 8 cm is melted to form a solid cylinder of radius 6 cm and height 6cm. In the whole process what percent of material is wasted?

(a) 48.5 (b) 37.5
(c) 57.8 (d) 64

[SSC CGL Tier-I CBE, 2017]

17. A solid cylinder having radius of base as 7 cm. and length as 20 cm. is bisected from its height to get two identical cylinders. What will be the percentage increase in the total surface area?

(a) 29.78 (b) 25.93
(c) 27.62 (d) 32.83

[SSC CAPFs ASI & Delhi Police SI, 2017]

18. Three sphereical balls of radius 2 cm, 4 cm and 6 cm are melted to form a new apherical ball. In this process there is a loss of 25% of the material. What is the radius (in cm) of the new ball?

(a) 6 (b) 8
(c) 12 (d) 16

[SSC CAPFs ASI & Delhi Police SI, 2017]

19. A solid cone of height 36 cm and radius of base 9 cm is melted to form a solid cylinder of radius 9 cm and height 9 cm. What percent of material is wasted in this process?

(a) 25 (b) 0
(c) 5 (d) 10

[SSC CGL Tier-I CBE, 2017]

20. The radius of a wheel is 3.5 cm. What is the distance (in cm) travelled by the wheel in 20 revolutions?

(a) 220 (b) 440
(c) 880 (d) 1320

[SSC CAPFs ASI & Delhi Police SI, 2017]

21. What is the length (in metre) of the longest rod that can be placed in a room which is 2 metre long 2 metre broad and 6 metre high?

(a) 8 (b) $2\sqrt{11}$
(c) $3\sqrt{11}$ (d) 10

[SSC CAPFs ASI & Delhi Police SI, 2017]

22. The radius of a wheel is 21 cm. What is the distance (in cm) travelled by the wheel in 10 revolutions?

(a) 660 (b) 1320
(c) 1980 (d) 2640

[SSC CAPFs ASI & Delhi Police SI, 2017]

23. A right circular cylindrical tunnel of diameter 5m and length 10m is to be constructed from a sheet of iron. The area of iron sheet required will be:

(a) 52π (b) 50π
(c) 51π (d) 49π

[SSC CAPFs (CPO) SI & ASI, Online, 2016]

24. A rectangular water tank is 80 metre x 40 metre. Water flows into it through a pipe of 40 sq. cm. at the opening at a speed of 10 km/hr. The water level will rise in the tank in half an hour by

(a) $\frac{3}{2}$ cm. (b) $\frac{4}{9}$ cm.

(c) $\frac{5}{9}$ cm. (d) $\frac{5}{8}$ cm.

[SSC CGL Tier-II, 2016]

25. A right circular cylinder having diameter 21 cm and height 38 cm is full of ice cream. The ice cream is to be filled in cones of height 12 cm and diameter 7 cm having a hemispherical shape on the top. The number of such cones to be filled with ice cream is

(a) 54 (b) 44
(c) 36 (d) 24

[SSC CGL Tier-II, 2016]

26. A spherical aquarium can accommodate 11 fishes, and each fish requires 1.54 cu. Metre of water. What is the volume of the aquarium?

(a) 11.14 cu. metre (b) 16.94 cu. Metre
(c) 10.25 cu. Metre (d) 17.84 cu. Metre

[SSC CPO, 2016]

27. A prism with a right triangular base is 25 cm high. If the shorter sides of the triangle are in the ration of 1:2 and the volume of the prism is 100 cm^3 what is the length of the longest side of the triangle?

(a) $\sqrt{5}$ cm (b) $2\sqrt{5}$ cm
(c) $5\sqrt{2}$ cm (d) 5 cm

[SSC CAPFs (CPO) SI & ASI, Delhi Police, 2016]

28. A hollow hemispherical bowl is made of silver with its outer radius 8 cm and inner radius 4 cm respectively. The bowl is melted to form a solid right circular cone of radius 8 cm. The height of the cone formed is

(a) 7 cm (b) 9 cm
(c) 12 cm (d) 14 cm

[SSC CGL Tier-I (CBE), 2016]

29. A solid sphere and a solid hemisphere have the same total surface area. The ratio of their volumes is $\left(\text{Taker}, \pi = \frac{22}{7}\right)$

(a) $3\sqrt{3} : 4$ (b) $4 : 3\sqrt{3}$
(c) $3 : 4\sqrt{3}$ (d) $1 : 12\sqrt{3}$

[SSC CGL Tier-II (CBE), 2016]

30. Three solid spheres have their radii r_1, r_2 and r_3. The spheres are melted to form a solid spheres of bigger radius, Then the radius of the new spheres is:

(a) $(r_1 + r_2 + r_3)$ (b) $(r_1^2 + r_2^2 + r_3^2)^{\frac{1}{2}}$
(c) $(r_1^3 + r_2^3 + r_3^3)^{\frac{1}{3}}$ (d) $(r_1^4 + r_2^4 + r_3^4)^{\frac{1}{4}}$

[SSC CGL Tier-I (CBE), 2016]

31. An inverted conical shaped vessel is filled with water to its brim. The height of the vessel is 8 cm and radius of the open end is 5 cm. When a few solid spherical metallic balls each of radius $\frac{1}{2}$ cm are dropped in the vessel.25% water is overflowed. The number of balls is:

(a) 100 (b) 400
(c) 200 (d) 150

[SSC CGL Tier-I (CBE), 2016]

32. The respective ratio of curved surface area and total surface area of a cylinder is 4 : 5. If the curved surface area of the cylinder is 1232 cm^2, what is the height? (in cm)

(a) 14 cm (b) 28 cm
(c) 7 cm (d) 56 cm
(e) 24 cm

[IBPS, 2015]

33. A plane divides a right circular cone into two parts of equal volume. If the plane is parallel to the base, then the ratio, in which the height of the cone is divided, is

(a) $1 : \sqrt[3]{2} - 1$ (b) $1 : \sqrt{2}$
(c) $1 : \sqrt[3]{2}$ (d) $1 : \sqrt[3]{2} + 1$

[SSC, 2015]

34. A right prism has a triangular base whose sides are 13 cm, 20 cm and 21 cm. If the altitude of the prism is 9 cm, then its volume is

(a) 1413 cm^3 (b) 1314 cm^3
(c) 1143 cm^3 (d) 1134 cm^3

[SSC, 2015]

35. A cylinder with base radius 8 cm and height 2 cm is melted to form a cone of height 6 cm. The radius of the cone will be

(a) 5 cm (b) 4 cm
(c) 8 cm (d) 6 cm

[SSC, 2015]

36. The portion of a ditch 48 m long, 16.5 m wide and 4 m deep that can be filled with stones and earth available during excavation of a tunnel, cylindrical in shape, of diameter 4 m and length 56 m is $\left(\text{Take}\pi = \frac{22}{7}\right)$

(a) $\frac{1}{2}$ Part (b) $\frac{1}{4}$ Part
(c) $\frac{2}{9}$ Part (d) $\frac{1}{9}$ Part

[SSC, 2015]

37. The numerical values of the volume and the area of the lateral surface of a right circular cone are equal. If the height of the cone be h and radius be *r*, the value of $\frac{1}{h^2} + \frac{1}{r^2}$ is

(a) $\frac{9}{1}$ (b) $\frac{1}{9}$
(c) $\frac{3}{1}$ (d) $\frac{1}{3}$

[SSC, 2015]

38. The radii of two solid iron spheres are 1 cm and 6 cm respectively. A hollow sphere is made by melting the two spheres. If the external radius of the hollow sphere is 9 cm, then its thickness (in cm) is

(a) 1.5 (b) 2
(c) 0.5 (d) 1

[SSC, 2015]

39. If a hemisphere is melted and four spheres of equal volume are made, the radius of each sphere will be equal to

(a) 1/2 of the radius of the hemisphere
(b) $1/6^{th}$ of the radius of the hemisphere
(c) radius of the hemisphere
(d) $1/4^{th}$ of the radius of the hemisphere

[SSC, 2015]

40. There is a wooden sphere of radius 6 $\sqrt{3}$ cm. The surface area of the largest possible cube cut out from the sphere will be

(a) 464 $\sqrt{3}$ cm^2 (b) 864 cm^2

(c) 462 cm^2 (d) 646 $\sqrt{3}$ cm^2

[SSC, 2015]

41. If 64 buckets of water are removed from a cubical shaped water tank completely filled with water, 1/3 of the tank remains filled with water. The length of each side of the tank is 1.2 m. Assuming that all buckets are of the same measure, then the volume (in litres) of water contained by each bucket is

(a) 15 (b) 18

(c) 16 (d) 12

[SSC, 2015]

42. Base of a right pyramid is a square of side 10 cm. If the height of the pyramid is 12 cm, then its total surface area is

(a) 460 cm^2 (b) 260 cm^2

(c) 360 cm^2 (d) 400 cm^2

[SSC, 2015]

43. The length of canvas 75 cm wide required to build a conical tent of height 14 m and the floor area 346.5 m^2 is:

(a) 770 m (b) 490 m

(c) 860 m (d) 665 m

[SSC, 2015]

44. If the area of the base, height and volume of a right prism be $\left(3\sqrt{3}/2\right)P^2 cm^2, 100\sqrt{3}$ and 7200 cm^3 respectively, then the value of P will be?

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{3}{2}$

(c) $\sqrt{3}$ (d) 4

[SSC, 2015]

45. If the volume of a sphere is numerically equal to its surface area then its diameter is:

(a) 4 cm (b) 2 cm

(c) 3 cm (d) 6 cm

[SSC, 2015]

46. 5 persons will live in a tent. If each person requires $16m^2$ of floor area and $100m^3$ space for air then the height of the cone of smallest size to accommodate these persons would be?

(a) 18.75 m (b) 16 m.

(c) 10.25 m (d) 20 m

[SSC, 2015]

47. The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base. If its volume be $\frac{1}{27}$th of the volume of the given cone, at what height above the base is the section made?

(a) 19 cm (b) 20 cm

(c) 12 cm (d) 15 cm

[SSC Examination, 2014]

48. If the surface area of a sphere is 346.5 cm^2, then its radius is $\left[\text{taking } \pi = \frac{22}{7}\right]$

(a) 7 cm (b) 3.25 cm

(c) 5.25 cm (d) 9 cm

[SSC Examination, 2014]

49. The height of the right pyramid whose area of the base is 30 m^2 and volume is 500 m^3, is:

(a) 50 m (b) 60 m

(c) 40 m (d) 20 m

[SSC Examination, 2014]

50. The base of a prism is a right-angled triangle with 2 sides 5 cm and 12 cm. The height of the prism is 10 cm. The total surface area of the prism is :

(a) 360 cm^2 (b) 300 cm^2

(c) 330 cm^2 (d) 325 cm^2

[SSC Examination, 2014]

51. The base of a right prism is an equilateral triangle. If the lateral surface area and volume is 120 cm^2 and $40\sqrt{3}$ cm^3, respectively, then the side of base of the prism is:

(a) 4 cm (b) 5 cm

(c) 7 cm (d) 40 cm

[SSC Examination, 2014]

52. A ball of lead, 4 cm in diameter, is covered with gold. If the volume of the gold and lead are equal, then the thickness of gold is approximately [given $\sqrt[3]{2} = 1.259$]

(a) 5.038 cm (b) 5.190 cm

(c) 1.038 cm (d) 0.518 cm

[SSC Examination, 2014]

53. A large solid sphere is melted and moulded to form identical right circular cones with base radius and height same as the radius of the sphere. One of

these cones is melted and moulded to form a smaller solid sphere. Then the ratio of the surface area of the smaller to the surface area of the larger sphere is:

(a) $1:3^{\frac{4}{3}}$ (b) $1:2^{\frac{3}{2}}$

(c) $1:3^{\frac{2}{3}}$ (d) $1:2^{\frac{4}{3}}$

[SSC Examination, 2014]

54. A conical cup is filled with ice cream. The ice cream forms a hemispherical shape on its open top. The height of the hemispherical part is 7 cm. The radius of the hemispherical part equals the height of the cone. Then the volume of the ice cream is $\left[\pi = \frac{22}{7}\right]$

(a) 1078 m^3 (b) 1708 m^3

(c) 7108 m^3 (d) 7180 m^3

[SSC Examination, 2014]

55. If each side of a cube is increased by 10%, the volume of the cube will increase by:

(a) 30% (b) 10%

(c) 33.1% (d) 25%

[SSC Examination, 2014]

56. A rectangular plot, 36 m long and 28 m broad, has two concrete roads 5 m wide running in the middle of the park,one parallel to the length and the other parallel to the breadth. What would be the total cost of gravelling the plot, excluding the area covered by the roads, at ₹3.60 per m^2?

(a) ₹2772.20 (b) ₹2466.60

(c) ₹2654.40 (d) ₹2332.60

(e) ₹2566.80

[IBPS PO/MT Examination, 2014]

57. A sphere is cut into two hemispheres. One of them is used as bowl. It takes 8 bowlfuls of this to fill a conical vessel of height 12 cm and radius 6 cm. The radius of the sphere (in centimeter) will be

(a) 3 (b) 2

(c) 4 (d) 6

[SSC, 2014]

58. The base of a right prism is a quadrilateral ABCD. Given that AB = 9 cm, BC = 14 cm, CD = 13 cm, DA = 12 cm and $\angle DAB = 90°$. If the volume of the prism be 2070 cm^3, then the area of the lateral surface is

(a) 720 cm^2 (b) 810 cm^2

(c) 1260 cm^2 (d) 2070 cm^2

[SSC, 2014]

59. The volumes of a right circular cylinder and a sphere are equal. The radius of the cylinder and the diameter of the sphere are equal. The ratio of height and radius of the cylinder is

(a) 3 : 1 (b) 1 : 3

(c) 6 : 1 (d) 1 : 6

[SSC, 2014]

60. A rectangular tin sheet is 12 cm long and 5 cm broad. It is rolled along its length to form a cylinder by making the opposite edges just to touch each other. Then the volume of the cylinder is

(a) $\frac{100}{\pi}\text{cm}^3$ (b) $\frac{60}{\pi}\text{cm}^3$

(c) $\frac{180}{\pi}\text{cm}^3$ (d) $\frac{120}{\pi}\text{cm}^3$

[SSC, 2014]

61. The area of the iron sheet required to prepare a cone 24 cm high with base radius 7 cm is $(Take \pi = \frac{22}{7})$

(a) 704 cm^2 (b) 408 cm^2

(c) 708 cm^2 (d) 804 cm^2

[SSC, 2014]

62. Some bricks are arranged in an area measuring 20 cu. m. If the length, breadth and height of each brick is 25 cm, 12.5 cm and 8 cm respectively, then in that pile the number of bricks are (suppose there is no gap in between two bricks)

(a) 4,000 (b) 10,000

(c) 6,000 (d) 8,000

[SSC, 2014]

63. A right circular cone is 3.6 cm high and radius of its base is 1.6 cm. It is melted and recast into a right circular cone with radius of its base as 1.2 cm. Then the height of the cone (in cm) is:

(a) 3.6 (b) 4.8

(c) 6.4 (d) 7.2

[SSC Examination, 2013]

64. If h, c, v are respectively the height, curved surface area and volume of a right circular cone, then the value of $3\pi vh^3 - c^2h^2 + 9v^2$ is:

(a) 2 (b) −1

(c) 1 (d) 0

[SSC Examination, 2013]

65. The volume of a conical tent is 1232 cu m and the area of its base is 154 m^2. Find the length of the canvas required to build the tent, if the canvas is 2 m in width. $\left(\text{Take } \pi = \frac{22}{7}\right)$

(a) 270 m (b)272 m
(c) 276 m (d)275 m

[SSC Examination, 2013]

66. Assume that a drop of water is spherical and its diameter is $\frac{1}{10}$ of a centimetre. A conical glass has a height equal to the diameter of its rim. If 32,000 drops of water fill the glass completely, then the height of the glass (in cm) is:

(a) 1 (b) 2
(c) 3 (d) 4

[SSC Examination, 2013]

67. The total number of spherical bullets, each of diameter 5 decimetre, that can be made by utilizing the maximum of a rectangular block of lead with 11 m length, 10 m breadth and 5 m width is (assume that π>3):

(a) Equal to 8800 (b) Less than 8800
(c) Equal to 8400 (d) Greater than 9000

[SSC Examination, 2013]

68. A rectangular block of metal has dimensions 21 cm, 77 cm and 24 cm. The block has been melted into a sphere. The radius of the sphere is $\left(\text{Take } \pi = \frac{22}{7}\right)$

(a) 21 cm (b) 7 cm
(c) 14 cm (d) 28 cm

[SSC Examination, 2013]

69. If a right circular cone of height 24 cm has a volume of 1232 cm^3, then the area (in cm^2) of curved surface is:

(a) 550 (b) 704
(c) 924 (d) 1254

[SSC Examination, 2013]

70. If each edge of a cube is increased by 50%, the percentage increase in surface area is:

(a) 125% (b) 50%
(c) 100% (d) 75%

[SSC Examination, 2013]

71. If the surface areas of two spheres are in the ratio 4:9, then the ratio of their volumes will be:

(a) 4:9 (b) 16:27
(c) 8:27 (d) 16:9

[SSC Examination, 2013]

72. If each edge of a cube is increased by 50%, the percentage increase in its surface area is:

(a) 150% (b) 75%
(c) 100% (d) 125%

[SSC Assistant Grade III, 2013]

73. The diameter of a copper sphere is 18 cm. The sphere is melted and is drawn into a long wire of uniform circular cross-section. If the length of the wire is 108 m, the diameter of the wire is:

(a) 1 cm (b) 0.9 cm
(c) 0.3 cm (d) 0.6 cm

[SSC Assistant Grade III, 2013]

74. A semicircular sheet of metal of diameter 28 cm is bent into an open conical cup. The capacity of the cup $\left(\text{taking } \pi = \frac{22}{7}\right)$ is:

(a) 624.26 cm^3 (b) 622.36 cm^3
(c) 622.56 cm^3 (d) 623.20 cm^3

[SSC Assistant Grade III, 2013]

75. If surface area and volume of a sphere are S and V, respectively, then value of $\frac{S^3}{V^3}$ is:

(a) 36π (b) 9π
(c) 18π (d) 27π

[SSC Assistant Grade III, 2013]

76. The edge of an ice cube is 14 cm. The volume of the largest cylindrical ice cube that can be formed out of it is

(a) 2200 cm^3 (b) 2000 cm^3
(c) 2156 cm^3 (d) 2400 cm^3
(e) None of these

[IBPS PO/MT Examination, 2013]

77. If the side of a cube is increased by 100%, its volume is increased by:

(a) 400% (b) 800%
(c) 200% (d) 100%

[UPPCS Examination, 2012]

78. A solid sphere of radius 1 cm is melted to convert into a wire of length 100 cm. The radius of the wire (using $\sqrt{3} = 1.732$) is:

(a) 0.08 cm (b) 0.09 cm
(c) 0.16 cm (d) 0.11 cm

[SSC Assistant Grade III, 2012]

79. A field is in the form of a rectangle of length 18 m and width 15 m. A pit, 7.5 m long, 6 m broad and 0.8 m deep, is dug in a corner of the field and the earth taken out is evenly spread over the remaining area of the field. The level of the field raised is:

(a) 12 cm (b) 14 cm
(c) 16 cm (d) 18 cm

[SSC Assistant Grade III, 2012]

80. The base of a right pyramid is an equilateral triangle of side 4 cm. The height of the pyramid is half of its slant height. Its volume is:

(a) $\frac{8}{9}\sqrt{2}$ cm^3 (b) $\frac{7}{9}\sqrt{3}$ cm^3

(c) $\frac{8}{9}\sqrt{3}$ cm^3 (d) $\frac{7}{9}\sqrt{2}$ cm^3

[SSC Assistant Grade III, 2012]

81. Water flows in a tank $150\text{ m} \times 100\text{ m}$ at the base, through a pipe whose cross-section is 2 dm by 1.5 dm, at the speed of 15 Km/h. In what time will the water be 3 m deep?

(a) 100 hours (b) 120 hours
(c) 140 hours (d) 150 hours

[SSC Assistant Grade III, 2012]

82. A tent is of the shape of a right circular cylinder upto a height of 3 m and then becomes a right circular cone with maximum height of 13.5 m above the ground. If the radius of the base is 14 m, the cost of painting the inner side of the tent at the rate of ₹2 per m^2 is:

(a) ₹2,050 (b) ₹2,060
(c) ₹2,068 (d) ₹2,080

[SSC Assistant Grade III, 2012]

83. If the diameter of a sphere is decreased by 25%, its curved surface area will be decreased by:

(a) 43.25% (b) 43.50%
(c) 43.75% (d) 44.25%

[SSC Assistant Grade III, 2012]

84. The radius of a cylinder is 10 cm and height is 4 cm. The number of centim that may be added either to the radius or to the height to get the same increase in the volume of the cylinder is:

(a) 5 (b) 4
(c) 25 (d) 16

[SSC Examination, 2012]

85. If a solid cone of volume 27 π cm^3 is kept inside a hollow cylinder whose radius and height are that of the cone, then the volume of water needed to the empty space is:

(a) 3π cm^3 (b) 18π cm^3
(c) 54π cm^3 (d) 81π cm^3

[SSC Examination, 2012]

86. Two cm of rain has fallen on a square Km of land. Assuming that 50% of the raindrops could have been collected and contained in a pool having a 100 m × 10 m base, by what level would he water level in the pool have increased?

(a) 1 Km (b) 10 m
(c) 10 cm (d) 1 m

[SSC Examination, 2012]

87. A cylindrical can whose base horizontal and is of internal radius 3.5 cm contains sufficient water so that when a solid sphere is placed inside, water just covers the sphere. The sphere fits in the can exactly. The depth of water in the can before the sphere was put is:

(a) $\frac{35}{3}$cm (b) $\frac{17}{3}$cm

(c) $\frac{7}{3}$cm (d) $\frac{14}{3}$cm

[SSC Examination, 2012]

88. The height of a circular cylinder is increased 6 times and the base area is decreased to $\frac{1}{9}$ of its value. The factor by which the lateral surface of the cylinder increases is:

(a) 2 (b) $\frac{1}{2}$

(c) $\frac{2}{3}$ (d) $\frac{3}{2}$

[SSC Examination, 2012]

89. The volume of a right circular cone is 1232 cm^3 and its vertical height is 24 cm. Its curved surface area is:

(a) 154 cm^2 (b) 550 cm^2
(c) 604 cm^2 (d) 704 cm^2

[SSC Examination, 2012]

90. The height of a right prism with a square base is 15 cm. If the area of the total surfaces of the prism is 608 cm^2, its volume is:

(a) 910 cm^2 (b) 920 cm^2
(c) 960 cm^2 (d) 980 cm^2

[SSC Examination, 2012]

91. The volume of a solid hemisphere is 19404 cm^3. Its total surface area is:

(a) 4158 cm^3 (b) 2858 cm^3
(c) 1738 cm^3 (d) 2038 cm^3

[SSC Examination, 2012]

Directions (Q. 92–96): Study the following information and answer the questions that follow:
The premises of a bank are to be renovated. The renovation is in terms of flooring. Certain areas are to be floored either with marble or wood. All rooms/halls and pantry are rectangular. The area to be renovated comprises a hall for customer transaction measuring 23 m by 29 m, the branch manager's room measuring 13 m by 17 m, a pantry measuring 14 m by 13 m, a record keeping-cum-server room measuring 21 m by 13 m and locker area measuring 29 m by 21 m. The total area of the bank is 2000 m^2. The cost of wooden flooring is ₹170 per m^2 and the cost of marble flooring is ₹190 per sm^2. The locker area, record keeping-cum-server room and pantry are to be floored with marble. The branch manager's room and hall for customer transaction are to be floored with wood. No other area is to be renovated in terms of flooring.

92. What is the ratio of the total cost of wooden flooring to the total cost of marble flooring?

(a) 1879:2527 (b) 1887:2386
(c) 1887:2527 (d) 1829:2527
(e) 1887:2351

[IBPS PO/MT Examination, 2012]

93. If the 4 walls and ceiling of the branch manager's room (the height of the room is 12 m) are to be painted at the cost of ₹190 per m^2, how much will be the total cost of renovation of the branch manager's room, including the cost of flooring?

(a) ₹1,36,800 (b) ₹2,16,660
(c) ₹1,78,790 (d) ₹2,11,940
(e) None of these

[IBPS PO/MT Examination, 2012]

94. If the remaining area of the bank is to be carpeted at the rate of ₹110 per m^2, how much will be the increment in the total cost of renovation of bank premises?

(a) ₹5,820 (b) ₹4,848
(c) ₹3,689 (d) ₹6,690
(e) None of these

[IBPS PO/MT Examination, 2012]

95. What is the percentage area of the bank that is not to be renovated?

(a) 2.2% (b) 2.4%
(c) 4.2% (d) 4.4%
(e) None of these

[IBPS PO/MT Examination, 2012]

96. What is the total cost of renovation of the hall for customer transaction and the locker area?

(a) ₹2,29,100 (b) ₹2,30,206
(c) ₹2,16,920 (d) ₹2,42,440
(e) None of these

[IBPS PO/MT, 2012]

97. Water is flowing at the rate of 5 Km/h through a pipe of diameter 14 cm into a rectangular tank which is 50 m long, 44 m wide. The time taken, in hours, for the rise in the level of water in the tank to be 7 cm is:

(a) 2 (b) $1\frac{1}{2}$
(c) 3 (d) $2\frac{1}{2}$

[SSC (GL) Examination, 2011]

98. The areas of three consecutive faces of a cuboid are 12 cm^2, 20 cm^2 and 15 cm^2, then the volume (in cm^3) of the cuboid is:

(a) 3600 (b) 100
(c) 80 (d) 60

[SSC (GL) Examination, 2011]

99. Water is flowing at the rate of 3 Km/h through a circular pipe of 20 cm internal diameter into a circular cistern of diameter 10 m and depth 2 m. In how much time will the cistern be filled?

(a) 1 hour (b) 1 hour 40 mins
(c) 1 hours 20 mins (d) 2 hours 40 mins

[SSC (GL) Examination, 2011]

100. Marbles of diameter 1.4 cm are dropped into a cylindrical beaker containing some water and are fully submerged. The diameter of the beaker is 7 cm. Find how many marbles have been dropped in it if the water rises by 5.6 cm.

(a) 50 (b) 150
(c) 250 (d) 350

[SSC (GL) Examination, 2011]

101. A hemisphere and a cone have equal bases. If their heights are also equal, the ratio of their curved surfaces will be:

(a) $1:\sqrt{2}$ (b) $2:\sqrt{2}$
(c) 1:2 (d) 2:1

[SSC (GL) Examination, 2011]

102. The heights of a cone, cylinder and hemisphere are equal. If their radii are in the ratio 2:3:1, then the ratio of their volumes is:

(a) 2:9:2 (b) 4:9:1
(c) 4:27:2 (d) 2:3:1

[SSC Examination, 2011]

103. Base of a right pyramid is a square, length of diagonal of the base is $24\sqrt{2}$ m. If the volume of the pyramid is 1728 cu. m, its height is:

(a) 7 m (b) 8 m
(c) 9 m (d) 10 m

[SSC Examination, 2011]

104. The height of a right circular cone and the radius of its circular base are 9 cm and 3 cm respectively. The cone is cut by a plane parallel to its base so as to divide it into two parts. The volume of the frustum (i.e., the lower part) of the cone is 44 cm^2. The radius of the upper circular surface of the frustum $\left(\text{taking } \pi = \frac{22}{7}\right)$ is:

(a) $\sqrt[3]{12}$ cm (b) $\sqrt[3]{13}$ cm
(c) $\sqrt[3]{6}$ cm (d) $\sqrt[3]{20}$ cm

[SSC Examination, 2011]

105. The ratio of radii of 2 right circular cylinders is 2:3 and their heights are in the ratio 5:4. The ratio of their curved surface area is:

(a) 5:6 (b) 3:4
(c) 4:5 (d) 2:3

[SSC Examination, 2011]

106. A solid cylinder has total surface area of 462 sq.cm. Curved surface area is $\frac{1}{3}$rd of its total surface area. The volume of the cylinder is:

(a) 530 cm^2 (b) 536 cm^2
(c) 539 cm^2 (d) 545 cm^2

[SSC Examination, 2011]

107. A cylinder and a cone have equal radii of their bases and equal heights. If their curved surface areas are in the ratio 8:5, the ratio of their radius and height is:

(a) 1:2 (b) 1:3
(c) 2:3 (d) 3:4

[SSC Examination, 2011]

108. A solid is hemispherical at the bottom and conical above. If the surface areas of the two parts are equal, then the ratio of radius and height of its conical part is:

(a) 1:3 (b) 1:1
(c) 3:1 (d) $1:\sqrt{3}$

[SSC Examination, 2011]

109. Base of a right prism is an equilateral triangle of side 6 cm. If the volume of the prism is $108\sqrt{3}$ cc, its height is:

(a) 9 cm (b) 10 cm
(c) 11 cm (d) 12 cm

[SSC Examination, 2011]

110. The number of coins, each of radius 0.75 cm and thickness 0.2 cm, to be melted to make a right circular cylinder of height 8 cm and radius 3 cm, is:

(a) 640 (b) 600
(c) 500 (d) 480

[SSC Examination, 2010]

111. If the radius of a sphere is increased by 2 m, its surface area is increased by 704 m^2. What isthe radius of the original sphere? $\left(\text{use } \pi = \frac{22}{7}\right)$

(a) 16 m^2 (b) 15 m^2
(c) 14 m^2 (d) 13 m^2

[SSC Examination, 2010]

112. A right circular cylinderis circumscribing a hemisphere such that their bases are common. The ratio of their volumes is.

(a) 1:3 (b) 1:2
(c) 2:3 (d) 3:4

[SSC Examination, 2010]

113. 3 spherical balls of radii 1 cm, 2 cm and 3 cm are melted to form a single spherical ball. In the process, the loss of material is 25%. The radius of the new ball is:

(a) 6 cm (b) 5 cm
(c) 3 cm (d) 2 cm

[SSC Examination, 2010]

114. The length of the diagonal of a cube is 6 cm. The volume of the cube (in cm^3) is:

(a) $18\sqrt{3}$ (b) $24\sqrt{3}$
(c) $28\sqrt{3}$ (d) $30\sqrt{3}$

[SSC Examination, 2010]

115. If a sphere of radius r is divided into 4 identical parts, then the total surface area of the 4 parts is:

(a) $4\pi r^2$ square units (b) $2\pi r^2$ square units
(c) $8\pi r^2$ square units (d) $3\pi r^2$ square units

[SSC Examination, 2010]

ANSWER KEYS

EXERCISE-1

1. (c)	**2.** (b)	**3.** (c)	**4.** (b)	**5.** (c)	**6.** (b)	**7.** (c)	**8.** (c)	**9.** (b)	**10.** (a)	**11.** (d)	**12.** (b)
13. (b)	**14.** (c)	**15.** (c)	**16.** (c)	**17.** (b)	**18.** (a)	**19.** (d)	**20.** (c)	**21.** (b)	**22.** (d)	**23.** (c)	**24.** (c)
25. (c)	**26.** (a)	**27.** (d)	**28.** (a)	**29.** (c)	**30.** (c)	**31.** (d)	**32.** (b)	**33.** (d)	**34.** (a)	**35.** (b)	**36.** (a)
37. (a)	**38.** (d)	**39.** (d)	**40.** (b)	**41.** (a)	**42.** (c)	**43.** (b)	**44.** (c)	**45.** (a)	**46.** (a)	**47.** (c)	**48.** (d)
49. (a)	**50.** (a)	**51.** (a)	**52.** (b)	**53.** (d)	**54.** (b)	**55.** (d)	**56.** (b)	**57.** (d)	**58.** (c)	**59.** (d)	**60.** (b)
61. (a)	**62.** (b)	**63.** (b)	**64.** (b)	**65.** (b)	**66.** (a)	**67.** (a)	**68.** (c)	**69.** (b)	**70.** (d)	**71.** (b)	**72.** (c)
73. (c)	**74.** (a)	**75.** (d)	**76.** (b)	**77.** (b)	**78.** (c)	**79.** (a)	**80.** (b)	**81.** (d)	**82.** (b)	**83.** (a)	**84.** (c)
85. (c)	**86.** (b)	**87.** (a)	**88.** (c)	**89.** (b)	**90.** (c)	**91.** (b)	**92.** (a)	**93.** (b)	**94.** (a)	**95.** (c)	**96.** (a)
97. (b)	**98.** (c)	**99.** (c)	**100.** (c)	**101.** (a)	**102.** (d)	**103.** (a)	**104.** (b)	**105.** (a)	**106.** (c)	**107.** (a)	**108.** (b)
109. (d)	**110.** (c)	**111.** (c)	**112.** (b)	**113.** (c)	**114.** (c)	**115.** (b)					

EXERCISE-2

1. (b)	**2.** (b)	**3.** (d)	**4.** (d)	**5.** (b)	**6.** (d)	**7.** (b)	**8.** (a)	**9.** (a)	**10.** (b)	**11.** (b)	**12.** (c)
13. (d)	**14.** (b)	**15.** (c)	**16.** (c)	**17.** (b)	**18.** (a)	**19.** (a)	**20.** (b)	**21.** (b)	**22.** (b)	**23.** (b)	**24.** (d)
25. (a)	**26.** (b)	**27.** (b)	**28.** (d)	**29.** (a)	**30.** (c)	**31.** (a)	**32.** (b)	**33.** (a)	**34.** (d)	**35.** (c)	**36.** (c)
37. (b)	**38.** (d)	**39.** (a)	**40.** (b)	**41.** (b)	**42.** (c)	**43.** (a)	**44.** (d)	**45.** (d)	**46.** (a)	**47.** (b)	**48.** (c)
49. (a)	**50.** (c)	**51.** (a)	**52.** (d)	**53.** (d)	**54.** (a)	**55.** (c)	**56.** (e)	**57.** (a)	**58.** (a)	**59.** (d)	**60.** (c)
61. (a)	**62.** (d)	**63.** (c)	**64.** (d)	**65.** (d)	**66.** (d)	**67.** (a)	**68.** (a)	**69.** (a)	**70.** (a)	**71.** (c)	**72.** (d)
73. (d)	**74.** (b)	**75.** (a)	**76.** (c)	**77.** (d)	**78.** (d)	**79.** (c)	**80.** (c)	**81.** (a)	**82.** (c)	**83.** (c)	**84.** (a)
85. (c)	**86.** (b)	**87.** (c)	**88.** (a)	**89.** (b)	**90.** (c)	**91.** (a)	**92.** (c)	**93.** (e)	**94.** (e)	**95.** (b)	**96.** (a)
97. (a)	**98.** (d)	**99.** (b)	**100.** (b)	**101.** (b)	**102.** (c)	**103.** (c)	**104.** (b)	**105.** (a)	**106.** (c)	**107.** (d)	**108.** (d)
109. (d)	**110.** (a)	**111.** (d)	**112.** (c)	**113.** (c)	**114.** (b)	**115.** (c)					

CHAPTER 37

Plane Geometry

SECTION I LINES AND ANGLES

Line A geometrical straight line is a set of points that extends endlessly in both the directions.

Axiom-1 A line contains infinitely many points.

Axiom-2 Through a given point, infinitely many lines pass.

Axiom-3 Given two distinct points A and B, there is one and only one line that contains both the points.

Parallel Lines If two lines have no point in common, they are said to be *parallel lines*

Intersecting Lines If two lines have a point in common, they are said to be *intersecting lines*. Two lines can intersect at the most at one point.

Line Segment and Ray A part (or portion) of a line with two end points is called a *line segment* and a part of a line with one end point is called a *ray*. A line segment $\overline{AB}$ and its length is denoted as AB. Ray AB (i.e., A towards B) is denoted as $\vec{AB}$ and ray BA (i.e., B towards A) is denoted as $\vec{BA}$.

Collinear Points Three or more than three points are said to be *collinear* if there is a line which contains them all.

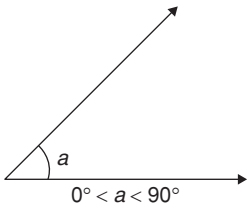
Concurrent Lines Three or more than three lines are said to be *concurrent* if there is a point which lies on all of them.

Angle An *angle* is a figure formed by two rays with a common initial point. The two rays forming an angle are called *arms* of the angle and the common initial point is called *vertex* of the angle.

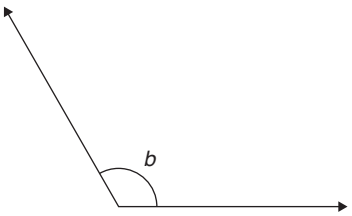
Types of Angles

An angle is said to be:

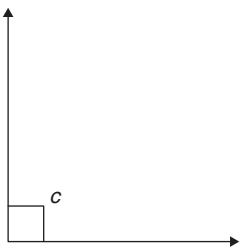
(i) *Acute*, if $a < 90^\circ$.



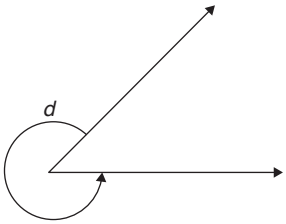
(ii) *Obtuse*, if $90^\circ < b < 180^\circ$.



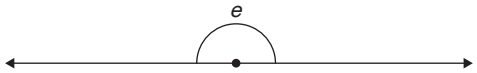
(iii) *Right angle*, if $c = 90^\circ$



(iv) *Reflex angle*, if $180^\circ < d < 360^\circ$



(v) *Straight angle*, if $e = 180^\circ$



(vi) *Complete angle*: An angle whose measure is 360º, is called a *complete angle*.

Complementary Angles Two angles, the sum of whose measures is 90º, are called *complementary angles,* e.g. 50º and 40º is a pair of complementary angles.

Supplementary Angles Two angles, the sum of whose measures is 180º, are called *supplementary angles*, e.g. 72º and 108º is a pair of supplementary angles.

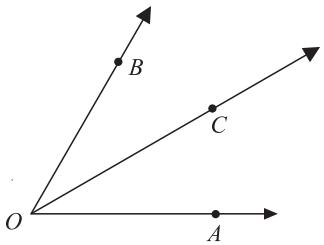
Adjacent Angles Two angles are called *adjacent angles* if

(i) they have the same vertex.

(ii) they have a common arm.

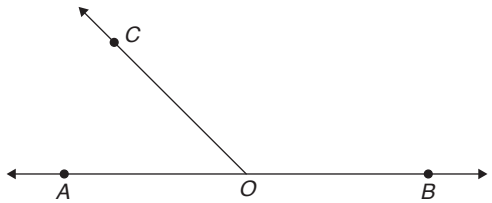
(iii) uncommon arms are on either side of the common arm.

E.g. $\angle AOC$ and $\angle BOC$ are adjacent angles.



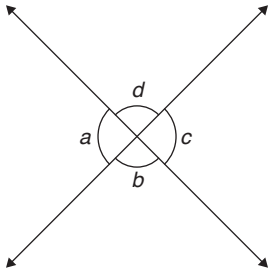
Linear Pair: Two adjacent angles are said to form a linear pair of angles if their non-common arms are two opposite rays.

E.g. $\angle AOC$ and $\angle BOC$ form a linear pair.



Linear Pair Axiom: If a ray stands on a line, then the sum of the two adjacent angles so formed is 180º. Conversely, if the sum of two adjacent angles is 180º; then the non-common arms of the angles are two opposite rays.

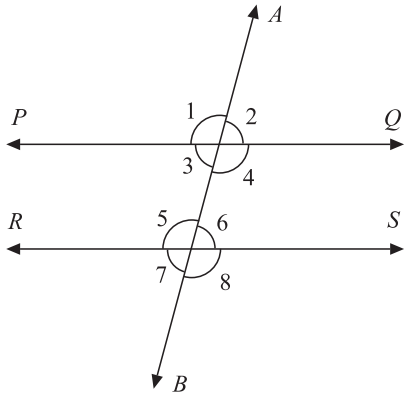
Vertically Opposite Angles: When two lines intersect, four angles are formed. The angles opposite to each other are called *vertically opposite angles*.



a and c are vertically opposite angles, $\angle a = \angle c$.

b and d are vertically opposite angles, $\angle b = \angle d$.

Angles made by a transversal* with two parallel lines Suppose $PQ \parallel RS$ and a transversal AB cuts them, then



(a) Pair of corresponding angles are
(1 and $\angle 5$), ($\angle 2$ and $\angle 6$),
($\angle 4$ and $\angle 8$) and ($\angle 3$ and $\angle 7$)

(b) Pair of alternate angles are
($\angle 3$ and $\angle 6$) and ($\angle 4$ and $\angle 5$)

(c) Pair of interior angles (consecutive interior angles or cointerior angles) on the same side of the transversal are
($\angle 3$ and $\angle 5$) and ($\angle 4$ and $\angle 6$)

KEY RESULTS TO REMEMBER

If two parallel lines are intersected by a transversal, then

(i) each pair of corresponding angles are equal.

(ii) each pair of alternate angles are equal.

(iii) interior angles on the same side of the transversal are supplementary.

Triangle A plane figure bounded by three lines in a plane is called a *triangle*.

*A line which intersects two or more lines at distinct points is called a transversal of the given lines.

SECTION 2 TRIANGLES

Types of Triangles (On the basis of sides)

Scalene triangle A triangle two of whose sides are equal is called a *scalene triangle*.

Isosceles triangle A triangle two of whose sides are equal in length is called an *isosceles triangle*.

Equilateral triangle A triangle all of whose sides are equal is called an *equilateral triangle*.

Types of Triangles (On the basis of angles)

Acute triangle A triangle, each of whose angle is acute, is called an *acute triangle* or *acute-angled triangle*.

Right triangle A triangle with one right angle is called a *right triangle* or a *right-angled triangle*.

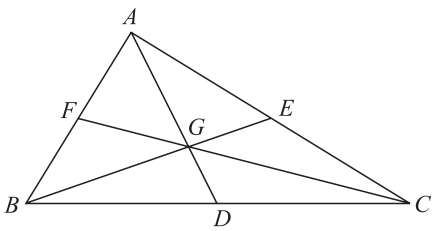
Obtuse triangle A triangle with one angle an obtuse angle, is known as *obtuse triangle* or *obtuse-angled triangle*.

Some Important Terms Related to a Triangle

1. **Median** The median of a triangle corresponding to any side is the line segment joining the midpoint of that side with the opposite vertex.

   In the figure given below, *AD, BE* and *CF* are the medians.

   The medians of a triangle are concurrent i.e., they intersect each other at the same point.

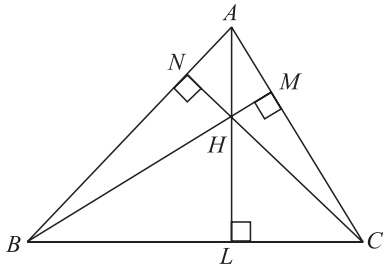


2. **Centroid** The point of intersection of all the three medians of a triangle is called its *centroid.*

   In the above figure G is the centroid of ΔABC.

   Note: The centroid divides a median in the ratio 2:1.

3. **Altitudes** The *altitude* of a triangle corresponding to any side is the length of perpendicular drawn from the opposite vertex to that side.



In the figure given above, *AL*, *BM* and *CN* are the altitudes.

Notes

The altitudes of a triangle are concurrent.

4. **Orthocentre** The point of intersection of all the three altitudes of a triangle is called its *orthocentre*. In the figure given above H is the orthocentre of ΔABC.

Notes

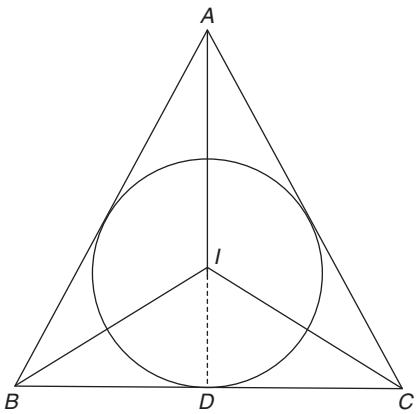
5. **Incentre of a triangle** The point of intersection of the internal bisectors of the angles of a triangle is called its *incentre*.

   In the figure given below, the internal bisectors of the angles of ΔABC intersect at I.

   $\therefore$ I is the Incentre of ΔABC.

   Let, $ID \perp BC$

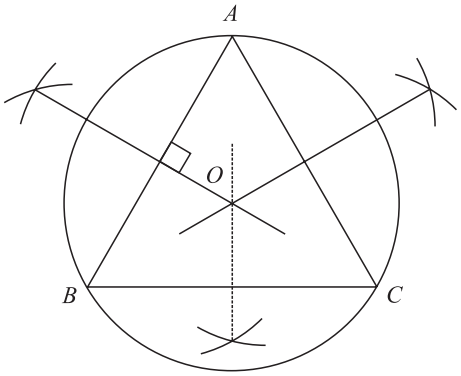
   Then, a circle with centre I and radius ID is called the *incircle* of ΔABC.



Notes

6. Circumcentre of a triangle The point of intersection of the perpendicular bisectors of the sides of a triangle is called its *circumcentre*.

In the figure given below, the right bisectors of the sides of ΔABC intersect at O.



$\therefore$ O is the *circumcentre* of ΔABC with O as centre and radius equal to $OA = OB = OC$. We draw a circle passing through the vertices of the given Δ. This circle is called the *circumcircle* of ΔABC.

Notes

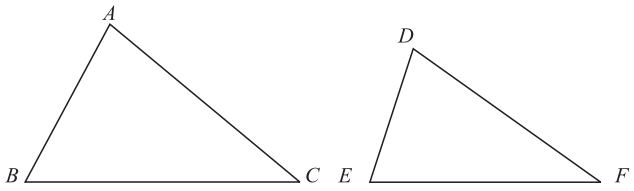
Note: The circumcentre of a triangle is *equidistant* from its *vertices*.

Congruent triangles

Two triangles are *congruent* if and only if one of them can be superposed on the other, so as to cover it exactly.

Thus, congruent triangles are exactly identical

For example, If $\Delta ABC \cong \Delta DEF$ then we have



$\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$;

and $AB = DE$, $BC = EF$ and $AC = DF$.

Similar Triangles

Congruent figures Two geometric figures having the same shape and size are known as *congruent figures*.

Similar figures Two figures (plane or solid) are said to be *similar* if they have the same shape irrespective of their sizes.

Note: Two similar figures may not be congruent as their size may be different.

For examples,

1. Any two lines segments are similar.
2. Any two equilateral triangles are similar.
3. Any two squares are similar.
4. Any two circles are similar.
5. Any two rectangles are similar.

Similar triangles Two triangles are similar if

(a) their corresponding angles are equal.

(b) their corresponding sides are proportional.

KEY RESULTS TO REMEMBER

1. The sum of all the angles round a point is equal to 360º.
2. Two lines parallel to the same line are parallel to each other.
3. The sum of three angles of a triangle is 180º.
4. If a side of a triangle is produced, the exterior angle so formed is equal to the sum of the two interior opposite angles. (Exterior Angle Theorem)
5. If two sides of a triangle are unequal, the longer side has greater angle opposite to it.
6. In a triangle, the greater angle has the longer side opposite to it.
7. The sum of any two sides of a triangle is greater than the third side.
8. If a, b, c denote the sides of a triangle then
   (i) If $c^2 < a^2 + b^2$, triangle is acute angled.
   (ii) If $c^2 = a^2 + b^2$, triangle is right angled.
   (iii) If $c^2 > a^2 + b^2$, triangle is obtuse angled.
9. Two triangles are congruent if:
   (i) Any two sides and the included angle of one triangle are equal to any two sides and the included angle of the other triangle.
   (SAS congruence theorem)
   (ii) Two angles and the included side of one triangle are equal to the corresponding two angles and the included side of the other triangle.
   (ASA congruence theorem)
   (iii) The three sides of one triangle are equal to the corresponding three sides of the other triangle.
   (SSS congruence theorem)

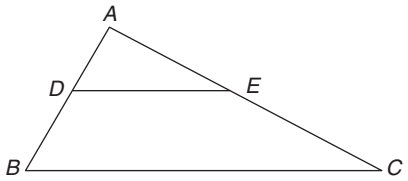
Notes

Two right triangles are congruent if the hypotenuse and one side of one triangle are respectively equal to the hypotenuse and the corresponding side of the other triangle.
(RHS Congruence theorem)

10. The line segments joining the mid-points of any two sides of a triangle is parallel to the third side and equal to half of it.

11. Basic Proportionality Theorem If a line is drawn parallel to one side of a triangle, to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.

In the figure given below, In a ΔABC



If $DE \parallel BC$

Then, $\frac{AD}{DB} = \frac{AE}{EC}$

Illustration 1: In the figure given above, D and E are the points on the AB and AC respectively such that $DE \parallel BC$. If $AD = 8$ cm, $AB = 12$ cm and $AE = 12$ cm. Find CE

Solution: In ΔABC, $DE \parallel BC$

$$\frac{AD}{DB} = \frac{AE}{EC} \text{ (Basic Proportionality Theorem)}$$

$$\Rightarrow \quad \frac{8}{12-8} = \frac{12}{EC}$$

$$\Rightarrow \quad \frac{8}{4} = \frac{12}{EC}$$

or $EC = 6$ cm

12. If a line divides any two sides of a triangle in the same ratio, the line is parallel to the third side.

Explanation In the above figure (given in point 11). In ΔABC

if $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$.

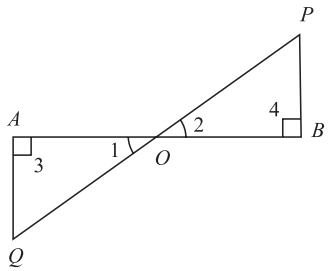
Similarity Theorems

13. AAA Similarity If in two triangles, corresponding angles are equal, then the triangles are similar.

Corollary (AA-similarity): If two angles of one triangle are respectively equal to two angles of another triangle then the two triangles are similar.

Illustration 2: In the figure given below, QA and PB are perpendiculars to AB. If $AO = 15$ cm, $BO = 9$ cm, $PB = 12$ cm, find AQ.

Solution:



In Δs AOQ and BOP

$\angle 1 = \angle 2$ [vertically opposite angles]

$\angle 3 = \angle 4$ [each 90º]

$\therefore \quad \Delta AOQ \sim \Delta BOP$ [AA Similarity Criterion]

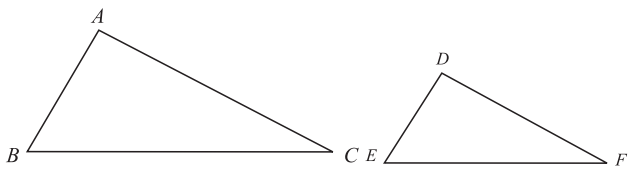
$$\therefore \quad \frac{AO}{BO} = \frac{AQ}{BP} \text{ (corresponding sides of } \sim\Delta\text{s)}$$

$$\text{or} \quad \frac{15}{9} = \frac{AQ}{12}$$

$$\text{or} \quad \frac{5}{1} = \frac{AQ}{4} \Rightarrow AQ = 20 \text{ cm.}$$

14. SSS-Similarity If the corresponding sides of two triangles are proportional then they are similar.

Explanation: In Δs ABC and DEF,

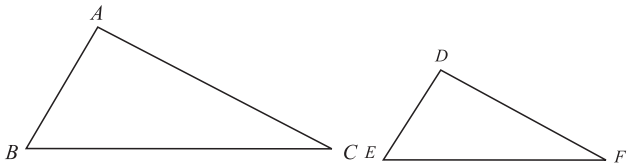


if $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$

Then $\Delta ABC \sim \Delta DEF$ [SSS Similarity]

15. SAS-Similarity If in two triangles, one pair of corresponding sides are proportional and the included angles are equal, then the two triangles are similar.

Explanation In Δs *ABC* and *DEF,*



if $\angle A = \angle D$ and $\frac{AB}{DE} = \frac{AC}{DF}$

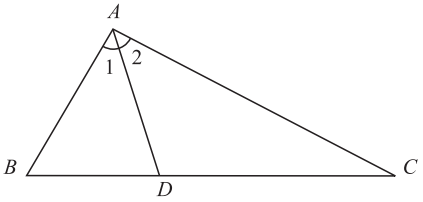
or $\angle B = \angle E$ and $\frac{AB}{DE} = \frac{BC}{EF}$

or $\angle C = \angle F$ and $\frac{AC}{DF} = \frac{BC}{EF}$,

then $\Delta ABC \sim \Delta DEF$ [SAS-Similarity]

16. Internal Bisector Property The internal bisector of an angle of a triangle divides the opposite side in the ratio of the sides containing the angle.

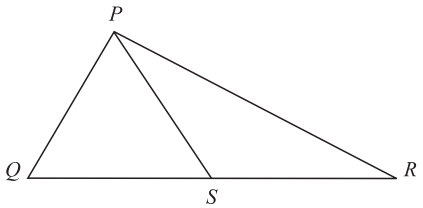
Explanation In ΔABC, if $\angle 1 = \angle 2$



Then $\frac{AB}{AC} = \frac{BD}{CD}$

17. If a line segment drawn from the vertex of an angle of a triangle to its opposite side divides it in the ratio of the sides containing the angle, then the line segment bisects the angle.

Illustration 3: In ΔPQR, $PQ = 6$ cm, $PR = 8$ cm,
Solution: $QS = 1.5$ cm, $RS = 2$ cm



$\therefore \quad \frac{PQ}{PR} = \frac{6}{8} = \frac{3}{4}$ and $\frac{QS}{RS} = \frac{1.5}{2} = \frac{3}{4}$

Thus, $\frac{PQ}{PR} = \frac{QS}{RS}$

$\therefore$ *PS* is the bisector of $\angle P$.

18. Pythagoras Theorem In a right triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

Explanation In a right ΔABC, right angled at *B*

$$AC^2 = AB^2 + BC^2$$

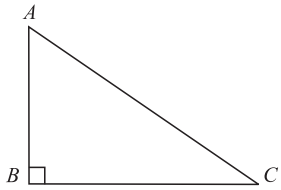
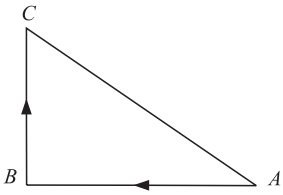


Illustration 4: A man goes 15 m west and then 8 m due north. How far is he from the starting point.

Solution: Let, the initial position of the man be *A*.



Let, $AB = 15$ m and $BC = 8$ m

$\therefore \quad AC^2 = AB^2 + BC^2$ (Pythagoras Theorem)

$= (15)^2 + (8)^2$

$= 225 + 64$

$= 289$

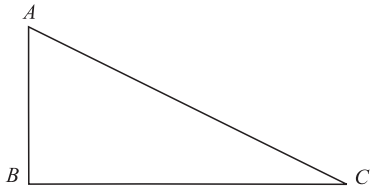
$AC = \sqrt{289}$

$= 17$ m

Hence, the man is 17 m away from the starting point.

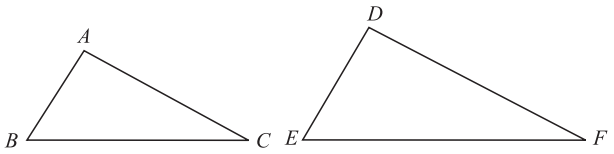
19. Converse of Pythagoras Theorem. In a triangle, if the square of one side is equal to the sum of the squares of the other two sides, then the angle opposite the first side is a right angle.

Explanation In a ΔABC if $AB^2 + BC^2 = AC^2$
Then, $\angle ABC = 90^\circ$



20. Area Theorem The ratio of the areas of two similar Ds is equal to the ratio of the squares of any two corresponding sides

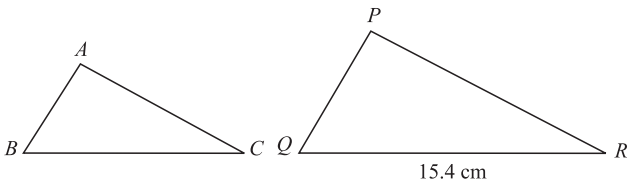
Explanation If $\Delta ABC \sim \Delta DEF$,



$$\text{then } \frac{ar(\Delta ABC)}{ar(\Delta DEF)} = \frac{AB^2}{DE^2} = \frac{AC^2}{DF^2} = \frac{BC^2}{EF^2}$$

Illustration 5: The areas of two similar Δs ABC and PQR are 64 cm^2 and 121 cm^2, respectively. If QR = 15.4 cm, find BC.

Solution: Since $\Delta ABC \sim \Delta PQR$



$$\therefore \quad \frac{ar(\Delta ABC)}{ar(\Delta PQR)} = \frac{BC^2}{QR^2} \text{ (Area Theorem)}$$

$$\text{i.e., } \frac{64}{121} = \frac{BC^2}{(15.4)^2} \Rightarrow \frac{8}{11} = \frac{BC}{15.4}$$

$\therefore \quad BC = 11.2$ cm

21. The ratio of the areas of two similar triangles is equal to the

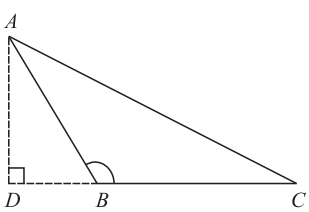
(i) ratio of the squares of the corresponding medians
(ii) ratio of the squares of the corresponding altitudes
(iii) ratio of the squares of the corresponding angle bisector segments

22. If two similar triangles have equal areas, then the Ds are congruent.

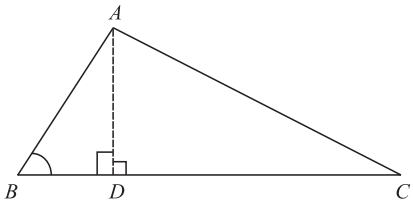
23. In two similar triangles, the ratio of two corresponding sides is same as the ratio of their perimeters.

24. Obtuse Angle Property in a ΔABC, if $\angle B$ is obtuse then $AC^2 = AB^2 + BC^2 + 2\ BC \times BD$

where $AD \perp BC$

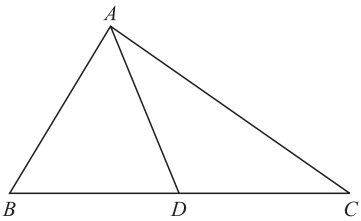


25. Acute Angle Property In a ΔABC, if $\angle C$ is acute, then $AB^2 = AC^2 + BC^2 - 2BC \times CD$ where $AD \perp BC$



26. Apollonius Theorem The sum of the squares on any two sides of a triangle is equal to the sum of twice the square of the median, which bisects the third side and half the square of the third side.

Explanation In the given ΔABC,



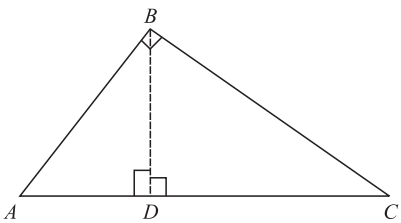
$$AB^2 + AC^2 = 2AD^2 + \frac{1}{2}BC^2$$

or $\quad AB^2 + AC^2 = 2[AD^2 + BD^2]$

27. If a perpendicular is drawn from the vertex of the right angle of a right triangle to the hypotenuse, the triangles on each side of the perpendicular are similar to the whole triangle and to each other. Also the square of the perpendicular is equal to the product of the lengths of the two parts of the hypotenuse.

Explanation In the figure given below,

ABC is a right triangle, right angled at B and $BD \perp AC$, then



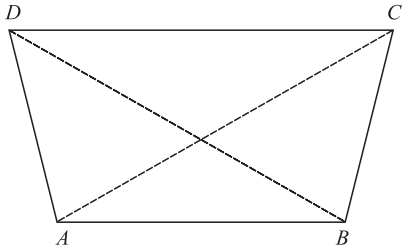
(i) $\Delta ADB \sim \Delta ABC$ (AA Similarity)

(ii) $\Delta BDC \sim \Delta ABC$ (AA Similarity)

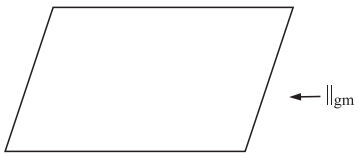
(iii) $\Delta ADB \sim \Delta BDC$ also $BD^2 = AD \times CD$

SECTION 3 QUADRILATERALS AND PARALLELOGRAMS

Quadrilateral A plane figure bounded by four line segments *AB*, *BC*, *CD* and *DA* is called a *quadrilateral*, written as quad. *ABCD* or $\angle ABCD$.

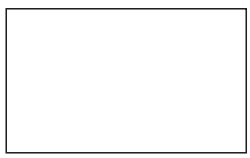


Various types of Quadrilaterals



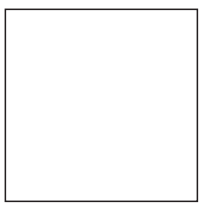
(i) Parallelogram A quadrilateral in which opposite sides are parallel is called *parallelogram*, written as $\|_{gm}$.

(ii) Rectangle A parallelogram each of whose angles is 90° is called a *rectangle*, written as rect. *ABCD*.



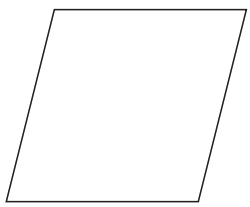
Rectangle

(iii) Square A rectangle having all sides equal is called a *square*.



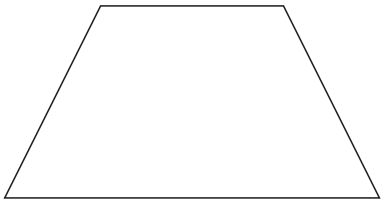
Square

(iv) Rhombus A quadrilateral having all sides equal is called a *rhombus*



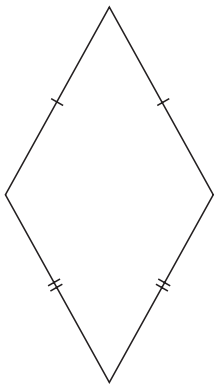
Rhombus

(v) Trapezium A quadrilateral in which two opposite sides are parallel and two opposite sides are non-parallel is called a *trapezium.*



Trapezium

(vi) Kite A quadrilateral in which pairs of adjacent sides are equal is known as *kite.*



KEY RESULTS TO REMEMBER

1. The sum of all the four angles of a quadrilateral is 360°.

2. In a parallelogram
(i) opposite sides are equal.
(ii) opposite angles are equal.
(iii) each diagonal bisects the parallelogram.
(iv) the diagonal bisect each other.

3. A quadrilateral is a ||gm
(i) if both pairs of opposite sides are equal.
or (ii) if both pairs of opposite angles are equal.
or (iii) if the diagonals bisect each other.
or (iv) if a pair of opposite sides are equal and parallel.

4. The diagonals of a rectangle are equal.

5. If the diagonals of a ||gm are equal, it is a rectangle.

6. Diagonals of a rhombus are perpendicular to each other.

7. Diagonals of a square are equal and perpendicular to each other.

8. The figure formed by joining the mid-points of the pairs of consecutive sides of a quadrilateral is a ||gm.
9. The quadrilateral formed by joining the mid-points of the consecutive sides of a rectangle is a rhombus.
10. The quadrilateral formed by joining the mid-points of the consecutive sides of a rhombus is a rectangle.
11. If the diagonals of a quadrilateral are perpendicular to each other, then the quadrilateral formed by joining the mid-points of its sides, is a rectangle.
12. The quadrilateral formed by joining the mid-points of the sides of a square, is also a square.

SECTION 4 POLYGONS

Polygon A closed plane figure bounded by line segments is called a *polygon.*

The line segments are called its *sides* and the points of intersection of consecutive sides are called its *vertices*. An angle formed by two consecutive sides of a polygon is called an *interior angle* or simply an *angle* of the polygon.

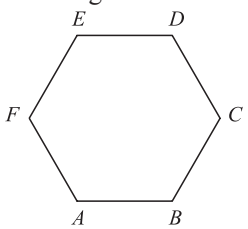
No. of sides	*Name*
3	Triangle
4	Quadrilateral
5	Pentagon
6	Hexagon
7	Heptagon
8	Octagon
9	Nonagon
10	Decagon

A polygon is named according to the number of sides, it has.

In general, a polygon of n sides is called *n-gon.* Thus, a polygon having 18 sides is called 18-gon.

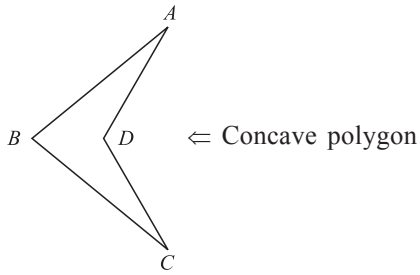
Diagonal of a Polygon Line segment joining any two non-consecutive vertices of a polygon is called its *diagonal*.

Convex Polygon If all the (interior) angles of a polygon are less than 180°, it is called a *convex polygon*. In the figure given below, *ABCDEF* is a convex polygon. In fact, it is a convex hexagon.

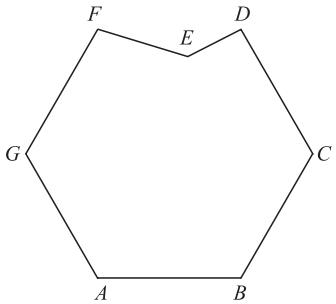


(In other words, a polygon is a convex polygon if the line segment joining any two points inside it lies completely inside the polygon).

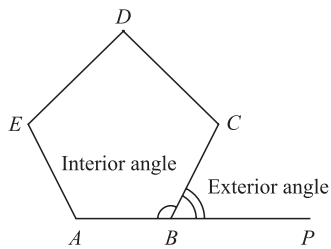
Concave Polygon If one or more of the (interior) angles of a polygon is greater than 180° i.e., reflex, it is called



concave (or re-entrant) *polygon* In the figure given below, *ABCDEFG* is a concave polygon. In fact, it is a concave heptagon.



Exterior Angle of Convex Polygon If we produce a side of polygon, the angle it makes with the next side is called an *exterior angle*. In the diagram given below, *ABCDE* is a pentagon. Its side *AB* has been produced to *P*, then $\angle CBP$ is an exterior angle.



Notes

Corresponding to each interior angle, there is an exterior angle. Also, as an exterior angle and its adjacent interior angle make a straight line, we have **an exterior angle + adjacent interior angle = 180°**

Regular Polygon A polygon is called regular polygon if all of its sides have equal length and all its angles have equal size.

Thus, in a regular polygon

(i) all sides are equal in length.

(ii) all interior angles are equal in size.

(iii) all exterior angles are equal size.

Notes

All regular polygons are convex.

KEY RESULTS TO REMEMBER

1. (a) If there is a polygon of n sides ($n \geq 3$), we can cut it into $(n - 2)$ triangles with a common vertex and so *the sum of the interior angles of a polygon of n sides would be*

$$(n - 2) \times 180° = (n - 2) \times 2 \text{ right angles}$$
$$= (2n - 4) \text{ right angles}$$

(b) If there is a regular polygon of n sides ($n \geq 3$), then its each interior angle is equal to

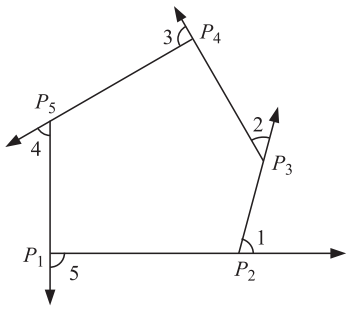
$$\left(\frac{2n-4}{n} \times 90\right)$$

(c) Each exterior angle of a regular polygon of n sides is equal to

$$= \left(\frac{360}{n}\right)^{\circ}$$

2. The sum of all the exterior angles formed by producing the sides of a convex polygon in the same order is equal to four right angles.

Explanation If in a convex polygon $P_1P_2P_3P_4P_5$, all the sides are produced in order, forming exterior angles $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$ and $\angle 5$, then $\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 = 4$ right angles.



3. If each exterior angle of a regular polygon is $x°$, then the number of sides in the polygon $= \dfrac{360°}{x}$.

Notes

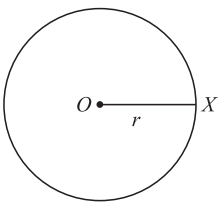
Greater the number of sides in a regular polygon, greater is the value of its each interior angle and smaller is the value of each exterior angle.

4. If a polygon has n sides, then the number of diagonals of the polygon

$$= \frac{n(n-1)}{2} - n.$$

SECTION 5 CIRCLES AND TANGENTS

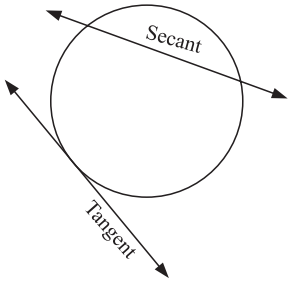
Circle A circle is a set of all those points in a plane, each one of which is at given constant distance from a given fixed point in the plane.



The fixed point is called the *centre* and the given constant distance is called the *radius* of the circle.

A circle with centre O and radius r is usually denoted by $C(O, r)$.

Tangent A line meeting a circle in only one point is called a *tangent* to the circle. The point at which the tangent line meets the circle is called the *point of contact*.

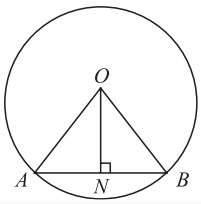


Secant A line which intersects a circle in two distinct points is called a *secant line*.

KEY RESULTS TO REMEMBER

1. The perpendicular from the centre of a circle to a chord bisects the chord.

Explanation If $ON \perp AB$, then $AN = NB$.

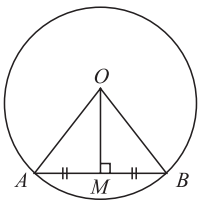


Notes

The converse of above theorem is true and can be stated as point 2.

2. The line joining the centre of a circle to the midpoint of a chord is perpendicular to the chord.

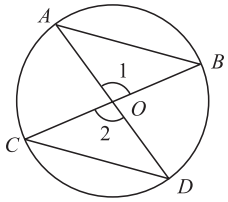
Explanation If $AM = MB$, then $OM \perp AB$.



Cor. The perpendicular bisectors of two chords of a circle intersect at its centre.

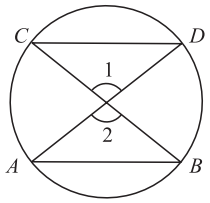
3. Equal chords of a circle subtend equal angles at the centre.

Explanation If $AB = CD$, then $\angle 1 = \angle 2$



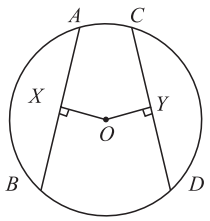
4. (Converse of above theorem) If the angles subtended by two chords at the centre of a circle are equal then the chords are equal.

Explanation If $\angle 1 = \angle 2$, then $AB = CD$



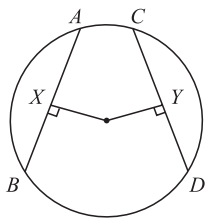
5. Equal chords of a circle are equidistant from the centre.

Explanation If the chords AB and CD of a circle are equal and if $OX \perp AB$ and $OY \perp CD$ then $OX = OY$.



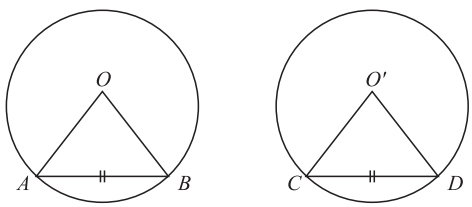
6. (Converse above theorem) Chords equidistant from the centre of the circle are equal.

Explanation If $OX \perp AB$ $OY \perp CD$ and $OX = OY$, then chords $AB = CD$



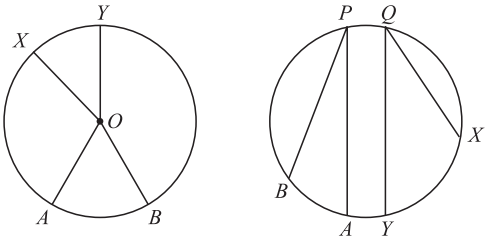
7. In equal circles (or in the same circle), equal chords cut of equal arcs.

Explanation If the chords $AB = CD$, then arc $AB =$ are CD.



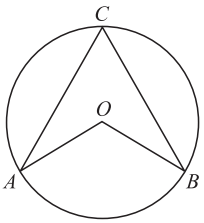
8. In equal circles (or in the same circle) if two arcs subtend equal angles at the centre (or at the circumference), the arcs are equal.

Explanation If $\angle BOA = \angle XOY$, then arc AB = arc XY or if $\angle BPA = \angle XQY$, then arc AB = arc XY.



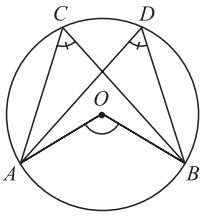
9. The angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the remaining part of the circle.
(The theorem is popularly known as Degree Measure Theorem).

Explanation A circle, centre O, with $\angle AOB$ at the centre, $\angle ACB$ at the circumference, standing on the same arc AB, then $\angle AOB = 2\angle ACB$

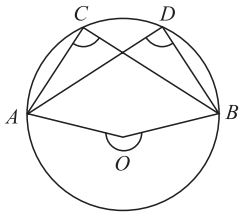


10. Angles in the same segment of a circle are equal.

Explanation A circle, centre O, $\angle ACB$ and $\angle ADB$ are angles at the circumference, standing on the same arc, then

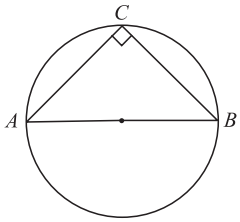


$\angle ACB = \angle ADB$
(angles in same arc)
or
(angles in same segment)



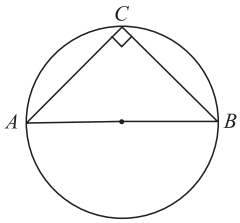
11. The angle in a semicircle is a right angle.

Explanation In the figure given below $\angle ACB = 90°$

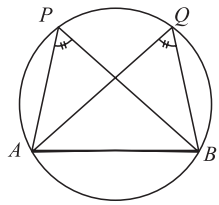


12. (Converse of above theorem) The circle drawn with hypotenuse of a right triangle as diameter passes through its opposite vertex.

Explanation The circle drawn with the hypotenuse AB of a right triangle ACB as diameter passes through its opposite vertex C.

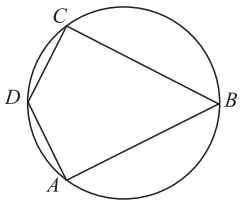


13. If $\angle APB = \angle AQB$, and if P, Q are on the same side of AB, then A, B, Q, P are concyclic i.e., lie on the same circle.



14. The sum of the either pair of the opposite angles of a cyclic quadrilateral is 180°.

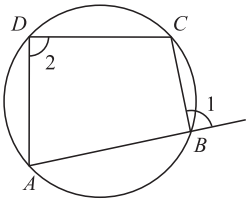
Explanation If $ABCD$ is a cyclic quadrilateral, then $\angle A + \angle C = \angle B + \angle D = 180°$



15. (Converse of above theorem) If the two angles of a pair of opposite angles of a quadrilateral are supplementary then the quadrilateral is cyclic.

16. If a side of a cyclic quadrilateral is produced then the exterior angle is equal to the interior opposite angle.

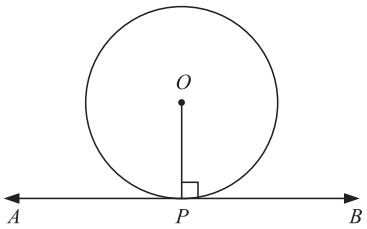
Explanation If the side AB of a cyclic quadrilateral $ABCD$ is produced then $\angle 1 = \angle 2$.



THEOREMS ON TANGENTS

17. A tangent at any point of a circle is perpendicular to the radius through the point of contact.

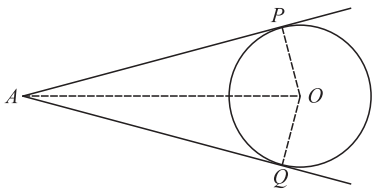
Explanation If AB is a tangent at a point P to a circle $C\ (O, r)$ then $PO \perp AB$



18. (Converse of above theorem) A line drawn through the end of a radius and perpendicular to it, is a tangent to the circle.

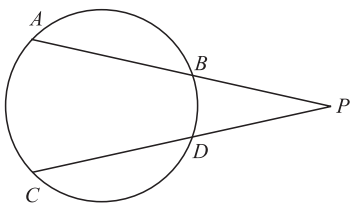
19. The lengths of two tangents drawn from an external point to a circle are equal.

Explanation If two tangents AP and AQ are drawn from a point A to a circle $C(O, r)$, then $AP = AQ$

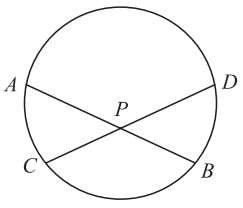


20. If two chords AB and CD intersect internally (ii) or externally (i) at a point P then

$$PA \times PB = PC \times PD$$



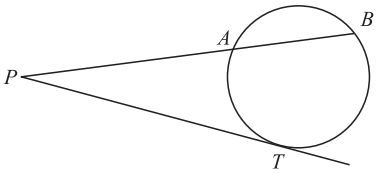
(i)



(ii)

21. If PAB is a secant to a circle intersecting the circle at A and B is a tangent segment then $PA \times PB = PT^2$ (refer the figure below).

(popularly known as Tangent-Secant theorem)



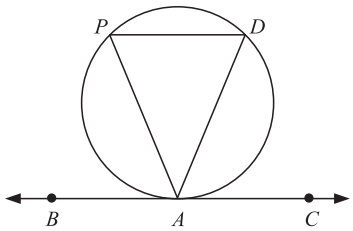
22. Alternate Segment Theorem:

In the figure below, if BAC is the tangent at A to a circle and if AD is any chord, then

$$\angle DAC = \angle APD \text{ and}$$

$$\angle PAB = \angle PDA$$

(Angles in alternate segment)

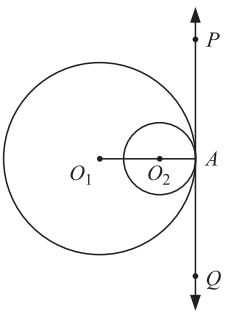


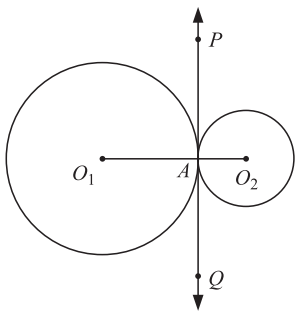
Notes

The converse of the above theorem is true.

23. If two circles touch each other internally or externally, the point of contact lies on the line joining their centres.

Explanation If two circles with centre O_1 and O_2 which touch each other internally (i) or externally (ii), at a point A then the point A lies on the line $O_1 O_2$, i.e., three points A, O_1 and O_2 are collinear.



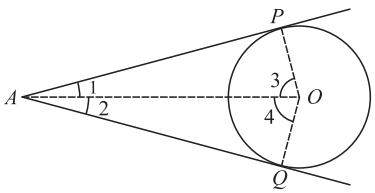


SOME USEFUL RESULTS

1. There is one and only one circle passing through three non-collinear points.
2. Two circles are congruent if and only if they have equal radii.
3. Of any two chords of a circle, the one which is greater is nearer to the centre.
4. Of any two chords of a circle, the one which is nearer to the centre is greater.
5. If two circles intersect in two points, then the line through the centres is the perpendicular bisector of the common chord.
6. Angle in a major segment of a circle is acute and angle in a minor segment is obtuse.
7. If two tangents are drawn to a circle from an external point then
   (i) they subtend equal angles at the centre.
   (ii) they are equally inclined to the segment, joining the centre to that point.

Explanation In a circle $C(O,r)$, A is a point outside it and AP and AQ are the tangents drawn to the circle

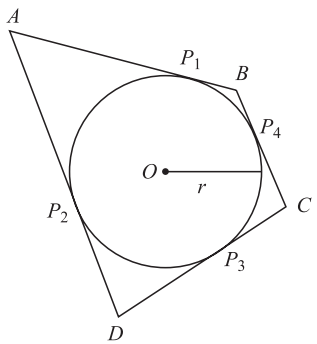
Then, $\angle 1 = \angle 2$ and $\angle 3 = \angle 4$



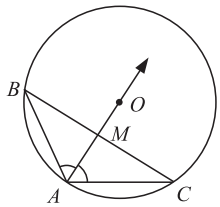
24. If a circle touches all the four sides of a quadrilateral then the sum of opposite pair of sides are equal.

Explanation If $ABCD$ is a circumscribed quadrilateral.

Then, $AB + CD = AD + BC$

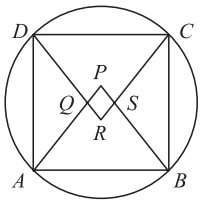


25. If two chords AB and AC of a circle are equal, then the bisector of $\angle BAC$ passes through the centre O of the circle.



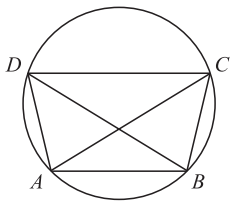
26. The quadrilateral formed by angle bisectors of a cyclic quadrilateral is also cyclic.

Explanation If $ABCD$ is a cyclic quadrilateral in which AP, BP, CR and DR are the bisectors of $\angle A$, $\angle B$, $\angle C$ and $\angle D$, respectively, then quadrilateral $PQRS$ is also cyclic.



27. A cyclic trapezium is isoceles and its diagonals are equal.

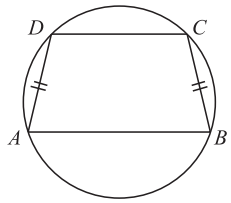
Explanation If ABC cyclic trapezium such that $AB \parallel DC$, then $AD = BC$ and $AC = BD$



28. If two opposite sides of a cyclic quadrilateral are equal, then the other two sides are parallel.

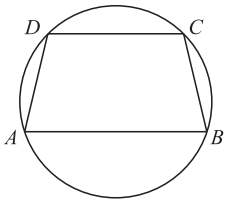
Explanation A cyclic quadrilateral $ABCD$ in which $AD = BC$

Then, $AB \parallel CD$



29. An isoceles trapezium is always cyclic.

Explanation A trapezium $ABCD$ in which $AB \parallel CD$ and $AD = BC$



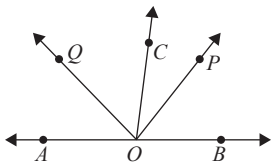
Then $ABCD$ is a cyclic trapezium.

EXERCISE-I

1. An angle is equal to one-third of its supplement. Its measure is equal to:
(a) 40° (b) 50°
(c) 45° (d) 55°

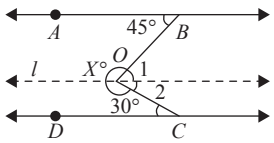
2. The complement of 30°20′ is:
(a) 69°40′ (b) 59°40′
(c) 35°80′ (d) 159°40′

3. In the given figure, OP bisect $\angle BOC$ and OQ bisects $\angle AOC$. Then $\angle POQ$ is equal to:



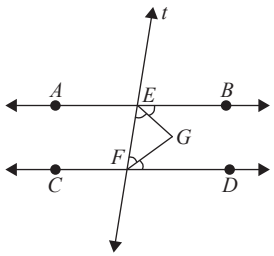
(a) 90° (b) 120°
(c) 60° (d) 100°

4. In the given, $AB \parallel CD$. Then X is equal to:



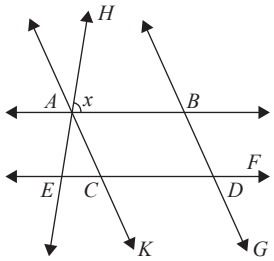
(a) 290° (b) 300°
(c) 280° (d) 285°

5. In the adjoining figure, $AB \parallel CD$, t is the transversal, EG and FG are the bisectors of $\angle BEE$ and $\angle DFE$ respectively, then $\angle EGF$ is equal to:



(a) 90° (b) 75°
(c) 80° (d) 110°

6. In the given figure, $AB \parallel CD$ and $AC \parallel BD$. If $\angle EAC = 40°$, $\angle FDG = 55°$, $\angle HAB = x$; then the value of x is:

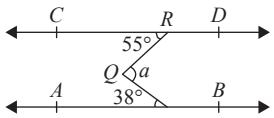


(a) 95° (b) 70°
(c) 35° (d) 85°

7. Find the measure of an angle, if six times its complement is 12° less than twice its supplement:
(a) 48° (b) 96°
(c) 24° (d) 58°

8. If two parallel lines are intersected by a transversal, then the bisectors of the two pairs of interior angles enclose a:
(a) Trapezium (b) Rectangle
(c) Square (d) none of these

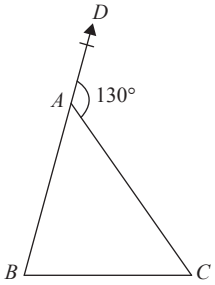
9. In fig., $AB \parallel CD$, $\angle a$ is equal to:



(a) 93° (b) 103°
(c) 83° (d) 97°

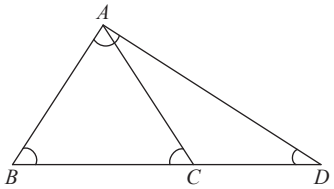
10. The complement of an angle exceeds the angle by 60°. Then the angle is equal to:
(a) 25° (b) 30°
(c) 15° (d) 35°

11. In the following figure, $\angle B:\angle C = 2:3$, find $\angle B + \angle C$.



(a) 120° (b) 52°
(c) 78° (d) 130°

12. In the given figure, $\angle B = \angle C = 55°$ and $\angle D = 25°$. Then:

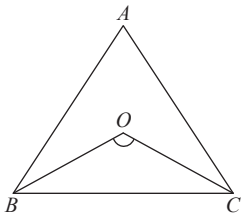


(a) $BC < CA < CD$
(b) $BC > CA > CA$
(c) $BC < CA$, $CA > CD$
(d) $BC > CA$, $CA < CD$

13. In a ΔABC, if $2\angle A = 3\angle B = 6\angle C$, Then $\angle A$ is equal to:
(a) 60° (b) 30°
(c) 90° (d) 120°

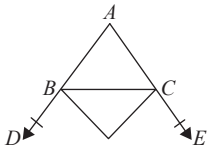
14. A, B, C are the three angles of a Δ. If $A - B = 15°$ and $B - C = 30°$. Then $\angle A$ is equal to:
(a) 65° (b) 80°
(c) 75° (d) 85°

15. In ΔABC, the angle bisectors of $\angle B$ and $\angle C$ meet at O. If $\angle A = 70°$, then $\angle BOC$ is equal to:



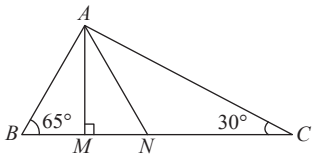
(a) 135° (b) 125°
(c) 115° (d) 110°

16. The sides AB and AC of ΔABC have been produced to D and E respectively. The bisectors of $\angle CBD$ and $\angle BCE$ meet at O. If $\angle A = 40°$, then $\angle BOC$ is equal to:

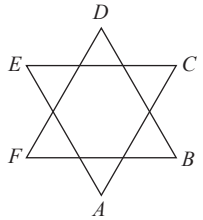


(a) 60° (b) 65°
(c) 75° (d) 70°

17. In the given figure, $AM \perp BC$ and AN is the bisector of $\angle A$. What is the measure of $\angle MAN$.
(a) 17.5° (b) 15.5°
(c) 20° (d) 25°

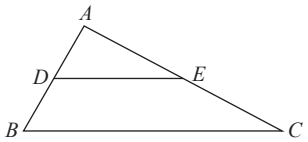


18. In the adjoining figure $\angle A + \angle B + \angle C + \angle D + \angle E + \angle F =$



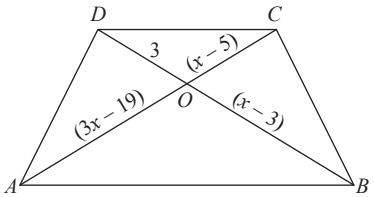
(a) 270° (b) 300°
(c) 360° (d) 330°

19. In the given figure, $DE \parallel BC$ if $AD = 1.7$ cm, $AB = 6.8$ cm and $AC = 9$ cm, find AE.



(a) 2.25 cm (b) 4.5 cm
(c) 1.25 cm (d) 2.5 cm

20. In the given figure, $AB \parallel DC$, find the value of x.

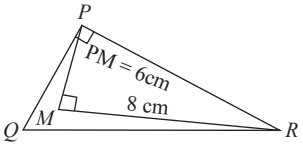


(a) $x = 8$ (b) $x = 9$
(c) $x = 8$ or 9 (d) $x = 10$

21. If the bisector of an angle of Δ bisects the opposite side, then the Δ is:

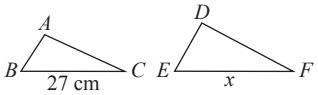
(a) Scalene (b) Isosceles
(c) Right triangle (d) None of these

22. In the given figure $\angle QPR = 90°$, $QR = 26$ cm, $PM = 6$ cm, $MR = 8$ cm and $\angle PMR = 90°$, find the area of ΔPQR.



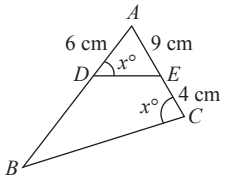
(a) 180 cm^2 (b) 240 cm^2
(c) 120 cm^2 (d) 150 cm^2

23. The areas of two similar Δs are 81 cm^2 and 144 cm^2. If the largest side of the smaller Δ is 27 cm, then the largest side of the larger Δ is:



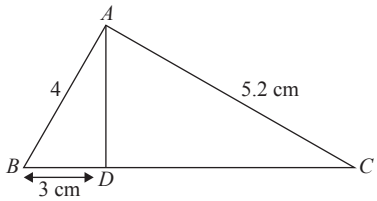
(a) 24 cm (b) 48 cm
(c) 36 cm (d) None of these

24. In the given figure, find the length of BD.



(a) 13.5 cm (b) 12 cm
(c) 14.5 cm (d) 15 cm

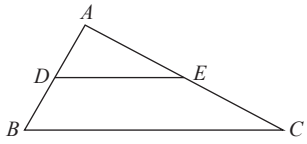
25. In the given figure $\angle BAD = \angle CAD$. $AB = 4$ cm, $AC = 5.2$ cm, $BD = 3$ cm. Find BC.



(a) 6.9 cm (b) 9.6 cm
(c) 3.9 cm (d) 9.3 cm

26. A ladder 15 m long reaches a window which is 9 m above the ground on one side of street. Keeping its foot at the same point, the ladder is turned to the other side of the street to reach a window 12 m high. What is the width of the street:
(a) 31 m (b) 12 m
(c) 30 m (d) 21 m

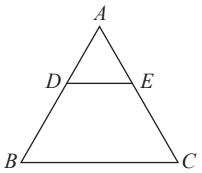
27. In, ΔABC, D and E are the mid-points of AB and AC respectively. Find the ratio of the areas of ΔADE and ΔABC.



(a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) $\frac{3}{4}$ (d) $\frac{1}{8}$

28. A vertical stick 12 cm long casts a shadow 8 cm long on the ground. At the same time a tower casts the shadow 40 m long on the ground. Find the height of the tower.
(a) 600 m (b) 160 m
(c) 60 m (d) 52 m

29. D and E are the points on the sides AB and AC respectively of ΔABC such that $AD = 8$ cm, $BD = 12$ cm, $AE = 6$ cm and $EC = 9$ cm. Then find BC/DE.



(a) $\frac{5}{2}$ (b) $\frac{2}{5}$
(c) $\frac{5}{7}$ (d) $\frac{5}{3}$

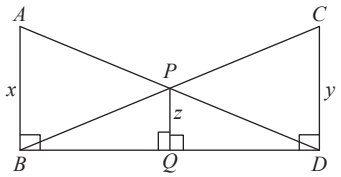
30. In an equilateral ΔABC, if $AD \perp BC$, then:
(a) $3AB^2 = 2AD^2$ (b) $2AB^2 = 3AD^2$
(c) $3AB^2 = 4AD^2$ (d) $4AB^2 = 3AD^2$

31. In a right angled ΔABC, rt. angled at A, $AD \perp BC$. Then:
(a) $AD^2 = BD \times CD$ (b) $AD^2 = AB \times AC$
(c) $AD^2 = BD \times AB$ (d) $AD^2 = CD \times AC$

32. If $ABCD$ is a ||gm and AC and BD be its diagonals, then:
(a) $AB^2 + BC^2 + CD^2 + DA^2 = AC^2 - BD^2$
(b) $AB^2 + BC^2 + CD^2 + DA^2 = AC^2 + BD^2$
(c) $4AD^2 = 2AC^2 + 2BD^2$
(d) $4AB^2 = 2AC^2 - 2BC^2$

33. In the given figure, $\angle ABD = \angle CBD = \angle PQB = 90°$. Then:

(a) $\frac{1}{x} - \frac{1}{y} = \frac{1}{z}$ (b) $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$
(c) $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$ (d) $\frac{1}{y} - \frac{1}{x} = \frac{1}{z}$

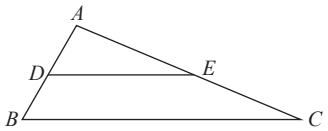


34. The area of two similar Δs are 121 cm^2 and 81 cm^2 respectively. What is the ratio of their corresponding heights (altitudes):

(a) $\frac{11}{9}$ (b) $\frac{22}{9}$

(c) $\frac{11}{18}$ (d) None of these

35. In the given figure, $DE \parallel BC$ and $DE : BC = 3:5$ the ratio of the areas of ΔADE and the trapezium $BCED$.



(a) $\frac{9}{25}$ (b) $\frac{12}{25}$

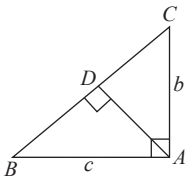
(c) $\frac{3}{4}$ (d) $\frac{9}{16}$

36. In a ΔABC, AD intersects ∠A and BC. If $BC = a$, $AC = b$ and $AB = c$, Then:

(a) $CD = \frac{b+c}{ab}$ (b) $CD = \frac{ab}{b+c}$

(c) $CD = \frac{bc+ab}{ac}$ (d) $CD = \frac{ac}{bc+ab}$

37. In the given figure, what is the length of AD in terms of b and c:



(a) $\frac{bc}{b^2+c^2}$ (b) $\frac{b^2+c^2}{bc}$

(c) $\frac{\sqrt{b^2+c^2}}{bc}$ (d) $\frac{bc}{\sqrt{b^2+c^2}}$

38. ABC is a Δ in which $AB = AC$ and D is a point on AC such that $BC^2 = AC \times CD$. Then:

(a) $BD = DC$ (b) $BD = BC$

(c) $BD = AB$ (d) $BD = AD$

39. Two poles of ht. a and b metres are p metres apart ($b > a$). The height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is:

(a) $\frac{a+b}{ab}$ (b) $\frac{p}{a+b}$

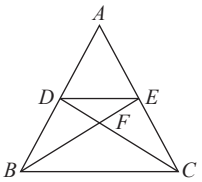
(c) $\frac{ab}{a+b}$ (d) $\frac{a+b}{p}$

40. ABC is a right Δ, right-angled at C. If $AB = c$, $BC = a$ and $CA = b$ and p is the length of the perpendicular from C on AB. Then:

(a) $\frac{1}{p^2}+\frac{1}{a^2}=\frac{1}{b^2}$ (b) $\frac{1}{p^2}+\frac{1}{b^2}=\frac{1}{a^2}$

(c) $\frac{1}{a^2}=\frac{1}{b^2}+\frac{1}{p^2}$ (d) $\frac{1}{p^2}=\frac{1}{a^2}+\frac{1}{b^2}$

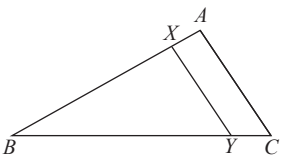
41. In the given figure $DE \| BC$ and $AD : DB = 5:4$, Then $\frac{ar(\Delta DFE)}{ar(\Delta CFB)}$



(a) $\frac{25}{81}$ (b) $\frac{25}{16}$

(c) $\frac{16}{25}$ (d) $\frac{16}{81}$

42. In the fig. $XY \| AC$ and XY divides triangular region ABC into two part equal in area. Then $\frac{AX}{AB}$ is equal to:



(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{\sqrt{2}+2}{\sqrt{2}}$

(c) $\frac{1}{2}$ (d) $\frac{\sqrt{2}-1}{\sqrt{2}}$

43. A point O in the interior of a rectangular $ABCD$ is joined with each of the vertices A, B, C and D. Then:

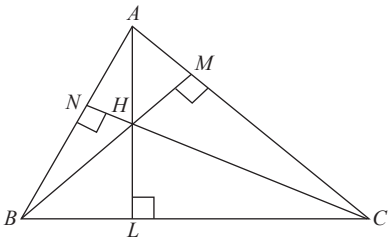
(a) $OA^2 + OC^2 = OB^2 - OD^2$

(b) $OA^2 + OC^2 = OB^2 + OD^2$

(c) $OA^2 = OB^2 = OC^2 + OD^2$
(d) $OA^2 + OD^2 = OB^2 + OC^2$

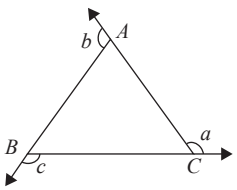
44. In ΔABC, the median BE intersects AC at E, if BG = 6 cm, where G is the centroid, then BE is equal to:
(a) 7 cm (b) 9 cm
(c) 8 cm (d) 10 cm

45. If H is the orthocentre of ΔABC, then the orthocentre of ΔHBC is (fig. given):



(a) N (b) M
(c) A (d) L

46. If the sides of a triangle are produced then the sum of the exterior angles i.e., $\angle a + \angle b + \angle c$ is equal to:



(a) 180° (b) 90°
(c) 360° (d) 270°

47. Incentre of a triangle lies in the interior of:
(a) an isosceles triangle only
(b) any triangle
(c) an equilateral triangle only
(d) a right triangle only

48. In a ΔABC, the bisectors of $\angle B$ and $\angle C$ *intersect* each other at apoint O. Then $\angle BOC$ is equal to:

(a) $90° - \frac{1}{2}\angle A$ (b) $120° + \frac{1}{2}\angle A$

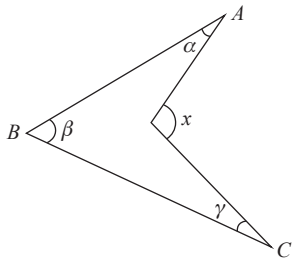
(c) $90° + \frac{1}{2}\angle A$ (d) $120° - \frac{1}{2}\angle A$

49. In a ΔABC, the sides AB and AC are produced to P and Q respectively. The bisectors of $\angle OBC$ and $\angle QCB$ intersect at a point O. Then $\angle BOC$ is equal to:

(a) $90° + \frac{1}{2}\angle A$ (b) $90° - \frac{1}{2}\angle A$

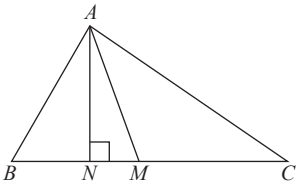
(c) $120° + \frac{1}{2}\angle A$ (d) $120° - \frac{1}{2}\angle A$

50. In the given figure, which of the following is true:



(a) $x = \alpha + \beta + \gamma$ (b) $x + \beta = \alpha + \gamma$
(c) $x + \gamma = \beta + \alpha$ (d) $x + \alpha = \beta + \gamma$

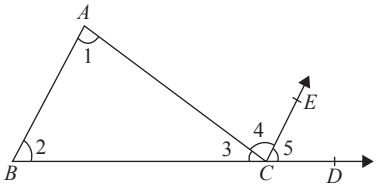
51. In the given figure, In a ΔABC, $\angle B = \angle C$. If AM is the bisector of $\angle BAC$ and $AN \perp BC$, then $\angle MAN$ is equal to:



(a) $\frac{1}{2}(\angle B + \angle C)$ (b) $\frac{1}{2}(\angle C - \angle B)$

(c) $\angle B + \angle C$ (d) $\frac{1}{2}(\angle B - \angle C)$

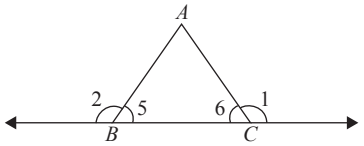
52. In the given figure, side BC of ΔABC is produced to form ray BD and $CE \parallel BA$. Then $\angle ACD$ is equal to:



(a) $\angle A - \angle B$ (b) $\frac{1}{2}(\angle A + \angle B)$

(c) $\angle A + \angle B$ (d) $\frac{1}{2}(\angle A - \angle B)$

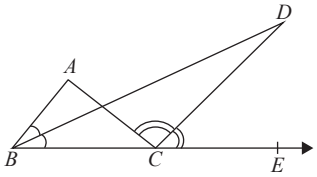
53. In the given figure, the side BC of a ΔABC is produced on both sides. Then $\angle 1 + \angle 2$ is equal to:



(a) $\angle A + 180°$ (b) $180° - \angle A$

(c) $\frac{1}{2}(\angle A + 180°)$ (d) $\angle A + 90°$

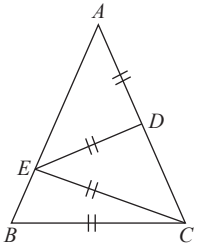
54. In the figure, BD and CD are angle bisectors of $\angle ABC$ and $\angle ACE$, respectively. Then $\angle BDC$ is equal to:



(a) $\angle BAC$ (b) $2\angle BAC$

(c) $\frac{1}{2}\angle BAC$ (d) $\frac{1}{3}\angle BAC$

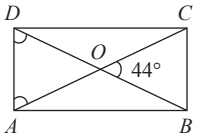
55. In fig, $AB = AC$, D is a point on AC and E on AB such that $AD = ED = EC = BC$. Then $\angle A:\angle B$:



(a) 1:2 (b) 2:1

(c) 3:1 (d) 1:3

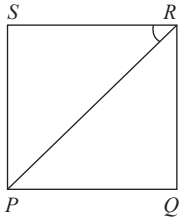
56. The diagonals of a rectangle $ABCD$ meet at 0. If $\angle BOC = 44°$, then $\angle OAD$ is equal to:



(a) 90° (b) 60°

(c) 100° (d) 68°

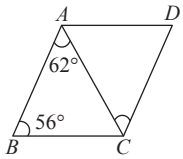
57. $PQRS$ is a square. The $\angle SRP$ is equal to:



(a) 45° (b) 90°

(c) 100° (d) 60°

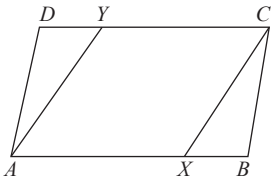
58. $ABCD$ is a rhombus with $\angle ABC = 56°$, then $\angle ACD$ is equal to:



(a) 90° (b) 60°

(c) 56° (d) 62°

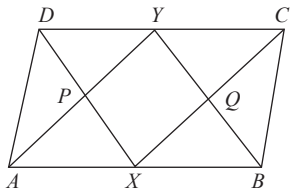
59. $ABCD$ is a parallelogram and X, Y are the mid-points of sides AB and CD respectively. Then quadrilateral $AXCY$ is a:



(a) parallelogram (b) rhombus

(c) square (d) rectangle

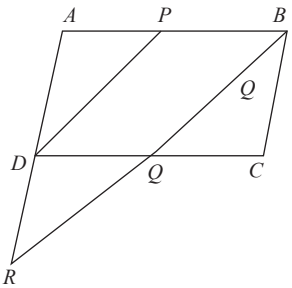
60. X, Y are the mid-points of opposite sides AB and DC of a parallelogram $ABCD$. AY and DX are joined intersecting in P; CX and BY are joined intersecting in Q. Then $PXQY$ is a:



(a) Rectangle (b) Rhombus

(c) Parallelogram (d) Square

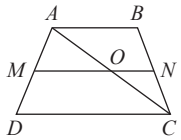
61. P is the mid-point of side AB to a parallelogram $ABCD$. A line through B parallel to PD meets DC at Q and AD produced at R. Then BR is equal to:



(a) BQ (b) $\frac{1}{2}$

(c) $2BQ$ (d) None of these

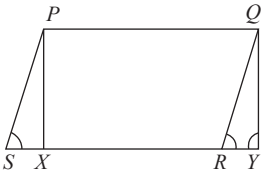
62. $ABCD$ is a trapezium in which $AB \parallel CD$. M and N are the mid-points of AD and respectively. If AB = 12 cm an MN = 14 cm. Find CD.



(a) 2 cm (b) 5 cm

(c) 12 cm (d) 16 cm

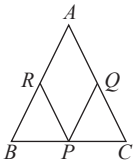
63. $PQRS$ is a parallelogram. PX and QY are, respectively, the perpendicular form P and Q to SR and SR produced. The PX is equal to:



(a) QY (b) $2QY$

(c) $\frac{1}{2}Q$ (d) XR

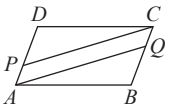
64. In a ΔABC, P, Q and R are the mid-points of sides BC, CA and AB respectively. If $AC = 21$ cm, $BC = 29$ cm and $AB = 30$ cm. The perimeter of the quad. $ARPQ$ is:



(a) 91 cm (b) 60 cm

(c) 51 cm (d) 70 cm

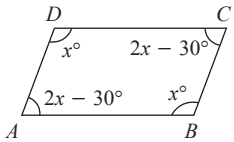
65. $ABCD$ is a parallelogram. P is a point on AD such that $AP = \frac{1}{3}AD$ and Q is a point on BC such that $CQ = \frac{1}{3}BC$. Then $AQCP$ is a:



(a) Parallelogram (b) Rhombus

(c) Rectangle (d) Square

66. Find the measure of each angle of a parallelogram, if one of its angles is 30° less than twice the smallest angle.



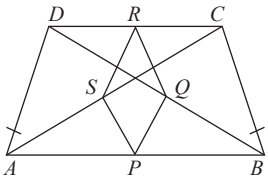
(a) 60°, 100°, 90°, 20°

(b) 80°, 40°, 120°, 90°

(c) 100°, 90°, 90°, 80°

(d) 70°, 110°, 70°, 110°

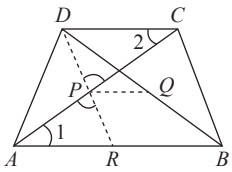
67. $ABCD$ is a trapezium in which $AB \parallel DC$ and $AD = BC$. If P, Q, R, S be respectively the mid-point of BA, BD and CD, CA. The $PQRS$ is a:



(a) Rhombus (b) Rectangle

(c) Parallelogram (d) Square

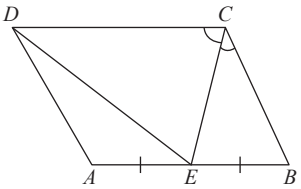
68. $ABCD$ is a trapezium and P, Q are the mid-points of the diagonals AC and BD. Then PQ is equal to:



(a) $\frac{1}{2}(AB)$ (b) $\frac{1}{2}(CD)$

(c) $\frac{1}{2}(AB - CD)$ (d) $\frac{1}{2}(AB + CD)$

69. $ABCD$ is a parallelogram, E is the mid-point of AB and CE bisects $\angle BCD$. The $\angle DEC$ is:



(a) 60° (b) 90°

(c) 100° (d) 120°

70. The angles of a quadrilateral are respectively 100°, 98° and 92°. The fourth angle is equal to:

(a) 90° (b) 95°

(c) 70° (d) 75°

71. The measure of each angle of a regular hexagon is:

(a) 110° (b) 130°

(c) 115° (d) 120°

72. The interior angle of a regular polygon is 108°. The number of sides of the polygon is:

(a) 6 (b) 7

(c) 8 (d) 5

73. The number of diagonals in a hexagon is:

(a) 9 (b) 8

(c) 10 (d) 7

74. If the number of diagonals of a polygon is 27, then the number of sides is:

(a) 10 (b) 9

(c) 11 (d) 6

75. One angle of a pentagon is 140°. If the remaining angles are in the ratio 1:2:3:4. The size of the greatest angle is:

(a) 150° (b) 180°

(c) 160° (d) 170°

76. The exterior angle of a regular polygon is $\frac{1}{3}$ of its interior angle. The number of the sides of the polygon is:
(a) 9 (b) 8
(c) 10 (d) 12

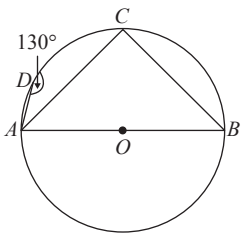
77. The ratio of the measure of an angle of a regular octagon to the measure of its exterior angle is:
(a) 3:1 (b) 2:1
(c) 1:3 (d) 1:2

78. $ABCDE$ is a regular pentagon. Diagonal AD divides $\angle CDE$ in to parts, then the ratio of $\frac{\angle ADE}{\angle ADC}$ is equal to:
(a) 3:1 (b) 1:4
(c) 1:3 (d) 1:2

79. The difference between an exterior angle of $(n-1)$ sided regular polygon and an exterior angle of $(n-2)$ sided regular polygon is 6°, then the value of n is:
(a) 14 (b) 15
(c) 13 (d) 12

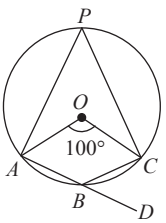
80. The radius of a circle is 13 cm and the length of one of its chords is 10 cm. What is the distance of the chord from the centre:
(a) 10 cm (b) 15 cm
(c) 12 cm (d) 9 cm

81. In the given figure, $ABCD$ is a cyclic quadrilateral whose side AB is a diameter of the circle through A, B and C. If $\angle ADC = 130°$, find $\angle CAB$.



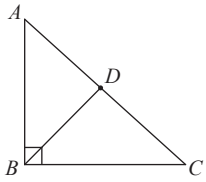
(a) 40° (b) 50°
(c) 30° (d) 130°

82. In the given figure, O is the centre of the circle, Find $\angle CBD$.



(a) 140° (b) 50°
(c) 40° (d) 130°

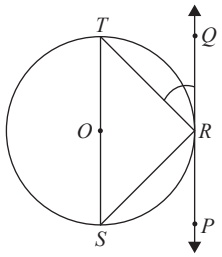
83. In a ΔABC, $\angle$B is a right angle, $AC = 6$ cm and D is the mid-point of AC. Find the length of BD.



(a) 4 cm (b) $\sqrt{6}$ cm
(c) 3 cm (d) 4.5 cm

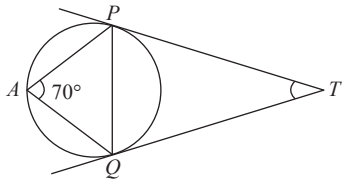
84. A cyclic quadirateral whose opposite angles are equal, is a:
(a) Parallelogram but not a rhombus
(b) Rhombus
(c) Rectangle
(d) Square but not a rectangle

85. In the given figure, ST is a diameter of the circle with centre O and PQ is the tangent at a point R. If $\angle TRQ = 40$, what is $\angle RTS$:



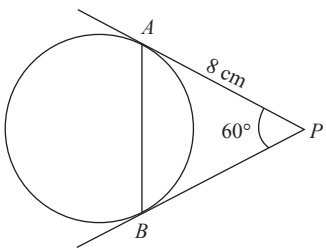
(a) 40° (b) 50°
(c) 60° (d) 30°

86. In the given figure, TP and TQ are tangents to the circle. If $\angle PAQ = 70°$, what is $\angle PTQ$?



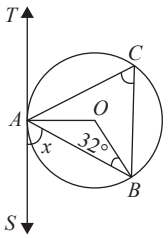
(a) 30° (b) 45°
(c) 60° (d) 40°

87. In the given figure, PA and PB are tangents from a point P to a circle such that $PA = 8$ cm and $\angle APB = 60°$. What is the length of the chord AB?



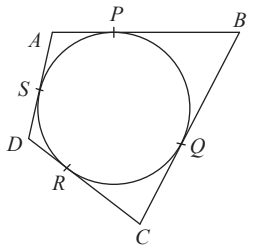
(a) 8 cm (b) 10 cm
(c) 6 cm (d) 12 cm

88. In the given figure, *TAS* is a tangent to the circle at the point *A*. If $\angle OBA = 32°$, what is the value of *x*:



(a) 64° (b) 40°
(c) 58° (d) 50°

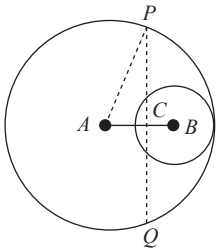
89. In the given figure, a circle touches all the four sides of quadrilateral *ABCD* whose sides *AB* = 6 cm, *BC* = 7 cm and *CD* = 4 cm. Find *AD*.



(a) 5 cm (b) 4 cm
(c) 3 cm (d) 2 cm

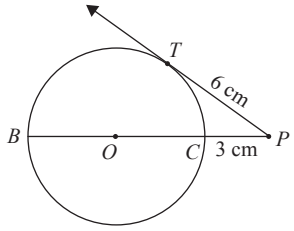
90. *AB* and *CD* are two parallel chords of a circle such that *AB* = 10 cm and *CD* = 24 cm. If the chords are on opposite sides of the centre and the distance between them is 17 cm, what is the radius of the circle:
(a) 14 cm (b) 10 cm
(c) 13 cm (d) 15 cm

91. In the given figure, two circle with centres *A* and *B* of radii 5 cm and 3 cm touch each other internally. If the perpendicular bisector of segment *AB* meets the bigger circle in *P* and *Q*, find the length of *PQ*.



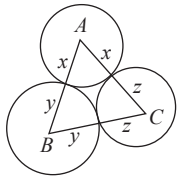
(a) $4\sqrt{6}$ cm (b) $\sqrt{24}$ cm
(c) $8\sqrt{3}$ cm (d) $4\sqrt{3}$ cm

92. In the given figure *O* is the centre of the circle and *PT* is a tangent at *T*. If *PC* = 3cm, *PT* = 6 cm, calculate the radius of the circle.



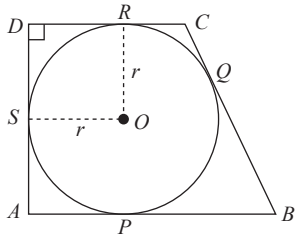
(a) 9 cm (b) 4.5 cm
(c) 8 cm (d) 12 cm

93. With the vertices of a ΔABC as centres, three circles are described, each touching the other two externally. If the sides of the Δ are 9 cm, 7 cm and 6 cm, find the radii of the circles:



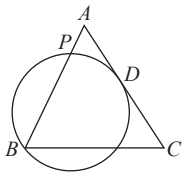
(a) 4cm, 7cm and 3cm
(b) 7cm, 5cm and 2cm
(c) 5cm, 4cm and 3cm
(d) 4cm, 5cm and 2cm

94. In the given figure, *ABCD* is a quadrilateral in which $\angle O = 90$. A circle *C*(0, *r*) touches the sides *AB*, *BC*, *CD* and *DA* at *P*, *Q*, *R*, *S* respectively. If *BC* = 38 cm, *CD* = 25 cm and *BP* = 27 cm, find the value of *r*.



(a) 14 cm (b) 15 cm
(c) 10 cm (d) 16 cm

95. In the given figure, *ABC* is an isosceles Δ in which *AB* = *AC*. A circle through *B* touches *AC* at its mid-point *D* and intersects *AB* at *P*. Then which of the following is correct:

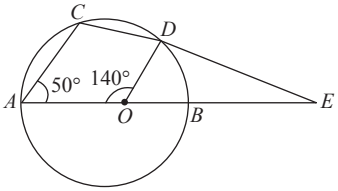


(a) $AP = \frac{3}{4}AB$ (b) $AP = \frac{2.5}{3}AB$
(c) $AP = \frac{4}{5}AB$ (d) $AP = \frac{1}{4}AB$

96. ABC is a right angled Δ with $BC = 6$ cm and $AB = 8$ cm. A circle with centre O is inscribed in ΔABC. The radius of the circle is:
(a) 4 cm (b) 3 cm
(c) 2 cm (d) 1 cm

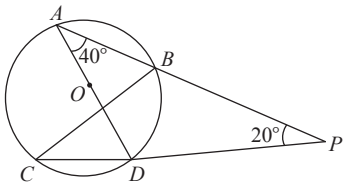
97. If all the sides of a parallelogram touch a circle, then the parallelogram is a:
(a) Rectangle (b) Rhombus
(c) Square (d) None

98. In the given figure, O is the centre of a circle. If $\angle AOD = 140°$ and $\angle CAB = 50°$, what is $\angle EDB$:



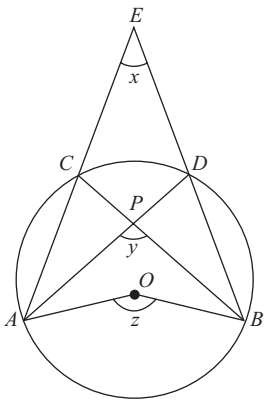
(a) 70° (b) 40°
(c) 60° (d) 50°

99. PBA and PDC are two secants. AD is the diameter of the circle with centre at O. $\angle A = 40$, $\angle P = 20°$. Find the measure of $\angle DBC$.



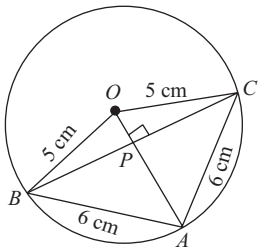
(a) 30° (b) 45°
(c) 50° (d) 40°

100. In the given figure, O is the centre of the circle. Then $\angle x + \angle y$ is equal to:



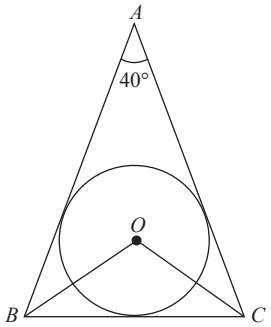
(a) $2\angle z$ (b) $\frac{\angle z}{2}$
(c) $\angle z$ (d) None

101. In a circle of radius 5 cm, AB and AC are two equal chords such that $AB = AC = 6$ cm. What is the length of the chord BC.



(a) 9.6 cm (b) 11 cm
(c) 12 cm (d) 9 cm

102. In the given figure O is the centre of incircle for ΔABC. Find $\angle BOC$ if $\angle BAC = 40°$.



(a) 100° (b) 120°
(c) 90° (d) 110°

103. If an equilateral triangle ABC be inscribed in a circle, then the tangents at their vertices will form another Δ.
(a) scalene (b) equilateral
(c) Isosceles (d) Right Δ

104. The radius of the incircle of a Δ is 4 cm and the segments into which one side is divided by the point of contact are 6 cm and 8 cm, then the length of the shortest side of the Δ is:
(a) 12 cm (b) 15 cm
(c) 13 cm (d) 14 cm

EXERCISE-2
(BASED ON MEMORY)

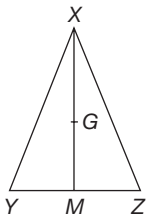
1. If the height of the equilateral triangle is $2\sqrt{3}$ cm, then determine the area (in cm^2) of the equilateral triangle.

(a) 6 (b) $2\sqrt{3}$

(c) $4\sqrt{3}$ (d) 12

[SSC CHSL (10+2) Tier-I CBE, 2018]

2. In triangle XYZ, G is the centroid. If XY = 11cm, YZ = 14cm and XZ = 7 cm, then what is the value (in cm) of GM?

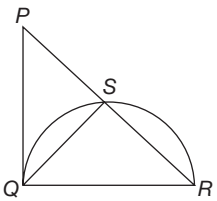


(a) 6 (b) 4

(c) 2 (d) 3

[SSC CGL Tier-II CBE, 2018]

3. In the given figure, triangle PQR is a right angled triangle at Q. If PQ = 35 cm and QS = 28 cm, then what is the value (in cm) of SR?

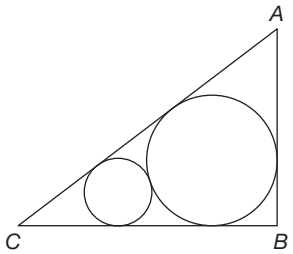


(a) 35.33 (b) 37.33

(c) 41.33 (d) 43.33

[SSC CGL Tier-I CBE, 2018]

4. In the given figure, ABC is a right angled triangle. $\angle ABC = 90°$ and $\angle ACB = 60°$. If the radius of the smaller circle is 2 cm, then what is the radius (in cm) of the larger circle?



(a) 4 (b) 6

(c) 4.5 (d) 7.5

[SSC CGL Tier-II CBE, 2018]

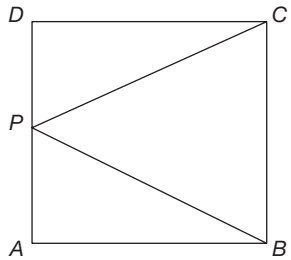
5. PQRS is a square whose side is 16 cm. What is the value of the side (in cm) of the largest regular octagon that can be cut from the given square?

(a) $8-4\sqrt{2}$ (b) $16+8\sqrt{2}$

(c) $16\sqrt{2}-\sqrt{16}$ (d) $16-\sqrt{2}$

[SSC CGL Tier-II CBE, 2018])

6. In the given figure, ABCD is a square whose side is 4 cm. P is a point on the side AD. What is the minimum value (in cm) of BP + CP?



(a) $4\sqrt{5}$ (b) $4\sqrt{4}$

(c) $6\sqrt{3}$ (d) $4\sqrt{6}$

[SSC CGL Tier-II CBE, 2018])

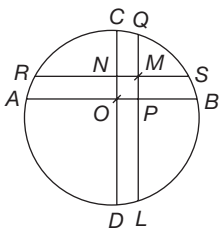
7. If a regular polygon has 5 sides then the measure of its interior angle is greater than the measure of its exterior angle by how many degrees?

(a) 60° (b) 36°

(c) 90° (d) 100°

[SSC CHSL (10+2) Tier-I CBE, 2018]

8. In the given figure, CD and AB are diameters of circle and AB and CD are perpendicular to each other. LQ and SR are perpendiculars to AB and CD respectively. Radius of circle is 5 cm, PB:PA = 2:3 and CN:ND = 2:3. What is the length (in cm) of SM?

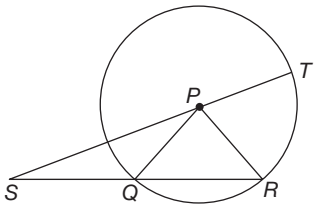


(a) $\left[(5\sqrt{3})-3\right]$ (b) $\left[(4\sqrt{3})-2\right]$

(c) $\left[(2\sqrt{5})-1\right]$ (d) $\left[(2\sqrt{6})-1\right]$

[SSC CGL Tier-II CBE, 2018]

9. In the given figure, P is the centre of the circle. If QS = PR, then what is the ratio of ∠RSP to the ∠TPR?

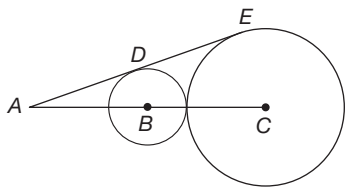


(a) 1:4 (b) 2:5

(c) 1:3 (d) 2:7

[SSC CGL Tier-II CBE, 2018]

10. In the given figure, B and C are the centres of the two circles. ADE is the common tangent to the two circles. If the ratio of the radius of both the circles is 3:5 and AC = 40, then what is the value of DE?



(a) $3\sqrt{15}$ (b) $5\sqrt{15}$

(c) $6\sqrt{15}$ (d) $4\sqrt{15}$

[SSC CGL Tier-II CBE, 2018])

11. The radius of two circles is 3 cm and 4 cm. The distance between the centres of the circles is 10 cm. What is the ratio of the length of direct common tangent to the length of the transverse common tangent?

(a) $\sqrt{51}:\sqrt{68}$ (b) $\sqrt{33}:\sqrt{17}$

(c) $\sqrt{66}:\sqrt{51}$ (d) $\sqrt{28}:\sqrt{17}$

[SSC CGL Tier-II CBE, 2018])

12. ΔABC is equilateral triangle. O is the point of intersection of altitudes AL, BM and CN. If OA = 16 cm, what is the semi-perimeter (in cm) of the triangle ABC?

(a) $8\sqrt{3}$ (b) $12\sqrt{3}$

(c) $16\sqrt{3}$ (d) $24\sqrt{3}$

[SSC CAPFs ASI & Delhi Police SI, 2017]

13. DEF is an isosceles triangle such that DE = DF = 60 cm and EF = 96 cm. DG is a median to base EF. What is the length (in cm) of DG?

(a) 22 (b) 36

(c) 24 (d) 32

[SSC CAPFs ASI & Delhi Police SI, 2017]

14. The side BC of ΔABC is produced to D. If ∠ACD = 114° and $\angle ABC = \left(\frac{1}{2}\right)\angle BAC$, what is the value (in degrees) of ∠BAC?

(a) 36 (b) 48

(c) 76 (d) 84

[SSC CGL Tier-I (CBE), 2017]

15. In ΔPQR, ∠P:∠Q:∠R = 1:3:5. What is the value (in degrees) of ∠R − ∠P?

(a) 30 (b) 80

(c) 45 (d) 60

[SSC CAPFs ASI & Delhi Police SI, 2017]

16. In triangle PQR, the sides PQ and PR are produced to A and B respectively. The bisectors of ∠AQR and ∠BRQ intersect at point O. If ∠QOR = 50, what is the value (in degree) of ∠QPR?

(a) 50 (b) 60

(c) 80 (d) 100

[SSC CGL Tier-I (CBE), 2017]

17. In a triangle PQR, ∠Q = 90°. If PQ = 12 cm QR = 5 cm, then what is the radius (in cm) of the circumcircle of the triangle?

(a) 5 (b) 6

(c) 6.5 (d) $6\sqrt{2}$

[SSC CGL Tier-I CBE, 2017]

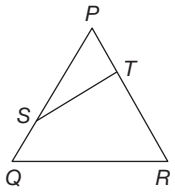
18. PQRA is a rectangle, AP = 22 cm, PQ = 8 cm. ΔABC is a triangle whose vertices lie on the sides of PQRA such that BQ = 2 cm and QC = 16 cm. Then the length of the line joining the mid points of the sides AB and BC is

(a) $4\sqrt{2}$ cm (b) 5 cm

(c) 6 cm (d) 10 cm

[SSC CGL Tier-II (CBE), 2017]

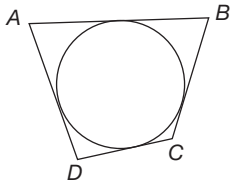
19. In the given figure, QRTS is a cyclic quadrilateral. If PT = 5 cm, SQ = 4 cm, PS = 6 cm and ∠PQR = 63°, then what is the value (in cm) of TR?



(a) 3 (b) 7
(c) 9 (d) 15

[SSC CHSL Tier-I CBE, 2017]

20. In the given figure, a circle touches quadrilateral ABCD. If AB = $2x + 3$, BC = $3x - 1$. CD = $x + 6$ and DA = $x + 4$, then what is the value of x?



(a) 3 (b) 4.5
(c) 6 (d) 6.5

[SSC CGL Tier-I (CBE), 2017]

21. The lengths of the sides of a triangle area a, b and c respectively. If $a^2 + b^2 + c^2 = ab + bc + ca$, then the triangle is:

(a) isosceles (b) equilateral
(c) scalene (d) right-angled

[SSC CGL Tier-I (CBE), 2016]

22. BD and CE are two medians of the triangle ABC. If EO = 7 cm. then the length of CE is

(a) 28 cm (b) 14 cm
(c) 21 cm (d) 35 cm

[SSC CGL Tier-I (CBE), 2016]

23. The lengths of side AB and side BC of a scalene triangle ABC are 12 cm and 8 cm respectively. The size of angle C is 90°. Find the approximate length of side AC.

(a) 12 (b) 9
(c) 14 (d) 16

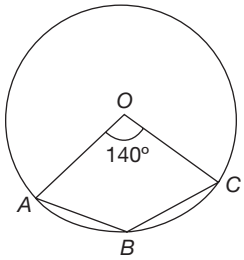
[SSC CGL Tier-I (CBE), 2016]

24. ΔABC is a triangle. PQ is line segment intersecting AB in P and AC in Q and PQ || BC. The ratio of AP:BP = 3:5 and length of PQ is 18 cm. The length of BC is

(a) 28 cm (b) 48 cm
(c) 84 cm (d) 42 cm

[SSC CGL Tier-II (CBE), 2016]

25. In the adjoining figure ∠AOC = 140 where O is the centre of the circle then ∠ABC is equal to:



(a) 110° (b) 100°
(c) 90° (d) 40°

[SSC CAPFs (CPO) SI & ASI, Delhi Police, 2016]

26. A square is inscribed in a quarter-circle in such a manner that two of its adjacent vertices lie on the two radii at an equal distance from the centre, while the other two vertices lie on the circular arc. If the square has sides of length x, then the radius of the circle is

(a) $\frac{16x}{\text{ð} + 4}$ (b) $\frac{2x}{\sqrt{\text{ð}}}$

(c) $\frac{\sqrt{5x}}{\sqrt{2}}$ (d) $\sqrt{2}x$

[SSC, 2015]

27. The measure of an angle whose supplement is three times as large as its complement is

(a) 30° (b) 45°
(c) 60° (d) 75°

[SSC, 2015]

28. Two chords of length a unit and b unit of a circle make angles 60° and 90° at the centre of a circle respectively, then the correct relation is

(a) $b = \sqrt{2}a$ (b) b = 2a

(c) $b = \sqrt{3}a$ (d) $b = \frac{3}{2}a$

[SSC, 2015]

29. In a parallelogram PQRS, angle P is four times of angle Q, then the measure of ∠R is

(a) 36° (b) 72°
(c) 130° (d) 144°

[SSC, 2015]

30. A tangent is drawn to a circle of radius 6 cm from a point situated at a distance of 10 cm from the centre of the circle. The length of the tangent will be

(a) 4 cm (b) 5 cm
(c) 8 cm (d) 7 cm

[SSC, 2015]

31. A ship after sailing 12 km towards south from a particular place covered 5 km more towards east. Then the straightway distance of the ship from that place is
(a) 18 km (b) 15 km
(c) 13 km (d) 11 km

[SSC, 2015]

32. A and B are centres of two circles of radii 11 cm and 6 cm, respectively. PQ is a direct common tangent to the circles. If $\overline{AB}$ = 13 cm, then length of $\overline{PQ}$ will be
(a) 8.5 cm (b) 17 cm
(c) 12 cm (d) 13 cm

[SSC, 2015]

33. Given that the ratio of altitudes of two triangles is 4 : 5, ratio of their areas is 3 : 2. The ratio of their corresponding bases is
(a) 5 : 8 (b) 15 : 8
(c) 8 : 5 (d) 8 : 15

[SSC, 2015]

34. AB and CD are two parallel chords of a circle of lengths 10 cm and 4 cm respectively. If the chords are on the same side of the centre and the distance between them is 3 cm, then the diameter of the circle is
(a) $2\sqrt{21}$ cm (b) $2\sqrt{29}$ cm
(c) $\sqrt{21}$ cm (d) $\sqrt{29}$ cm

[SSC, 2015]

35. In triangle ABC, DE || BC where D is a point on AB and E is a point on AC. DE divides the area of Δ ABC into two equal parts. Then DB : AB is equal to
(a) $(\sqrt{2}-1):\sqrt{2}$ (b) $(\sqrt{2}+1):\sqrt{2}$
(c) $\sqrt{2}:(\sqrt{2}+1)$ (d) $\sqrt{2}:(\sqrt{2}-1)$

[SSC, 2015]

36. If O is the circumcentre of a triangle ABC lying inside the triangle, the ∠OBC + ∠BAC is equal to
(a) 110º (b) 90º
(c) 60º (d) 120º

[SSC, 2015]

37. The interior angle of a regular polygon exceeds its exterior angle by 108º. The number of sides of the polygon is
(a) 14 (b) 12
(c) 16 (d) 10

[SSC, 2015]

38. In ΔABC, ∠BAC = 90° and AD ⊥ BC. If BD = 3 cm and CD = 4 cm, then the length (in cm) of AD is
(a) 6 (b) $2\sqrt{3}$
(c) 3.5 (d) 5

[SSC, 2015]

39. AD is perpendicular to the internal bisector of ∠ABC of ΔABC. DE is drawn through D and parallel to BC to meet AC at E. If the length of AC is 12 cm, then the length of AE (in cm.) is
(a) 6 (b) 8
(c) 3 (d) 4

[SSC, 2015]

40. The centroid of a Δ ABC is G. The area of Δ ABC is 60 cm^2. The area of ΔGBC is
(a) 10 cm^2 (b) 20 cm^2
(c) 40 cm^2 (d) 30 cm^2

[SSC, 2015]

41. Quadrilateral ABCD is circumscribed about a circle. If the lengths of AB, BC, CD are 7 cm, 8.5 cm and 9.2 cm respectively, then the length (in cm) of DA is
(a) 10.7 (b) 16.2
(c) 7.2 (d) 7.7

[SSC, 2015]

42. The measures of two angles of a triangle is in the ratio 4:5. If the sum of these two measures is equal to the measure of the third angle. Find the smallest angle.
(a) 50 (b) 10
(c) 40 (d) 90

[SSC, 2015]

43. If PQRS is a rhombus and ∠SPQ = 50°, then ∠RSQ is:
(a) 45° (b) 55°
(c) 65° (d) 75°

[SSC, 2015]

44. If two supplementary angles differ by 44°, then one of the angle is:
(a) 65° (b) 68°
(c) 72° (d) 102°

[SSC, 2015]

45. If two medians BE and CF of a triangle ABC, intersect each other at G and if BG = CG, ∠BGC = 60°, BC = 8cm, then area of the triangle ABC is:

(a) $48\sqrt{3}$ cm^2 (b) 48 cm^2

(c) $64\sqrt{3}$ cm^2 (d) $96\sqrt{3}$ cm^2

[SSC, 2015]

46. ABC is a cyclic triangle and the bisectors of $\angle$ BAC, $\angle$ ABC and $\angle$ BCA meet the circle at P, Q and R respectively. Then the angle $\angle$ RQP is:

(a) $90^\circ + \frac{C}{2}$ (b) $90^\circ - \frac{A}{2}$

(c) $90^\circ + \frac{B}{2}$ (d) $90^\circ - \frac{B}{2}$

[SSC, 2015]

47. ABC is a triangle and the sides AB, BC and CA are produced to E, F and G respectively. If $\angle$ CBE = $\angle$ ACF = 130°, then the value of $\angle$ GAB is:

(a) 80° (b) 130°

(c) 90° (d) 100°

[SSC, 2015]

48. In a rhombus ABCD, $\angle$ A = 60° and AB = 12cm. Then the diagonal BD is:

(a) 6 cm (b) 12 cm

(c) 10 cm (d) $2\sqrt{3}$ cm

[SSC, 2015]

49. If $x + \frac{1}{x} = 1$, then the value of $\frac{x^2+3x+1}{x^2+7x+1}$ is:

(a) $\frac{3}{7}$ (b) 2

(c) 1 (d) $\frac{1}{2}$

[SSC, 2015]

50. The ratio of each interior angle to each exterior angle of a regular polygon is 3:1. The number of sides of the polygon is:

(a) 7 (b) 8

(c) 9 (d) 6

[SSC, 2015]

51. O is the incentre of ΔPQR and $\angle$QPR = 50°, then the measure of $\angle$QOR is:-

(a) 130° (b) 100°

(c) 125° (d) 115°

[SSC, 2015]

52. O is the circumcentre of ΔABC. If $\angle$BAC = 85° $\angle$BCA = 75°, the $\angle$OAC is equal to:-

(a) 70° (b) 50°

(c) 40° (d) 60°

[SSC, 2015]

53. ABCD is a cyclic quadrilateral. Diagnals AC and BD meets at P. If $\angle$APB = 110° and $\angle$CBD = 30°, then $\angle$ADB measures-

(a) 55° (b) 30°

(c) 80° (d) 70°

[SSC, 2015]

54. The internal bisectors of the $\angle$B and $\angle$C of the $\angle$ABC, interest at O. If $\angle$A = 100°, then the measure of $\angle$BOC is:-

(a) 130° (b) 140°

(c) 110° (d) 120°

[SSC, 2015]

55. Given that: $\Delta ABC \sim \Delta PQR$, If $\frac{\text{area}(\Delta PQR)}{\text{area}(\Delta ABC)} = \frac{256}{441}$ and PR = 12cm. then AC is equal to?

(a) 16 cm (b) 15.75 cm

(c) 12 $\sqrt{2}$ cm (d) 15.5 cm

[SSC, 2015]

56. In ΔABC, D and E are two mid points of sides AB and AC respectively. If $\angle$BAC = 40° and $\angle$ABC = 65° then $\angle$CED is :-

(a) 75° (b) 125°

(c) 130° (d) 105°

[SSC, 2015]

57. Let C1 and C2 be the inscribed and circumscribed circles of a triangle with sides 3cm, 4cm, and 5cm then area of C1 is area of C2

(a) $\frac{9}{16}$ (b) $\frac{9}{25}$

(c) $\frac{4}{25}$ (d) $\frac{16}{25}$

[SSC, 2015]

58. G is the centrioid of ΔABC. The medians AD and BE intersect at right angles. If the lengths of AD and BE are 9 cm and 12 cm respectively; then the length of AB (in cm) is?

(a) 10 (b) 10.5

(c) 9.5 (d) 11

[SSC, 2015]

59. If the three angles of a triangle are: $(x+15^\circ), \left(\frac{6x}{5}+6^\circ\right)$ and $\left(\frac{2x}{3}+30^\circ\right)$ then the triangle is:

(a) Scalene (b) isocsceles

(c) right angled (d) equilateral

[SSC, 2015]

60. If the number of vertices, edges and faces of a rectangular parallelepiped are denoted by v, e and f respectively, the value of (v – e + f) is
(a) 4 (b) 2
(c) 1 (d) 0
[SSC, 2015]

61. If the measure of three angles of a triangle are in the ratio 2 : 3 : 5, then the triangle is:
(a) equilateral (b) isocsceles
(b) Obtuse angled (d) right angled
[SSC, 2015]

62. If the sum and difference of two angles are $\frac{22}{9}$ radian and 36° respectively, then the value of smaller angle in degree taking the value of Π as $\frac{22}{7}$ is:
(a) 60° (b) 48°
(c) 52° (d) 56°
[SSC, 2015]

63. Internal bisectors of ∠Q and ∠R of ΔPQR intersect at O. If ∠ROQ = 96° the the value of ∠RPQ is:
(a) 12° (b) 6°
(c) 36° (d) 24°
[SSC, 2015]

64. If D, E and F are the mid points of BC, CA and AB respectively of the ΔABC then the ratio of area of the parallelogram DEFB and area of the trapezium CAFD is:
(a) 1 : 3 (b) 1 : 2
(c) 3 : 4 (d) 2 : 3
[SSC, 2015]

65. a and b are two sides adjacent to the right angle of a right-angle triangle and p is the perpendicular drawn to the hypotenuse from the opposite vertex. Then p^2 is equal to:
(a) a^2+b^2 (b) $\frac{1}{a^2}+\frac{1}{b^2}$
(c) $\frac{a^2b^2}{a^2+b^2}$ (d) a^2-b^2
[SSC, 2014]

66. Two chords of lengths a metres and b metres subtend angles 60°and 90° at the centre of the circle, respectively. Which of the following is true?
(a) $b=\sqrt{2}a$ (b) $a=\sqrt{2}b$
(c) $a = 2b$ (c) $b = 2a$
[SSC, 2014]

67. In a $\Delta ABC, \angle A+\frac{1}{2}\angle B+\angle C=140°$, then ∠B is:
(a) 50° (b) 80°
(c) 40° (d) 60°
[SSC, 2014]

68. The radius of a circle is 6 cm. The distance of a point lying outside the circle from the centre is 10 cm. The length of the tangent drawn from the outside point to the circle is:
(a) 5 cm (b) 6 cm
(c) 7 cm (d) 8 cm
[SSC, 2014]

69. If $ABCD$ is a cyclic quadrilateral in which $\angle A=4x°,\ \angle B=7x°,\ \angle C=5y°,\ \angle D=y°$, then $x:y$ is:
(a) 3:4 (b) 4:3
(c) 5:4 (d) 4:5
[SSC, 2014]

70. G is the centroid of the equilateral ΔABC. If AB = 10 cm, then length of AG is:
(a) $\frac{5\sqrt{3}}{3}$ cm (b) $\frac{10\sqrt{3}}{3}$ cm
(c) $5\sqrt{3}$ cm (d) $10\sqrt{3}$ cm
[SSC, 2014]

71. Two chords AB and CD of a circle with centre O, intersect each other at P. If $\angle AOD=100°$ and $\angle BOC=70°$, then the value of $\angle APC$ is
(a) 80° (b) 75°
(c) 85° (d) 95°
[SSC, 2014]

72. $ABCD$ is a cyclic quadrilateral and AD is a diameter. If $\angle DAC=55°$, then value of $\angle ABC$ is:
(a) 55° (b) 35°
(c) 145° (d) 125°
[SSC, 2014]

73. In ΔABC a straight line parallel to BC intersects AB and AC at D and E, respectively. If $AB = 2AD$, then $DE:BC$ is:
(a) 2:3 (b) 2:1
(c) 1:2 (d) 1:3
[SSC, 2014]

74. ABC is an isosceles triangle such that $AB = AC$ and AD is the median to the base BC with $\angle ABC=35°$. Then $\angle BAD$ is:
(a) 35° (b) 55°
(c) 70° (d) 110°
[SSC, 2014]

75. A man goes 24 m due west and then 10 m due north. Now, the distance of him from the starting point is:
(a) 17 m (b) 26 m
(c) 28 m (d) 34 m
[SSC, 2014]

76. ABC is a right angled triangle, B being the right angle. Mid-points of BC and AC are respectively B' and A'. Area of $\Delta A'B'C$ is

(a) $\frac{1}{2} \times$ area of ΔABC

(b) $\frac{2}{3} \times$ area of ΔABC

(c) $\frac{1}{4} \times$ area of ΔABC

(d) $\frac{1}{8} \times$ area of ΔABC

[SSC, 2014]

77. 'O' is the centre of the circle, AB is a chord of the circle, OM $\perp$ AB. If AB = 20 cm, OM = $2\sqrt{11}$ cm, then radius of the circle is

(a) 15 cm (b) 12 cm
(c) 10 cm (d) 11 cm

[SSC, 2014]

78. If the angles of a triangle ABC are in the ratio 2 : 3 : 1, then the angles $\angle A, \angle B, \angle C$ is

(a) $\angle A = 60°, \angle B = 90°, \angle C = 30°$

(b) $\angle A = 40°, \angle B = 120°, \angle C = 20°$

(c) $\angle A = 20°, \angle B = 60°, \angle C = 60°$

(d) $\angle A = 45°, \angle B = 90°, \angle C = 45°$

[SSC, 2014]

79. In ΔABC, $\angle ABC = 70°$, $\angle BCA = 40°$. O is the point of intersection of the perpendicular bisectors of the sides, then the angle $\angle BOC$ is

(a) 100° (b) 120°
(c) 130° (d) 140°

[SSC, 2014]

80. If the measures of the sides of triangle are $(x^2 - 1)$, $(x^2 + 1)$ and $2x$ cm, then the triangle would be

(a) equilateral (b) acute-angled
(c) isosceles (d) right-angled

[SSC, 2014]

81. If angle bisector of a triangle bisect the opposite side, then what type of triangle is?

(a) Right angled (b) Scalene
(c) Similar (d) Isosceles

[SSC, 2014]

82. If each angle of a triangle is less than the sum of the other two, then the triangle is

(a) obtuse angled (b) right angled
(c) acute angled (d) equilateral

[SSC, 2014]

83. A, B, C are three points on the circumference of a circle and if $\overline{AB} = \overline{AC} = 5\sqrt{2}$ cm and $\angle BAC = 90°$, find the radius.

(a) 10 cm (b) 5 cm
(c) 20 cm (d) 15 cm

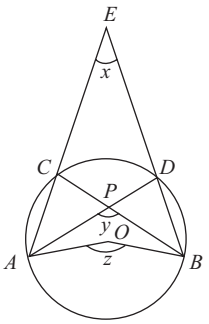
[SSC, 2014]

84. In ΔPQR, S and T are points on sides PR and PQ respectively such that $\angle PQR = \angle PST$. If PT = 5 cm, PS = 3 cm and TQ = 3 cm, then length of SR is

(a) 5 cm (b) 6 cm
(c) $\frac{31}{3}$ cm (d) $\frac{41}{3}$cm

[SSC, 2014]

85. The length of a tangent from an external point to a circle is $5\sqrt{3}$ unit. If radius of the circle is 5 units, then the distance of the point from the circle is



(a) 5 units (b) 15 units
(c) –5 units (d) –15 units

[SSC, 2014]

86. In ΔABC, $\angle C$ is an obtuse angle. The bisectors of the exterior angles at A and B meet BC and AC produced at D and E respectively. If AB = AD = BE, then $\angle ACB =$

(a) 105° (b) 108°
(c) 110° (d) 135°

[SSC, 2014]

87. ABC is an equilateral triangle and O is its circumcentre, then the $\angle AOC$ is

(a) 100° (b) 110°
(c) 120° (d) 130°

[SSC, 2014]

88. ABCD is a cyclic quadrilateral. The side AB is extended to E in such a way that BE = BC. If $\angle ADC = 70°$, $\angle BAD = 95°$, then $\angle DCE =$ is

(a) 140° (b) $120°$
(c) 165° (d) 110°

[SSC, 2014]

89. If the sum of interior angles of a regular polygon is equal to two times the sum of exterior angles of that polygon, then the number of sides of that polygon is

(a) 5 (b) 6
(c) 7 (d) 8

[SSC, 2014]

90. Two circles are of radii 7 cm and 2 cm, their centres being 13 cm apart. Then the length of direct common tangent to the circles between the points of contact is

(a) 12 cm (b) 15 cm
(c) 10 cm (d) 5 cm

[SSC, 2014]

91. In two similar triangles ABC and MNP, if AB = 2·25 cm MP = 4·5 cm and PN = 7·5 cm and m ∠ABC = m ∠MPN, then the length of side $\overrightarrow{BC}$, in cm, is

(a) 3·5 (b) 4·5
(c) 3·75 (d) 4·75

[SSC, 2014]

92. Given an equilateral ΔABC, D, E, F are the mid-points of AB, BC and AC respectively. Then the quadrilateral BEFD is exactly a

(a) trapezium (b) rhombus
(c) square (d) rectangle

[SSC, 2014]

93. If ABCD is a cyclic parallelogram, then the ∠A is

(a) 90° (b) 100°
(c) 60° (d) 80°

[SSC, 2014]

94. AC is a chord of circle whose centre is at O. If B is any point on the arc AC and ∠OCA = 20°, then the magnitude of ∠ABC is

(a) 140° (b) 110°
(c) 100° (d) 40°

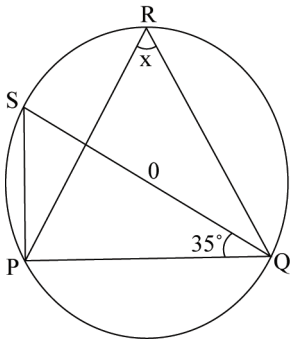
[SSC, 2014]

95. The co-ordinates of the vertices of a right-angled triangle are P(3, 4), Q(7, 4) and R(3, 8), the right-angle being at P. The co-ordinates of the orthocenter of Δ PQR are

(a) (3, 8) (b) 5,6)
(c) (3, 4) (d) (7, 4)

[SSC, 2014]

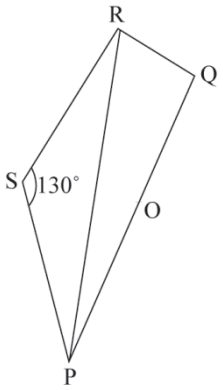
96. In the given figure, O is the centre. Then x is equal to



(a) 65° (b) 60°
(c) 45° (d) 55°

[SSC, 2014]

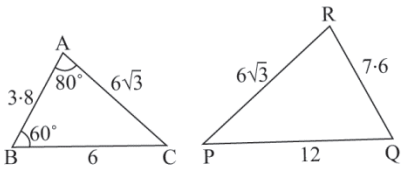
97. In the given figure, POQ is a diameter and PQRS is a cyclic quadrilateral. If ∠PSR = 130°, then the value of ∠RPQ is



(a) 35° (b) 30°
(c) 40° (d) 45°

[SSC, 2014]

98. In the given figures, the lengths of the sides of ΔABC and Δ PQR are given and they are given in same units. Also ∠A and ∠B are given. The value of ∠P is



(a) 40° (b) 42°
(c) 36° (d) 38°

[SSC, 2014]

99. In a $\Delta ABC, \angle A + \angle B = 180°$. Find the value of ∠ A.

(a) 30° (b) 34°
(c) 36° (d) 40°

[SSC, 2014]

100. In ΔABC, if AD $\perp$ BC, then $AB^2 + CD^2$ is equal to
(a) $2BD^2$ (b) $2AC^2$
(c) $BD^2 + AC^2$ (d) None of these

[SSC, 2014]

101. ABC is an equilateral triangle and CD is the internal bisector of $\angle C$. If DC is produced to E such that AC = CE, then $\angle CAE$ is equal to
(a) 30° (b) 15°
(c) 45° (d) 75°

[SSC, 2014]

102. $\angle ACB$ is an angle in the semicircle of diameter AB = 5 cm and AC : BC = 3 : 4. The area of the triangle ABC is
(a) 12 sq. cm (b) 6 sq. cm
(c) $6\sqrt{2}$ sq. cm (d) 4 sq. cm

[SSC, 2014]

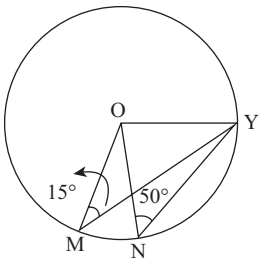
103. A, B and C are three points on a circle such that the angles subtended by the chords AB and AC at the centre O are 90° and 110° respectively. Further suppose that the center 'O' lies in the interior of $\angle$ BAC. Then $\angle BAC$ is
(a) 160° (b) 20°
(c) 40° (d) 80°

[SSC, 2014]

104. If the lengths of the sides AB, BC and CA of a triangle ABC are 10 cm, 8 cm and 6 cm respectively and if M is the mid-point of BC and MN ||AB to cut AC at N, then the area of the trapezium ABMN is equal to
(a) 12 sq. cm (b) 16 sq. cm
(c) 18 sq. cm (d) 20 sq. cm

[SSC, 2014]

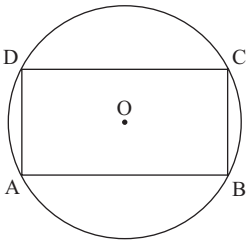
105. In the given figure, $\angle ONY = 50°$ and $\angle OMY = 15°$. Then the value of the $\angle MON$ is



(a) 20° (b) 70°
(c) 30° (d) 40°

[SSC, 2014]

106. In a cyclic quadrilateral
$\angle A + \angle C = \angle B + \angle D = ?$



(a) 90° (b) 180°
(c) 270° (d) 360°

[SSC, 2014]

107. $\angle A$, $\angle B$, $\angle C$ are three angles of a triangle. If $\angle A - \angle B = 15°$, $\angle B - \angle C = 30°$, then $\angle A$, $\angle B$ and $\angle C$ are:
(a) 80°, 60°, 40° (b) 70°, 50°, 60°
(c) 80°, 65°, 35° (d) 80°, 55°, 45°

[SSC, 2013]

108. If ABC is an equilateral triangle and D is a point on BC such that $AD \perp BC$, then:
(a) $AB:BD = 1:1$ (b) $AB:BD = 1:2$
(c) $AB:BD = 2:1$ (d) $AB:BD = 3:2$

[SSC, 2013]

109. ΔABC is an isosceles triangle and $\overline{AB} = \overline{AC} = 2a$ units, $\overline{BC} = a$ units. Draw $\overline{AD} \perp \overline{BC}$ and find the length of $\overline{AD}$.
(a) $\sqrt{15}a$ units (b) $\sqrt{\frac{15}{2}}a$ units
(c) $\sqrt{17}a$ units (d) $\sqrt{\frac{17}{2}}a$ units

[SSC, 2013]

110. All sides of a quadrilateral $ABCD$ touch a circle. If AB = 6 cm, BC = 7.5 cm, CD = 3 cm, then DA is:
(a) 3.5 cm (b) 4.5 cm
(c) 2.5 cm (d) 1.5 cm

[SSC, 2013]

111. In a right-angled triangle, the product of two sides is equal to half of the square of the third side, i.e., hypotenuse. One of the acute angles must be:
(a) 60° (b) 30°
(c) 45° (d) 15°

[SSC, 2013]

112. If two concentric circles are of radii 5 cm and 3 cm, then the length of the chord of the larger circle which touches the smaller circle is:

(a) 6 cm (b) 7 cm
(c) 10 cm (d) 8 cm

[SSC, 2013]

113. Inside a square $ABCD$, ΔBEC is an equilateral triangle. If CE and BD intersect at O, then $\angle BOC$ is equal to:
(a) 60° (b) 75°
(c) 90° (d) 120°

[SSC, 2013]

114. A point D is taken from the side BC of a right-angled triangle ABC, where AB is hypotenuse. Then:
(a) $AB^2 + CD^2 = BC^2 + AD^2$
(b) $CD^2 + BD^2 = 2AD^2$
(c) $AB^2 + AC^2 = 2AD^2$
(d) $AB^2 = AD^2 + BD^2$

[SSC, 2013]

115. Let C be a point on a straight line AB. Circles are drawn with diameters AC and AB. Let P be any point on the circumference of the circle with diameter AB. If AP meets the other circle at Q, then:
(a) $QC \parallel PB$
(b) QC is never parallel to PB
(c) $QC = \frac{1}{2}PB$
(d) $QC \parallel PB$ and $QC = \frac{1}{2}PB$

[SSC, 2013]

116. An isosceles triangle ABC is right-angled at B. D is a point inside the triangle ABC. P and Q are the feet of the perpendiculars drawn from D on the sides AB and AC, respectively of ΔABC. If $AP = a$ cm, $AQ = b$ cm and $\angle BAD = 15°$, then find sin 75°.
(a) $\frac{2b}{\sqrt{3}a}$ (b) $\frac{a}{2b}$
(c) $\frac{\sqrt{3}a}{2b}$ (d) $\frac{2a}{\sqrt{3}b}$

[SSC, 2013]

117. Each interior angle of a regular octagon in radians is:
(a) $\frac{\pi}{4}$ (b) $\frac{3\pi}{4}$
(c) $\frac{2\pi}{3}$ (d) $\frac{1}{3}\neq$

[SSC, 2013]

118. D and E are two points on the sides AC and BC, respectively of ΔABC such that DE = 18 cm, CE = 5 cm and $\angle DEC = 90°$. If tan ($\angle ABC$) = 3.6, then find AC:CD.
(a) BC:2CE (b) 2CE:BC
(c) 2CD:CE (d) CE:2BC

[SSC, 2013]

119. D is a point on the side BC of a triangle ABC such that $AD \perp BC$. E is a point on AD for which AE:ED = 5:1. If $\angle BAD = 30°$ and tan ($\angle ACB$) = 6 tan ($\angle DBE$), then find $\angle ACB$.
(a) 30° (b) 45°
(c) 60° (d) 15°

[SSC, 2013]

120. If the internal bisectors of $\angle ABC$ and $\angle ACB$ of ΔABC meet at O and also $\angle BAC = 80°$, then $\angle BOC$ is equal to:
(a) 50° (b) 160°
(c) 40° (d) 130°

[SSC Assistant Grade III, 2013]

121. O is the incenter of ΔABC. If $\angle BOC$ = 116°, then $\angle BAC$ is:
(a) 42° (b) 62°
(c) 58° (d) 52°

[SSC Assistant Grade III, 2013]

122. Inside a triangle ABC, a straight line parallel to BC intersects AB and AC at the points P and Q, respectively. If $AB = 3PB$, then PQ:BC is:
(a) 1:3 (b) 3:4
(c) 1:2 (d) 2:3

[SSC Assistant Grade III, 2013]

123. O is the circumcentre of ΔABC. If $\angle BAC = 85°$, $\angle BCA = 75°$, then $\angle OAC$ is equal to:
(a) 70° (b) 60°
(c) 80° (d) 100°

[SSC Assistant Grade III, 2012]

124. The distance between the centres of the two circles with radii 4 cm and 9 cm is 13 cm. The length of the direct common tangent (between two points of contact) is:
(a) 13 cm (b) $\sqrt{153}$ cm
(c) 12 cm (d) 18 cm

[SSC Assistant Grade III, 2012]

125. The external bisector of $\angle ABC$ of ΔABC intersects the straight line through A and parallel to BC at the point D. If $\angle ABC$ = 50°, then measure of $\angle ADB$ is:
(a) 65° (b) 55°
(c) 40° (d) 20°

[SSC Assistant Grade III, 2012]

126. AB is a diameter of a circle with centre at O. DC is a chord of it such that $DC \parallel AB$. If $\angle BAC$ = 20°, then $\angle ADC$ is equal to:

(a) 120° (b) 110°
(c) 115° (d) 100°

[SSC Assistant Grade III, 2012]

127. The tangents drawn at P and Q on the circumference of a circle intersect at A. If $\angle PAQ = 68°$, then the measure of the $\angle APQ$ is:
(a) 56° (b) 68°
(c) 28° (d) 34°

[SSC Assistant Grade III, 2012]

128. If the incentre of an equilateral triangle lies inside the triangle and its radius is 3 cm, then the side of the equilateral triangle is:
(a) $9\sqrt{3}$ cm (b) $6\sqrt{3}$ cm
(c) $3\sqrt{3}$ cm (d) 6 cm

[SSC, 2012]

129. Suppose ΔABC be a right-angled triangle where $\angle A = 90°$ and $AD \perp BC$. If Area $(\Delta ABC) = 40$ cm^2, Area $(\Delta ACD) = 10$ cm^2 and $AC = 9$ cm, then the length of BC is:
(a) 12 cm (b) 18 cm
(c) 4 cm (d) 6 cm

[SSC, 2012]

130. Two circles touch each other externally at P. AB is a direct common tangent to the two circles, A and B are points of contact and $\angle PAB = 35°$. Then $\angle ABP$ is:
(a) 35° (b) 55°
(c) 65° (d) 75°

[SSC, 2012]

131. The length of the common chord of two intersecting circles is 24 cm. If the diameters of the circles are 30 cm and 26 cm, then the distance between the centres in cm is:
(a) 13 (b) 14
(c) 15 (d) 16

[SSC, 2012]

132. In ΔABC, D and E are points on AB and AC respectively such the $DE \parallel BC$ and DE divides the ΔABC into two parts of equal areas. Then ratio $AD:BD$ is:
(a) 1:1 (b) $1:\sqrt{2}-1$
(c) $1:\sqrt{2}$ (d) $1:\sqrt{2}+1$

[SSC, 2012]

133. X and Y are centres of circles of radii 9 cm and 2 cm respectively, $XY = 17$ cm. Z is the centre of a circle of radius r cm which touches the above circles externally. Given that $\angle XZY = 90°$, the value of r is:
(a) 13 cm (b) 6 cm
(c) 9 cm (d) 8 cm

[SSC, 2012]

134. I is the incentre of a triangle ABC. If $\angle ABC = 65°$ and $\angle ACB = 55°$, then the value of $\angle BIC$ is:
(a) 130° (b) 120°
(c) 140° (d) 110°

[SSC, 2012]

135. If the radii of two circles be 6 cm and 3 cm and the length of the transverse common tangent be 8 cm, then the distance between the two centres is:
(a) $\sqrt{145}$ cm (b) $\sqrt{140}$ cm
(c) $\sqrt{150}$ cm (d) $\sqrt{135}$ cm

[SSC, 2012]

136. The ratio between the numbers of sides of two regular polygons is 1:2 and the ratio between their interior angles is 2:3. The number of sides of these polygons is respectively:
(a) 6 (b) 8
(c) 4 (d) 5

[SSC, 2012]

137. The angles of a triangle are in Arithmetic Progression. The ratio of the least angle in degrees to the number of radians in the greatest angle is $60:\pi$ .The angles in degrees are:
(a) 30°, 60°, 90° (b) 35°, 55°, 90°
(c) 40°, 50°, 90° (d) 40°, 55°, 85°

[SSC, 2012]

138. The ratio between the adjacent angles of a parallelogram are 2:3. The smallest angle of a quadrilateral is equal to the half of the smallest angle of a parallelogram. The highest angle of a quadrilateral is 4 times greater than its smallest angle. What is the sum of the highest angle of a quadrilateral and the smallest angles of a parallelogram?
(a) 252° (b) 226°
(c) 144° (d) 180°

[Union Bank of India PO, 2011]

139. One of the angles of a triangle is two-thirds angle of sum of adjacent angles of parallelogram. Remaining angles of the triangle are in ratio 5:7 respectively. What is the value of second largest angle of the triangle?
(a) 25°
(b) 40°
(c) 35°
(d) Cannot be determined

[Corporation Bank PO, 2011]

140. Smallest angle of a triangle is equal to two-thirds the smallest angle of a quadrilateral. The ratio between the angles of the quadrilateral is 3:4:5:6. Largest angle of the triangle is twice its smallest angle. What is

the sum of second largest angle of the triangle and largest angle of the quadrilateral?

(a) 160º (b) 180º
(c) 190º (d) 170º

[Bank of Baroda PO Examination, 2011]

141. A, B, C are three points on a circle. The tangent at A meets BC produced at T, $\angle BTA = 40°$, $\angle CAT = 44°$. The angle subtended by BC at the centre of the circle is:

(a) 84° (b) 92°
(c) 96° (d) 104°

[SSC, 2011]

142. If the length of a chord of a circle at a distance of 12 cm from the centre is 10 cm, then the diameter of the circle is:

(a) 13 cm (b) 15 cm
(c) 26 cm (d) 30 cm

[SSC, 2011]

143. In ΔABC, P and Q are the middle points of the sides AB and AC respectively. R is a point on the segment PQ such that $PR:RQ = 1:2$. If $PR = 2$ cm, then $BC =$

(a) 4 cm (b) 2 cm
(c) 12 cm (d) 6 cm

[SSC, 2011]

144. If O is the circumcentre of ΔABC and $\angle OBC = 35°$, then the $\angle BAC$ is equal to:

(a) 55° (b) 110°
(c) 70° (d) 35°

[SSC, 2011]

145. If I is the incentre of ΔABC and $\angle BIC = 135°$, then ΔABC is:

(a) acute angled (b) equilateral
(c) right angled (d) obtuse angled

[SSC, 2011]

146. The ratio of the adjacent angles of a parallelogram is 7:8. Also, the ratio of the angles of quadrilateral is 5:6:7:12. What is the sum of the smaller angle of the parallelogram and the second largest angle of the quadrilateral?

(a) 168º (b) 228º
(c) 156º (d) 224º
(e) None of these

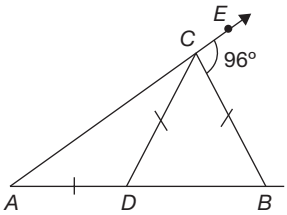
[IOB PO, 2011]

147. One of the angles of a triangle is two-thirds of the sum of the adjacent angles of parallelogram. Remaining angles of the triangle are in the ratio 5:7. What is the value of the second largest angle of the triangle?

(a) 25° (b) 40°
(c) 35° (d) Cannot be determined
(e) None of these

[Corporation Bank PO, 2011]

148. In the figure (not drawn to scale) give below, if AD = DC = BC and $\angle BCE = 96°$, then $\angle DBC$ is:



(a) 32° (b) 84°
(c) 64° (d) 96°

[SSC CGL Tier-I (CBE), 2011]

149. The angles of a quadrilateral are in the ratio of 2:4:7:5. The smallest angle of the quadrilateral is equal to the smallest angle of a triangle. One of the angles of the triangle is twice the smallest angle of the triangle. What is the second largest angle of the triangle?

(a) 80° (b) 60°
(c) 120° (d) Cannot be determined

[CBI (PO), 2010]

150. The ratio between the angles of a quadrilateral is 3:4:6:5. Two-thirds the largest angle of the quadrilateral is equal to the smaller angle of a parallelogram? What is the value of adjacent angle of the parallelogram?

(a) 120° (b) 110°
(c) 100° (d) 130°

[OBC PO, 2010]

151. The largest and the second largest angles of a triangle are in the ratio of 3:2, respectively. The smallest angle is 20% of the sum of the largest and the second largest angles. What is the sum of the smallest and the second largest angles?

(a) 80º (b) 60º
(c) 100º (d) 90º

[Bank of Baroda PO, 2010]

152. What is the length of the radius of the circum-circle of the equilateral triangle, the length of whose side is $6\sqrt{3}$ cm?

(a) $6\sqrt{3}$ cm (b) 6 cm
(c) 5.4 cm (d) $3\sqrt{6}$ cm

[SSC, 2010]

153. The two sides of a right triangle containing the right angle measure 3cm and 4cm. The radius of the incircle of the triangle is:

(a) 3.5 cm (b) 1.75 cm
(c) 1 cm (d) 0.875 cm

154. In a circular lawn, there is a 16 m long path in the form of a chord. If the path is 6 m away from the centre of the lawn, then find the radius of the circular lawn.

(a) 16 m (b) 6 m
(c) 10 m (d) 8 m

155. In a triangle ABC, the length of the sides AB, AC and BC are 3, 5 and 6 cm respectively. If a point D on BC is drawn such that the line AD bisects the angle A internally, then what is the length of BD?

(a) 2 cm (b) 2.25 cm
(c) 2.5 cm (d) 3 cm

156. In a triangle ABC, $\angle A = x°$, $\angle B = y°$ and $\angle C = (y + 20)°$. If $4x - y = 10$, then the triangle is:

(a) Right-angle (b) Obtuse-angled
(c) Equilateral (d) None of these

157. If the sides of a right triangle are x, $x + 1$ and $x - 1$, then the hypotenuse:

(a) 5 (b) 4
(c) 1 (d) 0

158. If P and Q are the mid points of the sides CA and GB respectively of a triangle ABC, right-angled at C. Then the value of $4\ (AQ^2 + BP^2)$ is equal to:

(a) $4\ BC^2$ (b) $5\ AB^2$
(c) $2\ AC^2$ (d) $2\ BC^2$

159. In a quadrilateral $ABCD$, $\angle B = 90°$ and $AD^2 = AB^2 + BC^2 + CD^2$, then $\angle ACD$ is equal to:

(a) 90° (b) 60°
(c) 30° (d) None of these

160. $ABCD$ is a square, F is mid point of AB and E is a point on BC such that BE is one-third of BC. If area of $\Delta FBE = 108\ m^2$, then the length of AC is:

(a) 63 m (b) $36\sqrt{2}$ m
(c) $63\sqrt{2}$ m (d) $72\sqrt{2}$ m

161. We have an angle of $2\frac{1^\circ}{2}$. How big it will look through a glass that magnifies things three times?

(a) $2\frac{1^\circ}{2} \times 4$ (b) $2\frac{1^\circ}{2} \times 3$

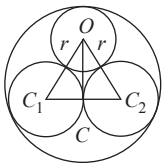
(c) $2\frac{1^\circ}{2} \times 2$ (d) None of these

162. Two circles with radii 'a' and 'b' respectively touch each other externally. Let 'c' be the radius of a circle that touches these two circle as well as a common tagent to the two circles. Then:

(a) $\frac{1}{\sqrt{a}} - \frac{1}{\sqrt{b}} = \frac{1}{\sqrt{c}}$ (b) $\frac{1}{\sqrt{a}} - \frac{1}{\sqrt{b}} = \frac{1}{\sqrt{c}}$

(c) $\frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}} = \frac{1}{\sqrt{c}}$ (d) None of these

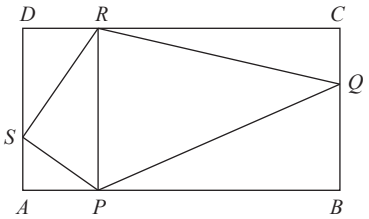
163. Two circles of unit radius touch each other and each of them touches internally a circle of radius two, as shown in the following figure. The radius of the circle which thouches all the three circles:



(a) 5 (b) $\frac{3}{2}$

(c) $\frac{2}{3}$ (d) None of these

164. $ABCD$ is a parallelogram P, Q, R and S are points on sides AB, BC, CD and DA respectively such that $AP = DR$. If the area of the parallelogram $ABCD$ is 16 cm^2, then the area of the quadrilateral $PQRS$ is:

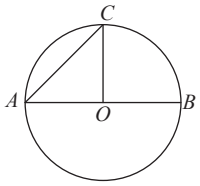


(a) 6 cm^2 (b) 6.4 cm^2
(c) 4 cm^2 (d) 8 cm^2

165. Let ABC be an acute-angled triangle and CD be the altitude through C. If $AB = 8$ and $CD = 6$, then the distance between the mid-points of AD and BC is:

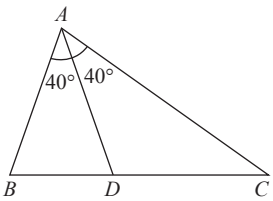
(a) 36 (b) 25
(c) 27 (d) 5

166. In the accompanying figure, AB is one of the diameters of the circle and OC is perpendicular to it through the centre O. If AC is $7\sqrt{2}$ cm, then what is the area of the circle in cm^2?



(a) 24.5 (b) 49
(c) 98 (d) 154

167. In the following figure, If $BC = 8$ cm, $AB = 6$ cm, AC 9 cm, then DC is equal to:



(a) 7 cm (b) 4.8 cm
(c) 7.2 cm (d) 4.5cm

168. With the vertices of a ΔABC as centers, three circles are described each touching the other two externally. If the sides of the triangle are 4, 6 and 8 cm respectively, then the sum of the radii of the three circles equals:

(a) 10 (b) 14
(c) 12 (d) 9

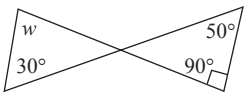
169. The sum of the interior angles of a polygon is 1620°. The number of sides of the polygon are

(a) 9 (b) 11
(c) 15 (d) 12

170. How many sides a regular polygon has with its interior angle eight times its exterior angle?

(a) 16 (b) 24
(c) 18 (d) 20

171. In the figure given below, what is the value of w?



(a) 100 (b) 110
(c) 120 (d) 130

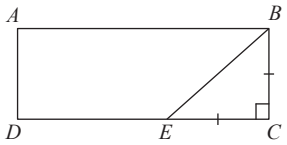
Notes

The diagram is not drawn to scale.

172. The radius of the circumcircle of an equilateral triangle of side 12 cm is:

(a) $\left(\frac{4}{3}\right)\sqrt{3}$ (b) $4\sqrt{2}$
(c) $4\sqrt{3}$ (d) 4
(e) None of above

173. In the diagram below, $ABCD$ is a rectangle. The area of isosceles right triangle BCE is 14, and $DE = 3EC$. What is the area of $ABCD$?



(a) 112 (b) 56
(c) 84 (d) $3\sqrt{28}$

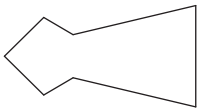
174. Find the distance of aperpendicular from the centre of a circle to the chord if the diameter of the circle is 30 cm and its chord is 24cm.

(a) 6 cm (b) 7 cm
(c) 9 cm (d) 10 cm

175. The medians AD, BE and CF of a triangle ABC intersect in G. Which of the following is true for any ΔABC?

(a) $GB + GC = 2GA$
(b) $GB + GC < GA$
(c) $GB + GC > GA$
(d) $GB + GC = GA$
(e) None of these

176. Find the sum of the degree measure of the internal angles in the polygon shown below.



(a) 600° (b) 720°
(c) 900° (d) 1080°
(e) 750°

177. A semi-circle is drawn with AB as its diameter. From C, a point on AB, a line perpendicular to AB is draw meeting the circumference of the semi-circle at D. Given that AC = 2 cm and CD = 6 cm the area of the semicircle (in cm^2) will be:

(a) 32π cm^2 (b) 50π cm^2
(c) 40.5π cm^2 (d) 81π cm^2
(e) undeterminable

178. An equilateral triangle BPC is drawn inside a square ABCD. What is the value of the angle APD in degrees:

(a) 75 (b) 90
(c) 12 (d) 135
(e) 150

179. The heights of two similar right-angled triangles Δ LMN and Δ OPQ are 48 cm and 36 cm. If OP = 12 cm, then LM is

(a) 12 cm (b) $\frac{10\sqrt{6}}{3}$
(c) 16 cm (d) 20 cm

ANSWER KEYS

EXERCISE–I

1. (c)	**2.** (b)	**3.** (a)	**4.** (d)	**5.** (a)	**6.** (d)	**7.** (a)	**8.** (b)	**9.** (a)	**10.** (c)	**11.** (b)	**12.** (d)
13. (c)	**14.** (b)	**15.** (b)	**16.** (d)	**17.** (a)	**18.** (c)	**19.** (a)	**20.** (a)	**21.** (b)	**22.** (c)	**23.** (c)	**24.** (a)
25. (a)	**26.** (d)	**27.** (b)	**28.** (c)	**29.** (a)	**30.** (c)	**31.** (a)	**32.** (b)	**33.** (c)	**34.** (a)	**35.** (d)	**36.** (b)
37. (d)	**38.** (b)	**39.** (c)	**40.** (d)	**41.** (a)	**42.** (d)	**43.** (b)	**44.** (b)	**45.** (b)	**46.** (c)	**47.** (b)	**48.** (c)
49. (b)	**50.** (a)	**51.** (d)	**52.** (c)	**53.** (a)	**54.** (c)	**55.** (d)	**56.** (d)	**57.** (a)	**58.** (d)	**59.** (a)	**60.** (c)
61. (c)	**62.** (d)	**63.** (a)	**64.** (c)	**65.** (a)	**66.** (a)	**67.** (a)	**68.** (c)	**69.** (b)	**70.** (c)	**71.** (d)	**72.** (d)
73. (a)	**74.** (b)	**75.** (c)	**76.** (b)	**77.** (a)	**78.** (d)	**79.** (c)	**80.** (c)	**81.** (a)	**82.** (d)	**83.** (c)	**84.** (c)
85. (b)	**86.** (d)	**87.** (a)	**88.** (c)	**89.** (c)	**90.** (c)	**91.** (a)	**92.** (b)	**93.** (d)	**94.** (a)	**95.** (d)	**96.** (c)
97. (b)	**98.** (d)	**99.** (a)	**100.** (c)	**101.** (a)	**102.** (d)	**103.** (b)	**104.** (c)				

EXERCISE–2

1. (c)	**2.** (c)	**3.** (b)	**4.** (b)	**5.** (d)	**6.** (a)	**7.** (b)	**8.** (d)	**9.** (c)	**10.** (d)	**11.** (b)	**12.** (d)
13. (b)	**14.** (c)	**15.** (b)	**16.** (c)	**17.** (c)	**18.** (d)	**19.** (b)	**20.** (c)	**21.** (b)	**22.** (c)	**23.** (b)	**24.** (b)
25. (a)	**26.** (c)	**27.** (b)	**28.** (a)	**29.** (d)	**30.** (c)	**31.** (c)	**32.** (c)	**33.** (b)	**34.** (b)	**35.** (a)	**36.** (b)
37. (d)	**38.** (b)	**39.** (a)	**40.** (b)	**41.** (d)	**42.** (c)	**43.** (c)	**44.** (b)	**45.** (a)	**46.** (d)	**47.** (d)	**48.** (b)
49. (d)	**50.** (b)	**51.** (d)	**52.** (a)	**53.** (c)	**54.** (b)	**55.** (b)	**56.** (d)	**57.** (c)	**58.** (a)	**59.** (d)	**60.** (b)
61. (d)	**62.** (c)	**63.** (a)	**64.** (d)	**65.** (c)	**66.** (a)	**67.** (b)	**68.** (d)	**69.** (b)	**70.** (b)	**71.** (d)	**72.** (c)
73. (c)	**74.** (b)	**75.** (b)	**76.** (c)	**77.** (b)	**78.** (a)	**79.** (d)	**80.** (d)	**81.** (d)	**82.** (c)	**83.** (b)	**84.** (c)
85. (a)	**86.** (b)	**87.** (c)	**88.** (a)	**89.** (b)	**90.** (a)	**91.** (c)	**92.** (b)	**93.** (a)	**94.** (b)	**95.** (c)	**96.** (d)
97. (c)	**98.** (a)	**99.** (b)	**100.** (c)	**101.** (d)	**102.** (b)	**103.** (d)	**104.** (c)	**105.** (b)	**106.** (b)	**107.** (c)	**108.** (c)
109. (b)	**110.** (d)	**111.** (c)	**112.** (d)	**113.** (b)	**114.** (a)	**115.** (d)	**116.** (c)	**117.** (b)	**118.** (c)	**119.** (c)	**120.** (d)
121. (d)	**122.** (d)	**123.** (a)	**124.** (c)	**125.** (a)	**126.** (b)	**127.** (a)	**128.** (b)	**129.** (b)	**130.** (b)	**131.** (b)	**132.** (b)
133. (b)	**134.** (b)	**135.** (a)	**136.** (c)	**137.** (a)	**138.** (d)	**139.** (c)	**140.** (b)	**141.** (d)	**142.** (c)	**143.** (c)	**144.** (a)
145. (c)	**146.** (a)	**147.** (c)	**148.** (c)	**149.** (b)	**150.** (c)	**151.** (d)	**152.** (b)	**153.** (c)	**154.** (c)	**155.** (b)	**156.** (a)
157. (a)	**158.** (b)	**159.** (a)	**160.** (b)	**161.** (d)	**162.** (c)	**163.** (c)	**164.** (d)	**165.** (d)	**166.** (d)	**167.** (b)	**168.** (d)
169. (b)	**170.** (c)	**171.** (b)	**172.** (c)	**173.** (a)	**174.** (c)	**175.** (c)	**176.** (c)	**177.** (b)	**178.** (e)	**179.** (c)	

EXPLANATORY ANSWERS

EXERCISE-I

1. (c) Let the measured of the required angle be x degree. Then, its supplement = $180 - x$

$$\text{Now, angle} = \frac{1}{3}\text{ (its supplement)}$$

$$x = \frac{1}{3}(180 - x)$$

$3x + x = 180° \Rightarrow x = 45°$.

2. (b) complement of $30°20' = 90° - (30°20') = 90° - (30° + 20')$

$= (89° - 30°) + (1° - 20')$

$= 59° + 60' - 20'$ [$\because$ $1° = 60'$]

$= 59° + 40' = 59°40'$.

3. (a) Since OP bisects $\angle BOC$,

$\therefore \angle BOC = 2\angle POC$

Again, OQ bisects $\angle AOC$, $\therefore$,

$\angle AOC = 2\angle QOC$

Since ray OC stands on line AB, $\therefore$,

$\angle AOC + \angle BOC = 180° \Rightarrow 2\angle QOC + 2\angle POC = 180°$

$\Rightarrow 2\angle QOC + \angle POC = 180°$

$\Rightarrow \angle QOC + \angle POC = 90°$

$\Rightarrow \angle POQ = 90°$.

The above sum can also be restated as follows; The angle between the bisectors of a linear pair of angles is a right angle.

4. (d) Through O, draw a line l parallel to both AB and CD. Then

$\angle 1 = 45°$ (alt. $\angle$S)

and $\angle 2 = 30°$ (alt. $\angle$S)

$\therefore \angle BOC = \angle 1 + \angle 2 = 45° + 30° = 75°$

So, $X = 360° - \angle BOC = 360° - 75° = 285°$

Hence $X = 285°$.

5. (a) AB || CD and t transversal intersects them at E and F

$\angle BEF + \angle EFD = 180°$ (co-interior angles)

$$\Rightarrow \frac{1}{2}\angle BEF + \frac{1}{2}\angle EFD = 180°$$

$\Rightarrow \angle BEF + \Delta EFD = 90°$

In $\angle EFG$

$\angle EFG + \angle FEG + \angle EGF = 180°$

$\therefore \angle EGF + 90° = 180°$

$\therefore \angle EGF = 90°$.

The above result can be restated as:

If two parallel lines are cut by a transversal, then the bisectors of the interior angles on the same side of the transversal intersect each other at right angles.

6. (d)

$\angle DCK = \angle FDG = 55°$ (corr. $\angle s$)

$\therefore \angle ACE = \angle DCK = 55°$ (vert. opp. $\angle s$)

So, $\angle AEC = 180° - (40° + 55°) = 85°$

$\therefore \angle HAB = \angle AEC = 85°$ (corr. $\angle s$)

Hence, $x = 85°$.

7. (a) Let, the measure of the required angled be $x°$.

Then, measure of its complement = $(90 - x)°$ measure of its supplement = $(180 - x)°$

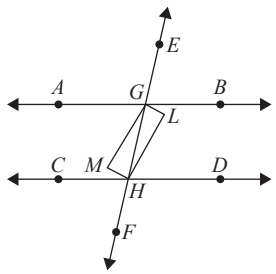
$6(90° - x) = 2(180° - x) - 12°$

$\Rightarrow 540° - 6x = 360° - 2x - 12°$

$\Rightarrow 4x = 192° \Rightarrow x = 48°$.

8. (b) $\angle AGH = \angle DHG$ (alt. int. angles)

$$\frac{1}{2}\angle AGH = \frac{1}{2}\angle DHG \Rightarrow \angle HGM = \angle GHL$$



Thus, lines GM and HL are intersected by a transversal GH at G and H respectively such that pair of alternate angles are equal, i.e., $\therefore \angle HGM = \angle GHL$

$\therefore GM \parallel HL$

Similarly, $GL \parallel HM$

So, $GMHL$ is a $\|_{gm}$.

Since $AB \parallel CD$ and EF is a transversal

$\therefore \angle BGH + \angle DHG = 180°$ [co-interior angles]

$$\therefore \frac{1}{2}\angle BGH + \frac{1}{2}\angle DHG = 90°$$

$\Rightarrow \angle LGH + \angle LHG = 90°$

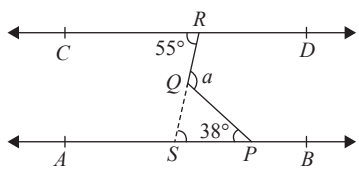
But $\angle LGH + \angle LHG + \angle GLH = 180°$

$\therefore 90° + \angle GLH = 180° \Rightarrow \angle GLH = 90°$

Thus, in $\|_{gm}$ $GMHL$, we have $\angle GLH = 90°$

Hence, $GMHL$ is a rectangle.

9. (a) CD || AB (Given)



CHAPTER 39

Data Interpretation

INTRODUCTION

Data interpretation deals with careful reading, understanding, organising and interpreting the data provided so as to derive meaningful conclusions. It is an important section today in all competitive examinations. The data representation can be broadly classified as tables, graphs and charts.

Types of Data Representation

1. **Tables It** is the systematic and scientific presentation of *numerical data*. It helps the person to make comparisons and draw quick conclusions. Tabular presentation makes complicated information easier to understand. In a table, data is arranged systematically in columns and rows.

Illustration 1: In the Table below are given daily wages of male and female workers (in rupees) of two different factories:

	FACTORY A		*FACTORY B*	
YEAR	*Male*	*Female*	*Male*	*Female*
1997	120	120	240	200
1998	100	108	170	160
1999	80	110	120	130
2000	90	108	140	160
2001	170	130	240	260
2002	120	170	250	270

2. **Line Graph** A line graph depicts the variation of a quantity with respect to the two parameters calibrated on the *x* and *y* axes, respectively. In most of the cases the quantity is measured as a function of time, that is, the variation in the quantity as time changes.

In the line graph:

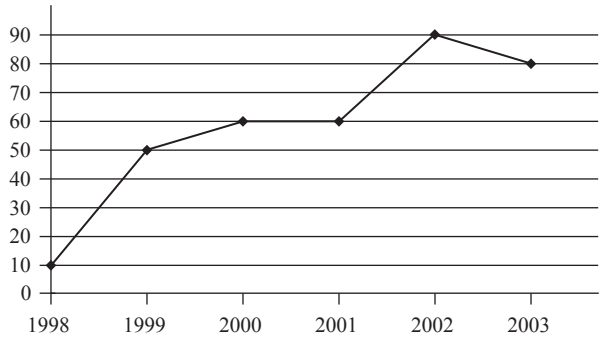
(a) *x*-axis represents the time parameter (may be year or month) and *y*-axis represents any other variable parameter which have different values with respect to time.

(b) the line going up indicates increase in the quantity with time.

(c) the line going down indicates decrease in the quantity with time.

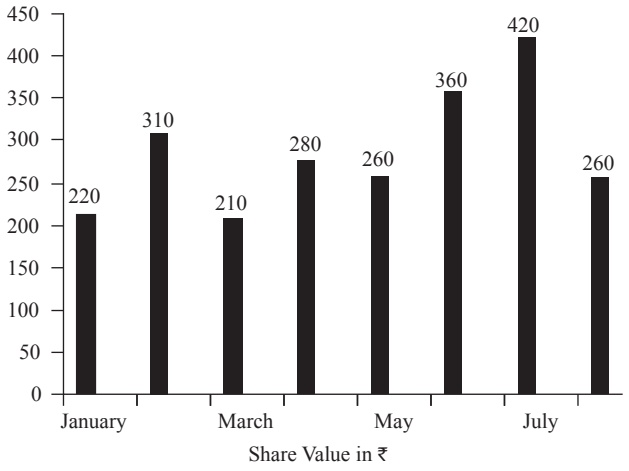
(d) a horizontal line indicates no change in the quantity over that period.

Illustration 2: The following 1 ne graph shows the number of children suffering from liver disorders in a State (in thousands).



3. **Bar Graph** A bar is a thick line whose width is shown merely for attention. A bar graph consists of bars. The height of the bar is a measure of the quantity that it represents. Therefore, quantities can be compared by the height of bars in the graph. Bars may be horizontal or vertical. They may be placed adjacent to each other or may be separated from each other by spaces depending upon the problem.

Illustration 3: The graph given below shows the share price (in rupees) of a company during first eight months of a year.



SHARE VALUE IN RUPEES

Illustration 4: The graph given below shows the yearly production and sale *of* a company in Lakh tons.

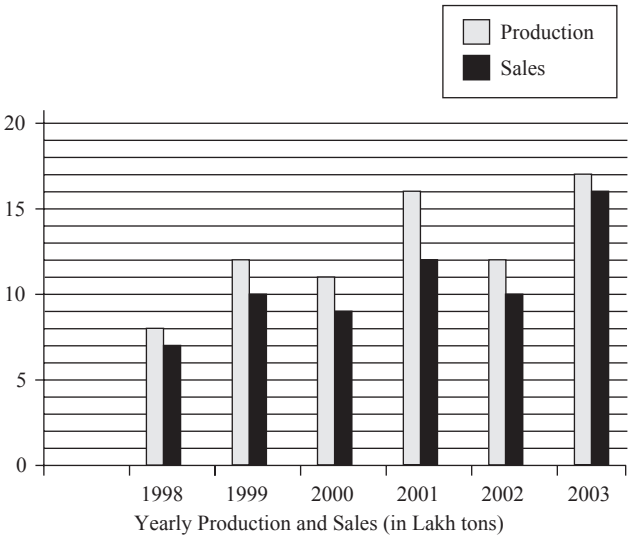
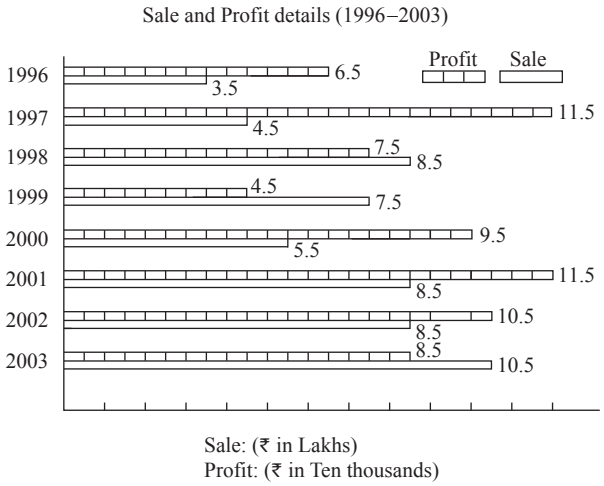


Illustration 5: The graph given below gives information about the sale and profit details of a departmental store during the years from 1996 to 2003.

Sale and Profit details (1996–2003)

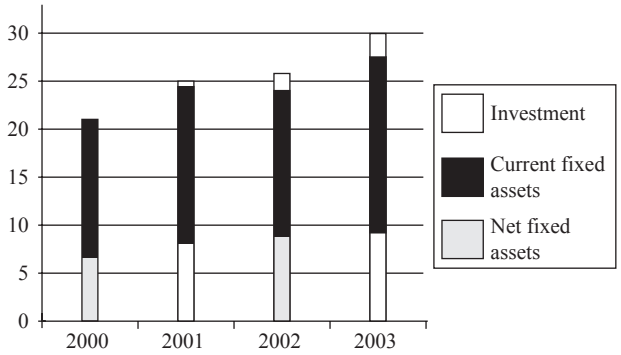


Sale: (₹ in Lakhs,)
Profit: ₹ in Ten thousands

4. **Cumulative Bar Graph** In a cumulative bar graph, the length of the bar is divided proportionately among various quantities represented in the graph. Thus, it may be conveniently used for making comparisons,

Illustration 6:

Crores rupees



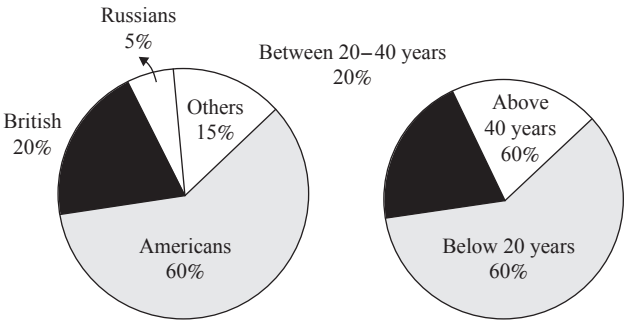
5. **Pie Charts** In a pie chart, the given data is distributed over a circle. Each part of the data makes a certain central angle.

For example, if from all the questions asked in a Bank PO exam, 25% are on data interpretation, then central angle made by this tern

$$= \left(\frac{25}{100} \times 360 \right) = 90°$$

Pie charts are useful for representing percentages or proportions of variou s elements with respect to the total quantity. They also represent shares of various parts of a particular quantity.

Illustration 7: The following pie charts describe the characteristics of foreign tourists visiting India in a particular year.

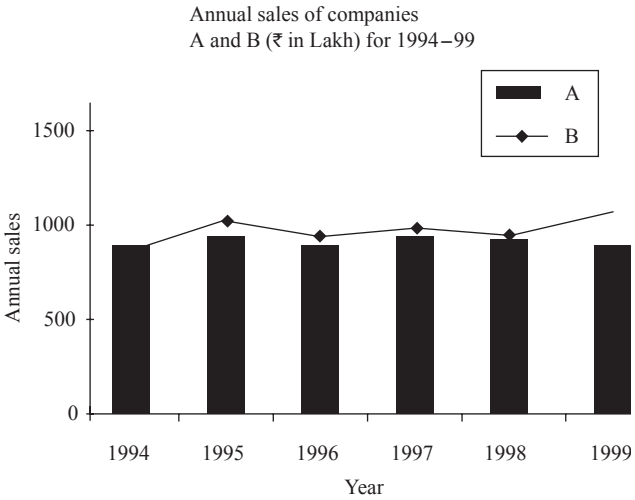


6. **Combination graph** Quite often, data interpretation question is based not on a single graph but on a combination of two or more graphs. It can be line and bar, line and pie, bar and pie, cumulative and pie charts.

Illustration 8: The following graph shows annual sales of companies *A* and *B* (₹ in lakh) for 1994–1999.

Annual sales of companies

A and B (₹ in lakh) for 1994–99



Line and Bar chart

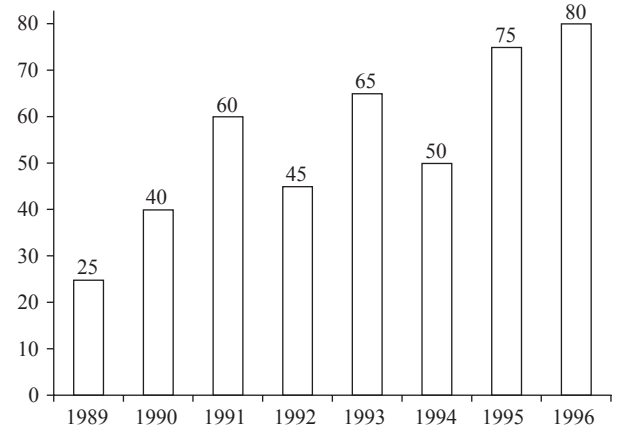
Some Useful Instructions

1. Do not waste time memorising a Table. You can refer to it as many times as you want during the examination.
2. Your answer should be accurate and based on precise data given in the Table.
3. Do not include your own information in answering questions, however, accurate you may be. Stick to the data i resented to you.
4. Be careful about minor details. Students often miss them and give wrong answers.

EXERCISE-I

1. Directions: Study the following graph carefully and answer the questions given below it.

Production of foodgrains by a State over the years (1000 tons)



(i) The average production of 1990 and 1991 was exactly equal to the average production of which of the following pairs of years?

(a) 1991 and 1992 (b) 1992 and 1994
(c) 1993 and 1994 (d) 1994 and 1995
(e) None of these

(ii) What was the difference in the production of foodgrains between 1991 and 1994?

(a) 10000 tons
(b) 15000 tons
(c) 500 tons
(d) 5000 tons
(e) None of these

(iii) In which of the following years was the percentage increase in production from the

previous year the maximum among the given years?

(a) 1991 (b) 1993
(c) 1995 (d) 1990
(e) None of these

(iv) In how many of the given years was the production of foodgrains more than average production of the given years?

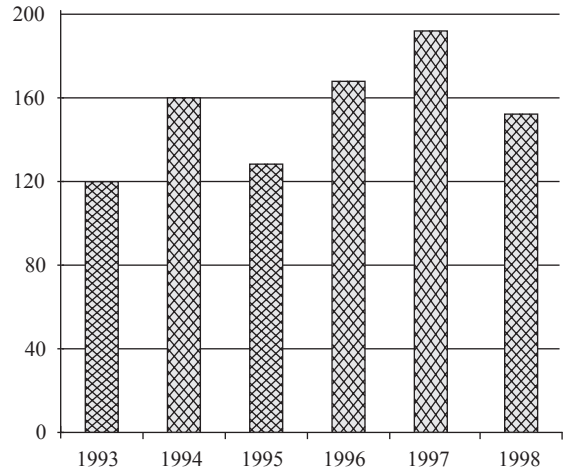
(a) 2 (b) 3
(c) 4 (d) 1
(e) None of these

(v) What was the percentage drop in the production of foodgrains from 1991 to 1992?

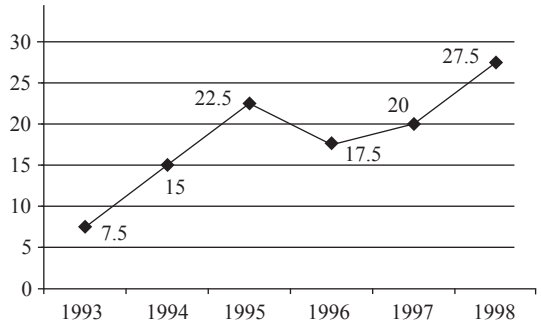
(a) 15 (b) 20
(c) 25 (d) 30
(e) None of these

2. Directions: Study the following graphs carefully and answer the questions given below:

INCOME OF A COMPANY (IN ₹ LAKHS)



PERCENTAGE PROFIT OVER THE YEARS



(i) In which of the following years was the amount of profit the maximum?

(a) 1997 (b) 1994
(c) 1993 (d) 1995
(e) None of these

(ii) **Approximately** what was the average expenditure of the given years?

(a) ₹110 Lakh (b) ₹130 Lakh
(c) ₹120 Lakh (d) ₹140 Lakh
(e) Data inadequate

(iii) In which of the following years was the increase/decrease in per cent profit from the previous year the minimum?

(a) 1994 (b) 1996
(c) 1997 (d) 1995
(e) None of these

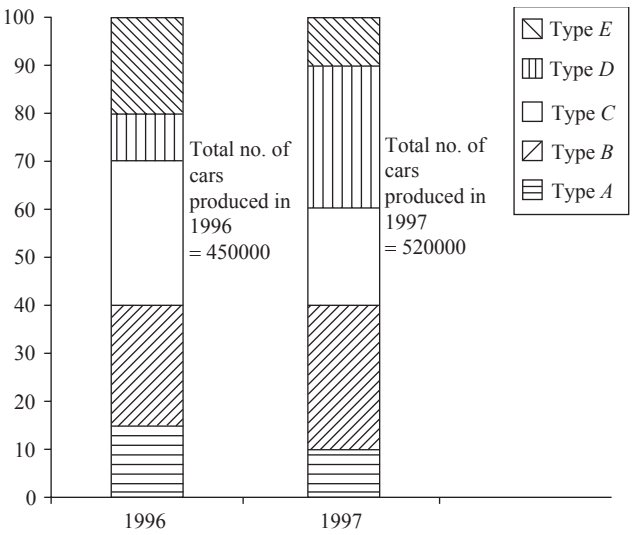
(iv) **Approximately** what was the expenditure in 1994?

(a) ₹120 Lakh (b) ₹160 Lakh
(c) ₹140 Lakh (d) ₹180 Lkh
(e) Data inadequate

(v) If the profit percentage in 1997 was 25, what would have been the expenditure in that year?

(a) ₹130 Lakh (b) ₹148 Lakh
(c) ₹120 Lakh (d) ₹152 Lakh
(e) None of these

3. Directions: Study the following graph carefully and then answer the questions based on it. The percentage of five different types of cars produced by a company during two years is given below.



(i) What was the difference in the production of *C* type cars between 1996 and 1997?

(a) 5000 (b) 7500
(c) 10000 (d) 2500
(e) None of these

(ii) If 85% of *E* type cars produced during 1996 and 1997 are being sold by the company, then how many *E* type cars are left unsold by the company?

(a) 142800 (b) 21825
(c) 29100 (d) 25200
(e) None of these

(iii) If the number of *A* type cars manufactured in 1997 was the same as that of 1996, what would have been its **approximate** percentage share in the total production of 1997?

(a) 11 (b) 13
(c) 15 (d) 9
(e) None of these

(iv) In the case of which of the following types of cars was the percentage increase from 1996 to 1997 the maximum?

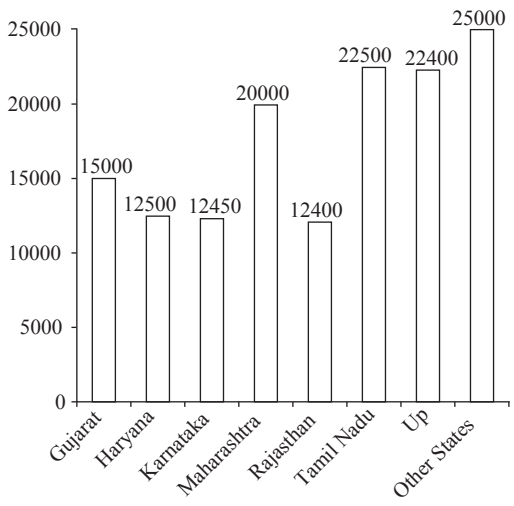
(a) A (b) E
(c) D (d) B
(e) C

(v) If the percentage production of *B* type cars in 1997 was the same as that of 1996, what would have been the number of cars produced in 1997?

(a) 112500 (b) 120000
(c) 130000 (d) Data inadequate
(e) None of these

4. Directions: Study the following graph carefully and answer the questions given below it.

Rose production



(i) Which of the following State(s) contribute(s) less than 10 per cent in the total rose production?

(a) Only Raj as than
(b) Rajasthan, Karnataka
(c) Rajasthan, Karnataka, Haryana
(d) Rajasthan, Karnataka, Haryana and Gujarat
(e) None of these

(ii) By what percentage rose production of other States is more than that of the Maharashtra?

(a) 25 (b) 30
(c) 20 (d) 15
(e) None of these

(iii) What is the **approximate** average production of roses (in thousands) across all the states?

(a) 21 (b) 20
(c) 19 (d) 18
(e) 17

(iv) Approximately what percentage of the total rose production is shared by the other States?

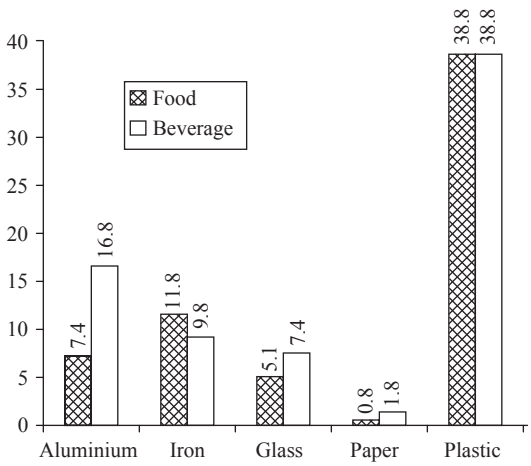
(a) 10 (b) 20
(c) 30 (d) 40
(e) 35

(v) If total percentage contribution of the States having production of roses below twenty thousand is considered, which of the following statements is true?

(a) It is little above 40%
(b) It is exactly 35%
(c) It is below 35%
(d) It is little below 30%
(e) None of these

5. Directions: Study the following graph carefully and answer the questions given below it.

Packaging Materials Used (In tonnes)



(i) What per cent of the total glass packaging material was used for packaging food items?

(a) 40.8 (b) 41.8
(c) 40.7 (d) 41.0
(e) None of these

(ii) **Approximately** how much per cent more plastic was used than iron for packaging food items?

(a) 32 (b) 320
(c) 33 (d) 325
(e) 225

(iii) In the case of which one of the following packaging materials used for packing food items and beverages respectively the ratio is 4:9?

(a) Glass (b) Paper
(c) Aluminium (d) Iron
(e) None of these

(iv) What is the ratio between the glass and aluminium packaging used for packing beverages?

(a) 17:56 (b) 56:17
(c) 84:37 (d) 37:84
(e) None of these

(v) **Approximately** what per cent of all the packaging materials used for packing food items was contributed by plastic?

(a) 60 (b) 65
(c) 70 (d) 55
(e) 50

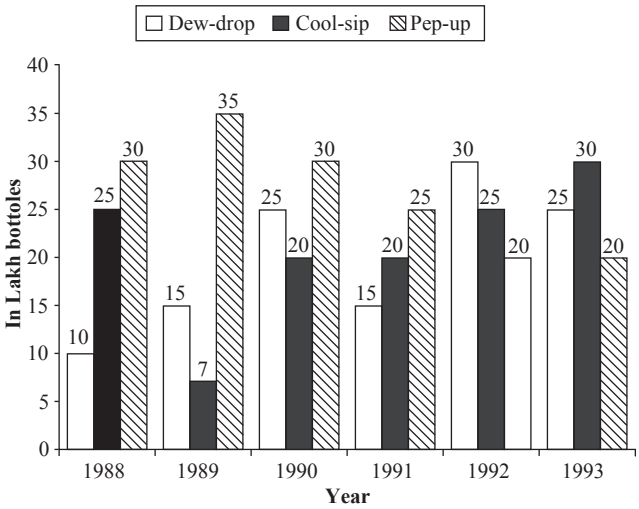
(vi) **Approximately** what per cent of all the packaging materials used for packing food items and beverages was contributed by plastic and aluminium together?

(a) 60 (b) 70
(c) 80 (d) 65
(e) 75

(vii) What per cent of all the packaging materials used for packing beverages was contributed by paper (Find the answer up to two decimal places).

(a) 2.42 (b) 3.41
(c) 2.41 (d) 3.42
(e) None of these

6. Directions: Study the following graph carefully and answer the questions given below.



(i) In which year was the sale of 'Pep-up' the maximum?

(a) 1990 (b) 1991
(c) 1992 (d) 1993
(e) None of these

(ii) In the case of which soft drink was the average annual sale maximum in the given period?

(a) Pep-up only (b) Cool-sip only
(c) Dew-drop only (d) Gool-sip and Dew-drop
(e) Pep-up and Dew-drop

(iii) In the case of Cool-sip drink, what was the approximate per cent increase in sale in 1992 over its sale in 1991?

(a) Less than 20 (b) 20–25
(c) 25 (d) 31–35
(e) 36–40

(iv) In the year 1990, what was the difference between the number of 'Pep-up' and 'Cool-sip' bottles sold?

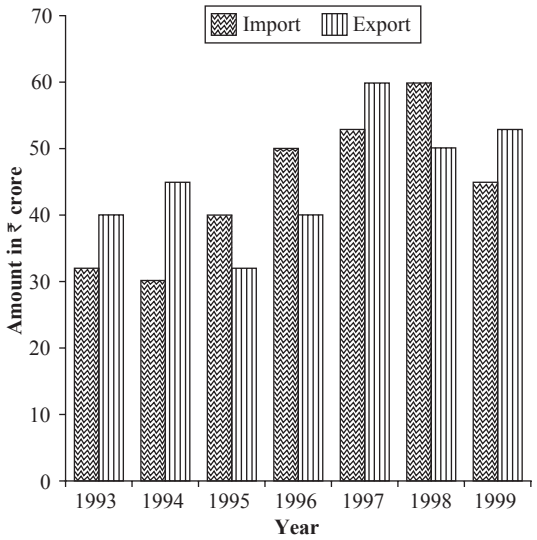
(a) 5000000 (b) 500000
(c) 50000 (d) 5000
(e) 1000000

(v) What was the approximate per cent drop in sale of Pep-up in 1990 over its sale in 1989?

(a) .5 (b) 12
(c) 14 (d) 20
(e) 28

7. Directions: Study the following graph carefully to answer the questions given below it.

Import and Export of spare parts by an automobile company over the given years



(i) During which year the percentage rise/fall in imports from the previous year is the lowest?
(a) 1994 (b) 1998
(c) 1997 (d) 1995
(e) None of these

(ii) What is the ratio of total imports to total exports for all the given years together?
(a) 31:35 (b) 35:31
(c) 65:63 (d) 63:65
(e) None of these

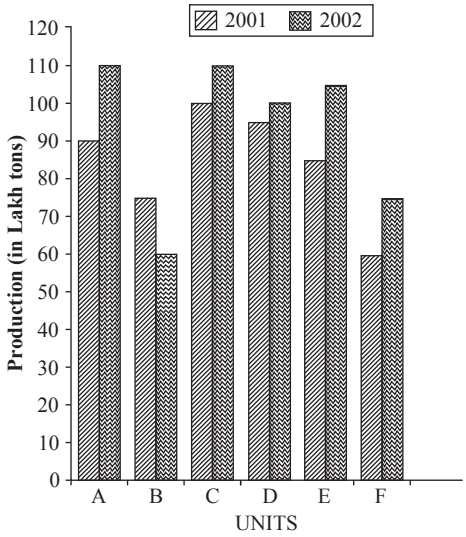
(iii) In which of the following pairs of years the total import is equal to total export in the same pair of years?
(a) 1996–1997 (b) 1993–1998
(c) 1998–1999 (d) 1995–1996
(e) None of these

(iv) The total exports in the years 1995, 1996 and 1999 together are what per cent of the total import during the same period? (up to two decimal places)
(a) 107.41 (b) 107.14
(c) 93.33 (d) 93.67
(d) None of these

(v) Which of the following pairs of years and the per cent increase in the export over the previous year is correctly matched?
(a) 1996–14.29 (b) 1997–10
(c) 1995–33.33. (d) 1994–11.11
(e) None of these

8. Directions: Study the following graphs carefully to answer these questions.

Production (in Lakh tons) of six units of a company 2001 and 2002



(i) What is the average production of all the units (in Lakh tons) for the year 2002?
(a) 89 (b) 92
(c) 87 (d) 95
(e) None of these

(ii) Average production of three units *A, B* and *C* in 2001 is what per cent of the average production of units *D, E* and *F* in 2002? (rounded off to two digits after decimal)
(a) 109.43 (b) 90.37
(c) 91.38 (d) 106.43
(e) None of these

(iii) What is the ratio of total production for two years together for unit *B* to that for *C?*
(a) 17:13 (b) 13:17
(c) 11:13 (d) 19:13
(e) None of these

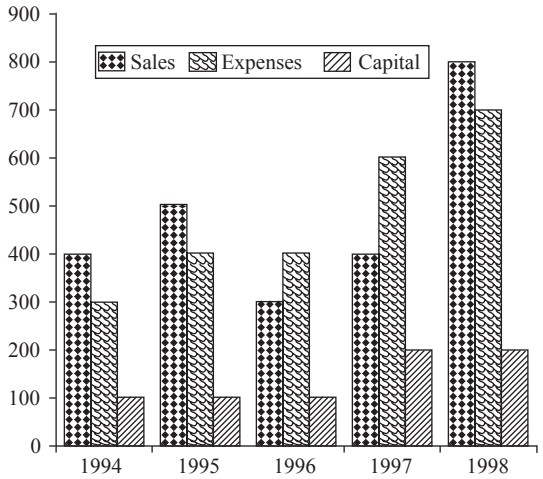
(iv) Total production for two years together by unit *F* is what per cent of the total production of the two years together by unit *D*? (rounded off to two digits after decimal)

(a) 79.49 (b) 78.49
(c) 78.47 (d) 79.29
(e) None of these

(v) What is the total production of units C, D_9 and E together for both the years? (in Lakh tons)

(a) 495 (b) 595
(c) 545 (d) 515
(e) None of these

9. The following graph gives Sales, Expense and Capital of a company for a period of five years–1994 to 1998. Read the graph and answer questions 140 to 144.



(i) What has been the simple average growth rate perannum of expense between 1994 and 1998?

(a) 25% (b) $33\frac{1}{3}\%$
(c) 40% (d) 130%

(ii) In which year was the sales-to-expense ratio the lowest?

(a) 199 (b) 1996
(c) 1997 (d) 1998

(iii) What was the average per annum increase in sales (in ₹ crore) from 1994 to 1998?

(a) 50 (b) 60
(c) 80 (d) 100

(iv) In which year was the ratio of profits to capital the highest?

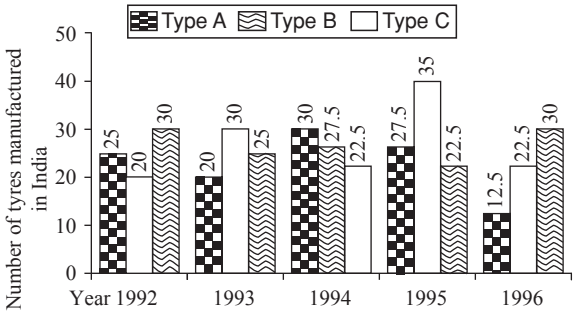
(a) 1998 (b) 1995
(c) 1996 (d) 1997

(v) In which year was the ratio of sales to capital the lowest?

(a) 1998 (b) 1997
(c) 1996 (d) 1995

10. Directions: Study the following graph carefully and answer the questions given below.

Production of three types of tyres by a company over the year (in Lakh)



(i) What was the percentage drop in the number of C type tyres manufactured from 1993 to 1994?

(a) 25 (b) 10
(c) 15 (d) 25
(e) None of these

(ii) What was the difference between the number of B type tyres manufactured in 1994 and 1995?

(a) 100000 (b) 2000000
(c) 1000000 (d) 1500000
(e) None of these

(iii) The total number of all the three types of tyres manufactured was the least in which of the following years?

(a) 1995 (b) 1996
(c) 1992 (d) 1994
(e) 1993

(iv) In which of the following years was the percentage production of B type to C type tyres the maximum?

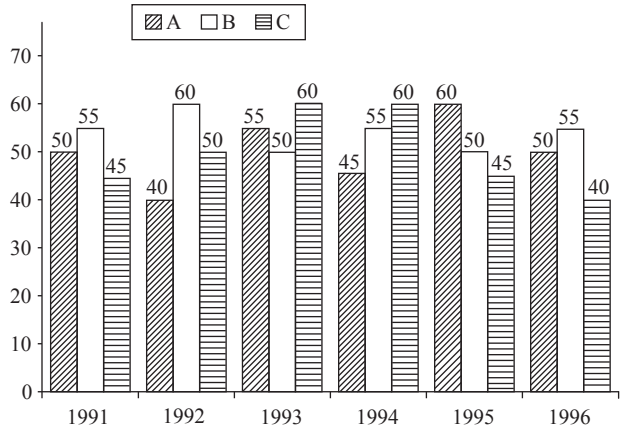
(a) 1994 (b) 1992
(c) 1996 (d) 1993
(e) 1995

(v) The total production of C type tyres in 1992 and 1993 together was what percentage of production of B type tyres in 1994?

(a) 50 (b) 100
(c) 150 (d) 200
(e) None of these

11. Directions: Study the following graph carefully to answer the questions given below it.

Production of paper (in Lakh tonnes) by 3 different companies A, B, and C over the years



(i) What is the difference between the production of company *C* in 1991 and the production of company *A* in 1996?

(a) 50000 tonnes (b) 50000000 tonnes
(c) 5000000 tonnes (d) 500000 tonnes
(e) None of these

(ii) What is the percentage increase in production of company *A* from 1992 to 1993?

(a) 37.5 (b) 38.25
(c) 35 (d) 36
(e) None of these

(iii) For which of the following years the percentage of rise/fall in production from the previous year the maximum for company *B*?

(a) 1992 (b) 1993
(c) 1994 (d) 1995
(e) 1996

(iv) The total production of company *C* in 1993 and 1994 is what percentage of the total production of company *A* in 1991 and 1992?

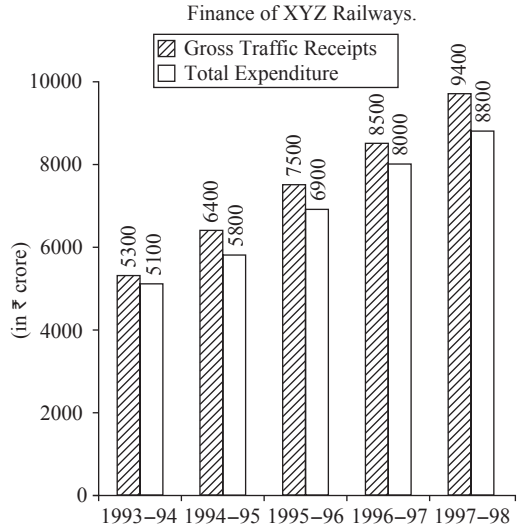
(a) 95 (b) 90
(c) 110 (d) 115
(e) None of these

(v) What is the difference between the average production per year of the company with highest average production and that of the company with lowest average production in Lakh tonnes?

(a) 3.17 (b) 4.33
(c) 4.17 (d) 3.33
(e) None of these

12. Directions: These questions are based on the following bar graph. Read the graph and answer the questions.

Finances of XYZ Railways



(i) What is the percentage increase in the gross traffic receipts in 1995–96 as compared to 1993–94?

(a) 33.9% (b) 41.5%
(c) 20.7% (d) 17%

(ii) If profit = gross traffic receipts – total expenditure, then in 1996–97, what percentage of gross traffic receipts is the profit made?

(a) 5.9% (b) 6.4%
(c) 7.2% (d) 8%

(iii) In which year was the profit as a percentage of gross traffic receipts the highest?

(a) 1997–98 (b) 1996–97
(c) 1995–96 (d) 1994–95

(iv) In order to make a profit of 10%, what should have been the gross traffic receipts (in ₹crore) in 1994–95, total expenditure remaining the same?

(a) 5667 (b) 5876
(c) 6444 (d) 7667

(v) By what amount (in ₹crore) has the expenditure increased over the period 1993–94 to 1997–98?

(a) 4100 (b) 3900
(c) 3850 (d) 3700

13. Directions: Study the following table carefully and answer the question given below it.

Number of candidates from different locations appeared and passed in a competitive examination over the years

	Rural		*Semi-urban*		*State capitals*		*Metropolises*	
Year	*App.*	*Passed*	*App.*	*Passed*	*App.*	*Passed*	*App.*	*Passed*
1990	1652	208	7894	2513	5054	1468	9538	3214

1991	1839	317	8562	2933	7164	3248	10158	4018
1992	2153	932	8139	2468	8258	3159	9695	3038
1993	5032	1798	9432	3528	8529	3628	11247	5158
1994	4915	1668	9784	4015	9015	4311	12518	6328
1995	5628	2392	9969	4263	1725	4526	13624	6419

(i) For the candidates from which of the following locations was there continuous increase both in appeared and passed?

(a) Semi-urban (b) State capital
(c) State capital and Rural (d) Metropolises
(e) None of these

(ii) In which of the following years was the percentage passed to appeared candidates from semi-urban area the least?

(a) 1991 (b) 1993
(c) 1990 (d) 1992
(e) None of these

(iii) What approximate value was the percentage drop in the number of semi-urban candidates appeared from 1991 to 1992?

(a) 5 (b) 10
(c) 15 (d) 8
(e) 12

(iv) In 1993 percentage of candidates passed to **appeared** was **approximately** 35 from which location?

(a) Rural
(b) Rural and Metropolises
(c) Semi-urban and Metropolises
(d) Rural and Semi-urban
(e) None of these

(v) The total number of candidates passed **from Rural** in 1993 and Semi-urban in 1990 was **exactly equal** to the total number of candidates passed from **State** capital in which of the following years?

(a) 1990 (b) 1993
(c) 1994 (d) 1992
(e) None of these

14. Directions: Study the following table carefully and answer the questions given below it.

Fare in rupees for three different types of vehicles

Fare for distance up to	*Vehicle*		
	Type *A*	Type *B*	Type *C*
2 Km	₹5.00	₹7.50	₹10.00
4 Km	₹900	₹14.50	₹19.00
7 Km	₹13.50	₹24.25	₹31.00
10 Km	₹17.25	₹33.25	₹41.50
15 Km	₹22.25	₹45.75	₹56.50
20 Km	₹26.00	₹55.75	₹69.00

Notes

Fare per Km for intermittent distance is the same.

(i) Shiv Kumar has to travel a distance of 15 Km in all. He decides to travel equal distance by each of the three types of vehicles. How much money is to be spent as fare?

(a) ₹51.75 (b) ₹47.50
(c) ₹47.25 (d) ₹51.25
(e) None of these

(ii) Ajit Singh wants to travel a distance of 15 Km. He starts his journey by Type *A* vehicle. After travelling 6 Km, he changes the vehicle to Type *B* for the remaining distance. How much money will he be spending in all?

(a) ₹42.25 (b) ₹36.75
(c) ₹40.25 (d) ₹42.75
(e) None of these

(iii) Mr *X* wants to travel a distance of 8 Km by Type *A* vehicle. How much more money will be required to be spent if he decides to travel by Type *B* vehicle instead of Type *A*?

(a) ₹16 (b) ₹12.50
(c) ₹14 (d) ₹13.50
(e) None of these

(iv) Rita hired a Type *B* vehicle for travelling a distance of 18 Km. After travelling 5 Km, she changed the vehicle to type A. Again, after travelling 9 Km by Type A vehicle, she changed the vehicle to Type *C* and completed her journey. How much money did she sp did in all?

(a) ₹50 (b) ₹45.50
(c) ₹55 (d) ₹50.50
(e) None of these

(v) Fare for 14th Km by Type *C* vehicle is equal to the fare for which of the following?

(a) Type *B*— 11th Km (b) Type *B*—9th Km
(c) Type *A*—4th Km (d) Type *C*— 8th Km
(e) None of these

15. Directions: Read the following table carefully and answer the questions given below it.

Details of leading openers' performance in 20 one-day cricket matches

Openers	Total Runs	Highest Runs	No. of matches with runs		
			100 or more	50–99	0's
A	994	141	5	3	1
B	751	130	1	8	2
C	414	52	—	2	2
D	653	94	—	4	1
E	772	85	—	7	—

(i) What is the difference between the **average runs** of top two openers in terms of **highest runs,** if matches having 0's were ignored?

(a) 4.7 (b) 13.7
(c) 11.1 (d) 16.62
(e) None of these

(ii) If matches having zero runs and the one with highest runs is ignored, what will be the average runs for opener *C*?

(a) 21.29 (b) 21.79
(c) 20.7 (d) 21.17
(e) 20.19

(iii) By how much the difference between the two highest total runs differs from the difference between the two lowest total runs?

(a) Lower by 18 (b) More by 18
(c) Lower by 4 (d) More by 4
(e) None of these

(iv) Which of the given pairs of openers have ratio 3:2 in their highest runs?

(a) *B* and *D* (b) *B* and *C*
(c) *A* and *D* (d) *D* and *C*
(e) None of these

(v) Excluding the match with the highest runs and matches with 50–99 runs, what will be the **approximate** average runs for opener *B*?

(a) 25 (b) 15
(c) 10 (d) 30
(e) None of these

16. Directions: Read the following Table carefully and answer the questions given below.

Highest marks and average marks obtained by students in subjects over the years.

The maximum marks in each subjects is 100.

	Subjects									
	English		Hindi		Maths		Science		History	
	High	Avg	High	Avg	High	Avg	High	Avg	High	Avg
1992	85	62	75	52	98	65	88	72	72	46
1993	80	70	80	53	94	60	89	70	65	55
1994	82	65	77	54	85	62	95	64	66	58
1995	71	56	84	64	92	68	97	68	68	49
1996	75	52	82	66	91	64	92	75	70	58
1997	82	66	81	57	89	66	98	72	74	62

(i) What was the grand average marks of the five subjects in 1996?

(a) 63 (b) 64
(c) 65 (d) 68
(e) None of these

(ii) The difference in the average marks in History between 1994 and 1995 was exactly equal to the difference in the highest marks in Hindi between which of the following pairs of years?

(a) 1992 and 1995 (b) 1993 and 1995
(c) 1992 and 1996 (d) 1993 and 1997
(e) None of these

(iii) What was the **approximate** percentage increase in average marks in History from 1992 and 1993?

(a) 20 (b) 25
(c) 24 (d) 16
(e) 18

(iv) The average highest marks in English in 1992, 1993 and 1996 was exactly equal to the highest marks in Hindi in which of the following years?

(a) 1996 (b) 1997
(c) 1994 (d) 1996
(e) 1993

(v) The difference between the highest marks and the average marks in Hindi was maximum in which of the following years?

(a) 1994 (b) 1997
(c) 1995 (d) 1996
(e) 1993

(vi) The highest marks in Hindi in 1993 was what per cent of the average marks in Mathematics in 1996?

(a) 135 (a) 130
(c) 125 (d) 140
(e) None of these

(vii) If there were 50 students in 1993, what was the total marks obtained by them in Mathematics?

(a) 2400 (b) 3000
(c) 2500 (d) 3200
(e) None of hese

(viii) The difference between the highest marks in Science was maximum between which of the following pairs of years among the given years?

(a) 1992 and 993 (b) 1992 and 1996
(c) 1996 and 997 (d) 1992 and 1995
(e) None of these

17. Directions: Read the following information carefully and answer the questions based on it.

In 6 educational years, number of students taking admission and leaving from the .5 different schools which are for nded in 1990 are given below.

School	*A*		*B*		*C*		*D*		*E*	
	Ad	L	Ad	L	Ad	L	Ad	L	Ad	L
1992	1025	—	950	—	1100	—	1500	—	1450	—
1993	230	120	350	150	320	130	340	150	150	125
1994	190	110	225	115	300	150	300	160	280	130
1995	245	100	185	110	260	125	295	120	310	120
1996	280	150	200	90	240	140	320	125	340	110
1997	250	130	240	120	310	180	360	140	325	115

In the above table shown Ad = Admitted, L = Left

(i) What is the average number of students studying in all the five schools in 1992?

(a) 1494 (b) 1294
(c) 1590 (d) 1640
(e) None of these

(ii) What was the number of students studying in school *B* in 1994?

(a) 2030 (b) 1060
(c) 1445 (d) 1150
(e) None of these

(iii) Number of students leaving school *C* from the year 1990 to 1995 is approximately what percentage of number of students taking admission in the same school and in the same year?

(a) 50% (b) 25%
(c) 48% (d) 36%
(e) 29%

(iv) What is the difference in the number of students taking admission between the years 1991 and 1995 in school *D* and *B*?

(a) 514 (b) 1065
(c) 965 (d) 415
(e) None of these

(v) In which of the following schools, percentage increase in the number of students from the year 1990 to 1995 is maximum?

(a) *A* (b) *B*
(c) *C* (d) *D*
(e) *E*

18. Directions: Study the following Table and answer the following questions carefully.

Following Table shows the percentage population of six States below poverty line and the proportion of male and female.

State	Percentage population below poverty line	Proportion of male and female: Below poverty line M:F	Above poverty line M:F
A	12	3:2	4:3
B	15	5:7	3:4
C	25	4:5	2:3
D	26	1:2	5:6
E	10	6:5	3:2
F	32	2:3	4:5

(i) The total population of state *A* is 3000, then what is the approximate no, of females above poverty line in state *A*?

(a) 1200 (b) 2112
(c) 1800 (d) 1950
(e) 2025

(ii) If the total population of C and *D* together is 18000, then what is the total no. of females below poverty line in the above stated states?

(a) 5000 (b) 5500
(c) 4800 (d) Data inadequate
(e) None of these

(iii) If the population of males below poverty line in State *A* is 3000 and that in State *E* is 6000, then .what is the ratio of the total population of State *A* and *E*?

(a) 3:4 (b) 4:5
(c) 1:2 (d) 2:3
(e) None of these

(iv) If the population of males below poverty line in State *B* is 500 then what is the total population of that State?

(a) 14400 (b) 6000
(c) 8000 (d) 7600
(e) None of these

(v) If in State *E* population of females above poverty line is 19800 then what is the population of males below poverty line in that State?

(a) 55000 (b) 30000
(c) 29700 (d) Date inadequate
(e) None of these

19. Direction the following Table carefully and answer the questions given below.

Production of main crops in India (in millionon tonnes)

Crops	91–92	92–93	93–94	94–95	95–96	96–97
Pulses	20.5	22.4	24.6	23.5	27.8	28.2
Oilseeds	32.4	34.6	40.8	42.4	46.8	52.4
Rice	80.5	86.4	88.2	92.6	94.2	90.8
Sugercane	140.8	150.2	152.2	160.3	156.4	172.5
Wheat	130.2	138.4	146.8	141.6	152.2	158.4
Coarse grain	45.6	52.8	60.4	62.4	58.2	62.8
Sum	**450**	**484.8**	**513.2**	**522.8**	**535.6**	**565.1**

(i) Production of sugarcane in 1993–94 was approximately what percentage of production of rice in 1992–93?

(a) 50 (b) 75
(c) 150 (d) 125
(e) 175

(ii) Production of what type of crop was going to increase in each year in the given years?

(a) Rice (b) Pulse
(c) Sugarcane (d) Oilseeds
(e) None of these

(iii) What was the average production of pulse in the given years?

(a) 26.8 million tones (b) 20.5 million tonnes
(c) 24.5 million tones (d) 22.5 million tonnes
(e) None of these

(iv) Production of oilseeds was what percentage of the total crops produced in the year 1991–92?

(a) 7.2 (b) 8.4
(c) 2.7 (d) 6.4
(e) None of these

(v) In which of the following years the total production *of* oil seeds in the years 1994–95, 1995–96 and 1996–97 was equal to the production of wheat?

(a) 1993–94 (b) 1994–95
(c) 1996–97 (d) 1992–93
(e) None of these

20. Directions: Study the following Tables carefully and answer the questions given below.

Number of Cars (in thousands) of Different Models and Colours sold in two Metro Cities in a year

Metro M

Type / Colour	Black	Red	Blue	White	Silver
A	40	25	55	75	15
B	20	35	60	80	20
C	35	30	50	90	35
D	45	40	45	85	40
E	50	35	35	60	30
F	55	42	40	65	52

Metro H

Type / Colour	Black	Red	Blue	White	Silver
A	45	32	40	60	20
B	30	37	39	81	35
C	40	42	41	6	37
D	35	39	37	90	42
E	50	44	43	77	22
F	47	34	45	87	17

(i) The difference between the white-coloured cars sold in the two metros of which of the following models is the **minimum**?

(a) *A* (b) *C*
(c) *D* (d) *F*
(e) None of these

(ii) The total number of blue-coloured cars of Model *E* and *D* sold in Metro *H* is exactly equal to the number of white-coloured cars of which model in Metro *M*?

(a) *B* (b) *F*
(c) *C* (d) *A*
(e) None of these

(iii) What is the difference between the number of blue-colour cars of model '*C*' sold in Metro *M* and number of red colour cars of model '*F*' sold in Metro *H*?

(a) 8,000 (b) 10,000
(c) 12,000 (d) 15,000
(e) None of these

(iv) The total number of silver-coloured cars sold in Metro *H* is approximately what percentage of that in Metro *M*?

(a) 130 (b) 140
(c) 90 (d) 100
(e) 110

(v) In Metro *M* the number of cars sold was maximum for which of the colour-model combinations?

(a) White-*C* (b) Blue-*B*
(c) Silver-*B* (d) White-*D*
(e) None of these

21. Directions: Study the following Table to answer the given questions.

Number of students of different classes of a school playing different games

Class → Games ↓	XII	XI	X	IX	VIII	VII	VI
Chess	11	12	5	4	2	2	1
Cricket	38	40	12	17	25	18	20
Basketball	11	9	7	6	0	0	0
Table Tennis	9	9	21	19	11	9	0
Football	40	27	18	19	12	16	14
Carrom	16	15	8	19	12	16	14
Tennis	8	9	11	5	6	0	0
Badminton	47	39	33	21	19	0	0

(i) **Approximately** what per cent of Class VIII students play Cricket out of the total students playing Cricket?

(a) 13 (b) 4
(c) 25 (d) 15
(e) 17

(ii) What is the ratio of the students playing Football in Class XI to those in Class *X*?

(a) 1:2 (b) 2:5
(c) 2:3 (d) 3:2
(e) None of these

(iii) Which game is the most popular?

(a) Badminton (b) Football
(c) Carrom (d) Table Tennis
(e) Cricket

(iv) **Approximately** what per cent of Class *X* students play Table Tennis out of the total Class *X* students playing the different given games?

(a) 20 (b) 21
(c) 27 (d) 26
(e) 18

(v) Which game has ascending number of students from class IX to XII?

(a) Basketball Only (b) Badminton Only
(c) Chess and Badminton (d) No game
(e) None of these

22. Directions: A Table showing the percentages of the total population of a State by age groups for the year 1991 is given below. Answer the questions given below it.

Age Group (in years)	*Per cent*
up to 15	30.00
16–25	17.75
26–35	17.25
36–45	14.50
46–55	14.25
56–65	5.12
66 and above	1.13
Total	100.0

(i) Which age group accounts for the maximum population in the State?

(a) 16–25 (b) 26–35
(c) 36–45 (d) 56–65

(ii) Out of every 4200 persons, the number of persons below 26 years is:

(a) 2006 approx (b) 1260 approx
(c) 746 approx (d) 515 approx

(iii) There are 200 million people below 36 years. How many million (approx) people are in the age group 56–65?

(a) 30.07 (b) 15.75
(c) 12.72 (d) 59.30

(iv) If there are 10 million people in the age group 56 years and above, what is the difference between the number of people in the age groups 16–25 and 46–55?

(a) 6.8 million (b) 5.6 million
(c) 28.4 million (d) 34.7 million

(v) If the difference between the number of people in the age groups 46–55 and 26–35 is 11.75 million, then total population of State is approximately

(a) 360.23 million (b) 391.67 million
(c) 400 million (d) 460.67 million

23. Directions: Study the following table to answer these questions based on it

***XYZ* Co. (Pvt.) Ltd. (in Lakh of ₹)**

Year	*Total Sales*	*Gross Profit*	*Net Profit*
1990	351.6	155.5	54.2
1991	407.9	134.3	42.6
1992	380.1	149.9	38.9
1993	439.7	160.5	50.3
1994	485.9	203.3	65.8

(i) In which year the difference between the total sales and the gross profit is the least?

(a) 1990 (b) 1991
(c) 1992 (d) 1993

(ii) The total sales in 1993 is approximately what per cent of the total sales in 1990?

(a) 75 (b) 85
(c) 110 (d) 125

(iii) Which years show increase in all the categories simultaneously, that is total sales, gross profit and net profit as compared to the previous year?

(a) 1993 and 1994 both
(b) 1994 and 1992 both
(c) 1992 and 1993 both
(d) 1990 and 1991 both

(iv) The net profit in 1991 is **approximately** what per cent of the total sales in 1993?

(a) 6.5 (b) 7
(c) 8 (d) 9.7

(v) The per cent increase in the gross profit was the largest in which year as compared to the previous one?

(a) 1991 (b) 1992
(c) 1993 (d) 1994

24. Directions: Refer to the following Table. Read the Table and answer the questions.

Foodgrain Production in a Country in 1999 (in Lakh tons)

State	*Rice*	*Wheat*	*Jowa*	*Pulses*	Others
P	45	103	—	27	29
Q	48	86	73	19	15
R	59	32	67	14	31
S	41	37	59	21	15
T	37	22	41	13	11
U	68	15	12	—	18
V	57	8	7	12	10
W	38	28	31	22	45

(i) Which State had the highest wheat production?

(a) *P* (b) *Q*
(d) *R* (d) *S*

(ii) What was the proportion of rice production to wheat production in the country?

(a) 1:1 (b) 0.8:1
(c) 1.2:1 (d) 2:1

(iii) Jowar was the most important foodgrain in the State/States;

(a) *Q, R, S* (b) *Q*
(c) *R, S* (d) *R, S, T*

(iv) States *P* alone accounted for **approximately** what percentage of wheat production in the country?

(a) 73% (b) 50%
(c) 41% (d) 30%

(v) If the average per hectare yield of rice in the country was 30 tons, then the area (approx.) under rice cultivation during the year was (in Lakh hectares)

(a) 1.5 (b) 8
(c) 13 (d) 40

25. Directions: Following Table gives the population of a locality from 1988 to 1992. Read the Table and answer the questions.

Year	*Men*	*Women*	*Children*	*Total*	*Increase (+) or decrease (–) over preceding year*
1988	65104	60387	—	146947	—
1989	70391	62516	—	—	+ (11630)
1990	—	63143	20314	152922	—
1991	69395	—	21560	—	– (5337)
1992	71274	65935	23789	160998	—

(i) The number of children in 1988 is:

(a) 31236 (b) 125491
(c) 14546 (d) 21456

(ii) The number of children in 1989 is:

(a) 144537 (b) 158577
(c) 146947 (d) 149637

(iii) The number of women in 1991 is:

(a) 25670 (b) 14040
(c) 13970 (d) 15702

(iv) The number of women in 1991 is:

(a) 57630 (b) 56740
(c) 52297 (d) 62957

(v) Increase or decrease of population in 1992 over 1991 is:

(a) – (12413) (b) + (12413)
(c) + 155661 (d) + 7086

26. Directions: The Table given below shows production of five types of cars by a company in the years 1989 to 1994. Study the Table and answer questions.

Production of cars by a company

Year → / Type ↓	1989	1990	1991	1992	1993	1994	Total
P	8	20	16	17	21	6	88
Q	16	10	14	12	12	14	18
R	21	17	16	15	13	8	90
S	4	6	10	16	20	31	87
T	25	18	19	30	14	27	133
Total	74	71	75	90	80	86	476

(i) In which year the production of cars of all types taken together was approximately equal to the average of the total production during the period?

(a) 1989 (b) 1991
(c) 1993 (d) 1994

(ii) In which year the total production of cars of types *P* and *O* together was equal to the total production of cars of types *R* and *S* together?

(a) 1990 (b) 1991
(c) 1994 (c) None of the above

(iii) During the period 1989–94, in which type of cars was a continuous increase in production?

(a) *P* (b) *Q*
(c) *R* (d) *S*

(iv) The production of which type of cars was 25% of the total production of all types of cars during 1993?

(a) *S* (b) *R*
(c) *Q* (d) *P*

(v) The per cent increase in total production of all types of cars in .1993 to that in 1991 was?

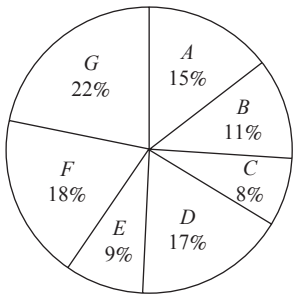
(a) 15 (c) 20
(c) 25 (d) 30

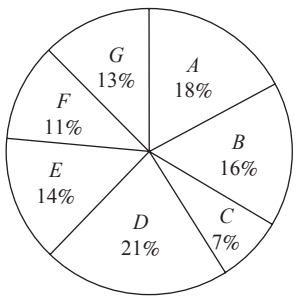
27. Directions: These questions are based on the following graphs:

Classification of appeared candidates in a competitive test from different States and qualified candidates from those States.

Appeared candidates = 45000.

Qualified candidates = 9000.





(i) What is the ratio of the number of appeared candidates from States *C* and *E* together to that of the appeared candidates from States *A* and *F* together?

(a) 17:33 (b) 11:13
(c) 13:27 (d) 17:27
(e) None of these

(ii) In which State, the percentage of qualified candidates with respect to that of appeared candidates is minimum?

(a) *C* (b) *F*
(c) *D* (d) *E*
(e) *G*

(iii) What is the difference between the number of qualified candidates' of States *D* and those of *G*?

(a) 690 (b) 670
(c) 780 (d) 720
(e) None of these

(iv) What is the percentage of qualified candidates with respect to appeared candidates from States *B* and *C* taken together? (rounded to two decimal places)

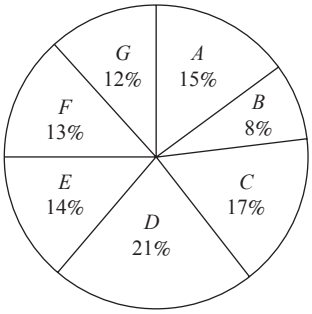
(a) 23.11 (b) 24.21
(c) 21.24 (d) 23
(e) None of these

(v) What is the ratio between the number of candidates qualified from States *B* and *D* together to the number of candidates appeared from State '*C*', respectively?

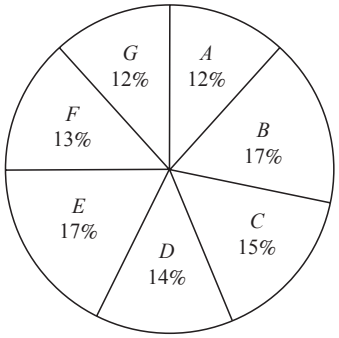
(a) 8:37 (b) 11:12
(c) 37:40 (d) 7:37
(e) None of these

28. Directions: These questions are based on the following graphs:

Distribution of candidates studying Arts and Commerce from seven different institutes *A*, *B*, *C*, *D*, *E*, *F* and *G*. Total Number of Students Studying Arts = 3800



Total Number of Students Studying Commerce = 4200



(i) What is the ratio between the number of students studying Arts from Institute *E* and the number of students studying Commerce from Institute *D*, respectively?

(a) 17:19 (b) 19:27
(c) 14:19 (d) 19:21
(e) None of these

(ii) What is the total number of students studying Arts from Institutes *A* and *G*. together?

(a) 1102 (b) 918
(c) 966 (d) 1130
(e) None of these

(iii) How many students are studying Commerce from Institutes *B* and *D* together?

(a) 1158 (b) 1302
(c) 1232 (d) 1272
(e) None of these

(iv) How many students are studying Art and Commerce from Institute '*B*'?

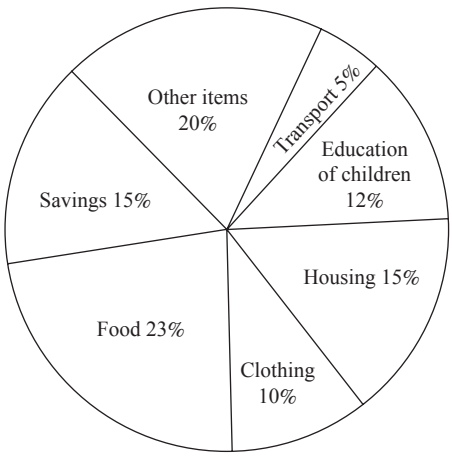
(a) 1418 (b) 2000
(c) 1018 (d) 1208
(e) None of these

(v) What is the ratio between the number of students studying Arts and Commerce, respectively from Institute '*E*'?

(A) 19:27 (b) 17:29
(c) 19:29 (d) 17:27
(e) None of these

29. Directions: The pie-chart drawn below shows the expenses of a family on various items and its savings during the year 2001. Study the graph and answer the questions given below:

Percentage of Money Spent on Various Items and Savings by a Family during 2001



(i) Maximum expenditure of the family was on

(a) Food (b) Housing
(c) Education of Children (d) Other items

(ii) The total savings of the family for the year were equal to the expenditure on:

(a) Food
(b) Clothing
(c) Housing
(d) Other items including transport

(iii) What per cent of the income was spent on transport and other items together?

(a) 25% (b) 20%
(c) 30% (d) 32%

(iv) If the total income of the family was ₹100000, how much money was spent on the eduction of children?

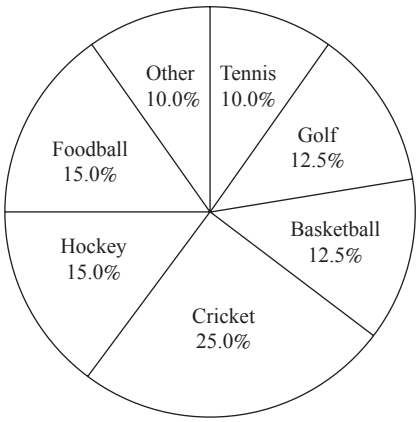
(a) ₹10000 (b) ₹12000
(c) ₹15000 (d) ₹23000

(v) If the total income for the year was ₹100000, the difference of the expenses (in rupees) between housing and transport was:

(a) 15000 (b) 12000
(c) 7000 (d) 10000

30. Directions: The circle graph given here shows the spendings of a country on various sports during a year. Study the graph carefully to answer these questions.

Percent of Money Spent on Various Sports for One Year



(i) If the total amount spent on sports during the year was ₹15000000, then the amount spent on Cricket and Hockey together was:

(a) ₹6000000 (b) ₹5000000
(c) ₹3750000 (d) ₹7500000

(ii) If the total amount spent during the year was ₹1,20,00,000 how much was spent on basketball?

(a) ₹12,50,000 (b) ₹10,00,000
(c) ₹12,00,000 (d) ₹15,00,000

(iii) The ratio of the total amount spent on football to that spent on Hockey was:

(a) 1:15 (b) 1:1
(c) 15:1 (d) 3:2

(iv) The graph shows that the most popular game is:

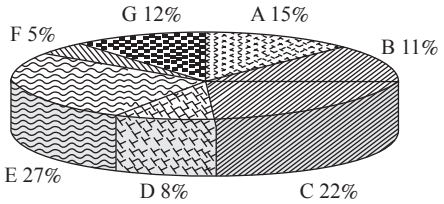
(a) Hockey (b) Football
(c) Cricket (d) Basketball

(v) The country spent the same amount of money on:

(a) Hockey and Tennis (b) Golf and Basketball
(c) Cricket and Football (d) Hockey and Golf.

31. Directions: Seven companies *A*, *B*, *C*, *D*, *E*, *F* and *G* are engaged in production of two items I and II. The comparative data about production of these items by the seven companies is given in the following graph and Table. Study them carefully and answer the questions given below.

Percentage of the total production produced by the seven companies



Cost of the total production (both items together) by seven companies = ₹25 crores

Ratio of production between items I and II and the per cent profit earned for the two items.

Company	Ratio of Production		Per cent profit earned	
	Item I	Item II	Item I	Item II
A	2	3	25	20
B	3	2	32	35
C	4	1	20	22
D	3	5	15	25
E	5	3	28	30
F	1	4	35	25
G	1	2	30	24

(i) What is the total cost of the production of item I by companies *A* and *C* together in ₹crore?

(a) 9.25 (b) 5.9
(c) 4.1625 (d) 4.9
(e) None of these

(ii) What is the amount of profit earned by company *D* on item II?

(a) ₹3.125 Crore (b) ₹31.25 Crore
(c) ₹3.125 Lakhs (d) ₹31.25 Lakhs
(e) None of these

(iii) Cost of production of item I by company *F* is what per cent of the cost of production of item II by company *D*?

(a) 16% (b) 33.33%
(c) 66.67% (d) 12.5%
(e) None of these

(iv) What is total profit earned by company *G* for items I and II together?

(a) ₹78 Lakhs (b) ₹1.62 Crore
(c) ₹7.8 Crore (d) ₹16.2 Lakhs
(e) None of these

(v) What is the ratio of the cost of production of item I by company *A* to the cost of production of item I by company *D*?

(a) 3:5 (b) 1:2
(c) 2:1 (d) 2:3
(e) None of these

(vi) What is the total of the profit earned by company *B* on production of item I and the profit earned by company *A* on production of item II?

(a) ₹9.78 Crore (b) ₹97.8 Lakhs
(c) ₹52.8 Lakhs (d) ₹5.28 Crore
(e) None of these

(vii) The cost of production of both items together by company *E* is equal to the total cost of production of both items together by which of the two companies?

(a) *C* and *D* (b) *B* and *G*
(c) *A* and *D* (d) *C* and *F*
(e) *A* and *B*

(viii) What is the total of the cost of production of item I by company *A* and the cost of production of item II by company *B*?

(a) ₹2.6 Crore (b) ₹26 Lakhs
(c) ₹3. 35 Crores (d) ₹33. 65 Lakhs
(e) None of these

32. Directions: Study the following information to answer the given questions.

Percentage of students in various courses (*A*, *B*, *C*, *D*, *E*, *F*) in pie chart I and percentage of girls in pie chart II.
Total students: 1200 (800 girls + 4000 boys)

Chart I

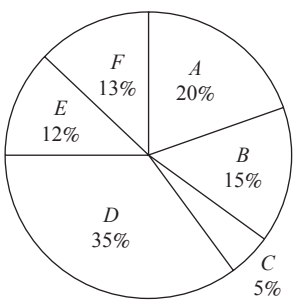
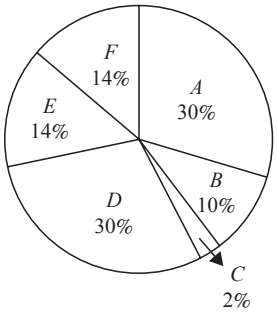


Chart II



(i) For course *D*, what is the respective ratio of boys and girls?

(a) 3:4 (b) 4:5
(c) 3:5 (d) 5:6
(e) None of these

(ii) For which pair of courses is the number of boys the same?

(a) *E* and *F* (b) *A* and *D*
(c) *C* and *F* (d) *B* and *D*
(e) None of these

(iii) For course *E,* the number of girls is how much per cent more than the number of boys for course *E*?

(a) 250 (b) 350
(c) 150 (d) 80
(e) None of these

(iv) For which course is the number of boys the minimum?

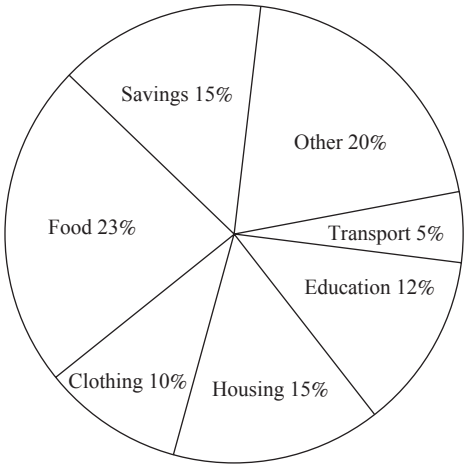
(a) *E* (b) *F*
(c) *C* (d) *A*
(e) None of these

(v) How many girls are there in course *C*?

(a) 44 (b) 16
(d) 40 (e) 160
(e) None of these

33. Directions: The circle graph given here shows the spendings by a family on various items during the year 1999 Study the graph and answer these questions.

Per cent of money spent by a family on various items during 1999



(i) If the total amount spent during the year 1999 was ₹46000, the amount spent on food was:

(a) ₹2000 (b) ₹10580
(c) ₹23000 (d) ₹2300

(ii) If the total amount spent was ₹46000, how much money was spent on clothing and housing together?

(a) ₹11500 (b) ₹1150
(c) ₹10000 (d) ₹15000

(iii) The ratio of the total amount of money spent on housing to that spent on education was:

(a) 5:2 (b) 2:5
(c) 4:5 (d) 5:4

(iv) Graph shows that the maximum amount was spent on:

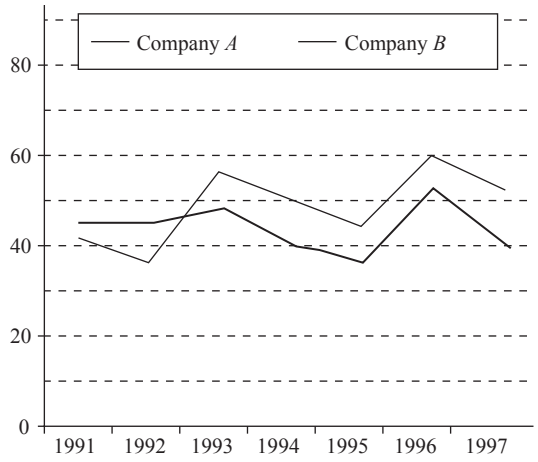
(a) Food (b) Housing
(c) Clothing (d) Others

(v) If the total expenditure of the family for the year 1999 was ₹46000, the family saved during the year:

(a) ₹1500 (b) ₹15000
(c) ₹6900 (d) ₹3067 approx

34. Directions: Study the following graph carefully and answer the questions given below.

Percentage net profit of two companies over the years



(i) If the total income in 1992 for Company *B* was 140 crores, what was the total expenditure in that year?

(a) 100 Crores (b) 110 Crores
(c) 98 Crores (d) Data inadequate
(e) None of these

(ii) If the total expenditure of 1993 and 1994 together of Company *B* was ₹279 crore, what was the total income in these years?

(a) ₹121.5 Crores (b) ₹135 Crores
(c) ₹140 Crores (d) Data inadequate
(e) None of these

(iii) In how many of the given years the percentage of expenditure to the income of Company *A* was less than fifty?

(a) One (b) Two
(c) Three (d) Four
(e) None of these

(iv) If the total expenditue of Company *B* in 1994 was ₹200 crore, what was the total income?

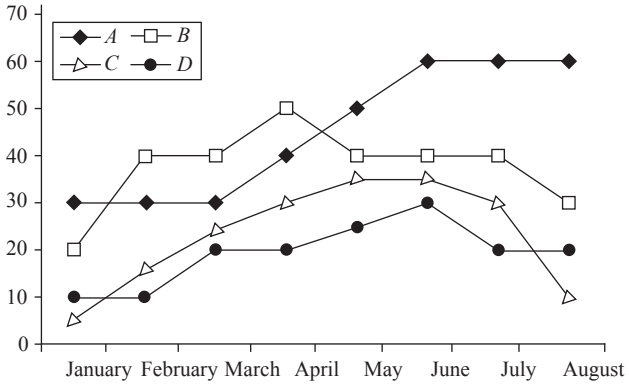
(a) ₹160 Crores (b) ₹240 Crores
(c) ₹260 Crores (d) Data inadequate
(e) None of these

(v) In which of the following years was the total income more than double the total expenditure in that year for Company *B*?

(a) 1995 (b) 1993
(c) 1997 (d) 1992
(e) None of these

35. Directions: Study the following graph carefully and answer the questions given below

Percentage increase in the sale of four commodities *A*, *B*, *C* and *D* for the given months



(i) For which months did the sale of commodities *A* and *D* show increase?

(a) April only
(b) May and June only
(c) May, June and July only
(d) April, May and June only
(e) None of these

(ii) In which month is the average percentage increase for the four commodities the lowest? (Those months where decrease rook place to be ignored)

(a) May (b) March
(c) January (d) June
(e) None of these

(iii) In which month(s) did all the commodities show decline or no increase from the previous months?

(a) July only
(b) August only
(c) April and July only
(d) July and August only
(e) None of these

(iv) If the sale of *C* was 100 in May, what was its sale in July?

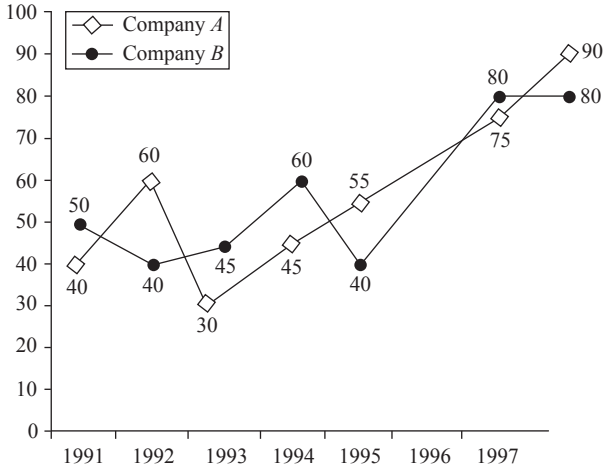
(a) 195 (b) 100
(c) 90 (d) Cannot be determined
(e) None of these

(v) For which commodity is the per cent increase in sale the highest in May from January?

(a) *C* (b) *A*
(c) *B* (d) *A* and *B*
(e) None of these

36. Directions: Study the following graph carefully and answer the questions given below it.

Per cent profit earned by two companies *A* and *B* over the years 1991 to 1997



(i) Investment of company '*B*' in 1997 is more by 40% than that in the previous year. Income in 1997 was what per cent of the investment in 1996?

(a) 280% (b) 252%
(c) 242% (d) 52%
(e) None of these

(ii) Average investment of company '*A*' over the years was ₹26 Lakhs. What was its average income over the years?

(a) ₹40.56 Lakhs (b) ₹41.60 Lakhs
(c) ₹50.26 Lakhs (d) Data inadequate
(e) None of these

(iii) Income of company '*A*' in 1995 was ₹21.7 Lakh. What was the investment?

(a) ₹14.5 Lakhs (b) ₹15.4 Lakhs
(c) ₹15.8 Lakhs (d) ₹14.6 Lakhs
(e) None of these

(iv) Income of company '*A*' in 1995 is equal to the investment of the company '*B*' in 1996. What is the ratio of the investment of company '*A*' in 1995 to the investment of company '*B*' in 1996?

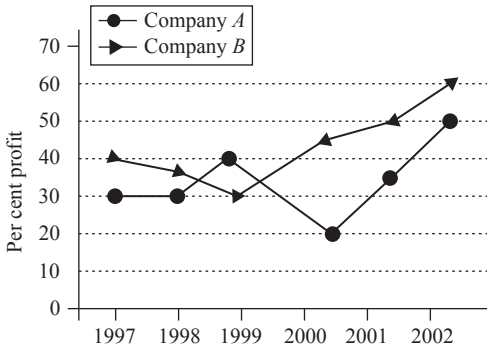
(a) 31:36 (b) 31:20
(c) 20:31 (d) Data inadequate
(e) None of these

(v) Investment of company '*B*' in 1993 was ₹1540000. What was its income in that year?

(a) ₹23.33 Lakhs (b) ₹22.33 Lakhs
(d) ₹22.23 Lakhs (d) ₹23.23 Lakhs
(e) None of these

37. Directions: Study the following graph to answer the given questions.

Per cent profit earned by two companies over the given years



(i) If the expenditure of Company *B* in 2000 was ₹200 crore, what was its income?
(a) ₹240 crore (b) ₹220 crore
(c) ₹160 crore (d) Cannot be determined
(e) None of these

(ii) If the income of Company *A* in 2002 was ₹600 crore, what was its expenditure?
(a) ₹360 crore (b) ₹480 crore
(c) ₹375 crore (d) Cannot be determined
(e) None of these

(iii) If the income of Company *B* in 1998 was ₹200 crores, what was its profit in 1999?
(a) ₹21.5 crore (b) ₹153 crore
(c) ₹46.15 crore (d) Cannot be determined
(e) None of these

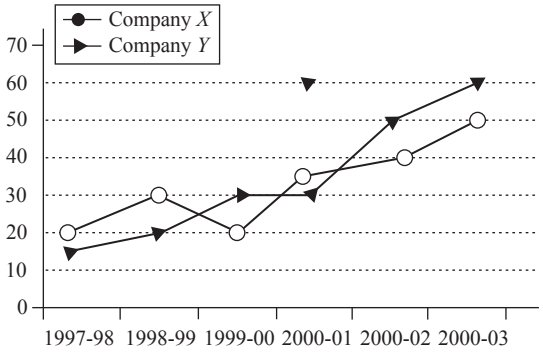
(iv) If the income of the two companies in 1998 were equal, what was the ratio of their expenditures?
(a) 1:2 (b) 26:27
(c) 100:67 (d) Cannot be determined
(e) None of these

(v) What is the percent increase in profit for company *B* from year 2000 to 2001?
(a) 75 (b) 175
(c) 42.86 (d) Cannot be determined
(e) None of these

38. Directions: Study the following graph to answer the given questions.

Per cent profit earned by two companies over the given years

$$\textbf{\% profit} = \frac{\text{Income} - \text{Expenditure}}{\text{Expenditure}} \times 100$$



(i) If the income of Company *X* in 1998–99 was equal to the expenditure of Company *Y* in 2001–2002, what was the ratio of their respective profits?
(a) 13:15 (b) 15:26
(c) 13:26 (d) Cannot be determined
(e) None of these

(ii) For Company *X,* its income in 2001–2002 was equal to its expenditure in 2002–2003, what was the ratio of its respective incomes in these two years?
(a) 4:5 (b) 3:4
(c) 2:3 (d) Cannot be determined
(e) None of these

(iii) For Company *Y*, in which year is the per cent of increase in per cent profit over that of previous year the highest?
(a) 2002–03 (b) 1999–2000
(c) 2001–02 (d) Cannot be determined
(e) None of these

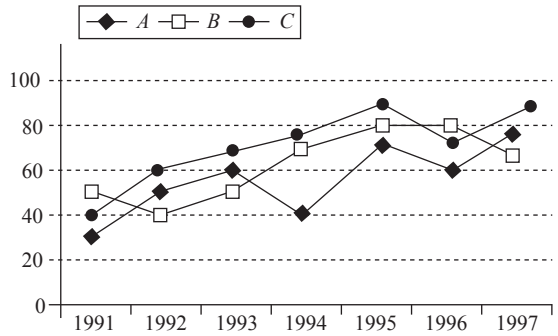
(iv) In 1997–98, the expenditure of Company *X* was ₹40 crores. What was its income in that year?
(a) ₹50 crore (b) ₹48 crore
(c) ₹46 crore (d) Cannot be determined
(e) None of these

(v) What was the difference in the expenditures of the two companies in 1999–2000?
(a) 10 (b) 100
(c) 1000 (d) Cannot be determined
(e) None of these

(vi) In 2002–03 the income of Company *Y* was ₹128 crores. What was its expenditure in that year?
(a) ₹76.8 crore (b) ₹64 crore
(c) ₹48 crore (d) Cannot be determined
(e) None of these

39. Directions: Study the following graph carefully and answer the questions given below it.

Imports of 3 companies over the years (₹ in crore)



(i) In which of the following years, the imports made by Company *A* was exactly equal to average imports made by it over the given years?

(a) 1992 (b) 1993
(c) 1994 (d) 1995
(e) None of these

(ii) In which of the following years was the difference between the imports made by Company *B* and *C* the maximum?

(a) 1995 (b) 1994
(c) 1991 (d) 1992
(e) None of these

(iii) In which of the following years was the imports made by Company *A* exactly half of the total imports made by Company *B* and *C* together in that year?

(a) 1992 only (b) 1993 only
(c) 1992 and 1993 (d) 1995 only
(e) None of these

(iv) What was the percentage increase in imports by Company *B* from 1992 to 1993?

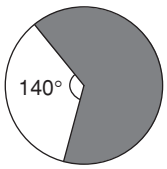
(a) 10 (b) 25
(c) 40 (d) 20
(e) None of these

(v) In which of the following years was the total imports made by all the three companies together the maximum?

(a) 1996 only (b) 1997 only
(c) 1995 only (d) 1995 and 1997 only
(e) None of these

EXERCISE-2
(BASED ON MEMORY)

1. In The figure given below, the perimeter of the circle is 220 cm. What is the area of the shaded portion in cm^2?



(a) $2542\frac{7}{9}$ (b) $2584\frac{1}{3}$

(c) $2447\frac{1}{9}$ (d) $2352\frac{7}{9}$

(e) $2376\frac{2}{3}$

[IBPS PO/MT, 2014]

Directions (Question 2 to 6): Study the table to answer the given questions.

	Percentage of people (male and female) who watch the TV Series out of the total population of the city								
	Total population of	Big Bang Theory		Arrow		Breaking Bad		Mentalist	
City	the city	Male	Female	Male	Female	Male	Female	Male	Female
P	40000	12	14	22	18	18	20	12	10
Q	20000	10	20	20	16	14	10	15	30
R	50000	18	12	10	22	16	12	16	22
S	30000	16	20	10	20	12	30	18	12
T	50000	22	30	12	14	20	12	15	20

2. What is the difference between the total number of people living in City R, Q and T together who do not watch Arrow and the total number of people living in these three cities together who watch Arrow?

(a) 47200 (b) 45300
(c) 47400 (d) 47600
(e) 45600

[IBPS PO/MT, 2014]

3. What is the average number of males who watch Big Bang Theory in all the cities together?

(a) 6320 (b) 6380
(c) 6340 (d) 6350
(e) 6360

[IBPS PO/MT, 2014]

4. The ratio of the total number of males to the total number of females in City P is 5:3. What per cent of the female population watches Breaking Bad in City P?

(a) $55\frac{1}{3}$ (b) $55\frac{2}{3}$
(c) $58\frac{1}{3}$ (d) $53\frac{1}{3}$
(e) $53\frac{2}{3}$

[IBPS PO/MT, 2014]

5. The total population (males and females) of City R watching Mentalist is what per cent more than the total population (male and female) of City T watching the same TV Series?

(a) $8\frac{3}{7}$ (b) $8\frac{5}{7}$
(c) $8\frac{4}{7}$ (d) $7\frac{3}{7}$
(e) $7\frac{4}{7}$

[IBPS PO/MT, 2014]

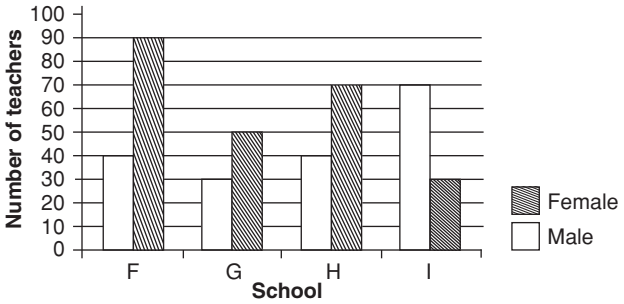
6. What is the ratio of the number of females who watch Breaking Bad in City Q and City S together to the number of females who watch Mentalist in the same cities together?

(a) 59:47 (b) 55:48
(c) 59:42 (d) 55:43
(e) 59:45

[IBPS PO/MT, 2014]

Directions (Question 7): This question is the graph below:

Number of male and female teachers in four schools

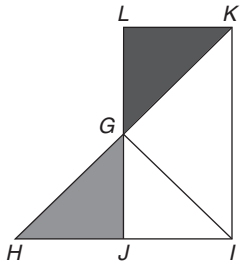


7. What is the difference between the average number ofmale and female teachers in the given schools?

(a) 10 (b 20
(c) 5 (d) 25
(e) 15

[IBPS PO/MT, 2014]

8. In the figure given below GHI is an equilateral triangle with side 14 cm. G is themidpoint of JL. What is the area of the shaded portion(in cm^2)?

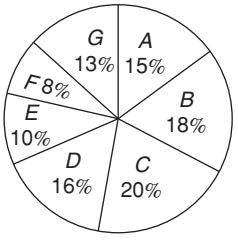


(a) $56\sqrt{3}$ (b) $70\sqrt{3}$
(c) $35\sqrt{3}$ (d) $49\sqrt{3}$
(e) $42\sqrt{3}$

Directions (Question 9 to 13): Refer to the pie-chart and answer the given questions:

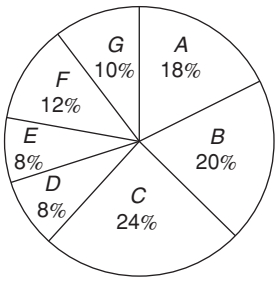
Distribution of the total number of novels (Romantic and Horror) sold by 7 stores

Total number = 63000



Distribution of the total number of Romantic novels sold by 7 stores

Total number = 36000



[IBPS PO/MT, 2014]

9. What is the ratio of the number of novels (Romantic and Horror) sold by store E to the total number of Horror novels sold by stores C and F together?

(a) 35:32 (b) 45:32
(c) 35:24 (d) 35:26
(e) 45:34

[IBPS PO/MT, 2014]

10. What is the average number of Horror novels sold by stores B, C, E and F together?

(a) 2960 (b) 3060
(c) 2680 (d) 3240
(e) 3180

[IBPS PO/MT, 2014]

11. What is the central angle corresponding to the number of novels (Romantic and Horror) sold by store B?

(a) 68.2° (b) 72.6°
(c) 62.4° (d) 64.8°
(e) 70.8°

[IBPS PO/MT, 2014]

12. The number of novels (Romantic and Horror) sold by store F is what percent less than the total number of Romantic novels sold by stores B and G together?

(a) $51\frac{2}{3}$ (b) $53\frac{1}{3}$
(c) $55\frac{2}{3}$ (d) $58\frac{1}{3}$
(e) $56\frac{1}{3}$

[IBPS PO/MT, 2014]

13. What is the difference between the total number of romantic novels sold by stores A, D and G together and the total number of Horror novels sold by the same stores together?

(a) 2000 (b) 1600
(c) 2400 (d) 1800
(e) 2200

[IBPS PO/MT, 2014]

Directions (Question 14–15): Study the following table to answer the given questions.

Number of girls studying IT and Electronics Engineering from Five colleges

College	IT	Electronics
A	240	315
B	350	285
C	260	225
D	325	255
E	275	220

14. The total number of girls studying IT Engineering from college B, C and D together is by what per cent more than the total number of girls studing Electronics Engineering from these three colleges?

(a) $22\frac{2}{9}$ (b) $23\frac{1}{9}$
(c) $22\frac{2}{3}$ (d) $23\frac{5}{9}$
(e) $23\frac{1}{3}$

[IBPS PO/MT, 2014]

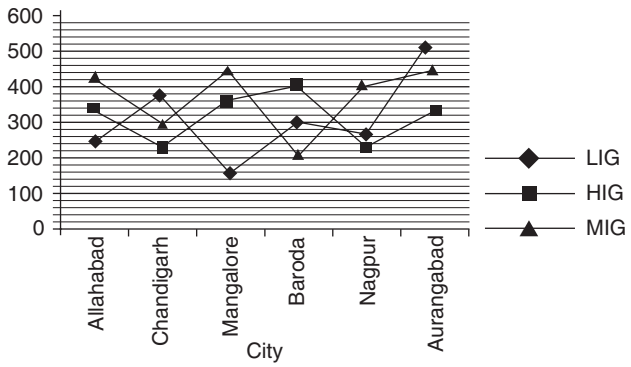
15. What per cent of the girls in college C study Electronics Engineering out of the girls studying IT and Electronics Engineering? (Rounded off to the nearest integer)

(a) 46 (b) 52
(c) 51 (d) 42
(e) 49

[IBPS PO/MT, 2014]

Directions (Question 16–20): Study the following graph carefully to answer the given questions.

Number of the flat booked in HIG, MIG and LIG categories from different cities in 2004.



16. If for Aurangabad the number of HIG flats booked in 2005 was more than that in 2004 by 15%, the number of MIG flats booked in 2005 was more than that in 2004 by 10% and the number of LIG flats booked in 2005 was more than that in 2004 by 20% then what was the total number of flats booked in Aurangabad in 2005?

(a) 1565 (b) 1521
(c) 1625 (d) 1642
(e) 1544

[IBPS PO/MT, 2014]

17. Out of the LIG flats booked from Chandigarh, 35% were by employees of a Financial Institution and out of the remaining flats, those booked by officers from a software company and HRM department of Government of India were in the ratio of 6:7. What was the total no. of LIG flats booked by officers from the software company?

(a) 130 (b) 120
(c) 160 (d) 140
(e) 150

[IBPS PO/MT, 2014]

18. The total number of MIG flats booked in Mangalore, Baroda and Nagpur is by what per cent more than the total number of LIG flats booked from these three cities together? (Rounded off to the nearest integer)

(a) 37 (b) 35
(c) 39 (d) 32
(e) 34

[IBPS PO/MT, 2014]

19. What is the difference between the total number of MIG flats booked in Allahabad, Mangalore, Nagpur and Aurangabad together and the total number of LIG flatsbooked in these four cities together?

(a) 420 (b) 480
(c) 460 (d) 360
(e) 260

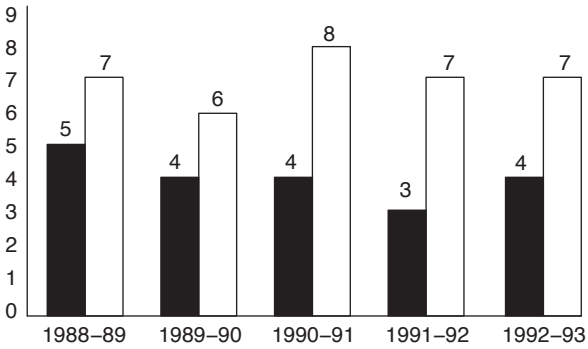
[IBPS PO/MT, 2014]

20. What is the ratio of the total number of flats (all three types) booked in Allahabad to that in Baroda?

(a) 54:49 (b) 51:46
(c) 54:47 (d) 58:49
(e) 55:48

[IBPS PO/MT, 2014]

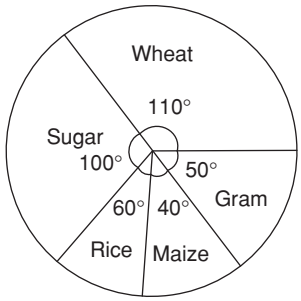
21. The average Kharif production of the given years is production of pulses in Rabi and Kharif season (in million tonnes)



(a) 4 million tonnes (b) 5 million tonnes
(c) 4.5 million tonnes (d) 5.5 million tonnes

[SSC, 2013]

Directions (Question 22): The annual agricultural production (in tonnes) of an Indian state is given in the pie chart. The total production is 9000 tonnes. Read the pie chart and answer the question.

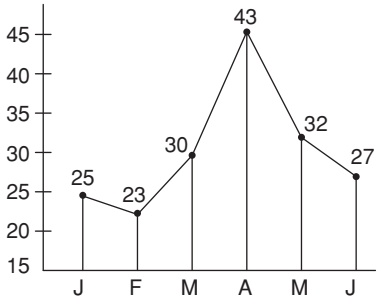


22. What is the annual production of wheat?

(a) 2750 tonnes (b) 3000 tonnes
(c) 3540 tonnes (d) 3500 tonnes

[SSC, 2013]

23. Given is a line graph showing the number of accidents in a city during the first 6 months of 1999.



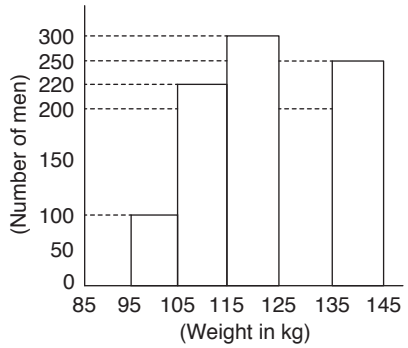
The decrease percentage of accidents from May to June is

(a) $15\frac{3}{8}\%$ (b) $15\frac{1}{8}\%$

(c) $15\frac{5}{8}\%$ (d) $15\frac{7}{8}\%$

[SSC, 2013]

Directions (Question 24): Study the histogram of weight distribution of different men and answer the question.

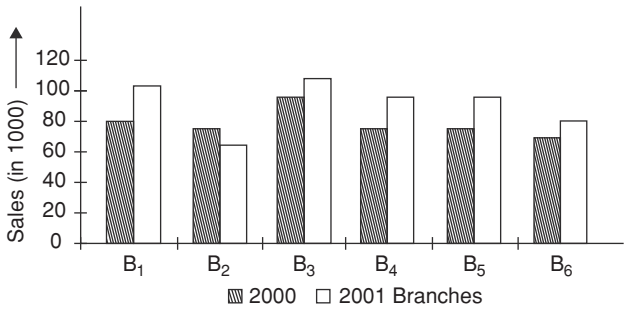


24. Average number of men per interval who participated in this survey is:

(a) 200 (b) 180
(c) 214 (d) 194

[SSC, 2013]

Directions (Question 25 to 28): Bar chart showing the sales of books (in 1000) from six branches B_1, B_2, B_3, B_4, B_5 and B_6 of a publishing company in 2000 and 2001 is given below. Study the chart and answer the questions.



[SSC Assistant Grade III, 2013]

25. Total sales of Branch B_6 for both the years is what per cent of the total sales of Branch B_3 for both the years?

(a) 71.11% (b) 73.17%
(c) 68.54% (d) 77.26%

[SSC Assistant Grade III, 2013]

26. What is the ratio of the total sales of Branch B_2 for both the years to the total sales of Branch B_4 for both the years?

(a) 2:3 (b) 3:5
(c) 5:7 (d) 7:9

[SSC Assistant Grade III, 2013]

27. What percent of the average sales of branches B_1, B_2 and B_3 in 2001 is the average sales of branches B_1, B_3 and B_6 in 2000?

(a) 107.28% (b) 104.28%
(c) 117.28% (d) 114.28%

[SSC Assistant Grade III, 2013]

28. What is the average sale of books from all the branches for the year 2000?

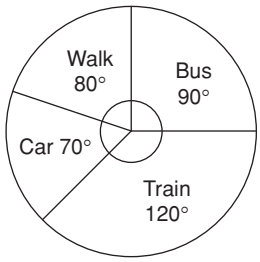
(a) 70 (b) 80
(c) 70.5 (d) 80.5

[SSC Assistant Grade III, 2013]

Directions (Question 29 to 33): The pie chart given below represents the number of students using different transport to a school in which total number of students is 2160.

Answer the questions based on the following diagram.

[SSC Assistant Grade III, 2013]



29. The total number of students who come to school by car is:

(a) 70 (b) 290
(c) 420 (d) 480

[SSC Assistant Grade III, 2013]

30. The ratio of the total number of students who come to school by car to the total number of students who come to school by bus is:

(a) 21:24 (b) 21:27
(c) 36:27 (d) 36:21

[SSC Assistant Grade III, 2013]

31. The total number of students coming to school either by walking or by bus is:

(a) 480 (b) 540
(c) 1020 (d) 170

[SSC Assistant Grade III, 2013]

32. The total number of students who don't come to school by train is:

(a) 720 (b) 1020
(c) 2040 (d) 1440

[SSC Assistant Grade III, 2013]

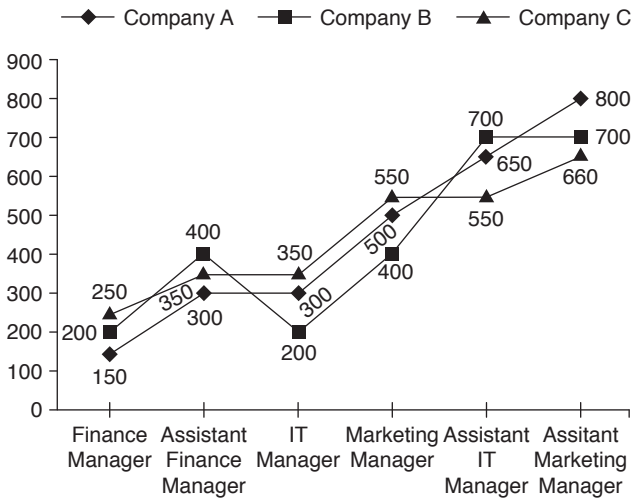
33. The total number of students coming to school by bus exceeds the total number of students coming to school by walking, by:

(a) 10% (b) 12.5%
(c) 11% (d) 11.5%

[SSC Assistant Grade III, 2013]

Directions (Question 34–38): Study the following graph carefully to answer the questions given below:

Number of selected employees in different grades/ ranks by three companies during 2012



34. What is the average number of selected employees by Company A in all grades taken together?

(a) 450 (b) 460
(c) 475 (d) 375
(e) None of these

[IBPS PO/MT, 2013]

35. What is the ratio of selected employees for the post of Marketing Managers by Companies A, B and C respectively?

(a) 8:10:11 (b) 10:8:11
(c) 11:10:8 (d) 10:11:8
(e) None of these

[IBPS PO/MT, 2013]

36. By what percent is the number of selected employees for Finance Managers by Company C more than that of the selected employees by Company B for the same post?

(a) 35% (b) 30%
(c) 25% (d) 40%
(e) None of these

[IBPS PO/MT, 2013]

37. What is the average number of selected employees for the post of Assistant Marketing Managers by all companies taken together?

(a) 570 (2) 520
(c) 620 (d) 720
(e) None of these

[IBPS PO/MT, 2013]

38. What is the ratio of selected employees for IT Managers by all Companies A, B and C?

(a) 6:4:7 (b) 5:3:7
(c) 4:7:9 (d) 8:7:6
(e) None of these

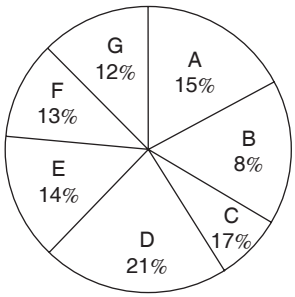
[IBPS PO/MT, 2013]

Directions (Question 39 to 43): The following questions are based on the pie-charts given below:

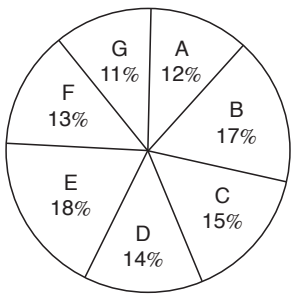
Percentage-wise distribution of students studying in Arts and Commerce in seven different institutions

Different institutions–A, B, C, D, E, F and G

Total number of students studying Arts = 3800



Total number of students studying Commerce = 4200



[IBPS PO/MT, 2013]

39. What is the total number of students studying Arts in Institutes A and G together?

(a) 1026 (b) 1126
(c) 1226 (d) 1206
(e) 1306

[IBPS PO/MT, 2013]

40. How many students from Institute B study Arts and Commerce?

(a) 1180 (b) 1108
(c) 1018 (d) 1208
(e) 1408

[IBPS PO/MT, 2013]

41. The ratio of the number of students studying Arts to that studying Commerce in Institute E is:

(a) 27:14 (b) 19:27
(c) 19:16 (d) 19:28
(e) None of these

[IBPS PO/MT, 2013]

42. The ratio of the number of students studying Arts in Institute E to that studying commerce in Institute D is:

(a) 12:17 (b) 12:7
(c) 19:21 (d) 17:19
(e) None of these

[IBPS PO/MT, 2013]

43. How many students in institutes B and D together study commerce?

(a) 1320 (b) 1302
(c) 1202 (d) 1220
(e) None of these

[IBPS PO/MT, 2013]

Directions (Question 44 to 48): Study the following table carefully to answer these questions.

Percentage of marks obtained by six students in six different subjects

Student ↓ Subject →	History (Out of 50)	Geography (Out of 50)	Maths (Out of 150)	Science (Out of 100)	English (Out of 75)	Hindi (Out of 75)
Amit	76	85	69	73	64	88
Bharat	84	80	85	78	73	92
Umesh	82	67	92	87	69	76
Nikhil	73	72	78	69	58	83
Pratiksha	68	79	64	91	66	65
Ritesh	79	87	88	93	82	72

44. What is the approximate integral percentage of marks obtained by Umesh in all the subjects?

(a) 80% (b) 84%
(c) 86% (d) 78%
(e) 77%

[IBPS PO/MT, 2013]

45. What is the average percentage of marks obtained by all students in Hindi? (Approximated to two places of decimal)

(a) 77.45% (b) 79.33%
(c) 75.52% (d) 73.52%
(e) None of these

[IBPS PO/MT, 2013]

46. What is the average mark of all the students in Mathematics?

(a) 128 (b) 112
(c) 119 (d) 138
(e) 144

[IBPS PO/MT, 2013]

47. What is the average mark obtained by all the students in Geography?

(a) 38.26 (b) 37.26
(c) 37.16 (d) 39.16
(e) None of these

[IBPS PO/MT, 2013]

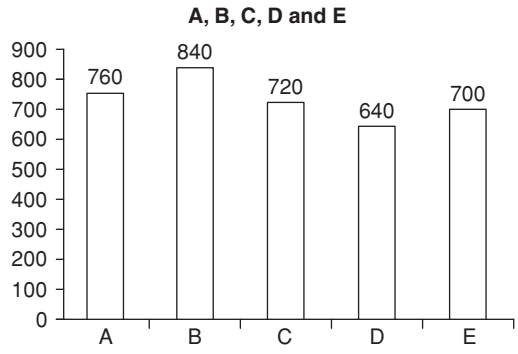
48. What is the total mark obtained by Ritesh in all the subjects taken together?

(a) 401.75 (b) 410.75
(c) 402.75 (d) 420.75
(e) None of these

[IBPS PO/MT, 2013]

Directions (Question 49 to 53): Study the following bar diagram and table carefully to answer the questions.

Number of employees working in five different companies. A,B,C,D and E



Company

Ratio of male to female employees

Company	Male:Female
A	13:6
B	4:3
C	7:8

D	9:11
E	23:12

49. What is the number of male employees, taking all the Companies together?
(a) 2084 (b) 2048
(c) 2064 (d) 2046
(e) 2066
[IBPS PO/MT, 2013]

50. What is the approximate average number of female employees, taking all the companies together?
(a) 340 (b) 315
(c) 335 (d) 325
(e) 321
[IBPS PO/MT, 2013]

51. By what percent is the number of male employees working in Company A and C more than that of female employees working in Company B and D?
(a) 164 (b) 146
(c) 144 (d) 154
(e) 184
[IBPS PO/MT, 2013]

52. What is the ratio of female employees working in Company D and E respectively?
(a) 17:22 (b) 22:17
(c) 15:22 (d) 22:15
(e) None of these
[IBPS PO/MT, 2013]

53. By what percent is the number of total employees of Company C more than that of Company D?
(a) 12.5% (b) 16.5%
(c) 21% (d) 20%
(e) 16%
[IBPS PO/MT, 2013]

Directions (Question 54 to 57): Study the following table and answer the questions.

Number of students from various schools playing various games (one student playsone game only)

	School				
Games	**A**	**B**	**C**	**D**	**E**
Cricket	150	200	250	230	200
Football	250	125	175	100	250
Basketball	200	195	245	200	225
Badminton	100	130	60	40	65
Tennis	120	180	150	130	165

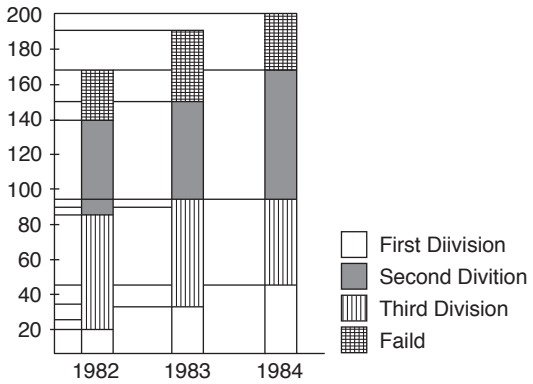
54. The difference between the total number of students playing Basketball from all the school and the total number of students playing Cricket from all the schools is:
(a) 27 (b) 35
(c) 28 (d) 26
[SSC Assistant Grade III, 2012]

55. The number of students playing Football from School C is x per cent of the total number of students playing Football from all the schools. Then$x$ equals
(a) $19\frac{7}{9}$ (b) $19\frac{4}{9}$
(c) 18 (d) $20\frac{2}{9}$
[SSC Assistant Grade III, 2012]

56. Which school has the maximum number of players?
(a) A (b) B
(c) C (d) E
[SSC Assistant Grade III, 2012]

57. The number of students playing Badminton from School E is x % of the students playing Badminton from School B. Then x equals:
(a) 40 (b) 50
(c) 42 (d) 41
[SSC Assistant Grade III, 2012]

Directions (Question 58 to 62): The following bar graph depicts the result for BSc students of a college for three years. Read the graph and answer the questions based on this graph.



58. The number of students passed in third division in 1984 was
(a) 165 (b) 75
(c) 70 (d) 65
[SSC Assistant Grade III, 2012]

59. The percentage of students failed in 1984 was

(a) $18\frac{1}{2}\%$ (b) $17\frac{3}{4}\%$

(c) $17\frac{1}{2}\%$ (d) 17%

[SSC Assistant Grade III, 2012]

60. The aggregate pass percentage during the three years was

(a) $82\frac{44}{113}\%$ (b) $82\frac{55}{113}\%$

(c) $80\frac{60}{113}\%$ (d) $77\frac{29}{113}\%$

[SSC Assistant Grade III, 2012]

61. The percentage of students passed in first division in 1982 was

(a) 20% (b) 34%

(c) $14\frac{2}{7}\%$ (d) $11\frac{13}{17}\%$

[SSC Assistant Grade III, 2012]

62. The percentage of students passed in 1982 was

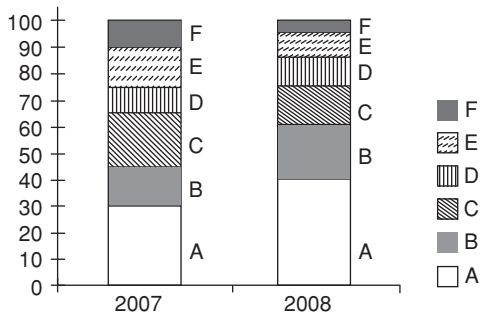
(a) 65% (b) 70%

(c) $74\frac{2}{17}\%$ (d) $82\frac{6}{17}\%$

[SSC Assistant Grade III, 2012]

Directions (Question 63 to 67): The bar chart given below shows the percentage distribution of the production of various models of a mobile manufacturing company in 2007 and 2008. The total production in 2007 was 35 Lakh mobile phones and in 2008 the production was 44 Lakhs. Study the chart and answer the following questions.

Percentage of six different types of mobiles manufactured by a company over two years



63. Total number of mobiles of models A, B and E manufactured in 2007 was

(a) 24,50,000 (b) 22,75,000

(c) 21,00,000 (d) 19,25,000

[SSC, 2012]

64. For which models was the percentage variation in production from 2007 to 2008 the maximum?

(a) B and C (b) C and D

(c) D and E (d) A and B

[SSC, 2012]

65. What was the difference in the number of B type mobiles produced in 2007 and 2008?

(a) 3,55,000 (b) 2,70,000

(c) 2,25,000 (d) 1,75,000

[SSC, 2012]

66. If the percentage production of A type mobiles in 2008 was same as that in 2007, then the number of A type mobiles produced in 2008 would have been

(a) 14,00,000 (b) 13,20,000

(c) 11,70,000 (d) 10,50,000

[SSC, 2012]

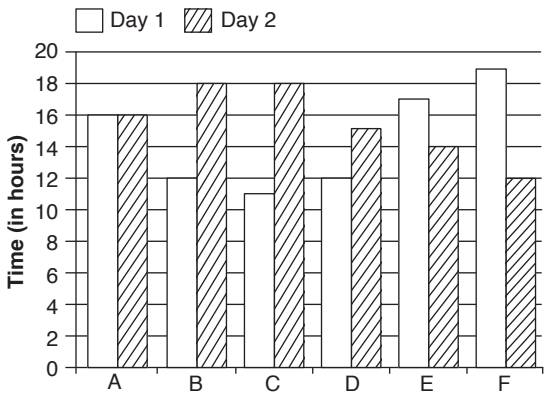
67. If 85% of the D type mobiles produced in each year were sold by the company, how many D type mobiles remained unsold?

(a) 76,500 (b) 93,500

(c) 1,18,500 (d) 1,22,500

[SSC, 2012]

Directions (Question 68 to 72): Study the following graph and table carefully and answer the questions given below:



Distance covered (in kilometres) by six vehicles on each day

[IBPS PO/MT, 2012]

Vehicle	Day 1	Day 2
A	832	864
B	516	774
C	693	810
D	552	765
E	935	546
F	703	636

68. Which of the following vehicles travelled at the same speed on both the days?

(a) Vehicle A (b) Vehicle C
(c) Vehicle F (d) Vehicle B
(e) None of these

[IBPS PO/MT, 2012]

69. What was the difference between the speed of Vehicle A on Day 1 and the speed of Vehicle C on the same day?

(a) 7 Km/h (b) 12 Km/h
(c) 11 Km/h (d) 8 Km/h
(e) None of these

[IBPS PO/MT, 2012]

70. What was the speed of Vehicle C on Day 2 in terms of metres per second?

(a) 15.3 (b) 12.8
(c) 11.5 (d) 13.8
(e) None of these

[IBPS PO/MT, 2012]

71. The distance travelled by Vehicle F on Day 2 was approximately what percent of the distance travelled by it on Day 1?

(a) 80 (b) 65
(c) 85 (d) 95
(e) 90

[IBPS PO/MT, 2012]

72. What is the ratio of the speeds of Vehicle D and Vehicle E on Day 2?

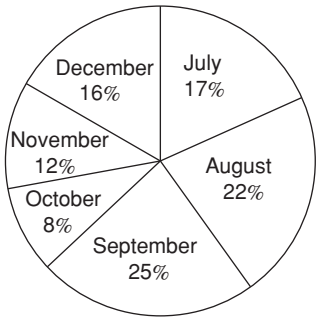
(a) 15:13 (b) 17:13
(c) 13:11 (d) 17:14
(e) None of these

[IBPS PO/MT, 2012]

Directions (Question 73 to 77): Study the following pie-chart and table carefully and answer the questions given below:

Percentage wise distribution of the number of mobile phones sold by a shopkeeper during six months

Total number of mobile phones sold = 45000



The ratio between the numbers of mobile phones sold of Company A and Company B during six months

[IBPS PO/MT, 2012]

Month	Ratio
July	8:7
August	4:5
September	3:2
October	7:5
November	7:8
December	7:9

73. What is the ratio of the number of mobile phones sold of Company B during July to those sold during December of the same company?

(a) 119:145 (b) 116:135
(c) 119:135 (d) 119:130
(e) None of these

[IBPS PO/MT, 2012]

74. If 35% of the mobile phones sold by Company A during November were sold at a discount, how many mobile phones of Company A during that month were sold without a discount?

(a) 882 (b) 1635
(c) 1638 (d) 885
(e) None of these

[IBPS PO/MT, 2012]

75. If the shopkeeper earned a profit of ₹433 one each mobile phone sold of Company B during October, what was his total profit earned on the mobile phones of that company during the same month?

(a) ₹6,49,900 (b) ₹6,45,900
(c) ₹6,49,400 (d) ₹6,49,500
(e) None of these

[IBPS PO/MT, 2012]

76. The number of mobile phones sold of Company A during July is approximately what per cent of the number of mobile phones sold of Company A during December?

(a) 110 (b) 140
(c) 150 (d) 105
(e) 130

[IBPS PO/MT, 2012]

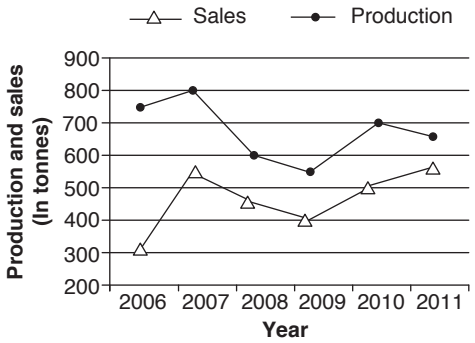
77. What is the total number of mobile phones sold of Company B during August and September together?

(a) 10000 (b) 15000
(c) 10500 (d) 9500
(e) None of these

[IBPS PO/MT, 2012]

Directions (Question 78 to 82): Study the following information and answer the questions that follow:

The graph given below represents the production (in tonnes) and sales (in tonnes) of a company from 2006–2011.



The table given below represents the ratio of the production (in tonnes) of Company A to the production (in tonnes) of Company B, and the ratio of the sales (in tonnes) of Company A to the sales (in tonnes) of Company B.

[IBPS PO/MT, 2012]

Year	Production	Sales
2006	5:4	2:3
2007	8:7	11:12
2008	3:4	9:14
2009	11:12	4:5
2010	14:13	10:9
2011	13:14	1:1

78. What is the approximate percentage increase in the production of Company A (in tonnes) from the year 2009 to the production of Company A (in tonnes) in the year 2010?

(a) 18% (b) 38%
(c) 23% (d) 27%
(e) 32%

[IBPS PO/MT, 2012]

79. The sales of Company A in the year 2009 was approximately what per cent of the production of Company A in the same year?

(a) 65% (b) 73%
(c) 79% (d) 83%
(e) 69%

[IBPS PO/MT, 2012]

80. What is the average production of Company B (in tonnes) from the year 2006 to the year 2011?

(a) 574 (b) 649
(c) 675 (d) 593
(e) 618

[IBPS PO/MT, 2012]

81. What is the ratio of the total production (in tonnes) of Company A to the total sales (in tonnes) of Company A?

(a) 81:64 (b) 64:55
(c) 71:81 (d) 71:55
(e) 81:55

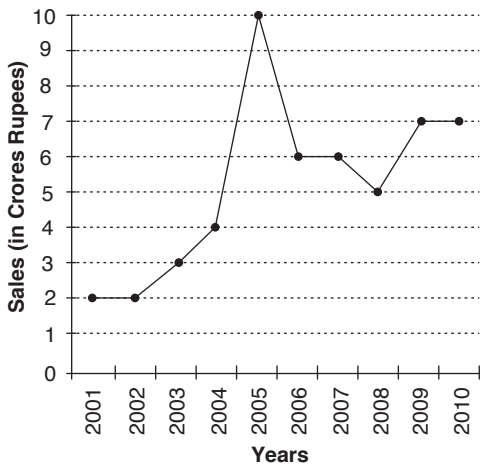
[IBPS PO/MT, 2012]

82. What is the ratio of production of Company B (in tonnes) in the year 2006 to production of Company B (in tonnes) in the year 2008?

(a) 2:5 (b) 4:5
(c) 3:4 (d) 3:5
(e) 1:4

[IBPS PO/MT, 2012]

Directions (Question. 83 to 87): The following line-diagram represents the yearly sales figures of a company in the years 2001–2010. Examine the diagram and answer the questions 81 to 85.



83. By what per cent did the sales in 2008 decrease in comparison to the sales in 2006?

(a) 20 (b) 18

(c) $16\frac{2}{3}$ (d) $15\frac{2}{3}$

[SSC, 2011]

84. The ratio of sales in 2002 to that in 2007 is:

(a) 2:3 (b) 1:3

(c) 1:1 (d) 3:2

[SSC, 2011]

85. Average sale (in crores) of the company during the period 2003–2007 is:

(a) 5.8 (b) 5

(c) 6 (d) 5.5

[SSC, 2011]

86. The percentage increase in sales in the year 2005 with respect to the previous year is:

(a) 80 (b) 100

(c) 120 (d) 150

[SSC, 2011]

87. Total sales (in crores of) from 2005 to 2008 is:

(a) 17 (b) 27

(c) 22 (d) 31

[SSC, 2011]

Directions (Question 88 to 92): The following table shows the number of students of seven colleges participating in extra-curricular activities:

Extra-Curricular Activity	Colleges						
	A	B	C	D	E	F	G
I	200	300	500	100	400	300	200
II	100	200	200	100	100	100	100
III	65	130	420	75	540	220	153
IV	317	155	438	105	385	280	120

Read the table and answer the questions given below:

88. The difference of the range of number of students in activity IV and the average is of number of students in activity III per college is:

(a) 111 (b) 153

(c) 104 (d) 217

[SSC, 2011]

89. Percentage of the number of students in activity II to that of IV is:

(a) 37 (b) 42

(c) 48 (d) 50

[SSC, 2011]

90. The median of data pertaining to activity III is:

(a) 540 (b) 229

(c) 153 (d) 75

[SSC, 2011]

91. The college in which minimum number of students participate in extra-curricular activities is:

(a) D (b) G

(c) F (d) A

[SSC, 2011]

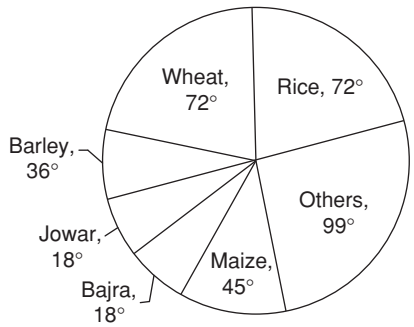
92. The ratio of total number of students in II and I is:

(a) 1:2 (b) 9:20

(c) 19:7 (d) 21:10

[SSC, 2011]

Directions (Question 93 to 97): The pie-chart provided below gives the distribution of land (in a village) under various food crops. Study the pie-chart carefully and answer the questions that follow:



93. If the total area under bajra was three hundred acres, then the total area (in hundred acres) under rice and barely together is:

(a) 18 (b) 12

(c) 15 (d) 20

[SSC, 2011]

94. The combination of three crops which contribute to more than 50% of the total area under the food crops is:

(a) Wheat, rice and maize

(b) Wheat, rice and jowar

(c) Wheat, rice and bajra

(d) Rice, barley and maize

[SSC, 2011]

95. The ratio of the land used for rice and barley is:

(a) 3:1 (b) 1:2

(c) 2:1 (d) 3:2

[SSC, 2011]

96. If 10% of the land reserved for rice be distributed to wheat and barley in the ratio 2:1, then the angle corresponding to wheat in the new pie-chart will be:

(a) 38.4° (b) 76.8°
(c) 75.6° (d) 45.5°

[SSC, 2011]

97. If the production of rice is 5 times that of jowar and the production of jowar is 2 times that of bajra, then the ratio between the yield per acre of rice and bajra is:

(a) 5:2 (b) 3:1
(c) 4:1 (d) 6:1

[SSC, 2011]

Directions (Question 98 to 102): The bar graph provided below represents the production of rice and wheat in different states of a country in a certain year. Answer the question given below based on the bar graph.



98. The total production of rice and wheat in all the mentioned states is minimum in the state _____.

(a) B (b) C
(c) D (d) E

[SSC, 2011]

99. The ratio of total production of rice in the mentioned states to that of wheat in those states, is:

(a) 15:16 (b) 12:13
(c) 13:14 (d) 14:15

[SSC, 2011]

100. The difference between the production in rice and wheat is maximum in:

(a) A only (b) All of A, B and E
(c) B and E both (d) A and B both

[SSC, 2011]

101. The state which is the largest producer of rice is:

(a) A (b) B
(c) C (d) D

[SSC, 2011]

102. The average of production of rice in the mentioned states (in lakh tonnes) is:

(a) 5.5 (b) 5.6
(c) 5.7 (d) 5.8

[SSC, 2011]

Directions (Question 103 to 107): Study the table carefully to answer the questions that follow:

The number of persons visiting six different Supermarkets and the percentage of Men, Women and Children visiting those Super markets

Names of the Super markets	Total Number of Persons	Percentage of		
		Men	Women	Children
A	34560	35	55	10
B	65900	37	43	20
C	45640	35	45	20
D	55500	41	26	33
E	42350	06	70	24
F	59650	24	62	14

103. The number of men visiting Super market D forms approximately what per cent of the total number of person visiting all the Super markets together?

(a) 11 (b) 5.5
(c) 13 (d) 9
(e) 7.5

[IBPS PO/MT, 2011]

104. The number of children visiting market C forms what per cent of the number of children visiting Supermarket F? (Rounded off two digits after decimal)

(a) 91.49 (b) 49.85
(c) 121.71 (d) 109.30
(e) None of these

[IBPS PO/MT, 2011]

105. What is the total number of children visiting Supermarket B and D together?

(a) 18515 (b) 28479
(c) 31495 (d) 22308
(e) None of these

[IBPS PO/MT, 2011]

106. What is the average of women visiting all the Supermarket together?

(a) 24823.5 (b) 22388.5
(c) 26432.5 (d) 20988.5
(e) None of these

[IBPS PO/MT, 2011]

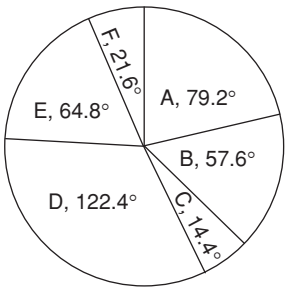
107. What is the ratio the number of women visiting Supermarket A to that of those visiting Supermarket C?

(a) 35:37 (b) 245:316
(c) 352:377 (d) 1041:1156
(e) None of these

[IBPS PO/MT, 2011]

Directions (Question 108 to 112): Study the following pie-chart and answer the questions given below:
Preferences of students of six beverages A, B, C, D, E and F in terms of degrees of angle in the pie-chart

Total No. of students = 6800



108. What is the difference between the total number of students who prefer beverage A and C together and the total number of students who prefer beverage D and F together?

(a) 959 (b) 955
(c) 952 (d) 954
(e) None of these

[IBPS PO/MT, 2011]

109. What is the ratio of the number of students who prefer beverage F to the number of students who prefer beverage A?

(a) 3:11 (b) 3:13
(c) 6:11 (d) 5:11
(e) None of these

[IBPS PO/MT, 2011]

110. The number of students who prefer beverage E and F together is what percent of the total number of students?

(a) 18 (b) 14
(c) 26 (d) 24
(e) None of these

[IBPS PO/MT, 2011]

111. The number of students who prefer beverage C is approximately what per cent of the number of students who prefer beverage D?

(a) 7 (b) 12
(c) 18 (d) 22
(e) 29

[IBPS PO/MT, 2011]

112. How many students prefer beverage B and Beverage E together?

(a) 2312 (b) 2313
(c) 2315 (d) 2318
(e) None of these

[IBPS PO/MT, 2011]

Directions (Question 113 to 117): Study the table carefully to answer the questions that follow:
Percentage of Marks Obtained by Different Students in Different Subject of MBA

	SUBJECTS (Maximum Marks)					
Students	**Strategic Management (150)**	**Brand Management (100)**	**Compensation Management (150)**	**Consumer Behaviour (125)**	**Service Markeing (75)**	**Training & Development (50)**
Anushka	66	75	88	56	56	90
Archit	82	76	84	96	92	88
Arpan	76	66	78	88	72	70
Garvita	90	88	96	76	84	86
Gunit	64	70	68	72	68	74
Pranita	48	56	50	64	64	58

113. How many marks did Anushka get in all the subject together?

(a) 369 (b) 463
(c) 558 (d) 496
(e) None of these

[IBPS PO/MT, 2011]

114. The marks obtained by Garvita in Brand Management is what per cent of the marks obtained by Archit in the same subject? (Rounded off to two digits after decimal)

(a) 86.36 (b) 101.71
(c) 115.79 (d) 133.33
(e) None of these

[IBPS PO/MT, 2011]

115. What is the average marks obtained by all students together in compensation Management?
(a) 116 (b) 120
(c) 123 (d) 131
(e) None of these

[IBPS PO/MT, 2011]

116. Who has scored the highest total marks in all the subjects together?
(a) Archit (b) Gunit
(c) Pranita (d) Garvita
(e) Arpan

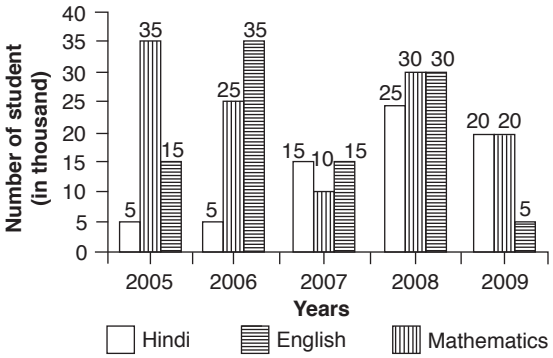
[IBPS PO/MT, 2011]

117. How many students have scored the highest marks in more than one subject?
(a) three (b) two
(c) one (d) none
(e) None of these

[IBPS PO/MT, 2011]

Directions (Question 118 to 122): Study the following graph and answer the questions that follow:

No. of students (in thousand) who opted for three different specializations during the given five years in a university



118. Out of the total number of students who opted for the given three subjects, in the year 2009, 38% were girls. How many boys opted for Mathematics in the same year?
(a) 1322 (b) 1332
(c) 1312 (d) Cannot be determined
(e) None of these

[IBPS PO/MT, 2011]

119. If the total number of student in the university in the year 2007 was 455030, the total number of students who opted for the given three subjects was approximately what per cent of the total students?
(a) 19 (b) 9
(c) 12 (d) 5
(e) 23

[IBPS PO/MT, 2011]

120. What is the total number of students who opted for Hindi and Mathematics in the years 2006, 2007 and 2009 together?
(a) 97000 (b) 93000
(c) 85000 (d) 96000
(e) None of these

[IBPS PO/MT, 2011]

121. The total number of students who opted for Mathematics in the years 2005 and 2008 together is approximately what per cent of the total number of students who opted for all three subjects in the same years?
(a) 38 (b) 28
(c) 42 (d) 32
(e) 48

[IBPS PO/MT, 2011]

122. What is the ratio of the number of students who opted for English in the years 2006 and 2008 together to the number of students who opted for Hindi in the year 2005 and 2009 together?
(a) 11:5 (b) 12:7
(c) 11:7 (d) 12:5
(e) None of these

[IBPS PO/MT, 2011]

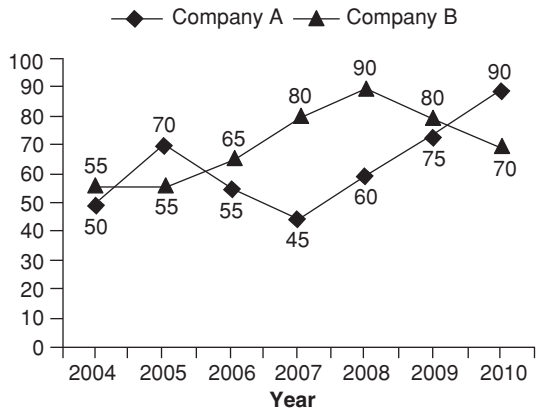
Directions (Question 123 to 130): Study the following graph carefully to answer these question.

Per cent profit earned by two companies producing electronic goods over the years

$$\% \text{ Profit} = \frac{\text{Profit Earned}}{\text{Total Investment}} \times 100$$

[SBI Associates Banks PO/MT, 2011]

Profit Earned = Total Income – Total Investment in the year



123. If the profit earned in 2006 by Company B was ₹8,12,500, what was the total income of the company in that year?
(a) ₹12,50,000 (b) ₹20,62,500
(c) ₹16,50,000 (d) ₹18,25,000
(e) None of these
[SBI Associates Banks PO, 2011]

124. If the amount invested by the two companies in 2005 was equal, what was the ratio of the total income of the Company A to that of B in 2055?
(a) 31:33 (b)33:31
(c) 34:31 (d) 14:11
(e) None of these
[SBI Associates Banks PO, 2011]

125. If the total amount invested by the two companies in 2009 was ₹27 Lakhs, while the amount invested by company B was 50% of the amount invested by Company A, what was the total profit earned by the two Companies together?
(a) ₹21.15 Lakhs (b) ₹20.70 Lakhs
(c) ₹18.70 Lakhs (d) ₹20.15 Lakhs
(e) None of these
[SBI Associates Banks PO, 2011]

126. If the income of Company A in 2007 and that in 2008 were equal and the amount invested in 2007 was ₹12 Lakhs, what was the amount invested in 2008?
(a) ₹10,87,500 (b) ₹10,85,700
(c) ₹12,45,000 (d) ₹12,85,000
(e) None of these
[SBI Associates Banks PO, 2011]

127. If the amount of profit earned by Company A in 2006 was ₹10.15 Lakhs, what was the total investment?
(a) ₹13.8 Lakhs (b) ₹14.9 Lakhs
(c) ₹15.4 Lakhs (d) ₹14.2 Lakhs
(e) None of these
[SBI Associates Banks PO, 2011]

128. If the amount invested by Company B in 2004 is ₹12 Lakhs and the income of 2004 is equal to the investment in 2005, what is the amount of profit earned in 2005 by Company B?
(a) ₹6.6 Lakhs (b) ₹18.6 Lakhs
(c) ₹10.23 Lakhs (d) ₹9.6 Lakhs
(e) None of these
[SBI Associates Banks PO, 2011]

129. If the investments of Company A in 2007 and 2008 were equal, what is the difference between the profits earned in the two years if the income in 2008 was ₹24 Lakhs?
(a) ₹2.25 Lakhs (b) ₹3.6 Lakhs
(c) ₹1.8 Lakhs (d) ₹2.6 Lakhs
(e) None of these
[SBI Associates Banks PO, 2011]

130. If each of the companies A and B invested ₹25 Lakhs in 2010, what was the average profit earned by the two companies?
(a) ₹18 Lakhs (b) ₹22.5 Lakhs
(c) ₹17.5 Lakhs (d) ₹20 Lakhs
(e) None of these
[SBI Associates Banks PO, 2011]

Directions (Question 131 to 136): Study the following table carefully and answer the questions which follow. Number of Candidates found Eligible and the Number of Candidates Short listed for Interview for a recent Recruitment Process for Six Posts form different states

Post	I		II		III		IV		V		VI	
State	E	S	E	S	E	S	E	S	E	S	E	S
A	2500	65	7200	240	5200	76	3600	200	4600	110	5400	380
B	3200	220	8500	420	8400	190	6200	320	5800	180	6200	430
C	2800	280	4500	350	7600	160	8200	440	7300	310	3700	250
D	2400	85	4800	200	2600	55	7500	350	3900	160	4800	360
E	3000	120	5600	280	3800	75	6800	280	6100	260	7800	520
F	4800	325	6400	320	4400	220	4700	180	4900	220	8800	640
G	6500	550	7000	140	6000	325	5500	220	8100	410	2700	200

E-Eligible **S**-Short listed

131. From State B, which post had the highest percentage of candidates short listed?

(a) V (b) IV
(c) VI (d) II
(e) None of these

[SBI Associates Banks PO, 2011]

132. What is the average number of candidates (approximately) found eligible for Past III form all states?

(a) 6700 (b) 6200
(c) 4200 (d) 4500
(e) 5500

[SBI Associates Banks PO, 2011]

133. What is the overall percentage (rounded off to one digit after decimal) of candidates short listed over the total number of candidates eligible for Post I form all the States together?

(a) 9.5% (b) 12.5%
(c) 7.2% (d) 6.52%
(e) None of these

[SBI Associates Banks PO, 2011]

134. What is the ratio of the total number of candidates shortlisted for all the posts together from State E to that from stateG?

(a) 307:369 (b) 73:79
(c) 6:5 (d) 9:7
(e) None of these

[SBI Associates Banks PO, 2011]

135. The total number of candidates found eligible for Post I from all states together is approximately what percent of total number of candidates found eligible for Post VI from all States together?

(a) 45% (b) 50%
(c) 60% (d) 55%
(e) 63.9%

[SBI Associates Banks PO, 2011]

136. What is the ratio of the total number of candidates short listed for post V to that for post VI from all states together?

(a) 6:7 (b) 55:96
(c) 165:278 (d) 16:25
(e) None of these

[SBI Associates Banks PO, 2011]

Directions (Question 137 to 140): These questions are based on following date. Study it carefully and answer the questions that follow.

In a school having 400 students, boys and girls are in the ratio of 3:5. The students speak Hindi, English or both the languages. 12% of the boys speak only Hindi. 22% of the girls speakonly English. 24% of the total students speak only Hindi and the number of boys speaking both the languages is six times the number of boys speaking only Hindi.

137. How many boys speak Hindi?

(a) 18 (b) 126
(c) 108 (d) 26
(e) None of these

[SBI Associates Banks PO, 2011]

138. How many students speak English?

(a) 304 (b) 79
(c) 225 (d) 117
(e) None of these

[SBI Associates Banks PO, 2011]

139. The number of girls speaking only Hindi is what per cent of the total number of students speaking only Hindi?

(a) 38.2% (b) 71.8%
(c) 31.2% (d) 78%
(e) None of these

[SBI Associates Banks PO, 2011]

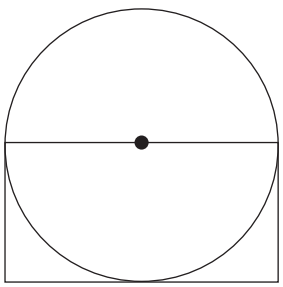
140. What is the ratio of the number of boys to the number of girls speaking both the languages?

(a) 23:25 (b) 12:25
(c) 12:13 (d) 25:13
(e) None of these

[SBI Associates Banks PO, 2011]

Directions (Question 141 to 144): Study the information given in eachof thesequestions and then answer the questions.

141. The area of the circle is 616 cm^2. What is the area of the rectangle?



(a) 784 cm^2 (b) 196 cm^2
(c) 392 cm^2 (d) Cannot be determined
(e) None of these

[SBI Associates Banks PO, 2011]

142. Population in Million

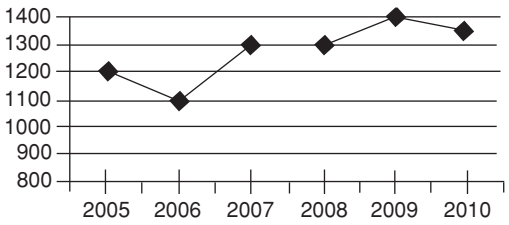
City	Total Population	Male Population
A	12	6.5
B	15	7.2
C	17	9.0
D	19	9.9
E	22	10.8

What is the average female population in million?

(a) 8.32 (b) 8.86
(c) 8.68 (d) 9.12
(e) None of these

[SBI Associates Banks PO, 2011]

143. What is the per cent rise in production in 2007 from 2006? (Round off to two digits after decimal.)

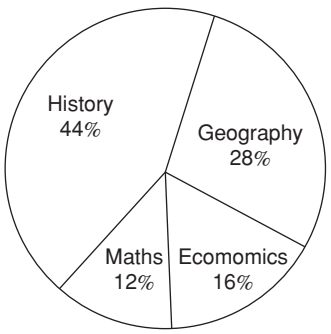


(a) 28.18% (b) 18.18%
(c) 16.28% (d) 26.18%
(e) None of these

[SBI Associates Banks PO, 2011]

144. Out of a total 550 students, how many students did not prefer Maths or Economics?

Break-up of students having preference for each subject

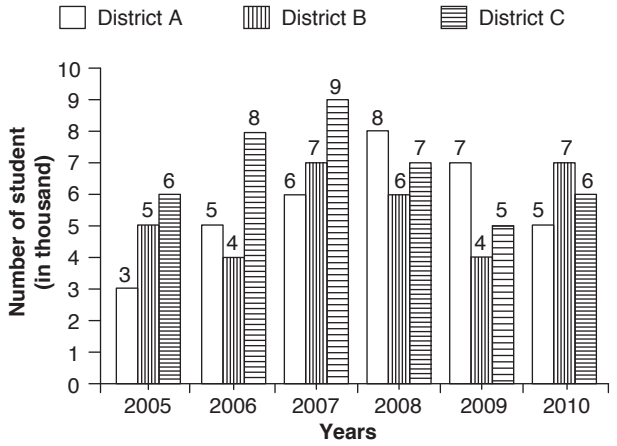


(a) 462 (b) 154
(c) 196 (d) 396
(e) None of these

[SBI Associates Banks PO, 2011]

Directions (Question 145 to 149): Study the following graph carefully to answer the questions that follow:

Number of students (in thousand) enrolled in three different districts in six different years



145. What was the percentage increase in enrolment in the number of students in District C in year 2007 as compared to that in the previous year?

(a) 115.5 (b) 112.5
(c) 15.5 (d) 12.5
(e) None of these

[IOB PO, 2011]

146. What was the difference between the number of students enrolled in all the three districts together in the year 2008 and the number of students enrolled in District Bover all the years together?

(a) 12000 (b) 11000
(c) 1100 (d) 1400
(e) None of these

[IOB PO, 2011]

147. What was the approximate average number of students enrolled in District A over all the years together?

(a) 5999 (b) 5666
(c) 5444 (d) 5333
(e) None of these

[IOB PO, 2011]

148. In which year was the number of students enrolled in all the three districts together the second highest?

(a) 2006 (b) 2007
(c) 2008 (d) 2009
(e) 2010

[IOB PO, 2011]

149. Total number of students enrolled in District A and District B together in the year 2010 was what percentage of the total number of students enrolled in District A in the year 2008?

(a) 150 (b) 120
(c) 250 (d) 220
(e) None of these

[IOB PO, 2011]

Directions (Question 150 to 154): Study the table carefully to answer the questions that follow:
Number of candidates appeared and qualified for a test (in hundred) in six different years from five different zones.

	Zone									
Year	*P*		*Q*		*R*		*S*		*T*	
	App.	*Qual.*	*App.*	*Qual.*	*App.*	*Qual.*	*App.*	*Qual.*	*App.*	*Qual.*
2005	3.2	2.5	3.5	1.4	3.8	2.2	4.2	2.4	6.2	2.6
2006	4.6	3.4	6.9	4.2	6.9	4.4	7.4	3.3	6.2	4.8
2007	6.5	4.9	7.7	4.5	5.9	4.8	8.3	5.6	6.4	4.2
2008	7.4	5.7	5.4	3.4	7.2	3.2	9.3	6.4	7.8	6.2
2009	8.8	4.8	6.6	5.2	8.6	6.8	11.4	5.2	9.9	6.9
2010	9.2	5.6	10.6	6.4	10.3	7.4	14.2	11.4	11.8	9.4

App. — Appeared; Qual. — Qualified

150. In which year was in Zone S the difference between the appeared candidates and qualified candidates the second lowest?

(a) 2005 (b) 2007
(c) 2008 (d) 2009
(e) 2010

[IOB PO, 2011]

151. The number of candidates who qualified the test from Zone R in the year 2010 was approximately what percentage of the number of candidates who appeared from Zone Q in the year 2008?

(a) 152 (b) 147
(c) 142 (d) 132
(e) 137

[IOB PO, 2011]

152. What was the average number of candidates appeared from Zone T over all the years together?

(a) 810 (b) 815
(c) 825 (d) 805
(e) 820

[IOB PO, 2011]

153. What was the ratio of the number of candidates appeared from Zone P in the year 2005 to the number of candidates qualified from Zone S in the year 2007?

(a) 4:7 (b) 4:9
(c) 9:4 (d) 8:13
(e) None of these

[IOB PO, 2011]

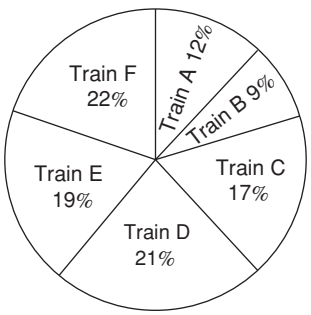
154. From which zone was the total number of candidates who qualified the test, the second highest in the year 2009 and 2010 together?

(a) P (b) Q
(c) R (d) S
(e) T

[IOB PO, 2011]

Directions (Question 155 to 159): Study the following pie-chart carefully to answer these questions:

Total number of passengers in six different trains = 4800
Percentage-wise distribution of passengers



155. What was the average number of passengers travelling in Train A, Train C and Train F together?

(a) 816 (b) 826
(c) 824 (d) 812
(e) None of these

[IOB PO, 2011]

156. If the cost of one ticket is ₹124, what is the total amount paid by passengers of Train B? (Assuming all the passengers purchased tickets and cost of each ticket is equal)

(a) ₹53,658 (b) ₹53,568
(c) ₹53,558 (d) ₹53,468
(e) None of these

[IOB PO, 2011]

157. The number of passengers in Train E is approximately what percentage of the total number of passengers in Train B and Train D together?

(a) 63 (b) 69
(c) 75 (d) 54
(e) 79

[IOB PO, 2011]

158. What is the difference between the number of passengers Train C and the number of passengers in Train A?

(a) 280 (b) 250
(c) 230 (d) 260
(e) None of these

[IOB PO, 2011]

159. What is the total number of passengers in Train D, Train E and Train F together?

(a) 2796 (b) 3225
(c) 2976 (d) 3125
(e) None of these

[IOB PO, 2011]

Directions (Question 160 to 164): Study the following table carefully to answer the questions that follow.

Semester fees (in ₹ thousand) for five different courses in six different years

	Course				
Years	*B Tech*	*M Sc*	*B Ed*	*M Phil*	*Diploma*
2005	11.5	5.8	7.5	4.7	1.8
2006	14.5	6.4	11.6	5.8	3.2
2007	20.0	10.2	13.9	8.6	4.8
2008	22.2	14.6	15.8	12.7	5.6
2009	35.8	17.7	18.5	25.1	12.5
2010	50.7	20.9	22.6	18.9	14.9

160. What was the approximate per cent increase in the semester fees of BEd course in the year 2007 as compared to the previous year?

(a) 26 (b) 30
(c) 20 (d) 16
(e) 10

[IO PO, 2011]

161. What was the average semester fee charged for MSc course over all the years together?

(a) ₹12,700 (b) ₹12,600
(c) ₹12,060 (d) ₹12070
(e) ₹13,140

[IOB PO, 2011]

162. What was the difference between the total semester fee charged for Diploma course over all the years together and the fee charged for BTech course in the year 2009?

(a) ₹8,500 (b) ₹8,000
(c) ₹6,500 (d) ₹7,000
(e) None of these

[IOB PO, 2011]

163. The semester fee charged for M Phil course in the year 2008 was approximately what percentage of the semester fee charged for MSc course in the year 2009?

(a) 67 (b) 84
(c) 80 (d) 76
(e) 72

[IOB PO, 2011]

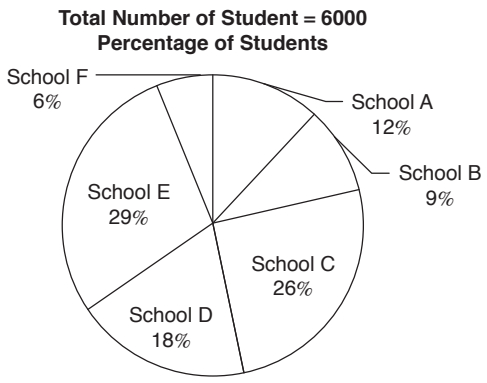
164. What was the total semester fee charged for all the courses together in the year 2006?

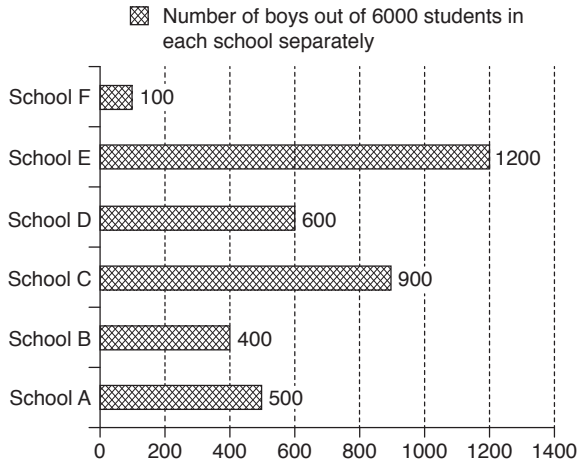
(a) ₹42,500 (b) ₹41,500
(c) ₹41,600 (d) ₹42,200
(e)None of these

[IOB PO, 2011]

Directions (Question 165 to 169): Study the following pie-chart and bar diagram and answer the following questions. Percentage-wise distribution of Students in six different Schools

Total number of Students = 6000 Percentage of students





165. What is the sum of the number of girls in School C, the number of girls in School E and the number of boys in School D together?

(a) 1700 (b) 1900
(c) 1600 (d) 1800
(e) None of these

[Allahabad Bank PO, 2011]

166. What is the ratio of the number of boys in School C, the number of girls in School B and the total number of students in School E?

(a) 45:7:97 (b) 43:9:97
(c) 45:7:87 (d) 43:9:87
(e) None of these

[Allahabad Bank PO, 2011]

167. What is the difference between the total number of students in School F and the number of boys in School E?

(a) 820 (b) 860
(c) 880 (d) 900
(e) None of these

[Allahabad Bank PO, 2011]

168. In which of the following schools is the total number of students equal to the number of girls in School E?

(a) A (b) B
(c) C (d) D
(e) F

[Allahabad Bank PO, 2011]

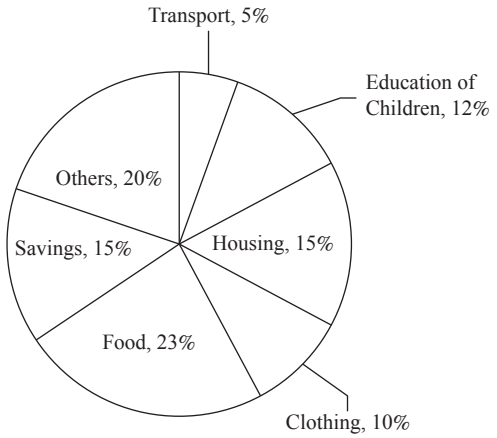
169. The number of girls in School A is approximately what percentage of the total number of students in School B?

(a) 55 (b) 50
(c) 35 (d) 45
(e) 41

[Allahabad Bank PO, 2011]

Directions (Question 170 to 174): The pie-chart given below, shows the expenditure on various items and savings of a family during the year 2009. Study the pie-chart and answer the questions based on it.

Percentage of money spent on various items and savings by a family during 2009



170. If the total income of the family for the year 2009 was ₹1,50,000 then the difference between the expenditures on housing and transport was:

(a) ₹15,000 (b) ₹10,000
(c) ₹12,000 (d) ₹7,500

[SSC, 2010]

171. Maximum expenditure of the family other than on food, was on:

(a) Housing (b) Clothing
(c) Others (d) Education of children

[SSC, 2010]

172. The savings of the family for the year were equal to the expenditure on:

(a) Food (b) Housing
(c) Education of children (d) Clothing

[SSC, 2010]

173. The percentage of the income which was spent on clothing, education of children and transport together is:

(a) 17 (b) 20
(c) 22 (d) 27

[SSC, 2010]

174. If the total income of the family was ₹1,50,000 then the money spent on food was:

(a) ₹20,000 (b) ₹23,000
(c) ₹30,000 (d) ₹34,500

[SSC, 2010]

Directions (Question 175 to 179): Study the bar diagram and answer questions based on it.

Persons killed in industrial accidents
Person killed in coal mines



175. The number of persons killed in coal mines in 2006 was what per cent of those killed in industrial accidents in that year?

(a) 4 (b) 25
(c) 36 (d) 300

[SSC, 2010]

176. In which year, minimum number of persons killed in industrial accidents and coal mines together?

(a) 2006 (b) 2007
(c) 2008 (d) 2009

[SSC, 2010]

177. In which year, maximum number of persons were killed in industrial accidents other than those killed in coal mines?

(a) 2006 (b) 2007
(c) 2008 (d) 2009

[SSC, 2010]

178. In which year, minimum number of persons were killed in coal mines other than those killed in industrial accidents?

(a) 2006 (b) 2007
(c) 2008 (d) 2009

[SSC, 2010]

179. In a year, on an average, how many persons were killed in industrial accidents and coal mines together?

(a) 121.25 (b) 1212
(c) 1212.5 (d) 1000

[SSC, 2010]

Directions (Question 180 to 184): Study the following table carefully and answer the questions given below:

Number of Tickets sold in a week of five movies in multiplexes in six different cities (Number in thousands)

Movie / City	A	B	C	D	E
Mumbai	20	15	35	26	18
Delhi	17	19	21	25	28
Kolkata	32	24	19	21	17
Chennai	18	21	32	28	34
Hyderabad	16	34	26	29	22
Lucknow	15	27	20	35	26

180. The number of tickets of Movie B sold in Hyderabad was approximately what percentage of the total numberof tickets of the same movie sold in all the cities together?

(a) 15 (b) 18
(c) 12 (d) 20
(e) 24

[Allahabad Bank PO, 2010]

181. What is the difference between the number of tickets of Movie D sold in Kolkata and the number of tickets of: Movie B sold in Lucknow?

(a) 700 (b) 7,000
(c) 14,000 (d) 9,000
(e) None of these

[Allahabad Bank PO, 2010]

182. What is the average number of tickets of Movie C soldin all the six cities?

(a) 15,500 (b) 2,550
(c) 24,000 (d) 25,500
(e) None of these

[Allahabad Bank PO, 2010]

183. The number of tickets of Movie E sold in Chennai is what percentage of the number of tickets of Movie A sold in Mumbai?

(a) 170 (b) 70
(c) 30 (d) 130
(e) None of these

[Allahabad Bank PO, 2010]

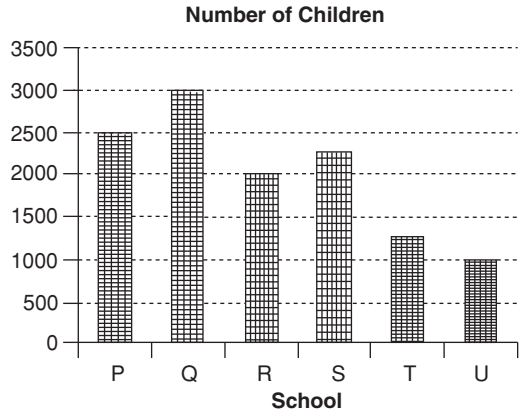
184. In which city was the total number of tickets of all the five movies together sold the minimum?

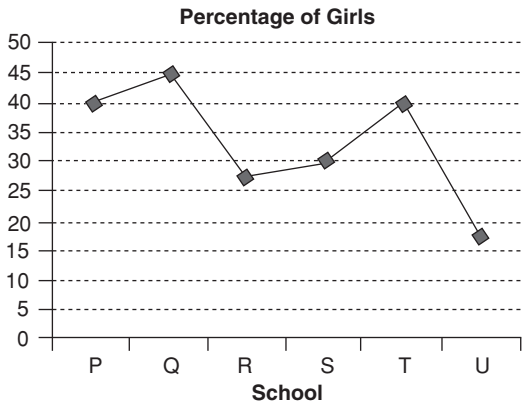
(a) Delhi (b) Chennai
(c) Lucknow (d) Kolkata
(e) None of these

[Allahabad Bank PO, 2010]

Directions (Question 185 to 189): Study the graphs carefully to answer the questions that follow:

Total number of children in 6 different schools and the percentage of girls in them





185. What is the total percentage of boys in schools R and U together? (Rounded off to two digits after decimal)
(a) 78.55 (b) 72.45
(c) 76.28 (d) 75.83
(e) None of these

[SBI Associate Banks PO, 2010]

186. What is the total number of boys in School T?
(a) 500 (b) 600
(c) 750 (d) 850
(e) None of these

[SBI Associate Banks PO, 2010]

187. The total number of students in school R is approximately what per cent of the total number of students in school S?
(a) 89 (b) 75
(c) 78 (d) 82
(e) 94

[SBI Associate Banks PO, 2010]

188. What is the average number of boys in schools P and Q together?
(a) 1425 (b) 1575
(c) 1450 (d) 1625
(e) None of these

[SBI Associate Banks PO, 2010]

189. What is the ratio of the number of girls in school P to the number of girls in school Q?
(a) 27:20 (b) 17:21
(c) 20:27 (d) 21:17
(e) None of these

[SBI Associate Banks PO, 2010]

Directions (Question 190 to 191): Study the table carefully to answer the questions that follow.
Percentage of marks obtained by six students in six different subjects

Subject→ Student↓	Hindi (Out of 175)	English (Out of 80)	Science (Out of 125)	Mathematics (Out of 100)	Social Studies (Out of 120)	Sanskrit (Out of 35)
A	87	84	91	66	39	84
B	58	68	87	74	57	79
C	63	71	81	94	44	86
D	48	57	70	79	68	44
E	83	83	49	77	55	50
F	74	68	42	63	61	58

190. What is the average marks obtained by student F in Hindi, English and Science subjects together?

(a) 78 (b) 82.4
(c) 78.8 (d) 84
(e) None of these

[Indian Bank PO, 2010]

191. What is the average marks obtained by all the students in Science?

(a) 87 (b) 86.5
(c) 90 (d) 87.5
(e) None of these

[Indian Bank PO, 2010]

Directions (Question 192 to 196): Study the following table carefully to answer the questions that follow:

Production of Sugar (in tonnes) of three different States over the years

Year→ State↓	2003	2004	2005	2006	2007	2008
P	4.3	4.9	5.6	5.8	6.7	7.4
Q	3.1	3.7	4.4	5.1	6.0	6.2
R	3.9	4.7	5.8	6.6	7.3	8.3

192. What is the approximate percentage increase in production of sugar in state Q from 2006 to 2007?

(a) 12 (b) 18
(c) 24 (d) 10
(e) 21

[Indian Bank PO, 2010]

193. What is the average production of sugar of all the three states in 2003 and 2004 together?

(a) 4.1 tonnes (b) 4.7 tonnes
(c) 5.1 tonnes (d) 4.8 tonnes
(e) None of these

[Indian Bank PO, 2010]

194. What is the ratio of the total production of sugar of all three states in the year 2006 to that in 2007?

(a) 7:9 (b) 6:7
(c) 8:7 (d) 7:8
(e) 11:12

[Indian Bank PO, 2010]

195. What is the average production of sugar of state R for all the years together?

(a) 24 tonnes (b) 6.3 tonnes
(c) 7.1 tonnes (d) 6.1 tonnes
(d) None of these

[Indian Bank PO, 2010]

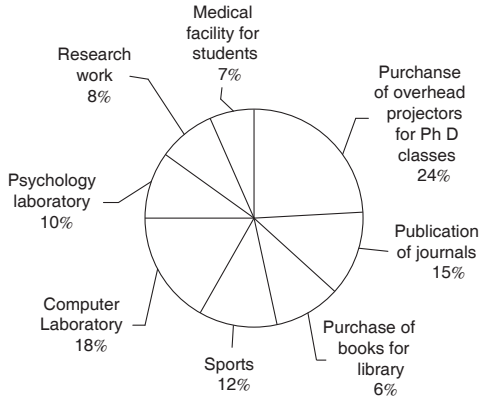
196. What is the difference between the total production of sugar of all the three states together in 2008 and that in 2005?

(a) 9 tonnes (b) 4.3 tonnes
(c) 6.1 tonnes (d) 5.1 tonnes
(e) None of these

[Indian Bank PO, 2010]

Directions (Question 197 to 200): Study the following pie-chart to answer these questions.

TOTAL EXPENDITURE: ₹60 Lakhs



197. What is the ratio of the expenditure made by the university on Research work and that on purchase of books for library?

(a) 4:5 (b) 5:4
(c) 8:3 (d) 8:5
(e) None of these

[Indian Bank PO, 2010]

198. What is the total sum of expenditure on Research work, Purchase of overhead projectors for PhD classes and Purchase of books for library together?

(a) ₹22.6 Lakhs (b) ₹22.8 Lakhs
(c) ₹23.4 Lakhs (d) ₹20.8 Lakhs
(e) None of these

[Indian Bank PO, 2010]

199. What is the difference between the expenditure made by the university for Publication of journals and for Psychology laboratory?

(a) ₹4 Lakhs (b) ₹3 Lakhs
(c) ₹4.2 Lakhs (d) ₹3. 8 Lakhs
(e) None of these

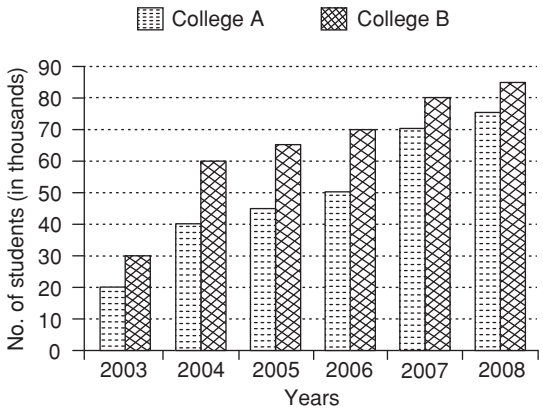
[Indian Bank PO, 2010]

200. If the expenditure on the Purchase of overhead projectors for PhD students is decreased by 7%, what will be the expenditure on the same after the decrease?

(a) ₹1,33,920 (b) ₹13,39,200
(c) ₹1,02,000 (d) ₹1,08,000
(e) None of these

[Indian Bank PO, 2010]

Directions (Question 201 to 205): Study the following graph carefully to answer these questions.
No. of students in College A and College B over the years



201. For which college(s) and in which year was the per cent rise in number of students from the previous year the highest?
(a) College A in year 2004 and College B in year 2005
(b) Only College B in year 2004
(c) College A in year 2004 and College B in year 2004
(d) College A in year 2007 and College B in year 2004
(e) None of these

[Indian Bank PO, 2010]

202. What is the ratio of the total number of students of College A in years 2004, 2006 and 2007 together and the total number of students of College B in years 2003, 2004 and 2008?
(a) 35:32 (b) 33.37
(c) 34:31 (d) 32:35
(e) None of these

[Indian Bank PO, 2010]

203. What is the average number of students in College A for all the years together?
(a) 45,000 (b) 50,000
(c) 52,000 (d) 48,000
(e) None of these

[Indian Bank PO, 2010]

204. What is the approximate percentage rise in the number of students of College B from 2005 to 2006?
(a) 8 (b) 12
(c) 4 (d) 15
(e) 20

[Indian Bank PO, 2010]

205. The number of students of College B in year 2008 is what per cent of the total students of College B in all the years together? (Round off to two digits after decimal)
(a) 20.61 (b) 23.79
(c) 21.79 (d) 17.29
(e) None of these

[Indian Bank PO, 2010]

RECENT YEARS' QUESTIONS

Direction (1–5): The table given below shows the number of applicants who have applied for exam at various centres as percentage of total number of applicants. The table also shows the number online applicants and absent applicants as a percentage of total applicants of each centre. Toatl number of applicants is 1200000.

Exam centre	Total applicant	Online applicant	Absent applicant
F	15%	30%	36%
G	25%	44%	25%
H	20%	52%	32%
J	24%	46%	18%
K	16%	38%	20%

1. If A equals to 15% of total applicants who are present at exam centre F and B equals to presents applicant at exam centre K, then A is what percent of B?
(a) 18.18 (b) 11.25
(c) 13.33 (d) 14.28

[SSC CGL Tier-II CBE, 2018]

2. Toatal number of offline applicants from exam centre H, K and F are how much less than the total number of present applicants from exam centre G and J?
(a) 111420 (b) 100920
(c) 127370 (d) 109990

[SSC CGL Tier-II CBE, 2018]

3. What are the total number of offline applicants from the exam centre F, H, J and G?

(a) 393720 (b) 963000
(c) 564720 (d) 428540

[SSC CGL Tier-II CBE, 2018]

4. What is the ratio of total number of present applicants from exam centre K to total number of offline applicants from exam centre J?

(a) 40:41 (b) 80:81
(c) 10:9 (d) 7:11

[SSC CGL Tier-II CBE, 2018]

5. What are the total number of present applicants from exam centre H and G together?

(a) 238200 (b) 151800
(c) 388200 (d) 442650

[SSC CGL Tier-II CBE, 2018]

6. Refer the following data table and answer the following question.

Day of the week	Distance jogged (in kms)
Monday	3
Tuesday	2
Wednesday	2.5
Thursday	5
Friday	1
Saturday	2.5
Sunday	4

If 400 calories are burnt by jogging 5 km how many calories were burnt in the given week?

(a) 1650 calories (b) 1550 calories
(c) 1500 calories (d) 1600 calories

[SSC CHSL (10+2) Tier-I CBE, 2017]

7. Refer the following data table and answer the following question.

Items Raw	Yearly Expense in ₹Lakhs
Materias	11
Labour	7
Rent	5
Interest	3
Taxes	3

Expenditure on raw materials and taxes is what percent of total expenses?

(a) 55.53 percent (b) 41.03 percent
(c) 33.78 percent (d) 48.28 percent

[SSC CHSL (10+2) Tier-I CBE, 2017]

8. Refer the following data table and answer the following question.

	Quantity of stock	Average Cost (₹)
Mobiles Phones	44	12000
Cameras	75	14000
TVs	55	56000
Refrigerators	29	53000
ACs	77	26000

What is the value of the total stock (In lakh rupees)?

(a) 81.97 (b) 819.7
(c) 161 (d) 280

[SSC CGL Tier-I CBE, 2017]

9. Refer the following data table and answer the question.

Year	Ratio:Import/Export
2011	1
2012	1.1
2013	1.5
2014	0.9
2015	1.1

If the imports in 2012 were ₹1000 crores and the total exports in the years 2018 and 2013 together were ₹4800 crores, them the imports in 2013 were (in ₹crore)?

(a) 3891 (b) 5836
(c) 909 (d) 2594

[SSC CGL (10+2) Tier-I CBE, 2017]

Direction (10-13): The table given below shows the production(In 000 tonnes) of five companies from 2012 to 2016.

[SSC CAPFs ASI & Delhi Police SI, 2017]

	Years				
Company	**2012**	**2013**	**2014**	**2015**	**2016**
P	302	386	412	503	470
Q	376	402	444	529	501
R	250	268	302	298	276
S	350	360	372	398	362
T	403	450	504	597	602

10. What is the total production (In 000 tonnes) of company R from year 2012 to 2016?

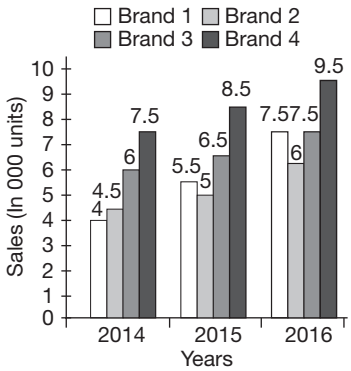
(a) 1276 (b) 1444
(c) 1394 (d) 1502
[SSC CAPFs ASI & Delhi Police SI, 2017]

11. What is the average (In 000 tonnes) of total production by all companies in the year 2013?
(a) 371.1 (b) 373.2
(c) 378.4 (d) 362.3
[SSC CAPFs ASI & Delhi Police SI, 2017]

12. The production of company R in year 2014 is how much per cent more than the production of company R in the year 2013?
(a) 12.68 (b) 11.56
(c) 14.42 (d) 15.31
[SSC CAPFs ASI & Delhi Police SI, 2017]

13. The production of all companies in the year 2014 is how much per cent less than the production off all companies in the year 2015?
(a) 8.25 (b) 11.3
(c) 14.4 (d) 12.5
[SSC CAPFs ASI & Delhi Police SI, 2017]

Directions (14–17): The bar chart given below shows the sales (In 000 units of 4 mobile brands for 3 years.



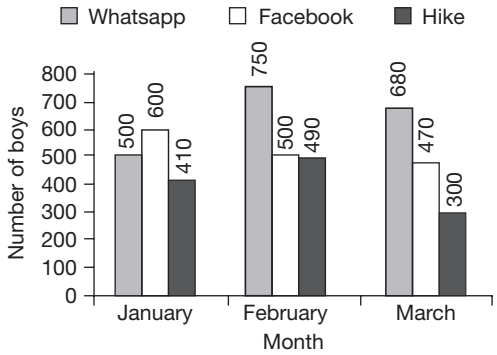
14. What is the percentage increase in the number of mobile phones of Brand 2 sold from 2014 to 2015?
(a) 8.33 (b) 33.33
(c) 37.5 (d) 11.11
[SSC CGL Tier-I CBE, 2017]

15. What is the percentage increase in the total number of mobiles sold by these four brands from 2014 to 2016?
(a) 42.16 (b) 38.63
(c) 32.43 (d) 30.16
[SSC CGL Tier-I CBE, 2017]

16. In 2017 the sales of each brand increased by the same percentage as it did in the year 2016. What will be the approximate average sales (in units) of mobiles per brand in year 2017?
(a) 9175 (b) 8360
(c) 9436 (d) 9678
[SSC CGL Tier-I CBE, 2017]

17. If for any year, the sales of a brand is more than average sales of these four brands in that year, then it gets a star. Which brand has the maximum stars?
(a) Brand 3 (b) Brand 3 and 4 both
(c) Brand 4 (d) All brands
[SSC CGL Tier-I CBE, 2017]

Directions (18–21): The bar graph given below represents the number of boys in a school using three apps for three months.
[SSC CAPFs ASI & Delhi St, 2017]



18. What is the total number of boys using the three apps in the month of March?
(a) 1420 (b) 1480
(c) 1450 (d) 1500
[SSC CAPFs ASI & Delhi St, 2017]

19. The number of boys using whatsapp in February is how much per cent more than the number of boys using Face book in February?
(a) 50 (b) 33.33
(c) 66.66 (d) 44.22
[SSC CAPFs ASI & Delhi St, 2017]

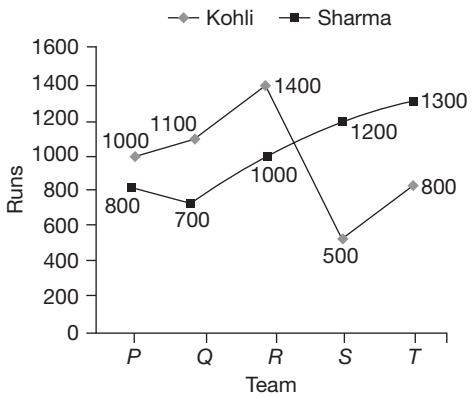
20. What is the percentage increase in the number of boys using Facebook from January to Febrauary?
(a) 26.66 (b) 20
(c) 21.33 (d) 16.66
[SSC CAPFs ASI & Delhi St, 2017]

21. The number of boys using whatsapp in March is what per cent of the number of boys using Facebook in Febuary?

(a) 126.47 (b) 136
(c) 128 (d) 131.3

[SSC CAPFs ASI & Delhi St, 2017]

Directions (22–25): The line chart below represents the runs scored by Kohli and Sharma against 5 teams.



22. What is the total runs scored by Kohli against 5 teams?

(a) 5000 (b) 4800
(c) 4700 (d) 4500

[SSC CAPFs ASI & Delhi Police SI, 2017]

23. What is the difference between the total runs scored by Kohli against the teams Q and R and total runs scored by Sharma against the teams Q and R?

(a) 200 (b) 400
(c) 900 (d) 800

[SSC CAPFs ASI & Delhi Police SI, 2017]

24. Runs scored by Kohli against team S is how much percent less than the runs scored by Sharma against team S?

(a) 70.21 (b) 68.33
(c) 58.33 (d) 41.66

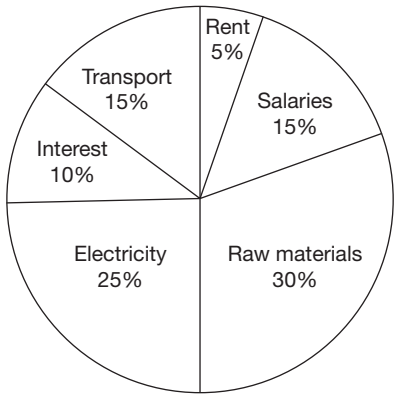
[SSC CAPFs ASI & Delhi Police SI, 2017]

25. Total runs scored by Sharma against 5 teams is what per cent total runs scored by Kohli against 5 teams?

(a) 104.16 (b) 96
(c) 98.13 (d) 108.24

[SSC CAPFs ASI & Delhi Police SI, 2017]

Direction (26–29): The pie-chart shows breakup in percentage of the various expenses of a Company. Study the diagram and answer the following questions.



26. Which is the second biggest expense of the company?

(a) Raw materials (b) Salaries
(c) Transport (d) Electricity

[SSC CGL Tier-I CBE (Exam), 2017]

27. The ratio of company's expenditure on raw material and transport to salaries is:

(a) 2 : 1 (b) 1 : 1
(c) 1 : 2 (d) 3 : 1

[SSC CGL Tier-I CBE (Exam), 2017]

28. The company's expenditure on interest is greater than expenditure on rent by:

(a) 100% (b) 50%
(c) 200% (d) 150%

[SSC CGL Tier-I CBE (Exam), 2017]

29. If the total expenses of the company are ₹50 crores, the total expenditure (In ₹ crores) on transport and electricity is :

(a) 7.5 (b) 12.5
(c) 20 (3) 10

[SSC CGL Tier-I CBE (Exam), 2017]

Directions (30–33): The pie-chart given below shows the percentage distribution of annual expenditure on various items of a company. The annual expenditure of the company is ₹72 crores.